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Sociedad Mexicana de Irradiación y Dosimetría

UFMG
UNIVERSIDADE FEDERAL DE MINAS GERAIS

XXV International Symposium on Solid State Dosimetry

Local Venue: Centro de Desenvolvimento da Tecnologia Nuclear (CDTN)
September 22-26, 2025 — Belo Horizonte, Brazil

Book of Proceedings: Volume 2

Editors
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Lucas F. Araújo
Ester M. R. Andrade

XXV ISSSD

BOOK OF PROCEEDINGS

Volume 2

XXV International Symposium on Solid State Dosimetry

Edited by

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XXV ISSSD

XXV International Symposium on Solid State Dosimetry

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CHAPTER 1

Preface

It is with great pleasure that we welcome all participants to the XXV International Symposium on Solid State Dosimetry (ISSSD) in September of 2025, for the first time, at Centro de Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN) and Universidade Federal de Minas Gerais (UFMG) in Belo Horizonte - Minas Gerais - Brazil. This milestone edition marks over two decades of continuous commitment to the advancement of radiation dosimetry through solid-state materials, techniques, and applications.

Since its inception, the ISSSD has served as a vital platform for scientists, engineers, and professionals to share knowledge, discuss new developments, and foster collaborations in the field of radiation detection and measurement. This 25^{th} edition continues this proud tradition, bringing together leading experts and emerging researchers from across the globe to present their latest findings and insights.

The program of this symposium reflects the diversity and depth of the field, covering topics ranging from fundamental studies of dosimetric materials to applied research in medical physics, environmental monitoring, and radiation protection. Special attention has been given to emerging technologies and interdisciplinary approaches that are shaping the future of dosimetry.

We are especially grateful to all contributors, authors, reviewers, session chairs, and members of the organizing, scientific, and support committees, whose dedication and effort have made this event possible. We also extend our sincere appreciation to our institutional partners and sponsors for their continued support.

We hope that this symposium will not only be an opportunity for scientific enrichment but also a space for meaningful exchanges, new collaborations, and the strengthening of our global community.

Welcome to the XXV ISSSD, and may this be a rewarding and inspiring experience for all.

Organizing Committee
XXV International Symposium on Solid State Dosimetry

Telma C. F. Fonseca
Chair at XXV ISSSD

Bruno M. Mendes
Chair at XXV ISSSD

Juan Azorin Nieto
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CHAPTER 2

Summary and Perspectives

The XXV International Symposium on Solid State Dosimetry (ISSSD-2025) brought together researchers, professionals, and students from across the globe, reinforcing the vitality and relevance of solid state dosimetry in advancing radiation science and applications.

The 250 submitted works for the event were distributed across eight thematic areas, reflecting a wide range of research interests. Dosimetry (G2) had the highest representation with 90 works (36% of total), followed by Medical Physics (G4) with 36 contributions (14.4%) and Applications of Thermoluminescence (G1) with 35 works (14%). Other areas included Luminescent Materials (G8, 24 works, 9.6%), Radiation Protection (G5, 21 works, 8.4%), Radiobiology (G7, 19 works, 7.6%), Radiation Sources (G6, 15 works, 6%), and Ionizing and Non-Ionizing Radiation (G3, 10 works, 4%).

Regarding the presentation format, a total of 99 oral communications were delivered, with 54 in person and 45 online, while 151 poster presentations were held, including 96 in person and 55 online. Dosimetry (G2) again showed strong engagement with 23 in-person oral presentations, 10 online oral presentations, 38 posters in person, and 19 online posters. Applications of Thermoluminescence (G1) accounted for 8 oral in person, 7 online, 12 posters in person, and 8 online, while Medical Physics (G4) registered 7 oral in person, 9 online, 11 posters in person, and 9 online. Other areas contributed more modest numbers, such as Luminescent Materials (G8) with 19 works, mainly presented as posters, and Radiobiology (G7) and Radiation Protection (G5) with 19 and 21 contributions, respectively, shared between oral and poster sessions. Radiation Sources (G6) and Ionizing and Non-Ionizing Radiation (G3) had smaller representations but maintained coverage of fundamental aspects of the field. Overall, the data demonstrate diverse scientific participation, with a healthy balance between oral and poster formats, and a substantial proportion of online presentations, highlighting the hybrid and inclusive nature of the event.

Reflecting the truly international character of the symposium, participants represented 18 countries, totaling 248 presentations, ensuring broad participation and accessibility despite geographical distances. An analysis of the data reveals a notable predominance from Brazil, with 175 presentations. Mexico followed with a significant participation of 37 presentations. Other Latin American countries also had a notable presence, including Colombia (7), Chile (4), Peru (4), and Argentina (3). The international representation was broadened by Saudi Arabia and Kazakhstan, both with 3 presentations, and Indonesia with 2. Finally, contributing to the event's diversity, Bolivia, Canada, Ecuador, Spain, Guatemala, Malaysia, the United Kingdom, Switzerland, and Uruguay each participated with a single presentation.

CHAPTER 3
Organization, Committees, and Reviewers

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CHAPTER 4

Event Organization, Supporting Institutions, and Sponsors

4.1 ORGANIZATION

4.2 SUPPORTING INSTITUTIONS

4.3 SPONSORS

CHAPTER 5

Schedule

XXV International Symposium on Solid State Dosimetry

22-26 SEPTEMBER 2025 - BELO HORIZONTE, BRAZIL

General Schedule

Pre-Symposium - Monday, 22 Sep

Hour: 8h, 9h, 9h30, 10h, 10h30, 11h, 11h30, 12h, 13h, 13h30, 14h, 14h30, 15h, 15h30, 16h, 16h30, 17h

	Spectrum	Lattice
Registration (8h) ***COFFE BREAK*** (10h30)	Development of PET Radiopharmaceuticals - **Dr. Marina Bicalho Silveira** **Dr. Isabel Dregely** **Dr. Emerson Bernardes Soares** (Spectrum)	Introduction to Radiation Detectors - **Dr. Marco A Polo Labarrios** (Lattice)
LUNCH (12h–13h)	***LUNCH***	***LUNCH***
COFFE BREAK (15h)	Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico - **Dr. Silvia S Hidalgo Tobón** (Spectrum)	Solid State Dosimetry: Principles and Applications & OSL Advanced Materials - **Dr. Ivonne Berenice Lozano Rojas** **Dr. Alida Tamar-Gabor** (Lattice)

Tuesday, 23 Sep

Hour: 8h, 8h40, 9h, 9h20, 10h, 10h30, 11h, 11h30, 12h, 12h30, 13h30, 14h, 14h30, 15h, 15h30, 16h, 16h30, 17h, 17h30, 18h00, 20h00

- ***Registration***
- **Opening Ceremony (Lumina)**
- ***COFFE BREAK***
- *Invited Speaker:* Wilson Calvo (Lumina)
- *Invited Speaker:* Yvone Maria Mascarenhas (Lumina)
- ***LUNCH***
- Oral Sessions (Lumina and Spectrum) | *Poster Session: **14h30 to 16h30** - Referees for evaluation (Hall)* | Roundtable: Radon in Environment: Tracer Applications and Public Health Risks (Lattice); Roundtable: AI's challenges and Opportunities: Connecting research, industry and society (Lumina)
- ***COFFE BREAK***
- *Poster Session: **14h30 to 16h30** - Referees for evaluation (Hall)* | *Invited Speaker:* Silvia Tobón (Lumina)
- Welcome & Honors Night (Lumina)

Wednesday, 24 Sep

- Oral Sessions (Spectrum and Lattice) | *Poster Session: **10h00 to 12h30** - Referees for evaluation (Hall, Virtual Room)*
- ***COFFE BREAK***
- Oral Sessions (Spectrum and Lattice) | *Poster Session: **10h00 to 12h30** - Referees for evaluation (Hall, Virtual Room)*
- ***LUNCH***
- ***Technical Visit*** | Oral Sessions (Virtual Room)

Thursday, 25 Sep

- *Invited Speaker:* Luciana Carvalheira (Lumina)
- *Invited Speaker:* Bernardo Maranhão Dantas Maria Jose Neves (Lumina)
- ***COFFE BREAK***
- *Invited Speaker:* Ademar Lugão (Lumina)
- *Invited Speaker:* Richard Hugtenburg (Lumina)
- ***LUNCH***
- Invited Speaker: Helio Yoriyaz (Lumina); *Invited Speaker:* Georgia Joana (Lumina) | *Poster Session: **14h30 to 17h** - Referees for evaluation - (Hall)* | Oral Sessions - (Spectrum and Lattice)
- ***COFFE BREAK***
- Helen Khoury Mentorship WIN-Brasil (Spectrum) | *Poster Session: **14h30 to 17h** - Referees for evaluation - (Hall)* | Roundtable: Leadership in Scientific and Technological Research (Lumina)
- Oral Sessions (Lumina and Lattice)

Friday, 26 Sep

- *Invited Speaker:* Rolf Behrens (Lumina)
- *Invited Speaker:* Susana Lalic (Lumina)
- *Invited Speaker:* Thiago Viana Miranda Lima (Lumina)
- ***COFFE BREAK***
- Oral Sessions (Lumina and Spectrum)
- *Invited Speaker:* Tarcísio Campos and Alberto Campos (Lumina)
- ***LUNCH***
- Oral Sessions (Lumina, Spectrum and Lattice)
- *Invited Speaker:* Carmen Bueno (Lumina)
- ***COFFE BREAK***
- **Closing Session:** Honor award & Helen Khoury award for best works (Lumina)

Monday, 09/22/2025

08:00 *Registration*

	Courses Pre-Symposium	
	Spectrum Auditorium **Chair: Marina Bicalho**	**Lattice Auditorium** **Chair: Álvaro M. L. Gómez**
10:30 COFFE BREAK	**08:00** Dr. Marina Bicalho (CDTN/CNEN-Brazil), Dr. Isabel Dregely (Fachhochschule Technikum Wien-Austria) and Dr. Emerson Bernardes (IPEN/CNEN-Brazil) *Development of PET Radiopharmaceuticals*	**08:00** Dr. Marco Antonio Polo Labarrios (UAM-Mexico) and Dr. Álvaro M. L. Gómez (DEN/UFMG-Brazil) *Introduction to Radiation Detectors*

12:00 LUNCH

	Courses Pre-Symposium	
	Spectrum Auditorium **Chair: Angel Ramirez**	**Lattice Auditorium** **Chair: Carlos E. Velasquez**
15:00 COFFE BREAK	**13:00** Dr. Silvia S. Hidalgo Tobón (UAM/Mexico) *Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico*	**13:00** Dr. Ivonne Berenice Lozano Rojas (CICATA-Mexico) *Solid State Dosimetry: Principles and Applications & OSL Advanced Materials*

Tuesday, 09/23/2025

08:00 *Registration*

Opening Cerimony - Lumina Auditorium

09:00 Opening Ceremony – **Amenônia Maria Ferreira Pinto** – Centro do Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN), **Bruno Melo Mendes (ISSSD 2025 – Chair)**, **Henrique Resende Martins** – Vice-director of Escola de Engenharia of Universidade Federal de Minas Gerais (UFMG), **Silvia S. Hidalgo Tobón (ISSSD 2025 – Chair)** – Universidad Autónoma Metropolitana (UAM), **Telma Cristina Ferreira Fonseca (ISSSD 2025 – Chair)** – Universidade Federal de Minas Gerais (UFMG), **Wilson Aparecido Parejo Calvo** – Comissão Nacional de Energia Nuclear (CNEN).

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium **Chair: Silvia Hidalgo**

10:30 *Invited Speaker* – Wilson Aparecido Parejo Calvo (CNEN-Brazil),
Current Status of Strategic Projects at CNEN.

11:20 *Invited Speaker* – Yvone Maria Mascarenhas (SAPRA Landauer-Brazil),
Personal Dosimetry: The Importance of Regulation, Technology, and Culture. Evaluation of Dosimetry Services in Latin America and the Caribbean.

12:30 LUNCH

Roundtable Session (Lattice Auditorium)

13:30 Radon in Environment: Tracer Applications and Public Health Risks
Chair: Talita Santos (CDTN/CNEN-Brazil),
Panelists: Elydio Soares, Laura Takahashi, and Nathan da Silva

Radiobiology Spectrum Auditorium Chair: Tarcísio Campos	Luminescent Materials/ Applications of Thermoluminescence Lumina Auditorium Chair: Susana Lalic

13:30 Jhon Brandon Felix Ferreira
G07-002 - *Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress*

13:45 Lucas Siqueira de Lima Neto
G07-013 - *Radiometric Analysis by gamma spectrometry of copaiba oil using a Hyper*

14:00 Iris Bomfim Martins Ferreira
G07-015 - *Radiometric analysis of the bivalve mollusk Perna*

13:30 João Pedro
VanellusRad - *The First Brazilian Real-Time IoT Electronic Dosimeter for Radiation Monitoring*

14:00 Ángel Ramírez Luna
G01-002 - *Afinando la Cronología Relativa de Cerámicas de la Zona de Playa Vicente, Veracruz, México Aplicando el Método de Datación Absoluta por Termoluminiscencia*

Roundtable Session (Lumina Auditorium)

14:30 AI's Challenges and Opportunities: Connecting Research, Industry, and Society
Chair: Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil),
Panelists: Ana Carolina Costa, Andre Henrique Alves Carneiro and Rita Sebastião

13:30 to 16:30 - Poster Session (Main Hall)

Note: 14:30 to 16:30 - Referees for evaluation
In-Person

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

G01-001 - Olania Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLI-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*
G01-011 - Samira Batista da Silva, *Enzymatic biocatalysis of alginate spheres and iodinated amino acids for applications in embolization guided by SPECT*
G01-026 - Roseli Künzel, *A low-cost integrated system for thermoluminescence and optically stimulated luminescence measurements*
G01-029 - Yuri Avelino Costa, *Analysis of the average geometric parameters of the closed pores of the fine aggregate matrix using X-ray micro-computed tomography and linear regression*
G01-032 - Danilo O. Junot, *Crab shell biowaste composites as luminescent radiation detectors*
G01-035 - Luiz Matheus de Souza Aguiar Leite, *Development of a radiochromic film using a polyacetylene compound*
G01-036 - Thalles Oliveira Campagnani, *Development of a test bench using a CsI/SiPM detector for mass flow measurement in industrial processes*
G01-038 - Ketrim Regina Souza, *Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership*
G01-040 - Raissa Alexia Camargo Guassu, *Estimation of absorbed dose in critical organs during interventional procedures: experimental and computational approach*
G01-041 - Danilo O. Junot, *Influence of precursor purity on the synthesis and TL/OSL response of $CaSO_4$-based radiation detectors*
G01-044 - Vinicius Costa Silva, *Radiological characterization of the waters feeding the Camorim Reservoir by alpha–beta – Pedra Branca State Park*
G01-045 - Margarete Cristina Guimarães, Thessa Cristina Alonso, *Standardization of gamma irradiation procedures for the preservation of bibliographic heritage*

G5 - Radiological Protection

G05-003 - Adriana de Souza Medeiros Batista, *Assessment of volcanic rock adsorbents for the remediation of acid mine drainage in uranium mining*
G05-006 - Daniel de Castro Pacheco, *Development of a distributed monitoring system for active radiological protection*
G05-007 - Cássio Miri Oliveira, *Dose evaluation and suggestion of typical values for head CT headache protocols: a preliminary study*
G05-015 - Thamyra Cybelle Vieira dos Santos, *Quality assurance in ^{90}Y radioembolization: an intercomparison protocol for activity optimization*
G05-017 - Fábio Lima Guimarães, *Radiometric survey at the Stone House Lookout, Saquarema-RJ*
G05-019 - Marcela Morais Freitas, *Structural and compositional changes in zeolitic materials exposed to uranium mine effluents: implications for environmental radioprotection*
G05-022 - Bianca Rossini Marques, *Validation of analytical methods using gamma spectrometry applied to the radiometric characterization of monazite sand*

G8 - Luminescent materials

G08-004 - Divanizia do Nascimento Souza, *Development of high-sensitivity $CaSO_4$:Tm,Mn detectors for thermoluminescent and optically stimulated luminescent dosimetry*
G08-007 - Roseli Künzel, *Effect of silver doping concentrations on the structural and luminescence properties of $CaMoO_4$ powders*
G08-008 - Paula Thays Santos Malachias, *Investigation of the luminescent properties of zirconia-enriched dental resins*
G08-009 - Bruno Araújo, *Optically stimulated luminescence (OSL) characteristics of seven different color types of quartz samples under 405 nm stimulation*
G08-010 - Rafael dos Santos Viana, *Promotion of the Compton effect of ^{137}Cs photons to enhance the effi-*

ciency of radioluminescent batteries
G08-011 - Anderson Bezerra, *Properties of $CaSO_4$:Eu,Mn for Luminescent Dosimetry*
G08-012 - Patrícia de Lara Antonio, *PTTL and PTOSL of TLD-100 exposed to ^{60}Co and lighting with LEDs*
G08-013 - Daniele Gonçalves Mesquita, *Radiation Detection of Lead-Free $BaZrO_3$ Perovskite Scintillators*
G08-014 - Claudete Roberta Evangelista, *Feasibility of Retrospective Dosimetry Using Ceramic Material from Spark Plugs*
G08-015 - Anderson Manoel Bezerra da Silva, *Structural, Morphological, Luminescent, and Dosimetric Properties of $CaSO_4$:Yb,Mn Phosphors*
G08-016 - André Felipe de Oliveira, *Tailored theranostic nanocomposites: mesoporous silica, gold nanoparticles, and gadolinium complexes for cancer diagnosis and treatment*
G08-020 - Jéssica Pauline Nunes Marinho, *Potential theranostic nanoparticles of HAp-Gd^{3+} with curcumin and folic acid for cancer therapy*

15:30 COFFEE BREAK

Invited Speaker - Lumina Auditorium **Chair: Ivonne B. L. Rojas**

16:00 *Invited Speaker* – Silvia S. Hidalgo Tobón (UAM-Mexico),
State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

17:00 - 20:00 Welcome & Honors Night - Lumina Auditorium

- For the Ceremony: Bruno M. Mendes, Marco Aurélio de Sousa Lacerda and Telma Cristina Ferreira Fonseca
- Award Ceremony for Honored Researchers: Dr. Tarcísio P. R. de Campos and Dr. Teógenes Augusto da Silva
- Cultural Performance by Aruanda Group
- Welcome Cocktail Gathering

Wednesday, 09/24/2025

Applications of Thermoluminescence **Spectrum Auditorium** **Chair: Andrea Mantuano**	**Dosimetry** **Lattice Auditorium** **Chair: Ana Carolina Costa**

08:00 Rafael Cogollo
G01-004 - *Dosimetric characteristics and kinetic analysis of pure and cerium-doped alumina (α-Al_2O_3) for low-dose thermoluminescent dosimetry applications*

08:15 Héctor Maya
G01-006 - *Kinetic analysis method for general order TL glow curves*

08:30 Natalia Barbosa Gonzaga Mendes
G01-024 - *Using gamma spectrometry as a tool for assessing residual dose in irradiated topaz*

08:45 Alberto Mizrahy Campos
G01-027 - *Random walk of neutrons in a rotational environment*

09:00 Enrique Alejandro Chávez Camacho
G01-030 - *Analysis of the effects of gamma radiation on amaranth seeds and sprouts: a design of experiments (DoE) approach*

08:00 Berenice Guadalupe Olmos Grimaldo
G02-006 - *Medición de radón en la zona del Peñón Blanco, Salinas de Hidalgo, San Luis Potosí, México*

08:15 Renato Monaco Nunes
G02-018 - *A Monte Carlo-based software for photon transport in water using the CUDA platform*

08:30 Iasmin Vieira Nishibayaski
G02-019 - *A new formulation of Fricke gel dosimeter development using salicylic acid*

08:45 Camilla da Silva Sampaio
G02-026 - *Assessing the applicability of TDCR counting for nonpure beta emitters: the case of ^{210}Pb in liquid scintillation cocktails*

09:00 Laura Cardoso Takahashi
G02-040 - *Development of an integrated national database for effective dose calculation from environmental radiation exposure in Brazil*

09:15 Laisa de Almeida Neves
G05-023 - *Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry*

10:00 COFFEE BREAK

Radiation Sources **Spectrum Auditorium** **Chair: Denison Santos**	**Medical Physics** **Lattice Auditorium** **Chair: Lucas Paixão**

10:30 Nayelly Nikole Díaz Gutiérrez
G06-002 - *Estimación del término fuente de un irradiador gamma de ^{60}Co utilizando el código MCNP*

10:45 Ismael Teixeira Vicente
G06-004 - *Alpha particle spectrometry and Monte Carlo evaluation of ^{241}Am sources for use in Elastic Recoil Detection Analysis (ERDA)*

11:00 Estela Maria de Oliveira
G06-006 - *Comparative study of gamma efficiency curves in HPGe detector with different calibration standards*

10:30 Sara Suely Ventura da Silva Lima
G04-004 - *Comparative Monitoring of Lens, Thyroid, and Whole Body in Interventional Cardiology Procedures*

10:45 Prycylla Gomes Creazolla
G02-089 - *Volumetric dose evaluation using 6 MV linear accelerators for blood irradiation*

11:00 Juraci Passos dos Reis Junior
G04-011 - *An AI revolution in radiotherapy in SUS: The example of Mário Kroeff Hospital*

11:15 Natasha Briggs e Silva
G04-020 - *Determination of the mass attenuation coefficient of thermoset and thermoplastic resins at energies used in conventional mammography*

11:15 Roos S. de Freitas Dam
G06-012 - *Neutron Source Applied for Eccentric Scale Prediction in Oil Pipelines using Deep Learning*

11:30 Roger Ferreira da Silva
G04-033 - *Mammography quality assurance based on phantom image with deep learning*

08:00 to 12:30 - Poster Session (Main Hall)

Note: 10:00 to 12:30 - Referees for evaluation
In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-003 - Emilia Suárez, *Development and Assessment of Perovskite-based X-Ray Detectors for In Vivo Dosimetry Applications*
G02-017 - Tales Noujaim Silveira, *^{68}Ga Production with the GE PETtrace-8 Cyclotron: a Monte Carlo study*
G02-020 - Leanderson Pereira Cordeiro, *AI-based segmentation performance and implications for organ-level dosimetry in radionuclide therapy*
G02-021 - José Marques Lopes, *An Environmental Radiometric Assessment Based on Airborne Gamma Spectrometry Data in the State of Rio de Janeiro*
G02-023 - Hugo Leonardo Lemos Silva, *Analytical and simulation methods in determining radiation exposure time of healthcare waste*
G02-027 - Alessandra Vaz dos Anjos Santos, *Automated System for Visual Detection of Personal Dosimeters Using Convolutional Neural Networks*
G02-028 - Tiago Macedo da Cunha e Silva, *Bayesian-Optimized Monte Carlo Modeling of CyberKnife Collimators and Source Parameters for Small-Field Dosimetry*
G02-030 - Heloísa Milena Caldeira Rhodes, *Comparison of Methods for Determining the Half-Value Layer (Hvl) in Computed Tomography Equipment*
G02-031 - Nicolle Marques Cardoso Ignacio, *Comparison of the Dosimetric Properties of EBT-3 and EBT-4 Radiochromic Films for Application in Hemocomponent Dosimetry*
G02-032 - Fernanda Rocha Cavalcante, *Comparison of X-ray spectral correction software for a CdTe detector*
G02-033 - Guilherme Gazolla Santana, *Computational Modeling of an Anthropomorphic Phantom for Applications in Aerospace Dosimetry*
G02-034 - José Marques Lopes, *Computational Simulation of a NaI(Tl) Detector and Definition of Its Main Parameters for Application in a Neutron Activation Analysis Laboratory*
G02-035 - Lucas Beloti Silva, *Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms*
G02-043 - João Vinícius Batista Valença, *Dose estimation in organs of a pediatric patient undergoing brain radiotherapy via Monte Carlo simulation*
G02-045 - Guilherme Cavalcante de Albuquerque Souza, *Dosimetric comparison of methods for obtaining dose profiles in head CT scans using 120 kV X-ray beams*
G02-046 - Frederico Possato Tagliaferro, *Dosimetry Of Electret Microphones Irradiated By X-Rays*
G02-047 - Josemary A. C. Gonçalves, *Energy Dependence of an Epitaxial Diode for Radioprotection Dosimetry*
G02-048 - Enzo Cortez, *Estimating the dose rate constant for Iodine-131 via Monte Carlo simulation: a first step towards pediatric radioiodine therapy shielding evaluation*
G02-050 - João Vinícius Batista Valença, *Evaluating Radiation Dose in Veterinary Professionals: MCNPX-Based Simulation of Equine Radiography Exposure*
G02-051 - Camila Nunes Carvalho Souza, *Evaluating the Performance of Bayesian Optimization in High-Dose-Rate Brachytherapy Planning*
G02-052 - Maria Rita Pascoal de Souza Barros, *Evaluation of Absorbed Dose Uniformity in Irradiated Foods with Different Packaging Geometries Using PMMA Dosimetry*
G02-054 - Gabriel Gomes do Nascimento, *Evaluation of the reproducibility of a new lens dosimeter holder*
G02-061 - João Victor Ramos Oliveira, *Influence of detector geometry on small-field radiotherapy simulations*
G02-066 - Daniel de Castro Pacheco, *Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6*

G02-070 - Greiciane de Jesus Cesário, *Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms*
G02-071 - Kyssylla Monnyelle Araujo Silva, *Multielectrode Array Devices for dosimetry and dose enhancement factor analysis in nanoparticle radiotherapy*
G02-072 - Mateus de Souza Resende, ^{99m}Tc *Radiolabeled Papain Nanoparticle: in silico Dosimetry Evaluation using DM BRA phantom and the experimental biodistribution data*
G02-074 - Henrique Chaves Gulino, *New Low Cost Process of PIN Diode Fabrication for Radiation Detection*
G02-077 - Carmen C. Bueno, *Online response of a homemade diode dosimetry system in a mobile 500-700 keV electron beam facility*
G02-078 - Anderson Manoel Bezerra da Silva, *Optimization of* Mn^{2+} *and* Tb^{3+} *Concentrations in* $CaSO_4$*:Mn,Tb and Advanced Investigation of its TL/OSL Dosimetric Properties*
G02-079 - Rômulo Monteiro Lopes, *Parametrization and Validation of a HPGe Detector by Monte Carlo Simulation*
G02-080 - Hugo Leonardo Lemos Silva, *Assessing Photon and Electron Ranges in Healthcare Waste for Radiation Sterilization*

Note: 10:00 to 12:30 - Referees for evaluation
On-line

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

G01-007 - Lenin Estuardo Cevallos Robalino, *Monte Carlo characterization of neutron personal dosimeter based on TLD-600 and TLD-700 irradiated with* ^{252}Cf *neutron source on the ISO-phantom at LPN-CIEMAT*
G01-010 - Abel Fernando Martínez Cruz, *Thermoluminescence of MgO Chemically Modified with Nd and Li Ions, Obtained by Glycine-based Solution Combustion Synthesis*
G01-013 - Hassan Salah, *Assessing Patient Exposure in PET/CT scans: A Head and Neck Dose Parameter Study*
G01-019 - Mohammed, *Evaluation of Radiation Dose Reduction Strategies in Whole-Body PET/CT Imaging*
G01-021 - Hassan Salah, *Optimizing Radiation Dose Parameters in PSMA PET/CT for Prostate Cancer: Balancing Diagnostic Efficacy and Patient Safety*
G01-022 - Abdelmoneim Sulieman, *Patient-Specific Dose Optimization in Breast PET/CT: Towards ALARA-Compliant Protocols*
G01-033 - Edwar Alonzzo Canaza Mamani, *Datación OSL-SAR y caracterización tecnológica de las cerámicas arqueológicas de los grupos de tradición ceramista Tupiguarani: Pirajuí I y Arataca II, Pernambuco, Brasil*
G01-034 - David Alexander Urbina Leal, *Design and Analysis of FDM Prototypes for Gold-198 Nanoparticles*

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-001 - Alejandro Ferreira Tapia, *Caracterización Física y Computacional de Dispositivos Modeladores de Radiación Electromagnética mediante Simulación Monte Carlo*
G02-008 - Whoody Alem Wanderley Araripe Farias, *A platform for creating computational exposure models: personalized dosimetry with Holmium-166*
G02-009 - Whoody Alem Wanderley Araripe Farias, *All about MARTIN: a BREP phantom with blood and lymphatic vessels applied to Yttrium-90 radioembolization*
G02-012 - Aline Fabiane Gonçalves de Oliveira, *Geospatial assessment of radionuclide dispersion and associated effective dose modeled with ARTM View: a case study at CDTN's radiopharmaceutical unit*
G02-015 - Jardel Lemos Thalhofer, *Radiometric Profile Analysis of Soils from the São José de Itaboraí Paleontological Natural Park*
G02-016 - Marcelo Franklin da Silva Santos, *Synthesis and Luminescent Characterization of* $CaSO_4$*:Tm,Li via Slow Evaporation, Co-precipitation, and Solid-State Diffusion Routes*
G02-022 - Akemi Yagui, *Organ Dose Estimation in Pediatric Abdomen and Pelvis Radiographs Using Caldose_X Monte Carlo Simulations*
G02-025 - Ranulfo da Silva Dias, *Assembly and Geant4-based computational simulation of a Helium-3 de-*

tector for ground-level cosmic radiation monitoring
G02-029 - Daniel Vieira Lamas, *Characterization of the Ubatuba Formation: Integration of Gamma Radiometry, Total Organic Carbon, and Well Logs from the Upper Cretaceous to Lower Paleogene Intervals of the Campos Basin*
G02-037 - Paulo Sérgio de Abreu Junior, *Cytogenetic assessment of ionizing radiation exposure in Allium cepa using micronucleus frequency: a comparative study between Antarctic and control conditions*
G02-038 - Ana Laura Burin, *Development of a Computational Phantom of the Vertebral Column for Dosimetric Investigations*
G02-039 - Leonardo da Silva Boia, *Development of a Dedicated System for Generating Input Files for the MCNP and Adaptation of a Phantom with Edema for Simulation of Prostate Brachytherapy with ^{125}I*
G02-044 - Wallacy Viana, *Dose Rate Estimation by Extracting and Analyzing Colors from an Image Generated by a CMOS Image Sensor*
G02-059 - Lucas Fabricio de Araujo, *In Silico Investigation of the Impact of Gold Nanoparticles on the Radiosensitivity of 3D HeLa Cells Using the Monte Carlo Method*
G02-060 - Gabriela Rodé de Assis da Silva, *In Situ Gamma-Ray Spectrometric Mapping of Beach Sand Radioactivity in Niterói, Brazil*
G02-067 - Pedro Henrique Avelino de Andrade, *Monte Carlo Modeling of the BPW34 Photodiode for X-ray Dosimetry*
G02-073 - Nadia Rodrigues dos Santos, *Natural Radionuclide Activity and Radiological Risk Assessment in Brazilian Coffee Samples*
G02-086 - Marcelo Luís dos Santos Filho, *Study of CaSO4:Tm,Li Based Luminescent Composites Applied to TL/OSL Dosimetry in Mammography*
G02-087 - Leonardo da Silva Boia, *MCNP simulation of the effects of prostatic edema in LDR brachytherapy with ^{125}I using the adapted ICRP 110 voxel male phantom*
G02-093 - Daniel V. Lamas, *Quantification of Gamma Radiation on Drilling Cuttings Through HPGe Detector: Comparative Study With Well Logs Spectrometry Data*

G3 - Ionizing and non-ionizing radiation

G03-005 - Samara Mendes Matos, *Effect of Neutron Activation in the Biodegradable Chitosan polymer associated with Gold Nanoparticles.*
G03-010 - Angel Garcia-Duran, *Simulador de detector de Germanio de alta pureza con fuentes radiactivas, embebido en FPGA*

G4 - Medical Physics

G04-001 - Nicolás Eugenio Martín, *Adaptation and performance of CdTe-based XFCT detection system using Ag-tacers for Biomedical Applications*
G04-007 - Priscila Santos Amorim, *Implementation of free software for managing imaging protocols*
G04-008 - Angelica Viridiana Roman Martinez, *Magnetic Resonance Imaging Assessment of Corpus Callosum Volume in Pediatric Epilepsy Patients*
G04-009 - Jaime Torres Juárez, *Optimization of the Pulse Sequence for Visualization of Myocardial Fiber Architecture Using Diffusion Tensor Imaging in the Study of Cardiomyopathies*
G04-012 - Gisell Ruiz Boiset, *Assessment of radiologically tissue-equivalent materials for breast imaging phantoms using X-ray spectrometry*
G04-024 - Jesús Emmanuel Mares Ponce, *Evaluation of elemental Iron content in pharmaceutical formulations using magnetic susceptibility technique*
G04-026 - Camila Hitomi Murata, *Evaluation of the affect on performance of a 3T MRI by the proximity of a subway station and an adjacent MRI scanner*
G04-027 - Carolina Salinas Domján, *Experimental characterization of magnetic fields for dosimetric modelling with Monte Carlo methods in MR-Linac radiotherapy*
G04-029 - Ana Carolina Costa da Silva, *Global Usability of Mobile CT Scanners in Rapid Stroke Diagnosis: A Literature Review*
G04-030 - Raissa Alexia Camargo Guassu, *Impact of Body Mass Index on Radiation Doses in Interventional Radiology*

G5 - Radiological Protection

G05-002 - Wallacy Viana, *Analysis of the characteristics of the Ingeo™ 3D850 polymer in the context of its use in visual inspections in the TRIGA IPR-R1 nuclear reactor*
G05-004 - Hericka O. Kenup-Hernandes, *Computational Modeling for Radiological Risk Assessment: Simulation of Plume Dispersion in Radioactive Waste Fires Using HotSpot*
G05-008 - Akemi Yagui, *Evaluation of radiation exposure in pediatric interventional cardiology procedures*
G05-009 - Hericka O. Kenup-Hernandes, *Gadolinium Adsorption by Functionalized Mesoporous Silicas (MCM- 41/SBA-15): A Proposal for Volume and Dose Reduction in Radioactive Waste*
G05-018 - Watila Lins Silva, *Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated* ^{6}*LiI (Eu) Detector at the Neutron Metrology Laboratory*
G05-020 - Fábio Sabará Dias, *Survey of radiation exposure from portable X-ray fluorescence*

G6 - Radiation Sources

G06-005 - Gláucia de Oliveira Barreto, *Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and* γ*-ray spectrometry techniques*
G06-008 - João Pedro Carvalho Andrade, *Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation*
G06-011 - Gabriel Tura Echeverria, *Identification and quantification of K, Na and Mn in textured soy protein sold in the city of Rio de Janeiro using the Argonauta research reactor of IEN/CNEN and the techniques of instrumental neutron activation analysis and* γ*-ray spectrometry*

G7 - Radiobiology

G07-004 - Ivana Mara Gomes Andersen Cavalcanti, *Determination of mechanical and physical properties of pig ear cartilage through modeling and simulation using DFT*
G07-014 - Yasmym Sarmento Pereira, *Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro.*
G07-016 - Gabriela Rodé de Assis da Silva, *Radiometric characterization of Erythrina verna Vell.: A study of natural radioactivity in plant tissues*

G8 - Luminescent materials

G08-003 - Marcelo Luis dos Santos Filho, *Development of* $CaSO_4$*:Li Composites by Different Synthesis Routes for Applications in TL and OSL Dosimetry*
G08-005 - Brunno Florencia de Oliveira, *Dosimetric Potential of Cowrie Shell Biomass under High Radiation Doses using TL and OSL Techniques*
G08-018 - Caroline Pereira dos Santos, *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy3+ y Eu3+) a dos concentraciones 0.1 M y 0.5 M*
G08-019 - André Luiz Tavares e Silva, *Unsupervised learning for pattern recognition in K-feldspar using cathodoluminescence emission and EDX composition mapping*
G08-021 - Joel Arturo Rivera-Garcia, *Estudio de las propiedades luminiscentes del aluminosilicato de estroncio dopado con Ce (*$SrAl_2Si_2O_8$*:Ce) sintetizado por el método de reacción en estado sólido*

12:30 LUNCH

Applications of Thermoluminescence / Luminescent Materials Virtual Room 1 Chair: Daniel Calheiro/Rodrigo Gadelha	Luminescent Materials / Dosimetry Virtual Room 2 Chair: Marco A. Labarrios/ Álvaro M. L. Gómez	Medical Physics / Ionizing and non-Ionizing / Radiobiology Virtual Room 3 Chair: Jaime T. Juárez/Bruno Gallotti

13:30 Katherine G. Delgado
G01-005 - *Effective dose received by Dentistry students in intraoral dental radiology using TLD dosimetry*

13:45 Nayely D. M. Ramirez
G01-008 - *Synthesis of Y_2O_3 nanophosphors and evaluation of their thermoluminescent response for use in radiation dosimetry*

14:00 Abel Fernando Martínez Cruz
G01-009 - *Thermally Stimulated Luminescence of Chemically Modified Magnesium Oxide with Gd^{3+} and Li^{+}, Obtained by a Glycine-based Combustion Process*

14:15 Radmila Sabitova
G01-016 - *Characterization of additional irradiation positions in the IVG.1M research reactor*

14:30 E. Cruz-Zaragoza
G01-020 - *Optically -and thermally stimulated luminescence detection of irradiated oregano (Lippia graveolens) polymineral*

14:45 Umme Muslima
G01-023 - *Thermoluminescence Characterization of Natural Flake Graphite Under 6 MeV Electron Beam Exposure from Medical LINAC for Dosimetric Applications*

15:00 André Luiz Tavares e Silva
G01-025 - *A Data Science Approach on the Study of Al_2O_3 Thermoluminescent Glow Curves Under Varying Heating Rates*

15:15 Gabriela Pontes Cardoso
G08-001 - *Cathodoluminescence behavior of PVDF films doped with $CuSO_4$ under high-dose gamma irradiation*

13:30 Jessica Paola C. Fraga
G02-005 - *La termoluminiscencia del Grafito para su posible aplicación dosimétrica*

13:45 José Edgardo Arellano Hernández
G02-004 - *Estudio de la respuesta termoluminiscente de NA2O-Y2O3-P2O5-SIO2 irradiados con Electrones*

14:00 Marcelo F. da Silva Santos
G02-014 - *New Approaches in the Synthesis of $CaSO_4$:Tm,Li Phosphors with Sugars as Chelating Agents for Dosimetry Applications*

14:15 Vinícius R. A. A. Martins
G02-055 - *Experimental Evaluation of the Mechanical and Dosimetric Properties of the Red Perspex 4034 Dosimeter After 15 Years of Aging*

14:30 Rayline Leite da Silva
G02-084 - *Standardized Protocol for the Analysis of Cutting Samples and Lithological Materials by Gamma Spectrometry*

14:45 Nadia Rodrigues dos Santos
G02-049 - *Estimation of the soil-to-plant transfer factor of natural radionuclides in organic crop from Pedra Branca State Park*

15:00 Leonardo da Silva Boia
G02-036 - *Construction of the Glioblastoma scenario in the right optic nerve region using DIP techniques for simulation in the MCNP code with conversion to lattice structure by SCAN2MCNP*

15:15 Leonardo Santiago Melgaço Silva
G02-041 - *Development of an optical sensor prototype for dose readings in Fricke gel dosimeters*

13:30 Erika Patricia Azorín Vega
G04-005 - *Dosimetric and Biological Characterization of a 3D Cell Culture Model for the Evaluation of Radiopharmaceutical Induced Bystander Effect*

13:45 Leticia González Zamora
G04-010 - *Study of Prefractal Antennas for Multiband Radiofrequency Applications*

14:00 Maria L. Miranda Maciel
G04-014 - *Calibration of radiochromic films EBT2, EBT3, EBT4 and RTQA2 and assessment of uncertainties for radiotherapy doses*

14:15 Elsa Bifano Pimenta
G04-022 - *Evaluating Lung CT Protocols Using the Detectability Index: An Anthropomorphic Phantom-Based Evaluation of Photon-Counting and Conventional CT*

14:30 Estefania Reyes Soto
G04-002 - *Automated Classification and Manual Segmentation of Pediatric Brain Tumors Using Deep Learning on MRI Data*

14:45 Giulia Rita de Souza Faés
G04-035 - *Radiotherapy Planning with VMAT in Halcyon 2.0: Impacts of Isocenter Configuration on Dosimetric Parameters and Treatment Efficiency*

15:00 Natalia Matos Orozco
G04-003 - *Diseño y cálculo de blindajes para garantizar la protección radiológica en una radiofarmacia productora de radiofármacos*

15:15 Kuanysh Samarkhanov
G05-016 - *Radiological Characterization and Dosimetric Assessment of Liquid Radioactive*

15:30 Júlia Carina Orfão Costa
G08-002 - *Characterizations of $SrMoO_4$:Dy^{3+} powders synthesized by hydrothermal method*

15:45 Daniele da Silva Machado
G08-006 - *Dosimetric Properties of nanoDot OSL Detectors for Radiotherapy Quality Control: An Initial Characterization and Uncertainty Analysis*

16:00 Carina Oliva Torres Cortés
G08-023 - *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy3+ y Eu3+) a dos concentraciones 0.1 M y 0.5 M*

16:15 Caroline Pereira dos Santos
G08-017 - *Uncovering the intrinsic nature of the ternary metallic alloys CuNiPd and Cu_2PPd_2: A DFT study*

16:30 Rodrigo Martínez Baltezar
G08-022 - *Point defects study on Aluminum nitride for applications in luminescence dosimetry*

16:45 Pedro R. González
G08-024 - *Thermoluminescent properties of MgO:Sm,Tb+PTFE dosimeters*

17:00 Sandrith M. Pérez Ramos
G01-003 - *Análisis cinético de la curva de brillo termoluminiscente del Óxido de Berilio (BeO) irradiado con Rayos X en un rango de baja dosis*

15:30 Carolina Salinas Domján
G02-090 - *Assessing Water-Equivalence in Gram-Negative Bacteria-Hydrogel Radiation Dosimeters: A Comparative Analysis with Gram-Positive Bacterial Systems*

15:45 Rayline Leite da Silva
G07-017 - *Study of Drinking Water Quality in Rio de Janeiro: Assessment of Gamma Radiation Levels and Physicochemical Parameters*

16:00 Teodoro Córdova Fraga
G04-019 - *COVID largo y fibrosis pulmonar: clasificación basada en tomografía en población mexicana*

16:15 Teodoro Córdova Fraga
G04-013 - *Automated detection and characterization of renal calculi using U-Net and Hounsfield unit analysis in computed tomography scans*

16:30 Priscila Santos Rodrigues
G04-036 - *Three-dimensional dose mapping of gold-198 nanoparticles: A comparative study of 3D-printed matrices using Fricke xylenol gel and Monte Carlo simulations*

16:45 Arthur Reis Martins
G06-015 - *Production of Radioisotopes in Nuclear Microreactors for Medical, Industrial, and Scientific Applications*

17:00 Jardel Lemos Thalhofer
G02-010 - *Assessment of the level of natural radioactivity in the region of the future Municipal Environmental Protection Area (APA) Ilha Verde, Muriqui, Mangaratiba - RJ*

Waste from BN-350 Reactor

15:30 Stanislav Svetachev
G06-007 - *Determination of detection limits of elements in sediment and rock standards for INAA using the IGR reactor*

15:45 Jonathan O. dos Santos
G06-017 - *Response Function Validation of HPGe Detector Through GADRAS and MCNP Simulation*

16:00 Roberto Méndez Villafañe
G06-003 - *Neutron measurements around the research reactor RA-6 at Centro Atómico Bariloche with a Bonner Sphere Spectrometer*

16:15 Roberto Méndez Villafañe
G06-001 - *Development of a detailed model with MCNP of the Neutron Standards Laboratory at CIEMAT and validation with a Bonner Sphere Spectrometer System*

16:30 Fatima Castañeda Ortiz
G07-001 - *Pulsed X-ray-Induced Nuclear Fragmentation in a Radioresistant Ovarian Cancer Cell Line*

16:45 Daniele G. Mesquita
G07-006 - *Gamma Radiation Modulates Genomic Maintenance Pathways in T-Lymphocytes: An Analysis of Clinical Doses*

17:00 Emilio Romero Muñoz
G07-007 - *Germination response of 'Micro-Tom' to controlled magnetization by using a Rodin coil*

17:15 Raúl Ken Delgado Velázquez
G07-010 - *Magnetic stimulation as a promoting factor for germination and development in Guazuma ulmifolia L. seeds*

13:30 to 16:00 TECHNICAL VISIT **Chair: Camila Engler**

Places to visit at CDTN:

- Gamma Irradiation Laboratory - More information, acess here
- Dosimeter Calibration Laboratory - More information, acess here
- TRIGA IPR-R1 Reactor - More information, acess here

Thursday, 09/25/2025

Invited Speaker - Lumina Auditorium **Chair: Bernardo Dantas**

08:00 *Invited Speaker* – Luciana Carvalheira (IEN/CNEN-Brazil),
Neutrons, Knowledge, and New Generations: How Brazil's Argonauta Reactor is Shaping the Future of Radiochemistry and Its Impact on Global Dosimetry Challenges.

09:00 *Invited Speaker* – Bernardo Maranhão Dantas (IRD/CNEN-Brazil),
Diseminación de técnicas de monitoreo in vivo ocupacional en una Red de Clínicas de Medicina Nuclear.

09:25 *Invited Speaker* – Maria Jose Neves (CDTN/CNEN-Brazil),
Range Radiation Exposure Indicator.

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium **Chair: Telma Cristina Ferreira Fonseca**

10:30 *Invited Speaker* – Ademar Lugão (IPEN/CNEN-Brazil),
Use of ionizing radiation for the production of nanoparticles and advanced curatives.

11:30 *Invited Speaker* – Richard Hugtenburg (Swansea University-United Kingdom),
New Technologies as a Route to Particle Therapy: Laser-Hybrid Acceleration for Radiobiology Applications – The LhARA Project.

12:30 LUNCH

Invited Speaker - Lumina Auditorium **Chair: Richard Hugtenburg**

13:30 *Invited Speaker* – Hélio Yoriyaz (IPEN/CNEN-Brazil),
Proton Beam Therapy: Main Characteristics and Challenges.

Dosimetry
Spectrum Auditorium
Chair: Peterson Squair

13:30 Divanizia do Nascimento Souza
G02-042 - *Dose Distribution in an Anthropomorphic Cervix Phantom Using Two Computer Modeling Approaches*

13:45 Francisco Harlley Dantas Hauradou Xavier
G02-057 - *Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP*

14:00 Francisco H. D. Hauradou Xavier
G02-094 - *Optimization of dose rate map acquisition time for internal dosimetry in nuclear medicine using 3D U-Net convolutional neural networks*

14:15 José Elizeu de Souza Junior
G02-062 - *Influence of Photon Emission Spectra on Air Kerma Rate Constants for Radionuclides: A Monte Carlo EGSnrc Study*

Medical Physics/Dosimetry
Lattice Auditorium
Chair: Lucas Paixão

13:30 Ana Beatriz Caldeira de Andrade
G04-025 - *Evaluation of spatial resolution in PET equipment with a Jaszczak-analog phantom built by additive manufacturing*

13:45 Crislane de Jesus Cesário
G04-028 - *Experimental Evaluation of 3D-Printed Materials (ABS and PETG) for Mammographic Phantom Applications Using $CaSO_4$:Dy Dosimetry*

14:00 Nathalia Luzia Aidar Alves
G04-032 - *Machine Learning in PSQA: Performance of a Random Forest Model on Radiotherapy Metrics*

14:15 André Luiz Espindola Fidelis
G04-037 - *Validation of a 3D-Printed Anthropomorphic Eye Phantom for Quality Assurance in External Beam Radiotherapy Using TLDs and Radiochromic Film*

14:30 Leonardo Gualberto Silva Macedo
G02-095 - *Mass Analysis of Computational Mouse Models Developed at CDTN: A Comparative Study of Male and Female Versions*

14:45 Beatriz Diniz de Oliveira Guedes
G02-082 - *Spectrophotometric Analysis of a 3D-Printed Photopolymerizable Resin Exposed to High-Dose Gamma Radiation*

15:00 Hirys Sales
G02-053 - *Evaluation of Polymeric Materials as Muscle Tissue Equivalents for Physical Phantoms Used in the Calibration of In Vivo Monitoring Systems*

13:30 - 17:00 - Poster Session (Main Hall)

Note: 14:30 to 17:00 - Referees for evaluation
In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-081 - Felipe Mamed de Souza, *SMS Notification System for Potential Radiological Risk Scenarios*
G02-085 - Alcilene Cristina da Silva, *Study and Characterization of the EPR Response of New Materials for High Dose Dosimetry*
G02-088 - Ana Letícia Almeida Dantas, *The IRD In Vivo Monitoring Laboratory - LABMIV: Instrumental Capabilities applied in routine and emergency situations*
G02-091 - Maryanna Regina de Souza Roberto, *OSL Tandem Dosimeter for ICRU 95 Operational Quantities*
G02-092 - Thalys Matheus Gama de Oliveira, *Photons Response of an Albedo Monitor*

G3 - Ionizing and non-ionizing radiation

G03-001 - Paula Regina Rodriguez Coelho, *"Analysis of spatial resolution in two different phantoms as a quality parameter in MRI pediatric pathology"*
G03-002 - José Marques Lopes, *Adjustment Factor for Gamma Spectra Due to the Influence of Sample Density on the Shielding Background Radiation*
G03-003 - Davyd Soares Dunda, *Application of Steam Methodology in Science Teaching: Prototyping a Cloud Chamber*
G03-004 - José Marques Lopes, *Comparison of environmental sampling results using portable and benchtop scintillation detectors*
G03-006 - Adriana de Souza Medeiros Batista, *Effect of repolymerization and gamma irradiation on the morphology of styrene-divinylbenzene composites with magnetite*
G03-007 - Paulo Cezar Rocha Silveira, *Gamma Irradiation Effects on the Mechanical and Ballistic Performance of 3D-Printed Polylactic Acid*
G03-008 - Luciana Carvalheira, *Gold nanoparticles with potential for use in multimodal combat of head and neck cancer*

G4 - Medical Physics

G04-006 - Otávio Gabriel Moreira Silva, *SNR and CTDIvol study as a function of the Noise Index in Computed Tomography exams*
G04-016 - Rute Cristina de Oliveira Soares, *Chest Computed Tomography in Oncology: Clinical Contributions and Dosimetric Assessment*

G04-017 - Gustavo Freire Pereira da Silva, *Comparative Validation of the DEEDZ Methodology and a Photon-Interaction-Based Model Implemented in TEMPy for Estimating Effective Atomic Number and Electron Density*
G04-018 - Rute Cristina de Oliveira Soares, *Contributions of Computed Tomography to Breast Cancer Staging and Follow-Up: Justification of Practice and Dosimetric Analysis*
G04-021 - Iasmin Vieira Nishibayaski, *Dosimetric evaluation of EBT3 and EBT4 in superficial HDR brachytherapy for keloids*
G04-023 - Camila Engler, *Evaluation of Breast Positioning Quality and Its Association with Technical Parameters in Screening Mammography*
G04-031 - Douglas Philipe Martinho Dos Santos, *Investigation of Diagnostic Reference Levels (DRL) in medical radiography: dose-area product (DAP)*
G04-034 - Kellen Adriana Curci Daros, *Radiation dose from pediatric coronary artery calcium scoring CT scan to evaluate chronic kidney disease-mineral and bone disorder: Observational study*
G04-038 - Arícia Ravane Pereira da Cruz, *Validation of an alanine-based postal dosimetry system for quality control in radiotherapy*

G6 - Radiation Sources

G06-013 - Kathleen Martins, *Neutronic Analysis of an Accelerator-Driven Subcritical Reactor Utilizing Thorium Dioxide and Reprocessed Nuclear Fuels*

G7 - Radiobiology

G07-003 - Christiana da Silva Leite, *Comparative Study of Organ Mass in Healthy and Tumor-Bearing Mice for Preclinical Dosimetry Purposes*
G07-005 - Camila Ramos Silva, *Development and validation of a body-shielding device for radiotherapy of breast cancer in mice using a panoramic gamma source*
G07-008 - Luciana Penedo de Melo Teixeira, *Impact of Physical Exercise on Irisin Expression in the Brain of Irradiated Mice: Effects on Behavior and Memory*
G07-009 - Olania Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLI-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*
G07-011 - Camila de Almeida Salvego, *Nitric Oxide-Donor Nanoparticles Potentiate Radiotherapy of TripleNegative Breast Cancer*
G07-018 - Bruno Gallotti, *Growth Inhibition of S. cerevisiae BY4741 Exposed to Gold Nanoparticles and Low-Energy Radiation*
G07-019 - Maurício Moacir da Silva Borges, *The Role of Caffeine in Reducing TNF-α and Improving Cognitive in Irradiated Mice in the Brain*
G07-020 - Luis Martín Garcia Ortiz, *Cellular response in human leukocytes (in vitro), due to exposure to UVB radiation using apoptotic biomarkers*

Invited Speaker - Lumina Auditorium — Chair: Teógenes da Silva

14:30 *Invited Speaker* – Georgia Joana (CNEN-Brazil) and Ketrim Souza (INB-Brazil),
WiN Brasil - Diversity & Inclusion Management.

15:30 COFFEE BREAK

16:00 - Mentorship Helen Khoury WiN Brasil (Spectrum Auditorium)

Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership
Moderator: Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil),
Panelists: Ketrim Souza and Winners (WiN Brasil)

Roundtable Session (Lumina Auditorium)

16:00 Leadership in Scientific and Technological Research
Moderator: Luciana Carvalheira (IEN/CNEN-Brazil),
Panelists: Adriana Marques and Anna Letícia Barbosa de Sousa

Radiation Protection
Lumina Auditorium
Chair: Teógenes da Silva

17:00 Mirella Maturo da Cruz
G05-021 - *Transfer Factors of Natural Radionuclides in Organic Foods from Rio de Janeiro, Brazil*

17:15 Elydio J. D. Soares
G05-010 - *Investigation of radon exhalation from mortars made from niobium tailings and the mitigating role of coating materials*

17:30 Raphael Francisco Gomes dos Santos
G05-005 - *Determination of Radiation Protection Procedures for an Offshore Plant Inspection in Brazil Using Radiotracer Techniques*

Radiation Sources / Radiation Protection
Lattice Auditorium
Chair: Jhonny A. Benavente Castillo

17:00 Gabriel do Nascimento Freitas
G06-014 - *Nuclear simulation workflow to correct gamma ray well logs for radioactive contaminated drilling fluids*

17:15 José Alípio dos Santos Filho
G06-016 - *Proposal for the construction of a neutron generator for application in research and industry*

17:30 Welen Nunes de Lima
G05-014 - *Proposal for the use of thorium as fuel in TRIGA® research reactors*

Friday, 09/26/2025

Invited Speaker - Lumina Auditorium | **Chair: Richard Hugtenburg**

08:00 *Invited Speaker* – Rolf Behrens (PTB-Germany),
From dosemeter development to routine use – Standards and Uncertainties.

08:40 *Invited Speaker* – Susana Lalic (DFI/UFS-Brazil),
Plant Stem Cells as an Ethical Alternative for Biodosimetry.

09:20 *Invited Speaker* – Thiago Viana Miranda Lima (University of Lucerne-Switzerland),
Current and Future Trends in Nuclear Medicine Dosimetry.

10:00 COFFEE BREAK

Radiological Sources
Spectrum Auditorium
Chair: Carmen C. Bueno

10:30 Matheus R. do Nascimento
G03-009 - *Speknife: open-source software for X-ray spectra correction in solid state dosimetry*

10:45 Eduardo B. de Paula
G05-011 - *Metrological Enhancement of Radiopharmaceuticals Half-Lives Determination via γ-ray Spectrometry and Algorithmic Data Processing*

11:00 Igor Andrade Machado
G02-083 - *SpecUnPy code: updates and what is coming*

Dosimetry
Lumina Auditorium
Chair: Peterson Squair

10:30 Jhonny A. Benavente Castillo
G02-007 - *Simulación del blindaje del búnker de un acelerador ciclotrón utilizando el código MCNP*

10:45 Natasha Briggs e Silva
G02-011 - *Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project*

Invited Speaker - Lumina Auditorium | **Chair: Bruno M. Mendes**

11:30 *Invited Speaker* – Tarcísio P. R. de Campos (DEN/UFMG and CDTN/CNEN-Brazil) and Alberto Mizrahy Campos (Augsburg-Germany),
Challenges in Radioisotope Decontamination and Production through Neutron Spectral Selection.

12:30 LUNCH

Dosimetry
Spectrum Auditorium
Chair: Maria da Penha A. Potiens

13:30 Isabela N. S. Ferreira
G02-064 - *Mapping of natural radionuclides U, Th and K in the city of Belo Horizonte/Minas Gerais*

13:45 Claudio A. Federico
G02-065 - *Measurements of Ionizing Radiation Dose Onboard*

Dosimetry
Lattice Auditorium
Chair: Bruno M. Mendes

13:30 Celso G. de Paulo
G02-069 - *Monte Carlo simulation of a HPGe detector*

13:45 Ana G. Leão Ferro
G02-075 - *Occupational exposure assessment in ROLL procedures with Tc-99m: A Monte Carlo simulation approach*

Applications of Thermoluminescence
Lumina Auditorium
Chair: Teógenes da Silva

13:30 Tarcísio P. R. Campos
G01-046 - *The reduction of the radiological impact of nuclear fuel waste*

14:00 Júlia B. Severo
G01-039 - *Environmental Gamma Radiation and Radon Exposure in Areas Surrounding Urban*

an Aircraft from Southeastern Brazil to the Antarctic Continent

14:00 Paulo J. Iack Silva
G02-068 - *Monte Carlo N-Particle (MCNP) Simulations for Scintillation Emission in Quantum Dot-Doped Systems*

14:00 Rafael S. Oliveira
G02-076 - *On the adaptation of a commercial Bonner Sphere Spectrometer to be used with thermoluminescent detectors*

Quarries in Belo Horizonte, Brazil

Invited Speaker - Lumina Auditorium **Chair: Telma Cristina Ferreira Fonseca**

14:30 *Invited Speaker* – Carmen Bueno (IPEN/CNEN-Brazil),
Radiation processing with electron beams: On the feasibility of using electronic dosimeters for real-time dosimetry.

15:30 COFFEE BREAK

16:00 CLOSING SESSION

- Honorary Award in Memory of Helen Jamil Khoury: Carmen Bueno
- Helen Khoury Award for Best Works: Andrea Mantuano & Ester Andrade
- Final Session: Summary and Perspectives: Telma Cristina Ferreira Fonseca & Bruno M. Mendes

CHAPTER 6

Pre-Symposium Courses

6.1 DEVELOPMENT OF PET RADIOPHARMACEUTICALS

Pre-symposium course speakers

- Dr. Marina Bicalho (CDTN/CNEN-Brazil)
- Special Guest: Dr. Isabel Dregely (Fachhochschule Technikum Wien-Austria)
- Special Guest: Dr Emerson Bernardes (IPEN/CNEN-Brazil)

Dr. Marina Bicalho Silveira is an Industrial Pharmacist (UFMG, 2007) with an M.Sc. (2010) and Ph.D. (2016) in Radiation Science & Technology (CNEN). Specialized in Radiopharmacy (SBRAFH, 2023), she works as a Senior Technologist at CDTN/CNEN, leading Quality Assurance for PET radiopharmaceutical production. Her expertise covers GMP, regulatory compliance, and R&D of novel radiopharmaceuticals, including preclinical studies. She also serves as a Professor in CDTN's Graduate Program, bridging industry and academia.

Dr. Isabel Dregely is Head of the Competence Center for Artificial Intelligence and Data Analytics at Fachhochschule Technikum Wien. She previously served as Senior Lecturer at King's College London and was a UKRI Future Leaders Fellow. Her research background includes advanced MRI techniques, particularly in prostate and cardiovascular imaging, with postdoctoral work at UCLA and experience at the Technical University of Munich. She holds a PhD in Physics from the University of New Hampshire, where she focused on hyperpolarized xenon MRI and RF coil development. Dr. Dregely combines expertise in biomedical imaging with AI and data analytics applications.

Dr. Emerson Soares Bernardes is a pharmacist and researcher with over two decades of experience in immunology, molecular biology, and nuclear medicine. He served as Manager of the Radiopharmacy Center at the Nuclear and Energy Research Institute (IPEN), where he led pioneering projects in the development of radiopharmaceuticals. He currently holds a position as a Visiting Researcher at Erasmus Medical Center in the Netherlands, contributing to innovation initiatives and patent development. His career is distinguished by the integration of scientific research, strategic management, and entrepreneurship.

Description: This course covers the basics of PET radiopharmaceuticals-production, clinical applications, and their impact on nuclear medicine-while showcasing CDTN'S cutting-edge technologies and their role in enhancing diagnostics and patient care in Brazil.
Dr. Marina Bicalho - Theoretical Aspects and State of the Art in PET Radiopharmaceuticals.
Dr. Emerson Bernardes - Advances in new PET Radiopharmaceuticals
Dr. Isabel Dregely - Artificial Intelligence Applied to PET Imaging

6.2 INTRODUCTION TO RADIATION DETECTORS

Pre-symposium course speakers

- Dr. Marco Antonio Polo Labarrios (UAM-Mexico)
- Dr. Álvaro M. L. Gómez (DEN/UFMG-Brazil)

Dr. Marco Antonio Polo Labarrios holds a Ph.D. and Master's degree in Energy Engineering from UNAM and a Bachelor's in Energy Engineering from UAM Iztapalapa. He teaches part-time at Universidad Iberoamericana and UNAM, focusing on Transport Phenomena and Renewable Energy. His research involves the simulation of energy processes based on mass, momentum, and energy transport. He has published 21 scientific articles, participated in multiple national and international conferences, and supervised four undergraduate theses. Dr. Polo Labarrios was part of the National Commission on Nuclear Safety and has been a Level 1 member of the National System of Researchers since 2021.

Dr. Álvaro M. L. Gómez holds a Ph.D. and M.Sc. in Nuclear Science and Technology from the Federal University of Minas Gerais – UFMG (2017; 2021). Holds a Bachelor's degree in Physics (diploma validated by UFMG) and two postgraduate specializations: Bioengineering – Universidad Distrital Francisco José de Caldas (2008; 2014) (Colombia), and University Teaching – Universidad Cooperativa de Colombia (2013). Has research experience in ionizing radiation dosimetry, development of tissue-equivalent materials for computed tomography phantoms, material characterization, and teaching experience at the Department of Nuclear Engineering (DEN) at UFMG, lecturing on radiation protection, radiation detection and nuclear instrumentation, radioisotope applications, and introduction to nuclear energy.

Description: Throughout the course, we will explore different types of radiation detectors, their operating principles, and criteria for their proper selection and use. The goal is to provide students with a solid foundation that enables them to identify the most suitable equipment for each situation and use it efficiently, maximizing both measurement accuracy and radiological protection.

6.3 IMPORTANCIA DE LA FÍSICA EN UNA IMAGEN POR RESONANCIA MAGNÉTICA EN EL DIAGNÓSTICO CLÍNICO.

Pre-symposium course speakers

- Dr. Silvia S Hidalgo Tobón (UAM-Mexico)
- Msc. Jaime Torres Juárez (UAM-Mexico)

Dr. Silvia Hidalgo earned her Bachelor's degree in Physics in 1998 from the University of the Americas-Puebla, Mexico. In 2000, she earned her Master of Science in Medical Physics from the Institute of Physics at the National Autonomous University of Mexico (the first Master of Science in Medical Physics from the UNAM program). In 2005, she earned her PhD in Physics, specializing in Magnetic Resonance Imaging from the Sir Peter Mansfield Magnetic Resonance Center at the University of Nottingham, England. She completed a postdoctoral fellowship in Magnetic Resonance Imaging at the Robarts Research Institute, University of Western Ontario, Canada. She also completed a postdoctoral fellowship at the Autonomous Metropolitan University, Iztapalapa. Her scientific interests are related to advanced functional magnetic resonance imaging, neural networks applied to neuroscience, and magnetic resonance safety in daily clinical practice. She has taught courses and supervised theses at the undergraduate, master's, and doctoral levels in magnetic resonance imaging at the UAM, UAEM, and UNAM. She has given lectures at various national and international forums on nuclear magnetic resonance imaging. She is a member of the International Society of Magnetic Resonance in Medicine, RSNA, the Mexican Physical Society, IEEE, and the Neuroscience Society, among others.

Descripción: Se presentarán las bases de la física de una imagen por resonancia magnética y su relación con el radio diagnóstico. La seguridad en resonancia magnética que evita complicaciones y accidentes que con la continua capacitación disminuyen notablemente. Así como las técnicas nuevas que están incluyendo inteligencia artificial para un mejor diagnóstico por imagen.

6.4 SOLID STATE DOSIMETRY: PRINCIPLES AND APPLICATIONS & OSL ADVANCED MATERIALS.

Pre-symposium course speakers

- Dr. Ivonne Berenice Lozano Rojas (IPN-Mexico)

Dr. Ivonne, a distinguished Mexican scientist holding a Ph.D. from Centro de Investigación en Ciencia Aplicada y Tecnología Avanzada, Unidad Legaria, conducts pioneering research in TL/OSL dosimetry, radiation detectors, and food irradiation detection, significantly advancing medical physics and safety.

Description: This minicourse focuses on quartz as a natural radiation dosimeter, exploring its defects and their role in optically stimulated luminescence (OSL). Advanced OSL materials, including synthetic and natural quartz, are analyzed for radiation dose assessment in dating and dosimetry applications.

CHAPTER 7

Symposium Speakers

7.1 DR. ADEMAR BENÉVOLO LUGÃO: USE OF IONIZING RADIATION FOR THE PRODUCTION OF NANOPARTICLES AND ADVANCED CURATIVES.

Dr. Ademar Lugão holds Bachelor's in Chemical Engineering from the Polytechnic School of the University of São Paulo – USP (1980), earned a Master's degree in Nuclear Technology from IPEN-USP (1985), and a Ph.D. in Nuclear Materials Technology, also from IPEN-USP (2004). With more than 30 years of experience as a research leader in polymer irradiation, he is currently the Manager of the Center for Chemistry and the Environment at the Nuclear and Energy Research Institute – IPEN. His main research focuses on the synthesis, processing, and characterization of polymers; structural modification of polymers through energetic processes, including crosslinking, grafting, branching, and chain scission; development of hydrogels for health and environmental applications; clean technologies for producing polyolefins with controlled rheology; starch foams for sustainable packaging; advanced low-cost dressings tailored to the needs of the Brazilian Public Health System (SUS); albumin and papain nanoparticles for encapsulating radiotherapeutics and chemotherapeutics; and radioactive gold nanoparticles (Au-198) for applications in radiodiagnosis and radiotherapy (theranostics). He currently serves as the Coordinator of the Nanomaterials and Nanocomposites Coordination Hub of SISNANO (National System of Nanotechnology Laboratories) of the Ministry of Science, Technology, and Innovation (MCTIC).

Description: Use of ionizing radiation for the production of nanoparticles and advanced curatives.

7.2 DR. BERNARDO DANTAS: DISEMINACIÓN DE TÉCNICAS DE MONITOREO IN VIVO OCUPACIONAL EN UNA RED DE CLINICAS DE MEDICINA NUCLEAR

Dr. Bernardo Maranhão Dantas holds a PhD in Sciences (Biology – Nuclear Biosciences) from the State University of Rio de Janeiro (1998). He is a Senior Technologist at the Brazilian National Nuclear Energy Commission (CNEN) and a faculty member of the Graduate Program at the Institute of Radiation Protection and Dosimetry, focusing on Radiation Biophysics. He has published 87 peer-reviewed articles and has supervised numerous students across undergraduate, master's, and doctoral levels. He has led five research and development projects and currently coordinates the CNPq-certified research group "Occupational Internal Dosimetry."

Description: Professionals who handle radiopharmaceuticals for diagnostic and therapeutic purposes in nuclear medicine are subject to the intake of radionuclides through inhalation and ingestion. The radiological protection aspects of this practice are regulated in Brazil by the National Nuclear Energy Commission (CNEN). The International Atomic Energy Agency (IAEA) recommends the implementation of occupational monitoring programs when there is a risk of annual effective doses exceeding 1 mSv. The control of radionuclide intake can be carried out using internal dosimetry techniques. Currently, in Brazil, there are not enough laboratories capable of providing internal monitoring services to meet the entire demand if this requirement were to be enforced by CNEN. This paper presents an overview of the current situation in Brazil and economically viable technical alternatives, aiming to make the implementation of routine internal monitoring programs for workers occupationally exposed to significant risks of radionuclide exposure in nuclear medicine feasible.

7.3 DR. CARMEN BUENO: RADIATION PROCESSING WITH ELECTRON BEAMS: ON THE FEASIBILITY OF USING ELECTRONIC DOSIMETERS FOR REAL-TIME DOSIMETRY

Dr. Carmen holds Bachelor's and Licentiate degrees in Physics from PUC/SP, as well as Master's and Doctorate degrees in Nuclear Instrumentation from the same university. She began her academic career at PUC/SP, where she was appointed as a Full Professor of Nuclear Physics. In 1995, she started working as a researcher at the Nuclear and Energy Research Institute (IPEN-CNEN/SP) and later completed her postdoctoral work at the Instrumentation and Experimental Particle Physics Laboratory (LIP) at the University of Coimbra. She also holds a second Doctorate in nuclear technology from IPEN-USP. She has published nearly 150 full-length articles in international journals and conference proceedings and has supervised numerous master's, doctoral, and postdoctoral students. Currently, she is a senior researcher at IPEN with over 30 years of experience in nuclear instrumentation for radiation measurement and dosimetry. Most of her research focuses on developing semiconductor-based dosimetry systems for medical and radiation processing fields. She currently serves on the Steering Committee of the National Institute of Science and Technology (INCT) project, "Nuclear Instrumentation and Applications in Industry and Health" (INAIS). Additionally, she is a member of the Advisory Boards of the Brazilian Society of Solid-State Dosimetry (SSD) and the Brazilian Association of Nuclear Energy (ABEN).

Description: In this lecture, we will briefly introduce the radiation processing of products, a relatively new industry with wide applications and significant commercial success, primly involving gamma rays and electron beams (EB). The main features of the irradiation facilities, the doses delivered, and the regulatory requirements for the processed products will be discussed to highlight the role of dosimetry in this field. Several passive solid-state dosimeters, which are well-established in radiation processing dosimetry, will be introduced to emphasize the need for online dosimeters, whether as routine tools or for monitoring the operating conditions of EB facilities. An overview of efforts to develop online dosimeters, with a focus on electronic devices, will be provided. Subsequently, the lecture will explore the operational characteristics and the physics behind the dose and dose rate response of diodes, explaining how our group has addressed issues related to their use as real-time dosimeters in fixed and mobile EB facilities. The latest data obtained with these diodes, both as routine dosimeters and as electron beam monitors, will be presented to assess the feasibility of their application. The current status of online high-dose dosimeters and the challenges of meeting the metrological standards required in radiation processing dosimetry will also be discussed.

7.4 DR. GEÓRGIA SANTOS JOANA: WIN BRASIL - DIVERSITY & INCLUSION MANAGEMENT

Bachelor's degree in Physics from the Federal University of Minas Gerais - UFMG (2003), master's degree in Radiation, Mineral, and Materials Science and Technology from the Nuclear Technology Development Center of the National Nuclear Energy Commission - CDTN/CNEN (2005), and a Ph.D. in Nuclear Science and

Technology from the Postgraduate Program of the Department of Nuclear Engineering at UFMG (2018). She is currently a technologist at the National Nuclear Energy Commission. She previously worked in the field of Medical Physics at the Superintendence of Sanitary Surveillance of the Minas Gerais State Health Secretariat, where she coordinated the State Quality Control Program in Mammography - PECQMamo. She has experience in Medical Physics and Health Surveillance, with an emphasis on Radiological Protection, Regulation, Safety, and Quality in health services. Her research focuses on Radiological Protection, Quality Control, Radiation Dosimetry and Metrology, and Risk Analysis and Management in the field of medical applications of radiation.

Description: WiN Brasil - Diversity & Inclusion Management

7.5 DR. HELIO YORIYAZ: PROTON BEAM THERAPY: MAIN CHARACTERISTICS AND CHALLENGES

Bachelor in Physics (University of São Paulo, 1979–1982), Master's (1984–1986) and PhD(1998–2000) in Nuclear Technology from the University of São Paulo. Former RHAE fellow atthe Paul Scherrer Institute (Switzerland, 1993–1994) and IAEA Fellow at Oak Ridge AssociatedUniversities (USA, 1996–1997). Researcher at the Nuclear and Energy Research Institute(IPEN), working in the field of Nuclear Engineering with an emphasis on Medical Physics, particularly in Monte Carlo simulation, dosimetry and nuclear medicine. Currently a faculty member of the Graduate Program in Nuclear Technology at USP and the Professional Master's Program in Radiation Technology in Health at IPEN. Bachelor's in Physics (University of São Paulo, 1979–1982), Master's (1984–1986) and PhD(1998–2000) in Nuclear Technology from the University of São Paulo. Former RHAE fellow atthe Paul Scherrer Institute (Switzerland, 1993–1994) and IAEA Fellow at Oak Ridge AssociatedUniversities (USA, 1996–1997). Researcher at the Nuclear and Energy Research Institute(IPEN), working in the field of Nuclear Engineering with an emphasis on Medical Physics, particularly in Monte Carlo simulation, dosimetry and nuclear medicine. Currently a faculty member of the Graduate Program in Nuclear Technology at USP and the Professional Master's Program in Radiation Technology in Health at IPEN.

Description: Proton therapy is an increasingly relevant form of radiotherapy because it enables areduction in the total dose compared to photons, preserving critical structures whilesimultaneously delivering higher and better-conformed doses to the tumor volume. Thiseffect is achieved through the superposition of several weighted Bragg curves with differentincident energies, resulting in a broadened Bragg peak, known as the Spread-Out BraggPeak (SOBP). Although already considered technologically mature, proton therapy still presents importantchallenges that need to be investigated. Among these, the precise calculation of the relativebiological effectiveness (RBE) stands out, defined as the ratio between the doses requiredfor two different types of radiation to produce the same biological effect. This aspect isdirectly linked to the physical interaction processes between the proton and matter. When a proton with kinetic energy passes through a medium, it interacts with electrons andnuclei of the target, causing its deceleration. These interactions are known as electronicstopping and nuclear stopping, respectively. The stopping power describes the force themedium exerts on the ion, corresponding to the average energy loss per unit path length. Inthe case of proton-matter interaction, electronic stopping is dominant. This energy loss dynamic is fundamental to understanding the dose deposition profile. Asthe proton slows down, the energy transferred to biological tissue per unit path lengthincreases, reaching a maximum at a specific depth. This region, known as the Bragg peak,coincides with the maximum stopping power and represents the area of greatest interest forclinical applications. Therefore, its precise positioning is critical during treatment planning.To study these processes in greater detail, advanced approaches have been employed, suchas real-time Time-Dependent Density Functional Theory (TDDFT). This is a non-perturbative method based on modern quantum simulations, which has proven particularlyuseful for investigating electronic stopping in complex systems like water and DNA,bringing computational modeling closer to the real conditions found in biological tissues.

7.6 DR. LUCIANA CARVALHEIRA: NEUTRONS, KNOWLEDGE, AND NEW GENERATIONS: HOW BRAZIL'S ARGONAUTA REACTOR IS SHAPING THE FUTURE OF RADIOCHEMISTRY AND ITS IMPACT ON GLOBAL DOSIMETRY CHALLENGES.

Dr. Luciana Carvalheira is a Black Brazilian scientist and singer from Rio de Janeiro, leading pioneering projects at the Institute of Nuclear Engineering and serving as a professor and vice-coordinator in postgraduate programs for Nuclear Sciences. She co-founded startups like Magtech Brasil and Radíon, driving innovation in radiochemistry and nanomaterials. Her research spans neutron activation, medical polymers, and environmental monitoring, alongside mentoring future scientists through WiN Brasil's Helen Khoury Program. As a member of MunaN, she advocates for the inclusion of young Black women in nuclear sciences. Honored by ALERJ for empowering women and recognized as a leader by ENAP's LideraGOV 4.0, she embodies audacity, joy, and a relentless pursuit of scientific freedom and innovation.

Description: For over 60 years, Brazil's Argonauta research reactor has addressed hands-on education in physics and engineering of reactors, and radiochemistry. As a hub for RD&I, Argonauta drives partnerships with institutions like IME, UFRJ and startups in radiotracer technologies. Modernization efforts, including fuel improvements and new radioisotopes production, exemplify its commitment to advancing radiation science nuclear applications worldwide.

7.7 DR. MARIA NEVES: RANGE RADIATION EXPOSURE INDICATOR

Dr. Maria José Neves is a Senior Researcher at the Brazilian National Nuclear Energy Commission (CNEN), based at the Centre for the Development of Nuclear Technology (CDTN), where she leads the Radiobiology Laboratory. She holds a Bachelor's degree in Medical Biological Sciences from the Faculty of Medicine of Ribeirão Preto, University of São Paulo (USP-RP), as well as a Master's and Ph.D. in Physiology from the same institution, with a focus on metabolism. She completed her postdoctoral training in Biochemistry at the Katholieke Universiteit Leuven (KUL), Belgium. Her main research interests include cancer cell metabolism and the biological effects of ionizing radiation.

Description: A small plastic or glass tube (with a final volume of 1.5mL or smaller) containing a solution, composed of sodium selenite and glycose, can be fixed to the surface of a material and will indicate whether it was itself exposed to moderate doses of range radiation (5 – 100 kGy) in a range irradiator contained in the source of ^{60}Co. This type of irradiator is widely used in the irradiation of foods such as dried grains, fruits, tubers, etc., sterilization of surgical material, among other activities. Given its low cost and ease of preparation, it can be frozen and kept in stock at the time of use, which is why we consider an alternative option to confirm exposure to high-range radiation.

7.8 DR RICHARD HUGTENBURG: NEW TECHNOLOGIES AS A ROUTE TO PARTICLE THERAPY: LASER-HYBRID ACCELERATION FOR RADIOBIOLOGY APPLICATIONS - THE LHARA PROJECT.

Dr Richard Hugtenburg is a medical physics expert and research academic at Swansea University, UK, specialising in radiotherapy physics. Dr Hugtenburg began his career in Christchurch, New Zealand, working as a medical physicist while studying for a PhD on the use of computational methods in radiation oncology. He moved to the UK in 1997 and continued to practice in radiotherapy physics; first at the Queen Elizabeth Medical Centre in Birmingham, then at Singleton Hospital in Swansea. His research examines dosimetry methods associated with emergent radiotherapy technologies, utilising Monte Carlo modelling for a diverse range of radiation physics problems, including the analysis of energy deposition in biological systems, the optimisation of dosimeters and detectors, nanoparticulates, shielding and therapy systems.

Description: Particle therapy is a form of external beam radiotherapy that utilises ions to achieve precise (sub-mm) physical distributions of physical dose with the benefits of high LET to combat radioresistant tumours. Particle therapies are also expected to offer better normal-tissue sparing and reduce long-term side-effects in patients, where the current state-of-the-art in radiotherapy has contributed to significant improvements in the survivability of many cancers in recent years. The LhARA project is a consortium of UK universities that considers three new technologies as a means to reduce the cost, physical footprint and flexibility of particle therapies, where currently treatments, such as carbon ion therapy, are restricted to a few laboratories world-wide. The laser-hybrid method uses high-powered lasers to generate ions with kinetic energies of 10s of MeV and at high dose-rates suitable to examine high dose-rate sparing effects such as FLASH radiotherapy.

7.9 DR. ROLF BEHRENS: FROM DOSEMETER DEVELOPMENT TO ROUTINE USE - STANDARDS AND UNCERTAINTIES

Dr. Rolf Behrens holds a diploma at Technical University of Braunschweig (Germany) and a doctorate (Dr. rer. nat.) in physics from the University of Jena (Germany). He is a senior scientist at the Physikalisch-Technische Bundesanstalt (PTB), Germany's national metrology institute, where he leads the Working Group on Dosimetry for Brachytherapy and Beta Radiation Protection. Dr. Behrens is responsible for quality assurance in the Ionizing Radiation division, serves as Radiation Protection Officer for brachytherapy and beta radiation, and acts as an auditor for the German Accreditation Service (DAkkS). He heads and participates in several national and international standardization committees, including the IEC and ISO working groups on dosimetry and uncertainties, and chairs the German Committee on Radiation Protection Dosimetry. His research focuses on dosimetry of beta and photon radiation for radiation protection and brachytherapy, Monte Carlo simulations, spectrometry, Bayesian data evaluation, measurement uncertainty, and dosimetry for the lens of the eye.

Description: Radiation protection uses protection quantities (not directly measurable) and operational quantities (measured to estimate protection quantities). Standards ensure the quality of dose measurements, established by IEC and ISO, and adopted by regional organizations (CEN, CENELEC, GCC, SARSO, ARSO). Type tests for photon dosemeters use ISO 4037 reference fields; beta and neutron fields follow ISO 6980 and ISO 8529. Standards like IEC 62387 and IEC 61526 address other influencing factors such as angle, dose rate, temperature, and electromagnetic immunity. Uncertainties are calculated by combining the effects of these influences, as described in IEC TS 62461 (updated 2025). Performance is regularly checked using comparison measurements following ISO 14146 (updated 2024, includes EURADOS). Click here to view the presentation slides.

7.10 DR. SILVIA HIDALGO TOBÓN: STATE OF THE ART OF MAGNETIC RESONANCE IMAGING: FROM FUNDAMENTAL PHYSICS TO CLINICAL APPLICATIONS.

Dr. Silvia Hidalgo earned her Bachelor's degree in Physics in 1998 from the University of the Americas-Puebla, Mexico. In 2000, she earned her Master of Science in Medical Physics from the Institute of Physics at the National Autonomous University of Mexico (the first Master of Science in Medical Physics from the UNAM program). In 2005, she earned her PhD in Physics, specializing in Magnetic Resonance Imaging from the Sir Peter Mansfield Magnetic Resonance Center at the University of Nottingham, England. She completed a postdoctoral fellowship in Magnetic Resonance Imaging at the Robarts Research Institute, University of Western Ontario, Canada. She also completed a postdoctoral fellowship at the Autonomous Metropolitan University, Iztapalapa. Her scientific interests are related to advanced functional magnetic resonance imaging, neural networks applied to neuroscience, and magnetic resonance safety in daily clinical practice. She has taught courses and supervised theses at the undergraduate, master's, and doctoral levels in magnetic resonance imaging at the UAM, UAEM, and UNAM. She has given lectures at various national and international forums on nuclear magnetic resonance imaging. She is a member of the International Society of Magnetic Resonance in Medicine, RSNA, the Mexican Physical Society, IEEE, and the Neuroscience Society, among others.

Description: State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

7.11 DR. SUSANA LALIC: PLANT STEM CELLS AS AN ETHICAL ALTERNATIVE FOR BIODOSIMETRY

Dr. Susana de Souza Lalic is Full Professor of Physics at the Federal University of Sergipe. She holds a B.Sc. (1997) and Ph.D. in Physics (2002) from the University of São Paulo and a postdoctoral fellowship at the University of Pisa (2011). She is a faculty member of the Ph.D. Program in Industrial Engineering (Nuclear Engineering and Industrial Safety) at the University of Pisa and on the Board of Directors of the Master Course in CBRNe Protection at Tor Vergata University of Rome. She is Counselor of the Brazilian Physical Society (SBF), Chair of its Medical Physics Committee, and Vice-chair of the International Solid State Dosimetry Organization (ISSDO). She previously served as Secretary-General of SBF and Coordinator of the Graduate Program in Physics at UFS. Her research focuses on medical physics, condensed matter physics, and applied nuclear physics, with emphasis on radiation detectors, material characterization, radiobiology, and radiation metrology.

Description: This study investigates an ethical approach to biodosimetry, utilizing plants as a sustainable alternative to traditional animal models. Plants, known for their sensitivity to environmental stressors such as radiation, provide a valuable platform for identifying genotoxic agents while reducing reliance on animal testing. The research highlights the utility of the onion (Allium cepa) as an in vivo model for assessing radiation-induced genetic and chromosomal damage. While Allium cepa is a widely recognized standard in environmental contamination studies, it remains underexplored as a tool for radiation detection. A comprehensive evaluation of radiation effects and their temporal progression is made possible by analyzing key variables, including linear energy transfer, dose, and dose rate. The study also examines the potential of photobiomodulation to enhance biological recovery in tissues damaged by radiation, offering a novel intervention strategy. To ensure robustness, findings from the Allium cepa model are cross-validated against other in vivo, in vitro, and in silico systems, establishing its efficacy as an ethical and effective alternative for radiation biodosimetry.

7.12 DR. TARCÍSIO CAMPOS AND DR. ALBERTO CAMPOS: CHALLENGES IN RADIOISOTOPE DECONTAMINATION AND PRODUCTION THROUGH NEUTRON SPECTRAL SELECTION

Dr. Tarcisio P. R. de Campos is retired full professor from the Federal University of Minas Gerais (UFMG), currently collaborating with both UFMG and the Nuclear Technology Development Center (CDTN). He holds a degree in Engineering from UFMG, an MSc in Nuclear Engineering from UFRJ, a PhD from the University of Illinois at Urbana-Champaign, and a postdoctoral fellowship at MIT as a Fulbright Fellow. His expertise spans nuclear engineering, radiobiology, radiochemistry, radiotherapy, and the development of radioisotopes, simulators, and medical software. He has published extensively, holds 14 patents, and has supervised over 75 graduate theses. He has coordinated over 28 research projects and actively collaborates with national and international research institutions.

Dr. Alberto M. Campos is Postdoctoral researcher in Augsburg, Germany (2024–present), with a PhD in Mathematics from the Institute for Pure and Applied Mathematics (IMPA, 2024). He completed a research stay at Instituto Superior Técnico (IST), Lisbon (2022), holds a Master's and Bachelor's degree in Mathematics from UFMG, and a technical degree in Electrotechnics from CEFET-MG. He has served as a teaching assistant in various probability and analysis courses at IMPA and UFMG, and has presented talks at the Erdős Institute (Budapest) and the Department of Nuclear Engineering at UFMG. His research interests lie broadly in Probability Theory, with a focus on random covering problems, percolation, and random walks.

Description: We present investigations into the production and decontamination of radioisotopes using a selective Spectral Resonant Selector (RSS), focusing on neutron radiative capture reactions in precursor nuclides and transmutation reactions of long-lived radioactive waste. The physical and mathematical aspects of biased Brownian motion are addressed, through which a tailored neutron spectrum can be generated to target specific nuclides. Reaction rates for both precursor nuclides and decontamination processes are presented. Two distinct neutron spectra are considered—one at low energy and another at resonance energy—allowing interaction with a broad range of target nuclides. The production of isotopes such as Sm-153, Lu-177, Re-188, Ho-166, and Yb-169 serves as representative examples of potential radionuclide generation. Additionally, the decontamination of long-lived actinides, including transuranic radionuclides and fission products, through fission or radiative capture reactions, illustrates a promising approach to nuclear waste mitigation. The specific activities and benchmark parameters of various radioisotopes under defined conditions demonstrate the effectiveness of the RSS in both radioisotope production and nuclear decontamination.

7.13 DR. THIAGO LIMA: CURRENT AND FUTURE TRENDS IN NUCLEAR MEDICINE DOSIMETRY

Dr. Thiago holds a BSc in Physics with Medical Physics from UFRJ and a PhD in Medical Physics from University College London. He is the chief diagnostic medical physicist at Lucerne University Hospitals (LUKS) and research physicist at Genolier Hospital. He also lectures at the University of Lucerne. He chairs the education committee of the Swiss Society of Radiobiology and Medical Physics and represents the society in EFOMP working groups. His work focuses on dosimetry, protocol optimization, radioprotection, and research in nuclear medicine and medical imaging.

Description: This lecture delves into the evolution of nuclear medicine dosimetry. It will explore the progression from early rudimentary methods, which relied on basic measurements and empirical estimates, to the sophisticated imaging techniques and computational models employed today. These advancements have significantly enhanced the accuracy of absorbed dose quantification, improving both safety and efficacy in treatments. The lecture will also discuss the future of dosimetry, highlighting the integration of artificial intelligence and personalized medicine. This forward-thinking approach promises even more precise dose calculations tailored to individual patient characteristics, reflecting a continuous effort to optimize therapeutic protocols and improve patient outcomes in nuclear medicine.

7.14 DR. YVONE MASCARENHAS: PERSONAL DOSIMETRY - THE IMPORTANCE OF REGULATION, TECHNOLOGY, AND CULTURE. EVALUATION OF DOSIMETRY SERVICES IN LATIN AMERICA AND THE CARIBBEAN

Dr. Yvone Maria Mascarenhas has a degree in Physics from the University of São Paulo (USP), and a master's, doctorate, and post-doctorate from the USP's Institute of Physics and Chemistry of São Carlos (IFSC), with research in basic physics as well as radiology and dosimetry. She has worked as a coordinator for research projects funded by agencies such as CNPq and FAPESP and has extensive experience in the field of Physics, working primarily in radiological protection, x-ray production, radiodiagnosis, and dosimetry.

Description: Personal dosimetry is a fundamental tool in radiation protection, ensuring that bestpractices in facilities using ionizing radiation effectively minimize risks to exposed workers.The use of radiation is highly regulated worldwide, and to make personal dosimetry widelyaccessible, technology plays a key role—not only as a scientific instrument but also as areliable and reproducible measurement technique. Accurate analysis of population-level dosimetry data is only possible with the correct andappropriate use of personal dosimeters. This can be achieved only when a strongradiation protection culture is promoted and adopted by all stakeholders involved inpersonal dosimetry. In Latin America and the Caribbean, the evaluation of Individual Monitoring Services(SMIE) varies by country, reflecting differences in regulations, technological infrastructure,and safety culture. This study presents an overview of the current state of personaldosimetry in the region, with emphasis on the 2022 intercomparison exercise organized byREPROLAM. A total of 26 services—both public and private—were evaluated usingdifferent irradiation beams and dose levels. The results show that 69.2% of the laboratories complied with ISO 14146:2018. However,some exhibited calibration deviations, particularly with X-ray beams. Notably, OSL(Optically Stimulated Luminescence) dosimetry has seen significant growth,accounting for 34% of the systems evaluated in 2022. In Brazil, the development of Personal Dosimetry Services has been shaped by twodecades of intercomparison exercises and over 30 years of a structured certificationprocess. Despite this progress, challenges remain in standardizing methodologies andenhancing laboratory training to improve dosimetry accuracy across the region.This analysis underscores the critical importance of regulation, technologicaladvancement, and the consolidation of a radiological safety culture as pillars forstrengthening occupational dosimetry in Latin America and the Caribbean.

7.15 DR. WILSON CALVO: CURRENT STATUS OF STRATEGIC PROJECTS AT CNEN

Dr. Wilson Calvo holds a diploma in Metallic and Ceramic Materials Engineering from the Federal University of São Carlos - UFSCar (1987) and holds a Master's (1997) and a PhD (2005) degrees in Nuclear Technology - Applications, from the University of São Paulo - USP. He began working at the Nuclear and Energy Research Institute (IPEN) in 1988 and went on to pursue a successful career in both administration and technology. He is a Senior Technologist, professor, and postgraduate advisor in the area of Nuclear Technology at IPEN/USP. He published nearly 100 full-length articles in international conferences and indexed magazines, and supervised more than 15 master's, doctoral, and postdoctoral students. He is a leading expert in Nuclear Engineering, with a focus on the applications of Nuclear Techniques in Industry, Health, Agriculture, and the environment. His emphasis is on radioisotope technology (radiotracers and sealed radioactive sources) and ionizing radiation (electron beams, X-rays, and gamma rays). Owing to his great contribution in the nuclear applications field, he was awarded the Nuclear Merit Award from ABDAN (2020). Regarding his administrative career, he served as a Manager of the Radiation Technology Center (2001-2013), Director of Administration and Infrastructure (2014-2016), and Superintendent of IPEN (2017-2023). He was also Vice President of the Board of Directors of the São Paulo Technology-Based Incubator at USP/IPEN-Cietec (2017-2023). He is a Member of the Deliberative Committee of CNEN (2023-), the Advisory Board of ABENDI, and the International Irradiation Association (IIA). Currently, he is the Director of Research and Development at the National Nuclear Energy Commission - CNEN.

Description: The National Nuclear Energy Commission's guidelines for technological innovation focus on three main areas: nuclear technologies—including reactors, instrumentation, materials, and the fuel cycle—applications of ionizing radiation in health, industry, agriculture, and the environment, as well as radioprotection and dosimetry. This lecture will highlight several strategic projects, including CENTENA, GraNioTer, the tracer technique for offshore oil and gas platforms and processing plants, mobile units equipped with electron accelerators, and a multipurpose reactor. The main features of each project, their contributions to addressing major national challenges in developing nuclear technology, and the current progress of each initiative will be discussed. In short, CENTENA aims to design, build, and commission the first nuclear and environmental technology facility in South America for disposing of low-level radioactive waste, thereby establishing a foundation of safety and sustainability for Brazil. GraNioTer's goal is to establish itself as a national technological hub, promoting research, development, and innovation based on graphene, niobium, rare earths, and other materials and minerals with increasing global demand. Another key project is the application of the radiotracer technique to offshore oil and gas platforms and processing plants, which seeks to develop a prototype and method for measuring gas flow in pipelines. Equally important is the Mobile Unit, equipped with an electron beam accelerator, which aims to expand the nation's capacity to treat industrial effluents and improve water quality in urban areas, addressing one of the primary challenges faced by modern society.

7.16 DR. BRUNO MELO MENDES

Dr. Bruno Melo Mendes holds a Bachelor's degree in Biological Sciences and a Master's and PhD in Nuclear Science and Technology from the Federal University of Minas Gerais (UFMG). He is currently a researcher at the Nuclear Technology Development Center (CDTN/CNEN), responsible for the Internal Dosimetry Laboratory, and a professor in the CDTN postgraduate program. He also coordinates the Monte Carlo Modelling Expert Group (MCMEG). His expertise lies in the nuclear field, with a focus on radiopharmacy, computational modeling (Monte Carlo), and internal dosimetry.

7.17 DR. TELMA CRISTINA FERREIRA FONSECA

Dr. Telma C. F. Fonseca is professor at the DEN/UFMG, faculty member of the PCTN/UFMG and member of the Diversity & Inclusion Management Committee (2024–2027) of WinBrasil – Women in Nuclear, and coordinates the Monte Carlo Modelling Expert Group (MCMEG). She holds degrees in the area of computational sciences, a Master's in Radiation Sciences from UFMG, and a PhD in Biomedical Sciences from KU Leuven (Belgium), where her research focused on internal dosimetry. She was a researcher at the Radiation Protection Dosimetry and Calibrations group (RDC) at SCK-CEN in Belgium. Additionally, she completed a postdoctoral fellowship at the Centre for Development of Nuclear Technology (CDTN/CNEN), where she also led the Internal Dosimetry Laboratory. Her expertise and research interests include dose calculation in radiotherapy, radiobiology, nuclear instrumentation, nanodosimetry, computational modeling, and the use of Monte Carlo methods in radiation protection and radiobiology. Website: www.telmafonseca.com.br

CHAPTER 8

Round Tables

8.1 AI'S CHALLENGES AND OPPORTUNITIES: CONNECTING RESEARCH, INDUSTRY, AND SOCIETY

Moderator

- Dr. Telma Fonseca (DEN/UFMG)

Panelists

- Dr. Rita Sebastião (ICEX/UFMG)
- Dr. Ana Carolina (Siemens Healthineers)
- Dr. Andre Henrique Alves Carneiro (USP)

Description: The roundtable is part of the Tech Area on Artificial Intelligence, an innovative space within ISSSD 2025 dedicated to the convergence of technology, healthcare, and innovation. Its main goal is to foster dialogue among startups, researchers, healthcare professionals, and investors, encouraging the use of AI in clinical practice. During the session, strategic topics will be discussed, such as: Challenges and opportunities of AI in clinical radiology Explainability and ethics of diagnostic algorithms Automation and integration of hospital workflows Success stories from innovative startups Trends for the next 5 years in medical AI The roundtable will feature multidisciplinary experts — including radiologists, data scientists, entrepreneurs, and regulators — and will be moderated by professionals with experience in healthcare innovation. There will also be room for audience interaction, with guided questions designed to stimulate debate.

Dr. Telma C. F. Fonseca is professor at the DEN/UFMG, faculty member of the PCTN/UFMG and member of the Diversity & Inclusion Management Committee (2024–2027) of WinBrasil – Women in Nuclear, and coordinates the Monte Carlo Modelling Expert Group (MCMEG). She holds degrees in the area of computational sciences, a Master's in Radiation Sciences from UFMG, and a PhD in Biomedical Sciences from KU Leuven (Belgium), where her research focused on internal dosimetry. She was a researcher at the Radiation Protection Dosimetry and Calibrations group (RDC) at SCK-CEN in Belgium. Additionally, she completed a postdoctoral fellowship at the Centre for Development of Nuclear Technology (CDTN/CNEN), where she also led the Internal Dosimetry Laboratory. Her expertise and research interests include dose calculation in radiotherapy, radiobiology, nuclear instrumentation, nanodosimetry, computational modeling, and the use of Monte Carlo methods in radiation protection and radiobiology. Website: www.telmafonseca.com.br

Dr. Rita C. O. Sebastião is a full professor at Universidade Federal de Minas Gerais (UFMG) with experience in Chemistry, working mainly on the topics: inverse problems, artificial neural networks, nuclear magnetic resonance, thermal decomposition, chemical kinetics, combustion processes, and thermodynamics. Extensive experience in education, research, and extension, especially in projects aiming for the popularization of science, educational innovation, and entrepreneurship, connecting the university to basic education. Experience in developing and managing projects on socio-environmental sustainability and education. She is currently a member of the Postgraduate Program in Chemistry (PPGQUI), Postgraduate Program in Technological Innovation (PPGIT) and Professional Master's in Technological Innovation and Intellectual Property (MPITPI) at UFMG.

Dr. André Carneiro is a Senior Data Scientist and Management Consultant with over 6 years of experience working with leading Brazilian companies. Skilled in developing mathematical, statistical, and machine learning models to deliver actionable insights and optimize business decisions. Currently, I am a Senior Data Scientist at Ambev Tech, one of Brazil's largest companies, where I focus on recommendation systems, clustering, predictive modeling, causal inference, and advanced techniques such as double machine learning (e.g., causal forests). In addition to my corporate experience, I have a strong academic background with over 5 years of research experience. I hold a PhD in Production Engineering from Escola Politécnica of the University of São Paulo (Poli-USP), as well as a double degree in Production Engineering from Poli-USP and a Master's degree in Management and Industrial Engineering from the Instituto Superior Técnico of the University of Lisbon. My research focuses on machine learning, statistical control, and project monitoring. I am also a collaborating researcher at LABDAPS (Laboratório de Big Data e Análise Preditiva em Saúde) of the Public Health School of the University of São Paulo, where I pursue postdoctoral research on fairness and social justice in machine learning models, aiming to enhance predictive performance for underrepresented populations.

8.2 RADON IN ENVIRONMENT: TRACER APPLICATIONS AND PUBLIC HEALTH RISKS

Moderator

- Dr. Talita Santos (CDTN/CNEN)

Panelists

- MSc. Elydio Soares (CDTN/CNEN)
- MSc. Laura Takahashi (CDTN/CNEN)
- Dr. Nathan da Silva (CDTN/CNEN)

Description: The two main aspects related to radon will be presented and discussed: the concern regarding the impacts on public health and the applications of radon isotopes as tracers of environmental phenomena.

Dr. Talita de Oliveira Santos holds a degree in Radiology Technology from the Federal Center for Technological Education of Minas Gerais (2007), a master's degree in Nuclear Sciences and Techniques from the Federal University of Minas Gerais (2010) and a PhD in Nuclear Sciences and Techniques from the Federal University of Minas Gerais (2015). She is currently an adjunct professor at the Federal University of Minas Gerais and a collaborator of the National Commission for Nuclear Energy. She has experience in Physics, with an emphasis on Radiation Physics, working mainly on the following topics: natural radioactivity, application of radiation to the environment, nuclear instrumentation and radioprotection.

8.3 LEADERSHIP IN SCIENTIFIC AND TECHNOLOGICAL RESEARCH

Moderator

- Dr. Luciana Carvalheira (IEN/CNEN)

Panelists

- Adriana Marques (Agência Nacional de Água - ANA)
- Anna Letícia Barbosa de Sousa (CNEN)

Description: This roundtable will convene leading experts to explore the pivotal role of leadership in advancing nuclear scientific and technological research. The discussion will address key challenges, including regulatory frameworks, ethical considerations, nuclear applications, diversity and inclusion, effective communication, and the integration of the next generation into the global nuclear sector. By examining how leadership drives innovation, navigates complex challenges, and shapes the future of nuclear science and technology, this session aims to foster meaningful dialogue. Participants will gain insights into emerging technologies, explore strategic leadership approaches, and strengthen collaboration among researchers, policymakers, and industry leaders.

Dr. Luciana Carvalheira is a Black Brazilian scientist and singer from Rio de Janeiro, leading pioneering projects at the Institute of Nuclear Engineering and serving as a professor and vice-coordinator in postgraduate programs for Nuclear Sciences. She co-founded startups like Magtech Brazil and Radíon, driving innovation in radiochemistry and nanomaterials. Her research spans neutron activation, medical polymers, and environmental monitoring, alongside mentoring future scientists through WiN Brazil's Helen Khoury Program. As a member of MunaN, she advocates for the inclusion of young Black women in nuclear sciences. Honored by ALERJ for empowering women and recognized as a leader by ENAP's LideraGOV 4.0, she embodies audacity, joy, and a relentless pursuit of scientific freedom and innovation.

CHAPTER 9

Mentorship Helen Khoury WIN-Brasil

The Helen Khoury Mentorship WinBrasil2025
Program: Advancing Diversity, Inclusion, and Female Leadership in the Nuclear Sector

The Helen Khoury Mentorship Program, promoted by Women in Nuclear Brazil (WiN Brasil), was established to foster the professional development of women in the nuclear sector, integrating diversity in age, educational background, and technical expertise. Over seven months, the program combined one-on-one mentoring sessions with collective workshops, encouraging both academic and professional growth.

Mentors and mentees co-developed final projects that reflected the program's multidisciplinary and inclusive approach. Outcomes included: the design of a critical meaningful learning model for future mentorship cycles; the creation of Radioarretadas, a local initiative to engage women from Pernambuco's countryside in nuclear discussions; research on gender and racial equity within Eletronuclear; studies on hybridization of nuclear and renewable energy; structured career development strategies in nuclear physics; participation in scientific proposal submissions (PPSUS/Innovation 2025); scientific dissemination projects on radiation and radioactivity; and professional milestones such as article submissions, approvals in competitive examinations, and postgraduate advancements.

The results demonstrated progress in strengthening both technical and behavioral competencies, expanding professional networks, and integrating diverse regional and cultural perspectives. By connecting institutions such as CNEN, IPEN, IEN, Eletronuclear, INMETRO, and universities across Brazil, the initiative reinforced the importance of knowledge sharing and inter-institutional collaboration. It is concluded that the Helen Khoury Mentorship has been consolidated as a model for professional development aligned with sectoral demands and individual expectations. The program not only contributes to participants' growth but also reinforces the presence and leadership of women in the nuclear sector, leaving a legacy of diversity, inclusion, and innovation.

Keywords: Mentorship, Diversity, Inclusion, Female Leadership, Nuclear Sector.

CHAPTER 10
Honors Awards

The International Symposium on Solid State Dosimetry (ISSSD) is proud to honor outstanding contributions in the field of radiation science and its applications in medicine, industry, and research. The 2025 Awards recognize excellence, innovation, and leadership that have advanced knowledge, safety, and education in nuclear science.

10.1 PROF. DR. TARCISIO PASSOS RIBEIRO DE CAMPOS

We are honored to recognize the exceptional contributions of Prof. Tarcísio Passos Ribeiro de Campos, a distinguished figure in nuclear engineering and radiation science in Brazil. With a career spanning over four decades, Prof. Campos has significantly advanced the fields of medical dosimetry, radiation protection, and nuclear technology.

As a retired full professor at the Federal University of Minas Gerais (UFMG), he continues to collaborate with the university and the Nuclear Technology Development Center (CDTN). Prof. Campos earned his undergraduate degree in Engineering from UFMG (1983), a master's in Nuclear Engineering from the Federal University of Rio de Janeiro (1985) with a focus on Reactor Physics and Nuclear Resonances, and a Ph.D. in Nuclear Engineering from the University of Illinois at Urbana-Champaign (UIUC, USA, 1993). At UIUC, he conducted research at the Beckman Institute in neuroscience, robotics, and visuo-motor coordination. He completed a postdoctoral fellowship at the Massachusetts Institute of Technology (MIT, USA, 1998) under the MIT-Harvard research program as a Fulbright Fellow, focusing on Boron Neutron Capture Therapy (BNCT) for brain tumors.

Prof. Campos has extensive experience across multiple areas of Engineering and Nuclear Energy, with current emphasis on materials, radiation science, radioisotopes, radiochemistry, and radiobiology. His research spans radioactive biomaterials, ionizing radiation effects, radiotherapy, brachytherapy, experimental and computational dosimetry, radiopharmaceutical production, development of physical radiology simulators, radiotherapy software, neutron capture therapy, radiobiology, radiochemistry, oncology, nuclear instrumentation, automation, and the design of particle accelerators, isotope transmuters, radioisotope generators, and nuclear reactor cores.

He holds 14 patents and 1 software registered with INPI, with 2 additional invention patents submitted to WIPO, and has developed 4 computational systems. Prof. Campos has published 148 articles in indexed journals, 340 full conference papers, and 34 abstracts, and has supervised 57 master's theses, 21 doctoral dissertations, 2 co-supervisions, and currently has 2 ongoing doctoral supervisions. He has also mentored 13 postdoctoral researchers, including 1 senior postdoc.

Prof. Campos has collaborated with numerous institutions, including departments of Surgery, Biology, Pharmacy, Veterinary, and Medicine at UFMG, as well as radiotherapy centers at Santa Casa de Misericórdia, Hospital Mário Pena, Hospital São Francisco, and CDTN. He has coordinated over 28 research projects funded by government agencies.

He is a member of the editorial boards of the Journal of Radiology (Austin Publishing Group) and the Revista Mineira de Radioterapia. Over 30 years of teaching, he has delivered undergraduate and graduate courses in applied radiobiology, applied radiochemistry, medical imaging concepts, biomedical radiation applications, particle accelerators applied to biomedicine, and radiation protection, within the Radiation Sciences research line focused on engineering and health applications.

Prof. Campos's dedication to education, research, and innovation has left a lasting legacy, inspiring generations of scientists and advancing nuclear science and technology both in Brazil and internationally.

10.2 PROF. DR. TEÓGENES AUGUSTO DA SILVA

We are honored to recognize the outstanding contributions of Prof. Teógenes Augusto da Silva, a distinguished researcher and educator in nuclear physics and medical physics in Brazil. With a career spanning over four decades, Prof. Silva has made significant advances in radiation metrology, radiological protection, and dosimetry.

He graduated in Physics from the Federal University of Rio de Janeiro (1976) and earned his Master's (1981) and Ph.D. in Nuclear Engineering from the Programa de Engenharia Nuclear – COPPE/UF (1996). During his doctoral studies, he was a CNPq sandwich fellow at the National Institute of Standards and Technology (NIST, Gaithersburg, USA) from December 1992 to June 1995.

Prof. Silva served as a senior researcher at the Comissão Nacional de Energia Nuclear (CNEN) from July 1977 until his retirement in September 2023. He was a faculty member in the graduate program at the Nuclear Technology Development Center (CDTN, Belo Horizonte) from March 2015 to September 2023 and served as an adjunct professor at the Postgraduate Program in Nuclear Science and Techniques, DEN/UFMG from March 2000 to June 2017.

His research has focused on Medical Physics, emphasizing radiation metrology, radiological protection, and dosimetry, particularly in the calibration of dosimeters, medical and dental radiodiagnostics, quality control, and radiation safety. He contributed as a Mineiro Researcher (2011–2019) and as a CNPq Productivity Fellowship (PQ 2) awardee (2009–2022). He also served as Vice-Coordinator of the CNPq INCT Radiation Metrology in Medicine project (2008–2017).

Prof. Silva held leadership positions at CDTN, including Chief of the Nuclear and Radiological Safety Division (1988–2010) and Chief of the Reactor and Radiation Division (2014–2018). After his retirement, he continues as a collaborator at CDTN (from October 2024) and currently serves as Executive Secretary and Administrative Director of the INCT for Nuclear Instrumentation and Applications in Industry and Health (INAIS, 2023–present). Through his extensive career, Prof. Silva has left a profound impact on radiation safety, dosimetry, and medical physics in Brazil, mentoring generations of scientists and promoting excellence in research and education.

10.3 PROF. DR. HELEN JAMIL KHOURY - IN MEMORIAM

Prof. Helen Khoury was a leading figure in radiological protection and scientific outreach in Brazil, dedicating over four decades to teaching, research, and development in the nuclear sector. She served as a full professor at the Department of Nuclear Energy (DEN) at the Federal University of Pernambuco (UFPE) and held a Ph.D. in Nuclear Physics from the Pontifical Catholic University of São Paulo.

Her work focused on nuclear dosimetry and instrumentation, with applications in medical imaging, radiation therapy, and radiation metrology. She led the Occupational Radiological Protection Network for Latin America and the Caribbean (Reprolam), fostering professional training and knowledge exchange, and founded the Nuclear Science Museum at DEN/UFPE, a pioneering initiative for popularizing nuclear science in Brazil.

Prof. Khoury's legacy lives on in the countless professionals she trained, the initiatives she helped establish, and the lives she inspired. Her dedication and vision continue to guide the field of radiological protection and nuclear science.

CHAPTER 11

Helen Khoury Award for Best Works

In recognition of excellence in research and innovation, the XXV International Symposium on Solid State Dosimetry is proud to present the Helen Khoury Award. This award honors the legacy of Professor Helen Khoury, whose significant contributions to the field of dosimetry continue to inspire generations of scientists worldwide.

The award is granted in two categories — Panel and Communication — highlighting outstanding contributions in each format. In addition to the winners, the Scientific Committee has selected the five best-ranked works in each category to receive an Honorable Mention, acknowledging the high quality and impact of their research.

11.1 BEST WORK IN COMMUNICATION

ID work code: G02-011
Title of Work: **Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project**
Authors: *Natasha Briggs e Silva, Alexander Camargo Firmino da Silva, Aneli Silva, Sandro Passos Leite, Alexandre Bernardino Pinto Jorge, Camila Salata, Maria Helena Marechal, Ademir Xavier da Silva, Andrea Mantuano Coelho da Silva.*
Presenters: *Natasha Briggs e Silva*

Mini-bio: Master in Nuclear Engineering, graduated from the Military Institute of Engineering (IME). Bachelor's degree in Physics with an emphasis in Medical Physics from Souza Marques College. Occupational Safety Technician from the Celso Suckow da Fonseca Federal Center for Technological Education (CEFET/RJ). Has experience as a teacher in the field of Occupational Safety Technology. Currently working as a Researcher in Dosimetry and Radioprotection at the Laboratory of Radiological Sciences (LCR/UERJ), and pursuing a Ph.D. in Nuclear Engineering at COPPE/UFRJ, focusing efforts on research and development in the field of Dosimetry for HDR brachytherapy.
More info:
`https://www.linkedin.com/in/natasha-briggs-2b722bb3/`
`http://lattes.cnpq.br/2115738852789471`

11.1.1 Honorable Mentions in Communication

ID work code: G05-023
Title of Work: **Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry**
Authors: *Laisa de Almeida Neves, Wander Nascimento de Araújo Júnior, Jonathan Oliveira, Leandro Barbosa*

da Silva, Ademir Xavier da Silva.
Presenters: *Laisa de Almeida Neves*

ID work code: G02-057
Title of Work: **Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP**
Authors: *Francisco Harlley Dantas Hauradou Xavier, Paula Selvatice Pereira Teles, Ademir Xavier da Silva, Mirta Bárbara Torres Berdeguez*.
Presenters: *Francisco Harlley Dantas Hauradou Xavier*

ID work code: G07-002
Title of Work: **Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress**
Authors: *Jhon Brandon Felix Ferreira, Luciana Penedo de Melo Teixeira, Maria Eduarda Gomes Moreira da Silva, Heitor gabriel de Souza Saraiva, Mauricio Moacir da Silva Borges, Bianca Torres Ciambarella, Carlos Eduardo Veloso de Almeida, Samara Cristina Ferreira Machado*.
Presenters: *Jhon Brandon Felix Ferreira*

ID work code: G02-019
Title of Work: **A new formulation of Fricke gel dosimeter development using salicylic acid**
Authors: *Iasmin Vieira Nishibayaski, Anna Luiza Fraga Silveira, Luiz Cláudio Meira Belo*.
Presenters: *Iasmin Vieira Nishibayaski*

11.2 BEST WORK IN PANEL

ID work code: G02-066
Title of Work: **Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6**
Authors: *Daniel de Castro Pacheco, Bruno Melo Mendes, Luiz Cláudio Meira Belo*.
Presenters: *Daniel de Castro Pacheco*

Mini-bio: Graduated in Mechanical Engineering (2020), with over 5 years of experience in the development and coordination of industrial engineering projects for refineries, mining, and steel industries, participating in the preparation of technical and managerial documents, as well as in the coordination and management of people and projects. Holds a Master's degree in Science and Radiation Technology (2023), focused on the development of a computational model for shielding studies in in vivo monitoring, using a CAD tool associated with a Monte Carlo code. Currently a Ph.D. candidate in Science and Radiation Technology at the Nuclear Technology Development Center (CDTN) / National Nuclear Energy Commission (CNEN), working on the development of the dosimetry monitoring and data collection system at CDTN, with an approach based on the Internet of Things (IoT), Internet of Everything, and Digital Twins. Also worked in supporting and developing projects in the field of Information and Communication Technology (ICT), focused on smart cities.
More info:
`https://www.linkedin.com/in/daniel-de-castro-pacheco`
`http://lattes.cnpq.br/0207136434699068`

11.2.1 Honorable Mentions in Panel

ID work code: G02-070
Title of Work: **Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms**
Authors: *Greiciane de Jesus Cesário, Crislane de Jesus Cesário, Linda Viola Ehlin Caldas, Edilaine Honório da*

Silva.
Presenters: *Greiciane de Jesus Cesário*

ID work code: G02-035
Title of Work: **Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms**
Authors: *Lucas Beloti Silva, Lucas Fabricio de Araujo, Telma Cristina Ferreira Fonseca*.
Presenters: *Lucas Beloti Silva*

ID work code: G04-038
Title of Work: **Validation of an alanine-based postal dosimetry system for quality control in radiotherapy**
Authors: *Aricia Ravane Pereira da Cruz, Lucas D. Albino, Ernesto Roesler, Claudio C. B. Viegas, Josemary Angélica Corrêa Gonçalves, Carmen Cecília Bueno, Vinicius Saito Monteiro de Barros, Viviane Khoury Asfora*.
Presenters: *Aricia Ravane Pereira da Cruz*

ID work code: G02-081
Title of Work: **SMS Notification System for Potential Radiological Risk Scenarios**
Authors: *Felipe Mamed de Souza, Alessandra Vaz dos Anjos Santos, Paulo Cézar Rocha Silveira, Wallace Vallory Nunes*.
Presenters: *Felipe Mamed de Souza*

CHAPTER 12

Symposium Papers

Evaluation of Polymeric Materials as Muscle Tissue Equivalents for Physical Phantoms used for Internal Dosimetry Calibration

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Abstract

Internal dosimetry aims to evaluate the activity of radionuclides incorporated by contaminated individuals and calculate the committed absorbed doses in the body tissues and organs. In vivo monitoring is one of the methods employed to estimate the incorporated activity, and it requires accurate calibration of the counting systems. Anthropomorphic physical phantoms containing known activity of radionuclides uniformly distributed in tissue-equivalent materials are used for these calibrations. These tissue-equivalent materials are specifically developed to simulate the interaction of radiation with biological tissues, reproducing key processes such as absorption and scatteringTheir physical properties, particularly mass density and the linear attenuation coefficient (μ), must closely match those of the target tissue to ensure equivalence. This study evaluates epoxy resin, body filler (material used to repair automotive bodywork), and PETG as potential muscle tissue-equivalent materials for producing calibration phantoms. Each material's mass density and linear attenuation coefficient were determined and compared to muscle tissue μ values provided in ICRU Report 44. Three samples with dimensions of 10×10×2 cm³ were prepared for each material. Using a NaI(Tl) detector and a Ra-226 source, measurements in an apparatus developed for μ evaluation were performed over three 24-hour acquisition cycles. Mean counts in characteristic photopeaks were analyzed to determine μ by linear regression. The epoxy resin sample showed a density of 1.12 g/cm³, with a difference of -6.7 % compared to the reference muscle tissue from ICRU (1989). The differences were 12,0 % and -24.4 % for PETG and body filler, respectively. The PETG printed with 85 % filling presented a density more compatible with adipose tissue, with a percentage difference of 2.90 %. The epoxy resin's linear attenuation coefficients (μ) were closer to the theoretical values for muscle tissue, ranging from -7.8 % to 11.9 % for the photon energies evaluated. These results, combined with physical properties such as physical strength, durability, and ease of shaping, indicate that epoxy resin is a promising material to simulate muscle tissue in developing physical phantoms.

Keywords: Tissue-equivalent materials; Linear attenuation coefficient; Phantom; Epoxy resin.

1 Introduction

Internal contamination by radionuclides requires internal dosimetry services to assess the activity incorporated by the contaminated individual. This assessment can be carried out using in vivo or in vitro monitoring. In vivo monitoring systems employ radiation detectors to estimate the incorporated activity at the time of monitoring. HPGe (high-purity Germanium) or NaI(Tl) (Thallium-doped Sodium Iodide) detectors are commonly adopted (IAEA, 2018). These detectors can be configured in different geometries to specifically target an organ or the whole body (IAEA, 1996; Li, 2018; Paquet, 2022).

The in vivo monitoring system requires accurate calibration to reduce uncertainties in calculating the incorporated activity and, consequently, in estimating committed absorbed doses (Dantas, 1998). The calibration must be performed in two steps: i) energy calibration, using point sources to correlate the photon energy with the detector channels; and ii) efficiency calibration, using physical models containing known activity of radionuclides homogeneously distributed in a tissue-equivalent material. The latter allows determining the calibration coefficients used to calculate the body (or organ) activity at the moment the individual was monitored (IAEA, 2018). Physical phantoms with anthropomorphic and anthropometric characteristics have been developed to calibrate in vivo systems. The aim is to minimize discrepancies between the geometry and composition of the phantom and those of the monitored individual, thus increasing the method's accuracy (Andrade et al., 2024; Dantas, 1998; IAEA, 1996).

Tissue-equivalent materials used to fill these models are characterized by their ability to mimic the interaction of radiation with biological tissues (ICRU, 1989; Khaidir et al., 2024). The linear attenuation coefficient and the density of the tissue-equivalent materials should be similar to those of the tissue of interest (ICRU, 1989). The linear attenuation coefficient (μ) is the probability of a photon interacting with a given material per unit distance. Thus, it is an essential parameter to characterize the interaction of radiation with the studied material (Knoll, 2010; Turner, 2007).

ICRU 44 introduces several polymer-based materials that can be used as tissue-equivalent materials. Studies on developing and characterizing these materials have been conducted in recent years. Andrade et al. (2025) studied ballistic gelatine as a muscle substitute, comparing the experimentally obtained linear attenuation coefficients (μ) with theoretical values provided by ICRU 44 (ICRU, 1989). Silva et al. (2021) evaluated the μ of several materials, including filaments used for 3D printing such as PLA, TPU, and ABS, comparing them with the μ values of muscle tissue provided by ICRU 44 and NIST/XCOM (National Institute of Standards and Technology, U.S. Department of Commerce, Gaithersburg, Maryland, USA) (ICRU, 1989; NIST, 2020).

Among various materials, epoxy resin has been studied as a tissue substitute in several areas of radiation metrology, usually applied in physical phantoms for calibration and dosimetry (ICRU, 1989; Alshipli et al., 2018; Ashour et al., 2024; Khaidir et al., 2024; Khallouqi et al., 2024; Shinde et al., 2025). Its mechanical strength and durability (Balguri et al., 2021; ICRU, 1989; Shinde et al., 2025) favor its use in phantom production, ensuring good durability of the models. The epoxy resin also has the advantage of high geometric accuracy, which is ideal for applications that require accurate reproduction of the body geometry (ICRU, 1989; White et al., 1977). Indeed, ICRU 44 (ICRU, 1989) recommends using epoxy resin to construct physical phantoms. Additionally, studies such as Jones et al., (2003) used this material to develop tissue substitutes with different densities, accurately representing specific tissues such as soft tissue and lung tissue for newborns, children, and adults. There are two types of epoxy resin based on the application's thickness: i) thin-pour epoxy, used for thin layers or small molds; and ii) thick-cast epoxy, designed for high volume applications. The latter is most suitable for producing anthropomorphic phantoms using large molds.

Body filler, in turn, is a modified resin composed of mineral fillers and additives (McNorton et al., 2008). This composition allows the modification of specific physical properties such as adhesion, mechanical strength, and density (Subramanian, 2017). The high viscosity of this material represents an interesting characteristic for avoiding the decantation of solid radioactive powders and maintaining the filling homogeneity. In this study, body filler will be evaluated as an alternative in constructing physical models, possibly representing a complementary option to epoxy resin.

Materials used in 3D printing have also been studied for manufacturing physical phantoms since many of these plastics have tissue-equivalence properties (Higgins et al., 2022; Jreije et al., 2021; Laurikaitiene et al., 2024; Silva et al., 2021a; Tino et al., 2019). The additive manufacture is being used in the production of 3D phantoms for quality control (Filippou and Tsoumpas, 2018; Laurikaitiene et al., 2024), as well as in the development of anthropomorphic and anthropometric phantoms for calibration of in vivo systems (Andrade et al., 2024; Sales et al., 2025; Silva et al., 2021b). Among the various materials used in 3D printing, the PETG (polyethylene terephthalate glycol) shows good mechanical strength. It also offers good chemical and heat resistance, ease of printing, and high-quality surface finish. This study will investigate this material to compare its tissue-equivalence properties with muscle tissue.

Thus, considering the need of tissue-equivalent materials for the production of phantoms for calibration of in vivo monitoring systems, the objective of this work was to study thick-cast epoxy resin, body filler, and PETG (polyethylene terephthalate glycol) as possible tissue-equivalent materials for this purpose. Each material's density and linear attenuation coefficient will be calculated and compared with the reference data for muscle tissue provided by ICRU 44 (ICRU, 1989).

2 METHODOLOGY

Sample preparation

Three samples measuring 10×10×2 cm^3 (**Figure 1**) were prepared for each material evaluated: body filler, thick-cast epoxy resin, and the 3D printing filament PETG. All samples were produced according to the manufacturer's specifications. The PETG samples were printed with 85% infill so that the density of the polymer (1.25 $g.cm^{-3}$) would be closer to the density of human soft tissues (~1.04 $g.cm^{-3}$).

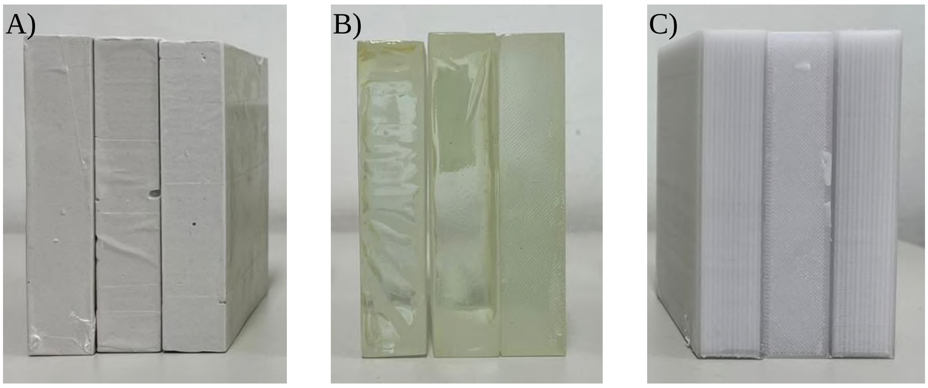

Figure 1 – Samples of the materials studied: A) body filler, B) epoxy resin, and C) PETG (85%).

Mass density calculation

The mass and volume of each sample were measured to allow the calculation of the materials' density. The dimensions of the samples were measured using a caliper. The sample was subsequently weighed with a precision balance. After these measurements, it was possible to calculate the density of the evaluated materials using the following equation:

$$\rho = \frac{Mean\ of\ the\ samples\ mass}{Mean\ of\ the\ samples\ volume} \tag{1}$$

Linear attenuation coefficient calculation

The Lambert-Beer Law (**Equation 2**) describes how the intensity of a radiation beam decreases exponentially as it passes through a material.

$$\frac{I}{I_0} = e^{-\mu x} \tag{2}$$

where, I_0 represents the intensity of the incident beam; I is the intensity of the transmitted beam after passing through the material, and the ratio (I/I_0) defines the transmittance; μ corresponds to the linear attenuation coefficient; and x to the thickness of the sample.

Thus, Lambert-Beer Law can be rewritten to calculate the linear attenuation coefficient, μ, according the **Equation 3**:

$$\mu = -\frac{ln\left(\frac{I}{I_0}\right)}{x} \quad (3)$$

The next step involved exposing the samples to a radiation beam generated by a Ra-226 source positioned 15 cm from a NaI(Tl), a scintillation detector, to obtain transmittance data. The setup (**Figure 2**) includes lead shielding to reduce the background radiation and an 8 mm hole for the source collimation to obtain the photon beam. The Ra-226 source has a broad energy range emission, with main peaks at 186.1, 241.9, 295.2, 351.9, 609.3, 1759, and 2204.1 keV. However, the 241.9 keV energy was excluded from the spectral evaluation, since backscattering peaks from higher energies interfere with the counts recorded in this range, compromising the analysis (Andrade et al., 2025). The Ra-226 wide energy range allows the evaluation of the behavior of the studied materials in the full scope of the photon emissions of the majority of the radionuclides possibly involved in human internal contamination.

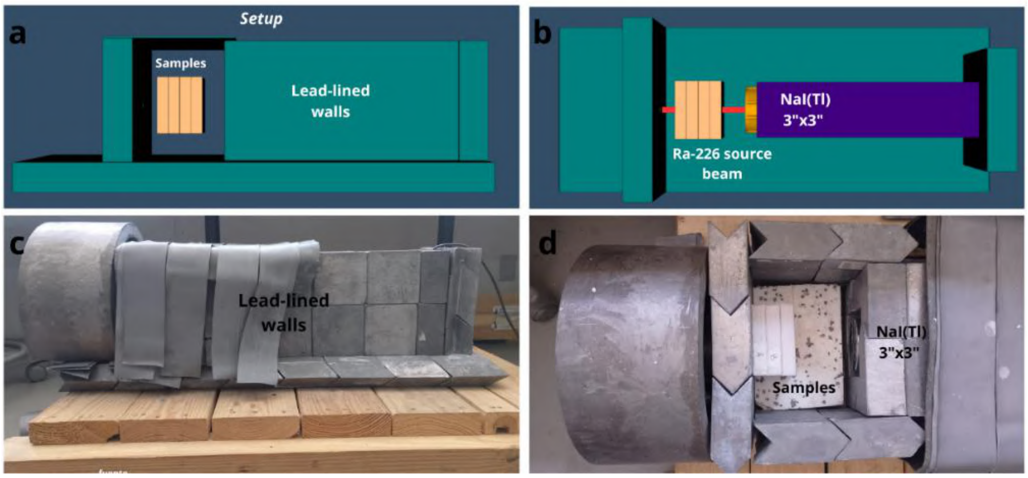

Figure 2 – Experimental setup. A and B) Schematic illustrations of the experimental setup show the samples' positioning, the detector, the Ra-226 source beam, and the surrounding lead shielding structure. C and D) Photographs of the experimental setup, highlighting the positioned samples and the lead shielding.

The NaI(Tl) detector was used in spectrometer mode to record the spectra of the Ra-226 beam (**Figure 3**) as the samples were added to the setup. The 3"x3" NaI(Tl) detector has high counting efficiency, although its energy resolution is relatively low, ranging between 5% and 10% (Dantas, 1998; Knoll, 2010). Since the Ra-226 emissions are well known, the low energy resolution does not considerably impair the analysis.

Initially, the spectrum was acquired without any sample to obtain I_0. Subsequently, acquisitions are made, adding samples individually, increasing the material thickness, and providing the corresponding I values. The acquisition time was set to 24 hours. This extended time improved the analysis of the spectrum, as it contributes to better counting statistics and lower noise. Three 24h acquisitions were made without any sample (I_0), which will be used in transmittance calculations for all samples. Three 24h acquisitions were made for each material thickness, totaling nine acquisitions for each material. The acquired spectra were analyzed with the Genie 2000 Spectroscopy Software (Canberra, 2006). Regions of interest (ROI) were positioned in the main photopeaks. The peak areas were registered. The photopeak area corresponds to the net counts in the ROI, i.e., background counts were removed. The peak areas obtained for spectra acquired with no samples in the setup represent I_0 values. The peak areas registered with different sample thickness, x, represent the beam intensity, I, related to the corresponding thickness of the analyzed sample. These values were used to calculate the mean transmittance (I/I_0) for each material thickness.

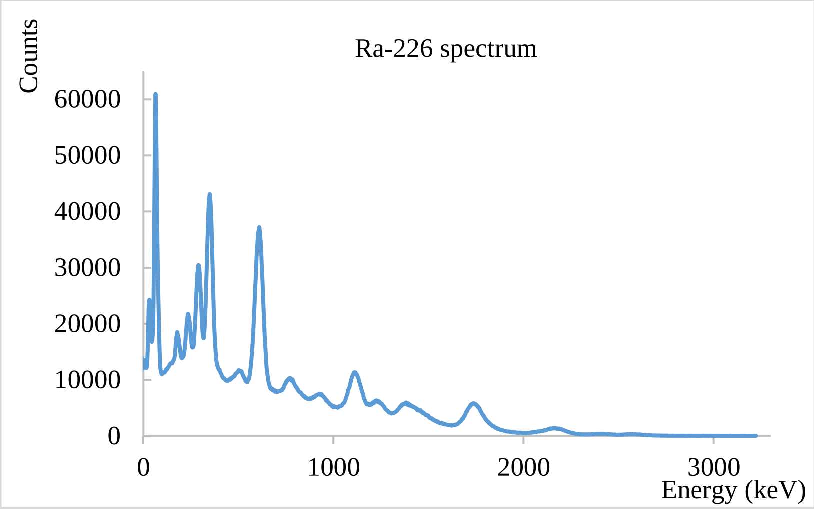

Figure 3 – Emission spectrum of radium-226

A curve of the natural logarithm of transmittance, $\ln(I/I_0)$, as a function of sample thickness was obtained for each material. A linear regression was made for each curve to estimate μ (which corresponds to the slope of the line) and its uncertainty. The linear attenuation coefficient values calculated for ICRU 44 muscle tissue composition were adopted as the theoretical reference for comparison with the experimental values obtained for the analyzed materials.

3 RESULTS

The density of the materials is a relevant factor in the process of beam hardening, since denser materials promote greater attenuation of ionizing radiation, thus reducing transmittance and increasing the values of μ (Turner, 2007). The density of the studied materials, calculated experimentally, was compared with that of muscle tissue, which is 1.05 g/cm³ according to the ICRU (1989). The ρ value calculated for the epoxy resin was 1.121 ± 0.003 g/cm³, resulting in a difference of -6%, being the closest to the density of the reference tissue. The density value of the epoxy resin was in agreement with the values described by M. Alshipli et al. (2018), who identified an average density of 1.11 g/cm³. Furthermore, the ICRU (1989) states that the density of epoxy resin can vary between 1.05 g/cm³ and 1.15 g/cm³, depending on its composition.

The differences were 12% for PETG (0.922 ± 0.008 g/cm³) and -24.36 % for body filler (1.305 ± 0.016 g/cm³). The body filler showed a density too high to be used as a muscle substitute tissue. On the other hand, PETG, printed with 85% infill, showed a density close to the ρ value of adipose tissue, which is 0.95 g/cm³ (ICRU, 1989). Under this condition, PETG had a percentage difference of 2.90 % compared to the density of adipose tissue.

The values of the linear attenuation coefficients obtained for the analyzed materials are presented in **Table 1**. It is noted that, for the energy range considered, the μ values of the epoxy resin are closer to the reference value, with a difference ranging from 12 % to -7.8 %. For the other materials, the difference ranged from 17 % to 3.2 % for body filler and from -18 % to -35 % for PETG. The linear attenuation coefficient depends on several properties of the material, including density, chemical composition, atomic number of the constituent elements, and the energy of the incident photons. This coefficient reflects the probability of occurrence of radiation interactions with matter, such as the photoelectric effect, Compton scattering, and pair production (Knoll, 2010; Turner, 2007).

Energy (keV)	Muscle	Body Filler		Epoxy Resin		PETG(85%)	
	μ (cm^{-1})	μ (cm^{-1})	Dif.	μ (cm^{-1})	Dif.	μ (cm^{-1})	Dif.
186.1	0.146	0.170 ± 0.002	17%	0.163 ± 0.003	12%	0.120 ± 0.005	-18%
295.2	0.124	0.136 ± 0.005	10%	0.130 ± 0.005	5.2%	0.096 ± 0.006	-22%
351.9	0.116	0.125 ± 0.001	7.9%	0.113 ± 0.001	-2.7%	0.088 ± 0.005	-24%
609.3	0.093	0.102 ± 0.001	11%	0.096 ± 0.001	4.4%	0.072 ± 0.001	-21%
1120.3	0.070	0.073 ± 0.001	6.1%	0.072 ± 0.001	4.9%	0.054 ± 0.001	-21%
1759	0.055	0.056 ± 0.002	3.2%	0.055 ± 0.002	1.5%	0.038 ± 0.003	-30%
2204.1	0.049	0.051 ± 0.002	5.8%	0.044 ± 0.002	-7.8%	0.031 ± 0.004	-35%

Table 1 – Values of the linear attenuation coefficient for the three studied materials compared to muscle tissue. The relative difference (Dif.) was calculated by: $(\mu_{material} - \mu_{Reference})/\mu_{Reference}$.

The graph in **Figure 2** illustrates the linear attenuation coefficient as a function of the energy obtained for the Ra-226 photons. It was observed that both epoxy resin and body filler show values close to ICRU muscle tissue curve. However, epoxy resin shows a more significant coincidence. Thus, thick-cast epoxy resin proved to be the best material as tissue equivalent among those studied in this work.

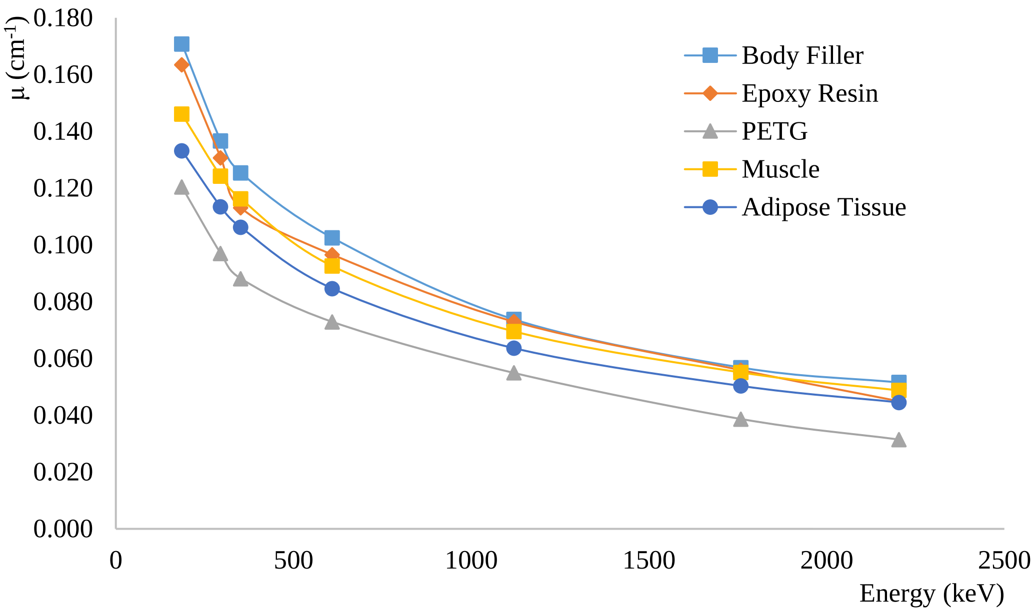

Figure 2 – Relationship of the linear attenuation coefficient for the energies of 186.1, 295.2, 351.9, 609.3, 1759, and 2204.1 keV, characteristic of Ra-226, comparing the studied materials to muscle tissue.

Considering the physical properties of epoxy resin, such as mechanical strength, durability, and high geometric precision, the results obtained from the analysis of the linear attenuation coefficient and density indicate that thick-cast epoxy resin could be used as a substitute for muscle tissue in the manufacturing of phantoms. Also epoxy resin presents itself as a good option when there is a need to produce anthropomorphic models, due to its ability to faithfully reproduce shapes and geometric details.

4 Conclusions

This study evaluated thick-cast epoxy resin, body filler, and PETG as potential substitute materials for muscle tissue, based on the analysis of density and linear attenuation coefficient. The results showed that epoxy resin presented the smallest percentage difference compared to the reference muscle tissue density (ICRU 44), in addition to being consistent with values previously reported in the literature. For the energy range considered, the μ values of epoxy resin were closer to the reference value. Its ability to replicate anthropomorphic shapes with high fidelity supports its potential as a tissue-equivalent material in clinical and research contexts.

PETG, printed with 85% infill, on the other hand, showed density close to that of adipose tissue, making it a viable alternative for phantoms that require equivalence to this type of tissue. Body filler, due to its significantly higher density, proved inadequate as a substitute for muscle tissue; however, the linear attenuation coefficient still showed good correspondence with that of muscle tissue.

In this way, the production of new physical phantoms for the calibration of *in vivo* systems and other applications within the field of ionizing radiation metrology depends on the study of new tissue-equivalent materials. Thus, the evaluation of these materials as tissue substitutes contributes to increasing the accuracy and precision of techniques that use physical phantoms. In the future, we intend to study the density and attenuation coefficient of PETG filaments with different infill percentages in order to obtain one that best resembles the behavior of soft tissues such as muscle. We also plan to study the radiological behavior of different resins and filaments.

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Evaluation of the reproducibility of a new lens dosimeter holder

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Abstract

This article presents the reproducibility assessment of a newly designed 3D-printed lens dosimeter holder, specifically developed to monitor radiation dose to the crystalline lens during interventional medical procedures. The motivation for this study arises from growing concern over the increased incidence of cataracts among radiology professionals—such as technologists, radiologists, and interventional cardiologists—raising suspicions of a direct link to the rising demand for fluoroscopy-guided procedures. Although best practice recommendations exist (e.g., proper use of personal protective equipment, adequate beam collimation, and minimizing unnecessary exposure), many professionals still lack adequate training in radiobiology and radiation safety, remaining unaware of the risks involved.Dosimeters are mandatory passive devices for occupationally exposed individuals (OEIs), traditionally placed at chest level, with monitors both above and below the lead apron. However, supplementary dosimeters, especially for the extremities and ocular lenses, have gained importance following the revision of dose limits in 2012, which reduced the annual allowable exposure to the lens from 150 mSv to 20 mSv. With increasing cataract cases among OEIs in Brazil, regulatory agencies are expected to make ocular dosimeters mandatory for interventional procedures. To ensure measurement reliability, the dosimeter must be calibrated to the Personal Dose Equivalent – Hp(3), measured at a depth of 3 mm. The dosimeter holder—designed using Fusion 360 and fabricated via additive manufacturing with PLA on a Raise Pro2 3D printer—was chosen for its density (~1.24 g/cm^3), which approximates that of soft biological tissue, enhancing physical representativity in simulated scenarios. Reproducibility tests employed two phantom models: a 3D-printed anthropomorphic head phantom and a cylindrical PMMA phantom filled with water (as per ISO 4037-3). Positioned 2.5 meters from a Cs-137 source, each phantom received a dose of 10 mSv over five irradiation cycles, maintaining a coefficient of variation (CV) below 7.5%. Results showed high reproducibility, with relative uncertainties of 16.60% for the anthropomorphic phantom and 15.63% for the cylindrical one. CVs ranged from 5.33% to 7.11% and 6.53% to 7.20%, respectively, confirming greater measurement stability with the anthropomorphic phantom. Overall, the 3D-printed holder offers a reliable, adaptable, and cost-effective solution for clinical implementation, with strong potential to enhance radiological protection in interventional settings.

Keywords: Dosimeter holder; Dosimetry; Hp(3); Printing 3D;

1 Introduction

It is questionable whether the increase in medical requests for interventional procedures is directly related to the rise in cases of cataract development among radiologic technologists (Fish et al., 2011), radiologists, and interventional cardiologists (Vano et al., 2010). Such uncertainty stems from good practice situations, such as: correct use of personal protective equipment (PPE), proper collimation, and avoiding unnecessary exposure. Considering best practices, a lead interventionalist may perform 970 angioplasty procedures and 140 abdominal aortic aneurysm treatments without exceeding the annual dose limits for the eye lens (Moura & Bacchim Neto, 2015). These interventional radiology techniques (guided by fluoroscopy) have been employed by professionals who often have not received adequate training in radiation safety or radiobiology, remaining unaware of the potential injury risks to which they and their teams may be exposed (Valentin, 2000).

The dosimeter is a passive device for personal monitoring and is mandatory for occupationally exposed individuals (OEIs). It should be positioned at chest level, with one monitor worn underneath the lead apron and another worn over the lead apron (Valentin, 2000), However, additional dosimeters can be used to monitor the doses received in the extremities and in the eye lenses (Whitby & Martin, 2005). In 2012, the annual dose limit for the eye lens was reduced from 150 mSv to 20 mSv (ICRP, 2012), This update was the result of epidemiological studies that suggest the development of cataracts can occur after exposure to doses significantly lower than those previously assumed (Barbosa et al., 2019).

With the considerable increase in cataract cases among occupationally exposed individuals (OEIs) in Brazil, the use of eye lens dosimeters in interventional applications will soon become mandatory. To ensure metrological reliability, these dosimeters must be calibrated to a relevant quantity, with the Personal Dose Equivalent – Hp(3) being used for lens dosimetry, where monitoring is performed at a depth of 3 mm (Petoussi-Henss et al., 2020). These metrological reliability tests consist of evaluations such as: response homogeneity, energy and angular dependence of the detectors, reproducibility of results, and dose-dependent linearity, for example. The purpose of these tests is to validate the use of the dosimeter in various scenarios.

The dosimeter holder is a device that forms part of the dosimetric system, consisting of the holder and the detector. This holder can come in various shapes to meet the needs of the specific field in which it will be used. A range of materials are suitable for manufacturing dosimeter holders, such as polylactic acid (PLA), a thermoplastic material commonly used in 3D printing (Diamantopoulos et al., 2018). 3D printing technology has been increasingly influential across various fields of engineering, medicine, and research, making it possible to develop anthropomorphic head phantoms with different densities and Hounsfield Unit (HU) scales that closely resemble those of a real person (M. B. M. B. Savi, 2022). This study aims to validate the reproducibility test of a 3D-printed dosimeter holder for eye lens monitoring.

2 Materials and Methods

2.1 LiF:Mg,Ti

A total of 32 thermoluminescent detectors (TLDs) made of lithium fluoride doped with magnesium and dysprosium (LiF:Mg,Ti) in chip format (3.2 × 3.2 × 0.89 mm), commercially known as TLD-100, were used. LiF:Mg,Ti is a material widely used in personal dosimetry. The detectors were divided into two groups with 16 samples each. Table 1 presents the organization of the groups.

Table 1: Organization of TLD Groups.

Groups	Number of Samples	Phantom Used
Group 1	16	Anthropomorphic
Group 2	16	Cylindrical

2.2 Thermoluminescent Reading and Thermal Treatment

The measurements were conducted using a Harshaw Model 4500 reader (Figure 1), which is equipped with a heating and cooling system capable of being adjusted to accommodate various types of materials.

Figure 1: Harshaw 4500 thermoluminescent reader.

For the thermal reuse treatment, a Vulcan Model 3-550PD furnace was employed (Figure 2). The thermoluminescent dosimeters (TLDs) were placed on an aluminum plate and inserted into the furnace. This thermal procedure was divided into two distinct cycles:

Figure 2: Vulcan Model 3-550PD furnace

I. 400 °C for 1 hour – rapid cooling by thermal shock: following removal from the furnace, the aluminum plate containing the TLDs was placed on a surface maintained at approximately -15 °C. Once the temperature had stabilized, the second treatment cycle was initiated; and

II. 100 °C for 2 hours – cooling at ambient temperature: after removal from the furnace, the aluminum plate with the TLDs was cooled under room-temperature conditions).

2.3 Sketch of the dosimeter holder

The design of the dosimeter holder was developed using Fusion 360 software, which offers functionalities for sketch creation (Figure 3) and 3D object modification. The main part of the holder features a rectangular shape with a curvature specifically designed to fit directly onto the eyeglass temple, positioning the detector near the crystalline lens. The holder includes two openings to accommodate up to two chip-format TLDs, with dimensions of up to 4.8 × 4.8 × 0.98 mm. A secondary component serves as a locking mechanism, securing the dosimeter firmly within the support and incorporating a slot that facilitates its removal.

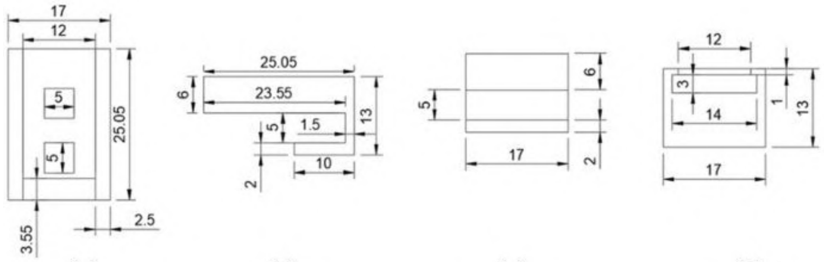

Figure 3: Sketches of the dosimeter holder — (a) top view, (b) side view, (c) rear view, and (d) front view.

2.4 3D Printing

The samples were fabricated through additive manufacturing using a Raise Pro2 3D printer (Figure 4). The operational parameters employed in the process are detailed in Table 1. The equipment operates based on the principle of Fused Filament Fabrication (FFF), a technique in which thermoplastic filament is heated in an extruder and subsequently deposited in a controlled manner onto the build platform. The deposition occurs in successive layers, which solidify under ambient conditions, resulting in the formation of the final geometry of the part.

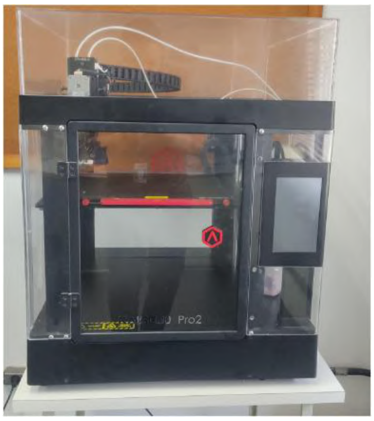

Figure 4: 3D printer.

Table 2: General printing settings for the dosimeter holder

Heated bed temperature	60 ºC
Extruder temperature	190 ºC
Printing speed	50.00 mm/s
Infill density	15 %
Infill speed	80.00 mm/s

The dosimeter holder (Figure 5) was fabricated using 3D printing technology with polylactic acid (PLA) thermoplastic, employing a filament diameter of 1.75 mm. The selection of PLA was based on its specific density (≈1.24 g/cm³), which closely approximates the properties of soft biological tissues. This characteristic enables enhanced physical representativeness in simulated and experimental applications designed for dosimetric purposes.

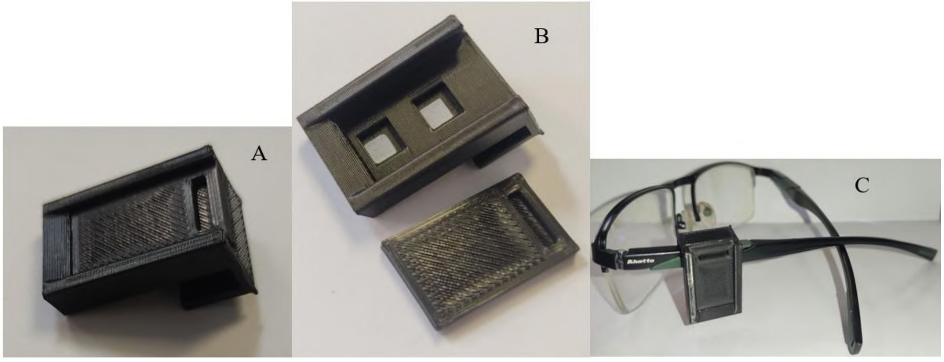

Figure 5: (a) dosimeter holder with fixation support; (b) open dosimeter holder with TLDs positioned and fixation support placed alongside; and (c) dosimeter holder mounted on the eyeglass temple.

2.5 Phantoms

Two phantoms were used during the reproducibility tests: the first is an anthropomorphic phantom of the head region, fabricated using 3D printing for educational purposes (Figure 6a); the second phantom is made of polymethyl methacrylate (PMMA), featuring a cylindrical shape filled with water (Figure 6b). This latter phantom is recommended by ISO 4037-3(ISO, 2019). The anthropomorphic head phantom was developed based on the reference anatomy provided by the CIRS 711 Atom Max model (M. Savi et al., 2024).

Figure 6: (a) Anthropomorphic phantom of the head region; and (b) cylindrical phantom made of PMMA.

The dosimeter holders were positioned at a distance of 2.5 meters from the radiation source (Figure 7).

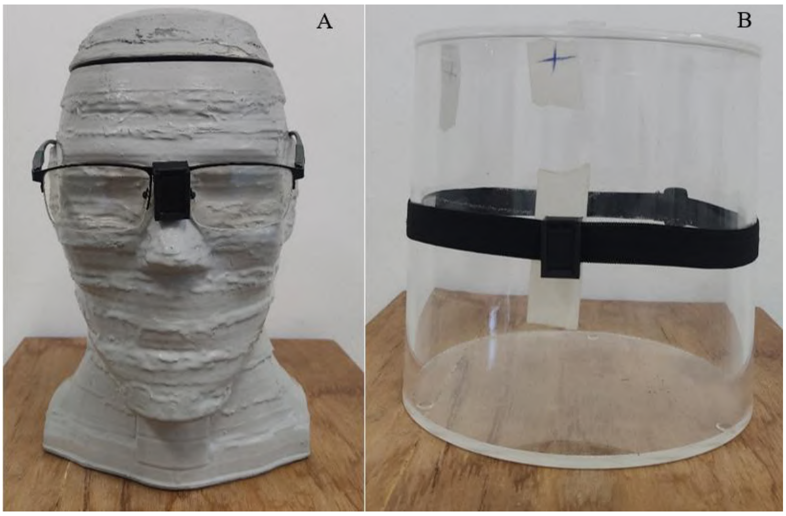

Figure 7: (a) Dosimeter holders mounted on eyeglasses and positioned on the anthropomorphic head phantom; (b) dosimeter holders secured to an elastic band and placed on the cylindrical phantom.

2.6 Reproducibility Test

Reproducibility testing is essential to ensure that the dosimetric system yields consistent results. In this evaluation, two dosimeter groups were positioned on their respective phantoms (Group 1: anthropomorphic phantom; Group 2: cylindrical phantom) and irradiated with a dose of 10 mSv using a Cs-137 irradiator system. The irradiation setup was carefully arranged to minimize uncertainties related to positioning factors. The procedures for irradiation, reading, and thermal treatment were carried out across five repeated cycles. The coefficient of variation (CV) must not exceed 7.5%, as defined in Equation 1.

$$CV = \frac{s_i}{\bar{G}_i} \leq 7{,}5\% \quad (1)$$

Where CV represents the coefficient of variation, s_i is the standard deviation, and $\bar{G}_i$ denotes the mean value corresponding to a 95% uncertainty level

3 Results and Conclusions

Figure 8 presents the data obtained from the reproducibility assessment of irradiation cycles using a new dosimeter holder model positioned on two types of phantom objects: anthropomorphic and cylindrical. The mean dose values (in mSv) for each cycle were compared. The relative uncertainties were 16.60% and 15.63% for the anthropomorphic and cylindrical phantoms, respectively.

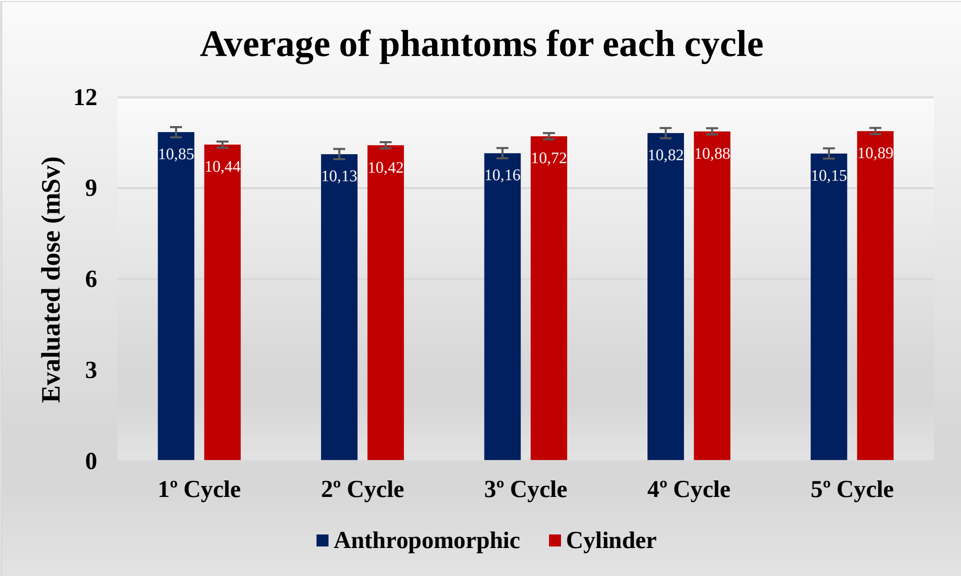

Figure 8: Evaluation of the mean dose per irradiation cycle for each phantom.

It can be observed that the variation across cycles is minimal, indicating good reproducibility of the results. In certain cycles (namely the third and fifth), the cylindrical phantom exhibited slightly higher dose values compared to the anthropomorphic phantom.

Figure 9 presents the results obtained for the coefficient of variation, as defined in Equation 1, over five repetitions. This metric served as an indicator of the relative dispersion among the measurements, reflecting the consistency of the values obtained in each cycle. The comparison between the anthropomorphic and cylindrical phantoms provides insight into the performance of the newly developed 3D-printed dosimeter holder.

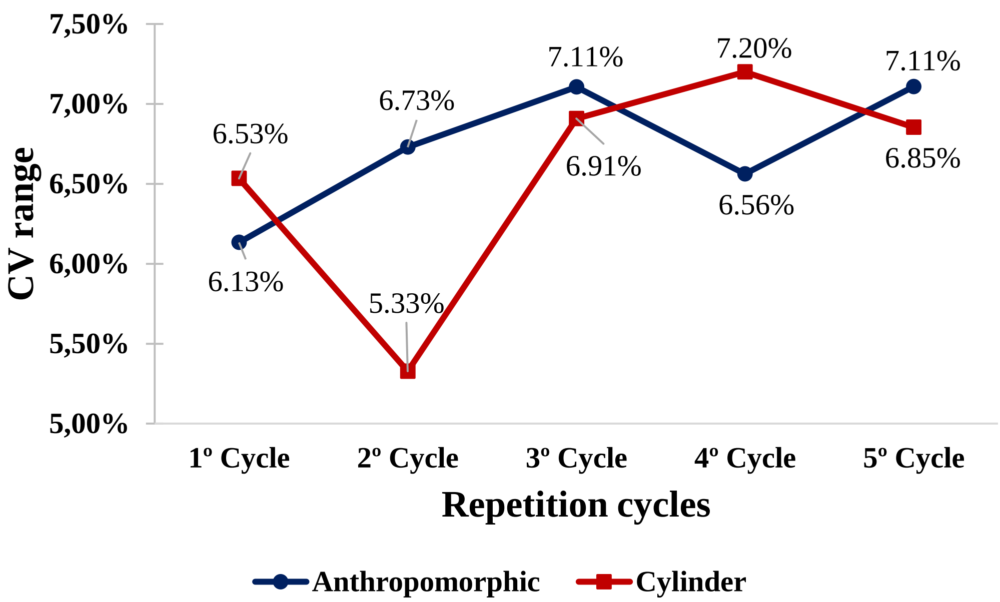

Figure 9: Results of the coefficients of variation

The data revealed that:

- The anthropomorphic phantom presented CV variations ranging from 5.33% to 7.11%, with greater dispersion observed in the third and fifth cycles;
- The cylindrical phantom exhibited values between 6.53% and 7.20%, with the highest dispersion occurring in the fourth cycle.

This difference suggests greater measurement stability with the anthropomorphic phantom, likely due to its geometry more accurately reflecting human anatomical structures, which facilitates dosimeter placement and reduces effects of geometric variability

The results demonstrate high reproducibility in dose measurements using the newly developed 3D-printed dosimeter holder, validating its application for crystalline lens monitoring. The consistency of mean values across cycles confirms the reliability of the process, while uncertainties, though present, remain within acceptable limits for experimental dosimetry studies.

Subtle differences observed between the phantoms can be attributed to their distinct geometric and material characteristics, which influence radiation distribution within the simulated structures. The use of 3D printing technology for the fabrication of the dosimetric support represents an accessible, reproducible, and adaptable alternative for personalized clinical settings, with potential to enhance radiological protection in interventional procedures.

Acknowledgements

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Experimental Evaluation of the Mechanical and Dosimetric Properties of the Red Perspex 4034 Dosimeter After 15 Years of Aging

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Abstract

The Red Perspex 4034 dosimeter is widely used in irradiation laboratories and industrial radiation sterilization processes (Ladeira, 2016; Olejnik, 1977). The shelf life of unirradiated dosimeters is ten years from date of release if stored at a temperature of 20±5°C and provided the sachet seal is intact (MDS Nordion 2021). Several tests have been proposed to investigate whether the dosimeters' shelf life can be extended under appropriate storage conditions. In the present study, dosimeters from a 15-year-old batch were evaluated for their radiation response and mechanical properties and compared with a more recent batch (five years old). Dosimetric tests were conducted in triplicate, using dosimeters irradiated with gamma radiation in 5 kGy increments throughout the calibrated range of 5 to 50 kGy. Readings were obtained by spectrophotometry at various post-irradiation time intervals. Mechanical tests assessed the structural integrity of the dosimeters through bending and compression tests along all axes. The resulting data were compared, and preliminary results indicate that even five years beyond their estimated shelf life (i.e., 15 years post-manufacture), the dosimeters exhibited no significant degradation in performance when compared to the five-year-old batch. This suggests that they can still be widely used without restrictions or adjustments, supporting potential cost savings and reduced disposal due to storage time limitations.

Keywords: Red Perspex 4034; Shelf life; Gamma-rays; Dosimetry;

1 Introduction

Ionizing radiation is widely used in various fields such as medicine, industry, and research due to its ability to modify materials, eliminate microorganisms, and detect structural failures in metallic and polymeric components. In industrial and medical contexts, radiation dosimetry is essential to ensure that irradiation processes are conducted safely and effectively. For this purpose, different types of dosimeters are employed, with polymethyl methacrylate (PMMA) polymers—such as Red Perspex 4034 dosimeters—being widely used for high-dose radiation measurements. (Mesquita, 2023)

Red Perspex 4034 dosimeters operate based on the alteration of their optical properties when exposed to ionizing radiation, becoming progressively darker as they absorb increasing doses. This color change is quantified through UV-VIS spectrophotometry, allowing a correlation to be established between absorbance and absorbed dose. However, the long-term stability of these dosimeters remains uncertain, as the manufacturer guarantees their validity for a period of ten years. During this timeframe, factors such as environmental storage conditions, exposure to temperature and humidity fluctuations, and potential variations between manufacturing batches may influence the dosimetric response, thereby affecting both accuracy and reliability (MDS NORDION, 2021).

In this context, it becomes necessary to evaluate the stability and efficiency of Red Perspex 4034 dosimeters over time, particularly to determine whether a batch acquired more than ten years ago maintains a response comparable to that of a recently manufactured batch. This investigation is crucial to ensure the reliability of routine dosimetry systems in industrial applications, thereby avoiding errors in absorbed dose quantification and ensuring that irradiation processes comply with safety and quality standards established by international regulations.

Additionally, the latency of the recorded information—i.e., the stability of absorbance over time after irradiation—is a critical factor for the reliability of measurements. Should the dosimetric response change significantly after a certain period, the accuracy of routine dosimetry could be compromised, necessitating the implementation of periodic calibration procedures and rigorous quality control strategies.

Another relevant aspect is the long-term resistance of the dosimeter material. Although PMMA is considered a stable polymer, it may undergo structural degradation due to environmental humidity, oxidation, and exposure to thermal fluctuations. These processes can affect the homogeneity of the dosimetric response and introduce uncertainties in measurements, particularly when dosimeters are used after prolonged storage periods.

Given this scenario, it is essential to investigate the long-term reliability of Red Perspex 4034 dosimeters, assessing whether batch-to-batch variation compromises their application in high-dose dosimetry, or whether a dosimeter, if properly stored, may still be reliably used beyond the expiration period stated by the manufacturer.

2 Bibliographic Review

2.1 GammaBeam-127 Irradiator

The Gamma Irradiation Laboratory (LIG) at the Nuclear Technology Development Center (CDTN) uses a multipurpose panoramic irradiator, classified as Category II, model IR-214, of the GammaBeam-127 (GB-127) type, developed by MDS Nordion. This irradiator operates with a dry-stored cobalt-60 (Co-60) source, consisting of twelve capsules, each containing approximately 183.2 TBq (5.0 kCi) of metallic Co-60. The capsules are enclosed in stainless steel tubes with double encapsulation and housed in a source holder, which is stored in a container model F-127. This container is licensed by Canadian regulatory authorities and complies with guidelines established by the International Atomic Energy Agency (IAEA), ensuring high safety and quality standards (Silva, 2023).

To ensure operational safety, when the source is in its rest position inside the container, the radiation dose rate in the irradiation room remains below 20 μSv/h at one meter from the container. The elevation of the source to the irradiation position is performed using a system with a stainless-steel cable, which moves along the ceiling of the irradiation chamber, driven by a pneumatic cylinder. This mechanism allows for precise control of source exposure, ensuring the irradiation process occurs safely and reliably (Silva, 2023).

2.2 Rotating Tables

During the irradiation process, items are placed on rotating tables arranged around the source to ensure uniform exposure. The distance between the products and the source can be adjusted to modify radiation dose rates, enabling simultaneous irradiation of different products with specific doses within the irradiation chamber (Silva, 2023).

2.3 Red Perspex 4034 Dosimeters

The Red Perspex 4034 dosimeter is a type developed in the 1960s, using a technology that remains efficient and widely applied today. This dosimeter is made of dyed polymethyl methacrylate (PMMA), a material that darkens upon radiation exposure due to a proprietary dye. The dosimeters are irradiated while still sealed in their individual pouches and are opened and read post-irradiation using a spectrophotometer. They are commercially available in boxes containing 1,000 units and are designed to measure high radiation doses in industrial irradiation processes. The color change of the Perspex enables precise radiation dose quantification (STERIS AST).

The dose range of the Red Perspex 4034 dosimeter varies depending on the measuring equipment and environmental conditions. Under controlled conditions, it can measure doses from 5 to 50 kGy, making it suitable for most industrial irradiation processes. Dose identification and measurement are carried out using an electron paramagnetic resonance (EPR) spectrometer. This dosimeter exhibits high reproducibility, with a coefficient of variation (CV) of up to 2.0% in response to irradiation, achievable within a uniform radiation field of ±1% across the calibrated 5–50 kGy range (STERIS AST).

Each batch of Red Perspex dosimeters undergoes third-party calibration before commercialization to ensure accuracy. When stored at 20±5°C with the pouch intact, unirradiated dosimeters have a shelf life of up to ten years. After irradiation, they should be read within two days to maintain measurement accuracy. Measurements are taken with a coverage factor of K=2, offering approximately 95% coverage probability. Each dosimeter has dimensions of 30 mm × 11 mm and a thickness of 3.00 ± 0.55 mm, and they are individually packaged in laminated pouches. In rare cases, a pouch may contain two dosimeters, which does not affect performance and allows one to be discarded or both readings to be averaged (STERIS AST).

2.3 UV-VIS Spectrophotometry

Spectrophotometry is an analytical technique based on the absorption of light by substances, allowing the measurement of transmittance and absorbance. The principle of this technique is governed by the Lambert-Beer law, which describes an inverse relationship between absorbance (A) and transmittance (T). This relationship indicates that absorbance is proportional to the concentration of the substance in the sample and the amount of light that passes through it. For dosimeter reading, a K37-UV-VIS spectrophotometer, manufactured by Kasvi, was used. Absorbance measurements were acquired using the UV Professional software, ensuring precise data collection (Macedo, 2024)

2.3 Universal Testing Machine – Instron 5882

The Instron 5882 is an electromechanical universal testing machine featuring a single-column load frame capable of performing tension and compression tests over a wide temperature range. The system was developed for high-precision mechanical characterization of metallic and advanced composite materials.

Load is monitored through a high-precision load cell integrated into the moving crosshead of the machine, while displacement can be recorded either by the crosshead's positional encoders or by extensometers, depending on experimental requirements. Force and displacement data are recorded in real time by Instron's own control and acquisition system. From these raw data, stress–strain curves are generated, from which mechanical properties such as Young's modulus, yield strength, ultimate tensile strength, elongation at break, as well as creep and relaxation behavior under temperature variation, can be determined (FZU).

3 Methodology

The study was conducted to evaluate the stability and effectiveness of Red Perspex 4034 dosimeters beyond the manufacturer's recommended shelf life (10 years). To this end, two batches of dosimeters were compared: one manufactured 15 years ago (PY) and another manufactured 5 years ago (LS). The methodology included dosimetric tests performed in triplicate, as well as mechanical tests, as described below:

To assess the stability and efficiency of Red Perspex 4034 dosimeters after the expiration date indicated by the manufacturer, a comparative study was carried out between two distinct batches: one with five years and another with fifteen years since manufacture. The tests included spectrophotometric analyses and mechanical testing.

Prior to irradiation, all dosimeters were subjected to absorbance readings using a Kasvi UV-VIS K37 spectrophotometer to establish background values (BG) and identify any pre-existing optical interferences. These initial measurements enabled correction and normalization of post-irradiation data.

The dosimeters were then irradiated in triplicate with gamma radiation using 5 kGy dose increments, ranging from 5 to 50 kGy, in a GammaBeam-127 (IR-214) irradiator operated at the Gamma Irradiation Laboratory of CDTN. During exposure, the dosimeters were placed on rotating tables to ensure dose uniformity.

Following irradiation, the dosimeters remained sealed until spectrophotometric readings were taken at various post-irradiation intervals (1 hour, 2 hours, 4 hours, 1 week, and 2 weeks), using the same spectrophotometer. Absorbance readings were normalized by dosimeter thickness, yielding specific absorbance values.

In addition to optical measurements, mechanical tests were conducted using an Instron 5882 universal testing machine, equipped with 5 kN and 100 kN load cells. The tests included lateral compression, longitudinal compression, and flexural strength assessments, allowing for comparison of the physical behavior of the materials from the two batches under different radiation doses.

All collected data were statistically analyzed, including calculation of standard deviations and percentage differences between batches, in order to evaluate reproducibility and possible variations due to dosimeter aging.

4 Results and Discussion

Based on the tests performed with Red Perspex 4034 dosimeters from different batches, the measured absorbance values and the respective thicknesses of the dosimeters were analyzed in order to evaluate the specific absorbance and identify possible variations in the efficiency of the batches, which were acquired approximately ten years apart.

4.1 Specific Absorbance

Initially, absorbance measurements were performed using the K37-UV-VIS spectrophotometer, following the procedure described in the Materials and Methods section. Three dosimeters from each batch were irradiated with the same dose to ensure consistency in the results. The mean specific absorbance was calculated by dividing the absorbance by the thickness of each dosimeter, then averaging the values for each batch and comparing them for better data analysis. Specific absorbance was measured at different post-irradiation intervals (1 h, 2 h, 4 h, 1 week, and 2 weeks). The standard deviation (SD) was calculated for each dose and batch, excluding the background (BG) to isolate the variability in the radiation response, as shown in Table 1.

Table 1: Percentage Standard Deviation of Red Perspex 4034 Dosimeter Fading with Intervals Ranging from One Hour to Two Weeks

Dose (kGy)	PY (%)	LS (%)
5	0,11	0,08
10	0,08	0,09
15	0,17	0,16
20	0,20	1,04
25	0,48	0,29
30	0,25	0,27
35	0,27	0,23
40	0,33	0,34
45	0,41	0,65
50	0,54	0,70

The data indicate minimal variability in fading for both dosimeter types, with values consistently below 1%. These findings demonstrate that both PY and LS Red Perspex 4034 dosimeters present stable fading behavior with low standard deviation values, thereby supporting their suitability and reliability for dosimetric applications.

Next, the percentage difference between batches at each dose was calculated, and the mean of the reading times was determined. The values were tabulated and are presented in Table 2.

Table 2: Mean specific absorbance

kGy	PY	LS	Δ%
BG	0,036	0,037	-3,368
5	0,126	0,126	-0,063
10	0,180	0,188	-4,153
15	0,218	0,234	-7,373
20	0,264	0,263	0,461
25	0,293	0,298	-1,806
30	0,322	0,327	-1,604
35	0,344	0,344	0,057
40	0,352	0,365	-3,825
45	0,357	0,370	-3,805

According to the collected data, the percentage difference between the batches remained below 5%, except for the dosimeters irradiated with 15 kGy, which may indicate an anomaly. However, this deviation does not compromise the dosimeter's quality, as it remained below the 10% limit established by the manufacturer, as shown in Figure 1.

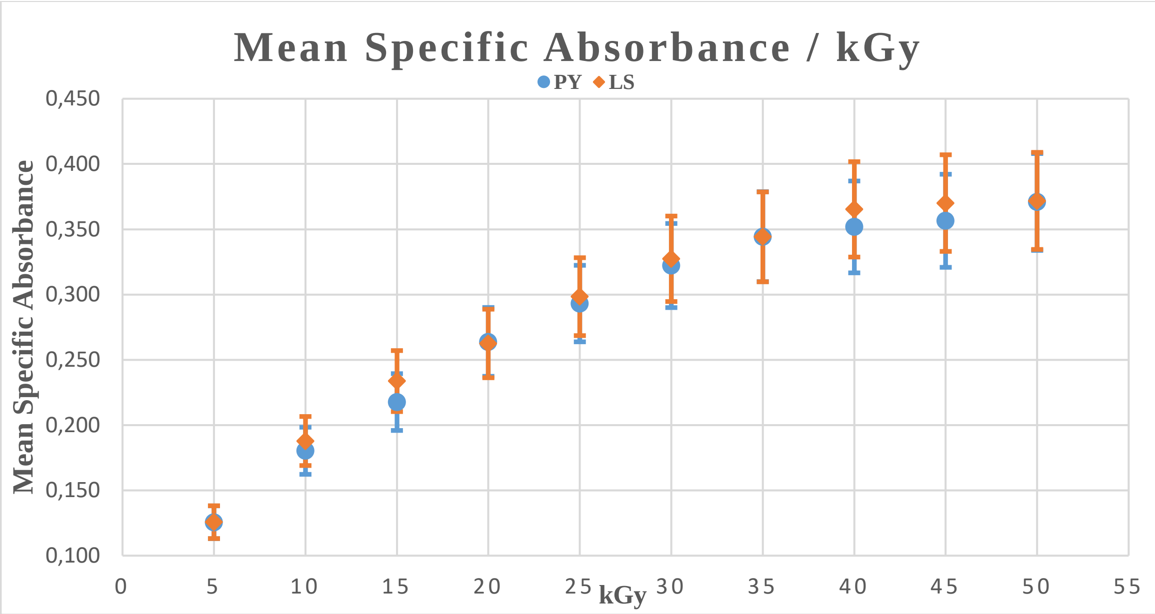

Figure 1: Graph of Mean Specific Absorbance vs. kGy.

The graph demonstrates the response of the older dosimeter compared to the new one, clearly showing that even after the 10-year expiration period, when properly stored, the dosimeter retains its dosimetric properties and can still be used for dosimetry in gamma irradiators.

4.1 Mechanical Testing

Regarding the mechanical tests performed to assess whether the polymer maintained its physical and mechanical properties, the Universal Testing Machine Instron 5882 was used with 100 kN and 5 kN load cells (Class 0.5 – ISO 7500-1:2018) for compression tests. Flexural tests were conducted using only the 5 kN load cell. The results were tabulated and compared among different batches and radiation doses, as shown in Tables 3, 4, and 5 and their respective graphs in Figures 2, 3, and 4..

Table 3: Strain per Dose in Lateral Compression (% / kGy)

	Maximum Load Strain (%)	
kGy	PY	LS
5	62,4	50,9
10	62,7	51,3
15	57,6	66
20	56,5	62,4
30	67,4	58,9
35	66,7	59,1
40	69,3	57
45	58,8	63,7
50	59,2	62,2

The lateral compression tests demonstrated that both dosimeters exhibit the same physical and mechanical behavior regarding lateral compression. Although the specimen irradiated with 25 kGy was removed due to an interruption in testing, the results from the other tests were conclusive and, as shown in the graph in Figure 2, remained between 50% and 70%, even after high radiation doses.

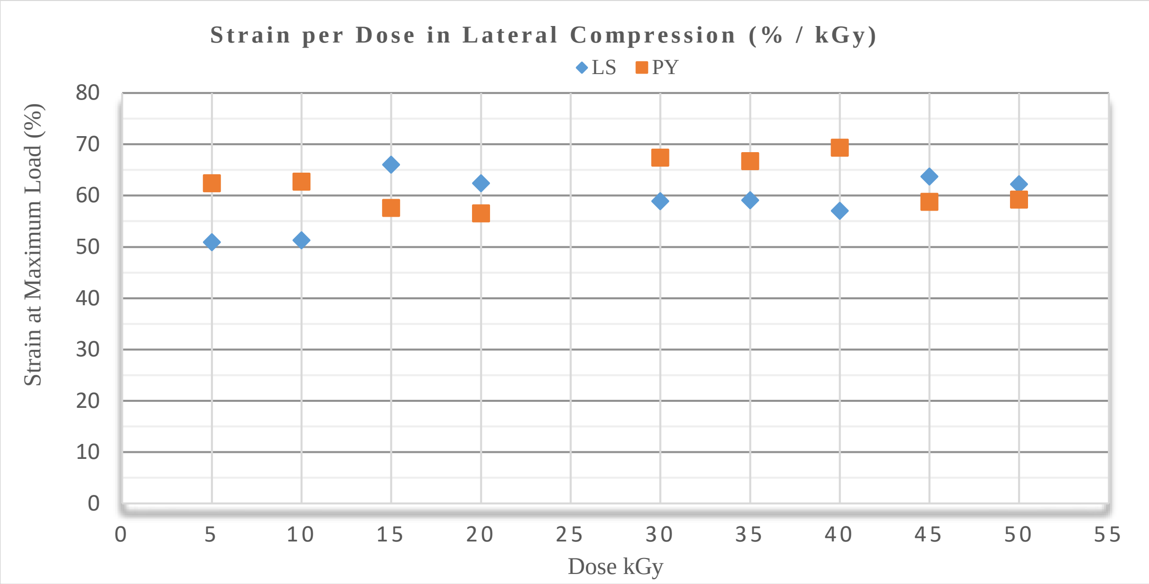

Figure 2: Strain per Dose in Lateral Compression (% / kGy).

Regarding the longitudinal compression tests, there were no interruptions, and the results were tabulated and plotted in the table and graph below.

Table 4: Strain per Dose in Longitudinal Compression (% / kGy)

	Maximum Load Strain (%)	
kGy	**PY**	**LS**
5	3,7	5,1
10	3,4	3,6
15	3,3	3,6
20	3,4	3,3
25	2,9	3,5
30	3,2	3,7
35	4,2	3,5
40	3,2	3,3
45	4,4	3,2
50	3,6	3,1

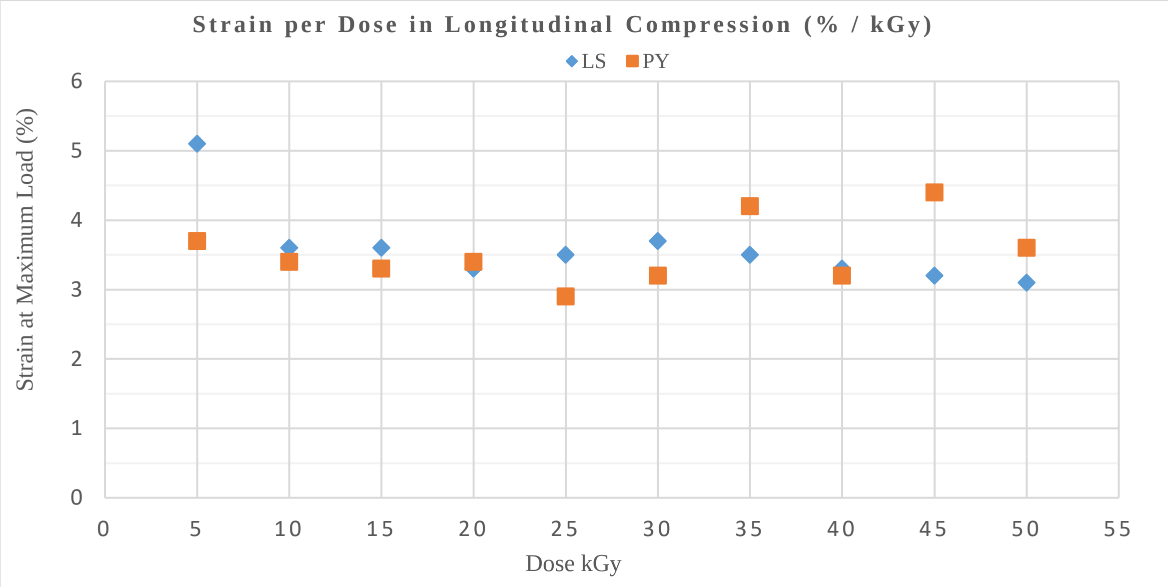

Figure 3: Graph of Strain per Dose in Longitudinal Compression (% / kGy).

The longitudinal compression tests demonstrated that both dosimeters exhibit similar physical and mechanical behavior as observed in lateral compression, maintaining values close to 3 to 5% even at high radiation doses, indicating the polymer's resistance to radiation.

Regarding the flexural tests, some considerations must be made due to the testing procedure: some specimens from the most recent batch (LS 5, 40, and 45 kGy) did not fracture but reached their load limits, as shown in Table 5 and the graph in Figure 4.

Table 5: Maximum Flexural Load per Dose (N / kGy)

	Maximum Flexural Load (N)	
kGy	**LS**	**PY**
5	377,92	330,93
10	335,14	392,56
15	367,17	304,73
20	383,74	368,26
25	360,8	343,04
30	340,52	294,11
35	398,69	439,94
40	407,22	436,49
45	363,75	296,97
50	350,45	322,03

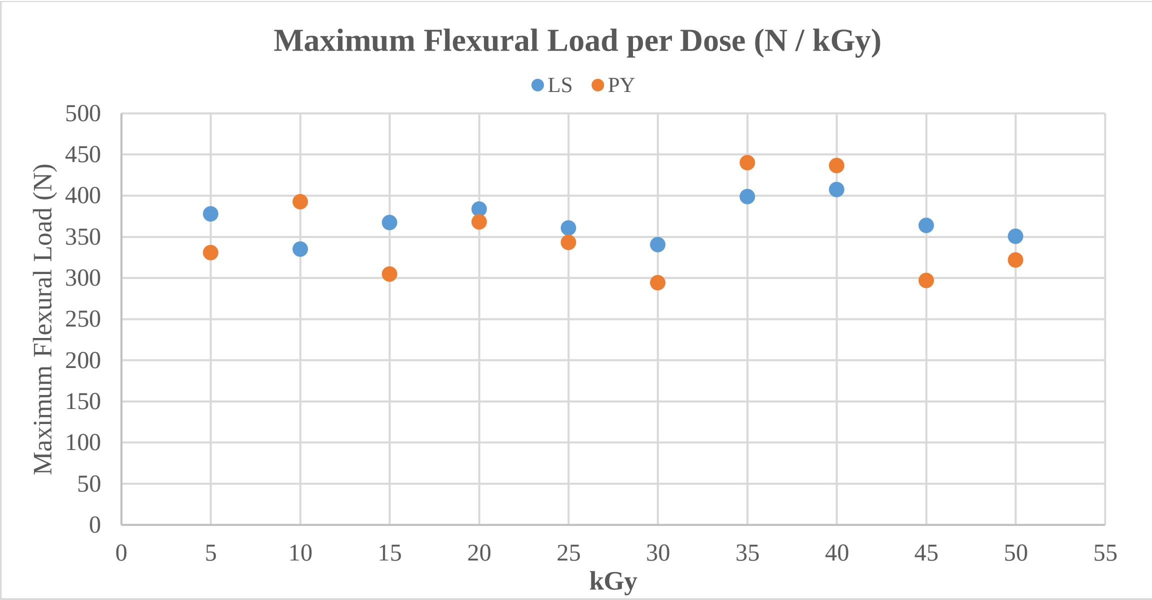

Figure 4: Graph of Mean Specific Absorbance vs. kGy.

The data obtained from the flexural tests demonstrated that there is no significant difference in the mechanical behavior between the dosimeters from the two batches (PY and LS), showing very similar performance in both tests. This clearly indicates that even after the manufacturer's recommended shelf life, if properly stored, the dosimeters maintain their properties.

5 Conclusions

The results obtained in this study demonstrate that Red Perspex 4034 dosimeters, even five years beyond the manufacturer's recommended shelf life, retain dosimetric and mechanical characteristics comparable to those of more recent batches, provided they are stored under appropriate conditions.

Spectrophotometric analysis, including background correction performed prior to irradiation, revealed percentage variations of less than 5% between batches for nearly all applied doses, remaining within the recommended tolerance limits. Additionally, mechanical testing indicated similar structural behavior between batches, with stability under high radiation doses and preserved mechanical strength.

These findings suggest that Red Perspex 4034 dosimeters can continue to be used safely and accurately for industrial dosimetry purposes even after the expiration of their formal shelf life. This observation may lead to significant cost savings and a reduction in waste, thereby contributing to operational sustainability in irradiation laboratories.

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Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP

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Abstract

Monte Carlo simulations are essential tools in dosimetry research for Nuclear Medicine and Radiotherapy, enabling high-precision calculations of radiation dose distribution. Among the available radiation transport codes, MCNP is widely recognized for its robustness and versatility enabling high-precision calculations of radiation dose distribution. However, the manual creation of MCNP input files, particularly for voxelized phantoms, remains a labor intensive and error prone task. This work presents the development of GHOST (Generator of Health Optimized Simulation Templates), an automated and efficient tool for generating MCNP input files with voxelized phantoms derived from 3D medical images. GHOST was developed as a plugin for 3D Slicer, written in Python. It constructs lattice format phantoms directly from segmented volumetric datasets, allowing users to assign material definitions based on segmentation labels and to incorporate custom materials beyond the predefined database. To validate GHOST, energy deposition simulations were performed using a 30×30×30 cm³ PMMA cube filled with water. The MCNP input file generated by GHOST from an image in MHD format, the result was compared with an equivalent ghost built using RPP cards. Additionally, cross-validation was conducted using GATE, another Monte Carlo simulation platform capable of generating phantoms from medical images. A maximum discrepancy of 2.3% was observed in dose calculations between GHOST generated inputs and both MCNP-RPP and GATE-based simulations. GHOST has proven to be an agile option for the automatic generation of voxelized ghosts to be used in MCNP.

Keywords: MCNP; 3D Slicer; Voxelized Phantoms; GHOST

1 Introduction

The Monte Carlo Method (MCM) is a probabilistic numerical approach employed to model radiation transport in matter through the simulation of random processes based on the generation of pseudo-random numbers. A key advantage of MCM simulations over physical experiments lies in their flexibility experimental conditions and governing rules can be altered, while still producing results that are physically meaningful. This capability enables detailed assessment of the impact of each event involved in the simulation process Berdeguez (2016); Yoriyaz (2009); Xavier (2022).

Monte Carlo simulations have become indispensable for estimating absorbed doses in cancer treatments involving ionizing radiation, encompassing applications in both nuclear medicine and external beam radiotherapy. They are also extensively used to assess radiation doses in diagnostic imaging and to investigate image quality. The MCNP code was developed to simulate radiation interactions with matter, yielding highly accurate results. Its versatility allows it to be applied across virtually all nuclear-related domains, including radiotherapy facility design, patient-specific treatment planning, and internal dosimetry in nuclear medicine Berdeguez (2016); Yoriyaz (2009).

Computational anatomical phantoms are typically categorized into three types: stylized (mathematical) models, voxel-based (tomographic) models, and hybrid models. Voxel-based phantoms are constructed from three-dimensional matrices of volumetric elements (voxels), which are commonly derived from the segmentation of tomographic imaging data. These phantoms provide a more anatomically realistic representation of internal body structures Berdeguez (2016); Xavier (2022).

The aim of this study is to develop a computational tool capable of generating MCNP input files from three-dimensional image data used to build voxel-based phantoms.

2 Methodology

2.1 Generator of High Optimized Simulation Templates (GHOST)

GHOST was developed in Python as a plugin for the open-source platform *3D Slicer* Fedorov et al. (2012). The generation of input files for MCNP using voxel-based phantoms follows the workflow illustrated in Figure 1:

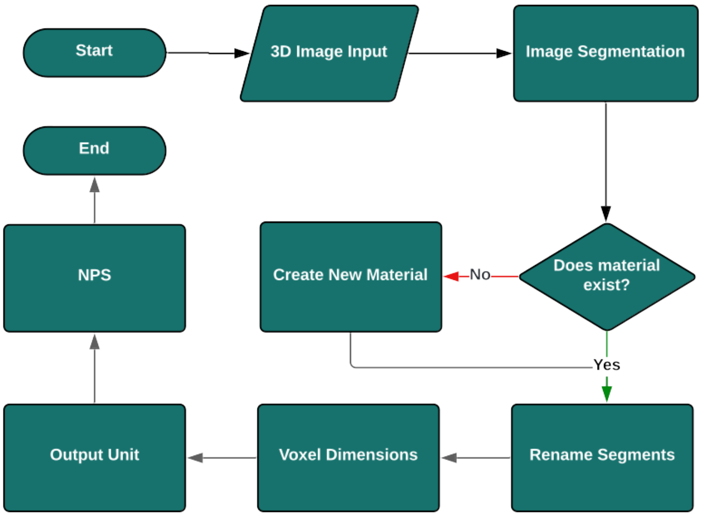

Figure 1: Flowchart for creating input with voxelized phantom for MCNP.
Source: Author

- **3D Image Input** – Supports the import of volumetric images in formats compatible with 3D Slicer (e.g., DICOM, MHD, MHA, STL, etc.).
- **Image Segmentation** – Allows the user to segment regions of interest using any preferred method. Multiple structures can be segmented without restriction.

- **Create New Material** – If the desired material is not listed in the database, the user may add it directly through the GHOST interface.
- **Rename Segments** – The names of the segmented structures must exactly match the material names defined in the `materials.db` file used by GHOST.
- **Voxel Dimensions** – Users can resample the image to change voxel spacing without altering the phantom's physical size. Lanczos interpolation is applied during this process.
- **Output Unit** – GHOST generates MCNP-ready models with `F6` tally (deposited dose) for each segmented structure. The user may select the output unit as either Gy (J/kg) or MeV/g. If Gy is chosen (default), a multiplier card with a value of 1.602×10^{-10} is added to convert MeV/g to Gy.
- **NPS** – Specifies the number of particles to be simulated at the source. The default is set to 1×10^6 histories.

All GHOST functionalities are accessible through a graphical interface integrated into 3D Slicer, making it intuitive and user-friendly.

The MCNP input generation process involves constructing a voxel matrix that encodes the material assignments and their densities according to the definitions in GHOST's internal database. Figure 2 illustrates the voxelization of a 3D image by GHOST for use in MCNP simulations.

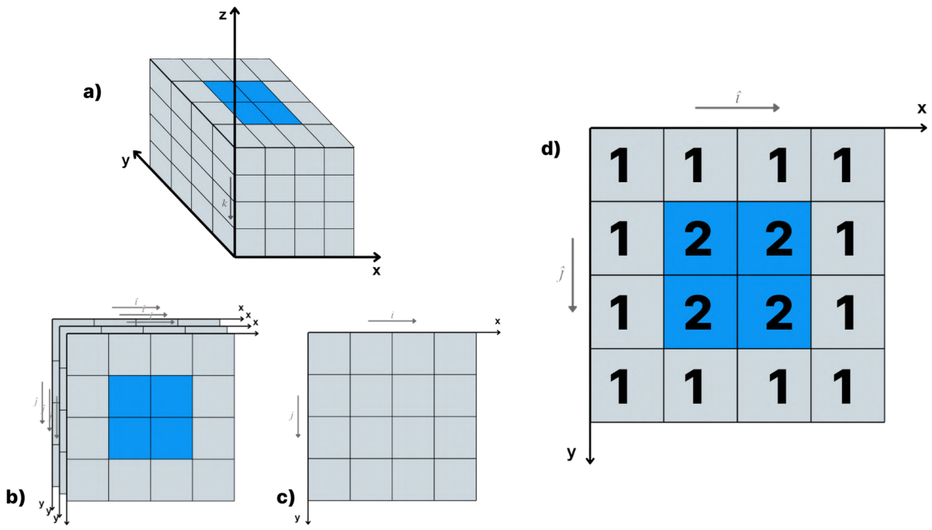

Figure 2: a) Direction of voxel, b) and c) visualization of slices in the k-direction, and d) how each material should be represented for MCNP: 1-PMMA, 2-Water.
Source: Author

2.2 Validation

To assess the performance of GHOST, a comparative validation study was performed using simulations carried out with both MCNP and GATE. Initially, a Python script was developed to generate an MHD image representing a cubic phantom with dimensions of $30 \times 30 \times 30$ cm^3, composed of voxels measuring $1 \times 1 \times 1$ cm^3. The phantom mimics a PMMA structure filled with water, featuring a wall thickness of 1 cm.

This configuration is analogous to the geometry depicted in Figure 2, though extended in size for validation purposes.

The MHD phantom image was processed entirely through GHOST's input generation pipeline, as outlined in the workflow diagram (Figure 1). In parallel, an equivalent input geometry was constructed manually using MCNP, consisting solely of `RPP` (rectangular parallelepiped) macrobodies with matching spatial dimensions to the voxelized structure.

For inter-code validation using voxelized phantoms derived from medical images, the same MHD file was employed in a GATE simulation. The phantom was imported using the `ImageNestedParametrisedVolume` feature, which allows direct voxel-based geometry generation from image data.

The evaluation of energy deposition was conducted using a monoenergetic point source emitting 10 MeV photons, positioned 1 meter from the phantom surface. In MCNP, the `*F8` tally was used to estimate the energy deposited in water. In GATE, the `DoseActor` was employed with the `Edep` (energy deposition) parameter enabled.

The outputs from the three simulation configurations voxelized MCNP, analytical MCNP, and voxelized GATE were compared to verify consistency and confirm the reliability of the GHOST generated MCNP input for voxel-based simulations.

3 Results

This section presents the outcomes obtained through the use of the GHOST plugin, emphasizing its intuitive graphical interface, the generation of voxelized phantoms, and the validation of its performance via comparative simulation studies.

Figure 3 displays the graphical user interface (GUI) of GHOST, which was designed to streamline the process of generating voxel-based phantoms. The interface is structured following a logical workflow, guiding the user through each step in an efficient and accessible manner. This design facilitates usability and enhances productivity during the phantom preparation and MCNP input generation processes.

The MHD image of the $30 \times 30 \times 30$ cm^3 cube, generated via Python scripting, was segmented into two distinct regions of interest, corresponding to PMMA and water materials. This segmentation step is depicted in Figure 3, and is fundamental to ensuring accurate material assignment within the three-dimensional phantom. Proper segmentation enables the correct association of voxel labels with physical materials, which is essential for reliable dosimetric calculations in Monte Carlo simulations.

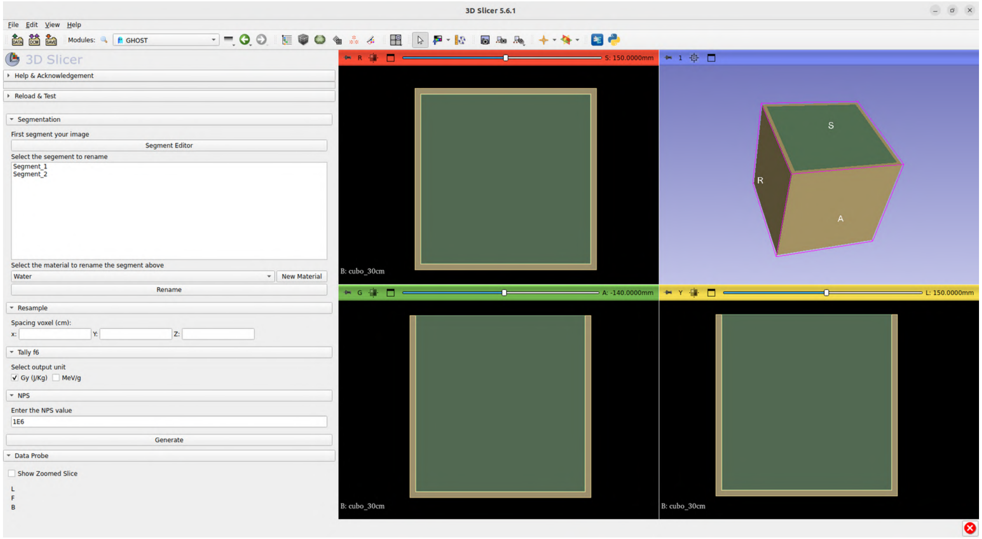

Figure 3: GUI show a MHD image of the 30x30x30 cube segmented into PMMA and Water.
Source: Author

The GHOST interface provides a dedicated functionality for renaming segmented regions. This tool enables users to visualize each segmented structure individually, facilitating the correct identification and assignment of materials within the phantom. By ensuring that each segment name exactly matches the corresponding entry in the material database (`materials.db`), this feature plays a critical role in guaranteeing compatibility and accuracy during the input generation for MCNP simulations.

GHOST also incorporates a feature that enables users to define and add new materials to its internal database. As shown in Figure 4, the material creation interface allows users to input the material's density (in g/cm^3) and specify its elemental composition. Elements can be selected directly from the periodic table, and their contributions can be defined either by weight fraction or atomic fraction.

This capability provides users with flexible customization of simulation parameters, making it possible to model a wide range of tissue-equivalent or shielding materials. As a result, the GHOST plugin can accommodate diverse simulation scenarios, thereby broadening its applicability in internal dosimetry and radiation transport studies.

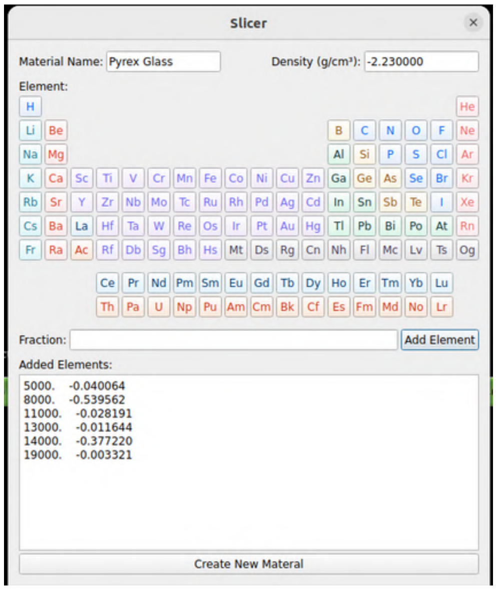

Figure 4: New material creation screen in GHOST, with input fields for density and elemental composition.
Source: Author

GHOST allows users to resample the voxel dimensions while preserving the original physical size of the phantom. This operation is particularly useful for optimizing the spatial resolution of simulations without altering the anatomical scale of the model. As shown in Figure 5, the interface includes controls for adjusting voxel size via resampling algorithms (e.g., Lanczos interpolation).

In addition, users can configure key simulation parameters such as the output unit for the `F6` tally choosing between MeV/g and Gy (J/kg) and define the number of particle histories (`NPS`) to be simulated. These settings provide enhanced control over the resolution, unit system, and statistical robustness of the resulting Monte Carlo simulation.

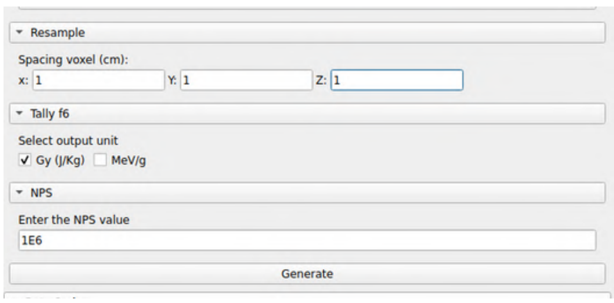

Figure 5: Simulation configuration sections.
Source: Author

Figure 6 shows two orthogonal views of the voxelized phantom generated by GHOST: These visualizations illustrate the spatial distribution of segmented regions and their respective material assignments.

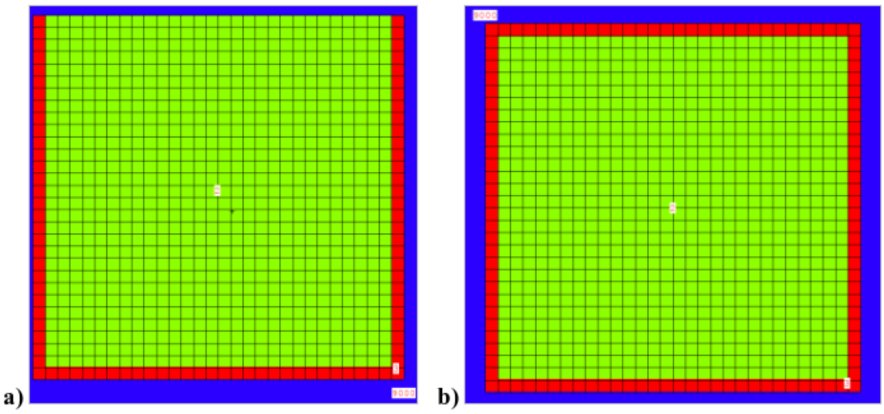

Figure 6: a) View in the XZ plane and b) view in the XY plane.

To validate the consistency and accuracy of GHOST's voxelization process, the energy deposition results from simulations were compared across different modeling approaches. Table 1 presents the energy deposition values (in MeV), their respective statistical errors, and the percentage difference between the results.

Table 1: Validation simulation results.

Phantom	**Energy Deposition [MeV]**	**Error [%]**	**Percentage Difference [%]**
RPP (MCNP)	1.74×10^{-2}	2.19	2.30%
Voxelized Image (GATE)	1.74×10^{-2}	2.18	
Voxelized Image (GHOST)	1.78×10^{-2}	2.16	

The comparative analysis indicates that the difference in energy deposition between the phantom generated using GHOST and those produced by traditional methods remains below 2.5%. Specifically, the variation of 2.30% between the RPP model and the GHOST voxelized image validates the fidelity of the voxelization procedure. These results confirm the reliability and robustness of the GHOST-generated input files for use in MCNP-based Monte Carlo simulations.

4 Discussion

The GHOST plugin presents a streamlined and intuitive interface, structured around a logical sequence of operations. A key advantage of the tool lies in its integration with *3D Slicer*, an open-source platform widely recognized for medical image processing and analysis.

Within the 3D Slicer environment, segmentation of regions of interest can be performed directly, serving as the foundation for generating voxelized phantoms. From this point, GHOST enables comprehensive customization of simulation parameters, including voxel spacing, material assignment, the number of histories (`NPS`), and the output unit for the `F6` tally (either Gy or MeV/g). Such configurability offers users the flexibility to tailor the simulation setup according to specific experimental or clinical objectives.

The validation tests carried out using voxelized phantoms created with GHOST demonstrated excellent agreement with reference results obtained from simpler geometrical models, such as those based on `RPP` volumes. This outcome confirms the tool's capability in accurately modeling complex three-dimensional structures. In addition, the consistency observed in simulations conducted with GATE further supports the robustness of the methodology implemented in GHOST, as both codes produced compatible energy deposition profiles for voxel-based phantoms.

The adoption of voxelized phantoms generated from three-dimensional imaging enhances the capacity to model anatomically realistic geometries, which are difficult to achieve using conventional mathematical models. This underscores the importance of employing GHOST in medical physics applications where both geometric complexity and modeling precision are essential.

5 Conclusions

GHOST was developed with the aim of streamlining the creation of voxelized phantoms derived from three-dimensional medical images for use in MCNP based Monte Carlo simulations. Its development was motivated by the demand for a user-friendly and accessible tool capable of rapidly generating such phantoms. By integrating seamlessly with the 3D Slicer platform, GHOST facilitates both the segmentation of anatomical structures and the configuration of simulation parameters in an intuitive and efficient manner.

Validation simulations demonstrated that phantoms generated with GHOST yield highly consistent results when compared to well-established approaches. The observed differences in energy deposition, on the order of 2.30%, relative to simplified geometries (e.g., RPP volumes in MCNP) and GATE based voxel simulations, confirm the accuracy and reliability of GHOST for dosimetric applications. These results support its suitability for use in medical physics scenarios where precise dose estimation is critical, such as in nuclear medicine and radiotherapy planning.

One of GHOST's key strengths is its high degree of flexibility. Users are able to customize voxel dimensions, define new materials with specific compositions, configure the number of simulated particle histories, and select the desired dose output unit (Gy or MeV/g). This level of control ensures that GHOST can be adapted to a variety of research and clinical contexts, accommodating the specific requirements of diverse studies.

In summary, GHOST emerges as a robust and versatile tool in the domain of medical physics, combining a straightforward interface with reliable performance and comprehensive customization capabilities, making it well-suited for complex simulation workflows.

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In Situ Gamma-Ray Spectrometric Mapping of Beach Sand Radioactivity in Niterói, Brazil

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Abstract

The main form of external exposure of human beings to radiation comes from natural radioactivity sources. The presence of radioactive elements contained in soils and rocks is a natural fact, existing throughout the Earth's crust, having as main elements from the ^{238}U, ^{232}Th and ^{40}K series, which are responsible for the greater part of natural terrestrial exposure. Brazil is among the countries that have the largest and most important deposits of these minerals in the world. The beaches of the Brazilian coast present varied concentrations of minerals, which can be associated with uranium and thorium. In this way, mineralogical associations may influence the exposure of beachgoers and other users. Thus, in order to promote public health, it becomes necessary to assess the levels of radionuclides found in beach sand, considering that beaches represent one of the main leisure options for residents living near the coast. TheIcaraí Beach in the city of Niterói is unsuitable for swimming but has a highly frequented stretch of sand, being one of the most popular and most intensively used beaches in the region. Therefore, the objective of this study is to perform an in situ mapping of radiation exposure levels in the sand of Icaraí Beach. The study used a portable spectrometer, the Atomtex AT-1145 to calculate the external gamma exposure risk index, the estimated absorbed dose rate in air, the annual effective dose, and the additional lifetime cancer risk due to this exposure. The absorbed dose rate in air ranged from 20 to 35 nSv/h, with an overall mean of 26.3 nSv/h. The highest activity levels were predominantly recorded under low tide conditions, suggesting an influence of sediment dynamics on the distribution of natural radionuclides along the beach strip. All values found are within international reference limits for public exposure, according to reports, indicating no significant radiological risk to beachgoers and the local population.

Keywords: Beach sand; monitoring; In situ gamma-ray spectrometry; Natural radioactivity.

1 Introduction

Living organisms are exposed to ionizing radiation due to natural sources, given the presence of radionuclides in water, soil, and air, as well as in their derivatives and in cultivated food. The radionuclides originate from various radioactive isotopes of the natural decay series of thorium (^{232}Th) and uranium (^{238}U and ^{235}U), as well as from the potassium radioisotope (^{40}K), with radionuclides from these series accounting for a significant portion of terrestrial natural exposure (Mehra, 2009).

The surface region of the soil is one of the main sources of natural radiation, and the radionuclide profile it contains depends essentially on the geographic and geological conditions of each region. Therefore, it is important to understand the distribution, composition, and background of the various soil types (Veiga et al., 2006), since the presence of natural radionuclides significantly contributes to the levels of environmental dose received by humans. In this context, the behavior of radium, particularly the isotope ^{226}Ra and its decay product ^{222}Rn, is of particular interest, as they are responsible for approximately 50% of the human exposure dose to radiation (National Council on Radiation Protection, 1987 apud Greeman et al., 1999).

Beach sands are mineral deposits formed from the erosion of rocks. Among their various constituents are natural radionuclides from the U, Th, and K series, which contribute to terrestrial radiation exposure (Veiga, 2006).

Brazil has an extensive coastal region, particularly the state of Rio de Janeiro, which features a long shoreline with world-renowned beaches. In the Metropolitan Region, the city of Niterói stands out, hosting several beaches that can be categorized into bay beaches (such as Flechas, Adão e Eva, Icaraí, São Francisco, Charitas, and Jurujuba) and oceanic beaches (Itacoatiara, Camboinhas, Piratininga, Itaipu, and Sossego) (Niterói, 2023). Currently, bathers use these areas differently depending on water quality and pollution levels. The oceanic beaches present water quality suitable for bathing, while the bay beaches are polluted and are mainly used for activities on the sand.

Icaraí Beach, one of the most popular in Niterói's coastal landscape, is located within Guanabara Bay. It stretches for approximately 1,200 meters, with a wide strip of sand that supports various sporting activities, walking, and landscape contemplation (offering views of Pedra do Índio, Pedra de Itapuca, the Museum of Contemporary Art—MAC—and parts of the city of Rio de Janeiro, notably Corcovado and Sugarloaf Mountain). The sandy area is bordered by a promenade and vegetation composed of almond and coconut trees and is also used as a venue for city-hosted events. Consequently, the sandy area is continuously used by the population, which may lead to exposure to radionuclides present in the beach sand, such as ^{232}Th, ^{238}U, ^{40}K, and their respective decay series radionuclides.

Thus, the study of natural radioactivity in the sands of Icaraí Beach—particularly the concentrations of the radionuclides ^{226}Ra, ^{232}Th, and ^{40}K—is of great relevance, considering that these radionuclides play a fundamental role in the population's external exposure due to their gamma emission and potential radiological implications (UNSCEAR, 2008). Therefore, it is necessary to assess the natural radioactivity of the sands at Icaraí Beach to evaluate the increase in annual dose resulting from the population's presence on the beach and their consequent external exposure to gamma radiation.

2 Materials & Methods

2.1 Study area

Icaraí Beach is located in the Icaraí neighborhood, in the city of Niterói. In addition to its natural beauty, with a view of some of the iconic landmarks of Rio de Janeiro, such as Sugarloaf Mountain and Corcovado, Icaraí Beach offers various leisure options, with sports being one of its main features. The site hosts both public and private lessons in different modalities, including volleyball, functional training circuits, beach tennis, jiu-jitsu, stand-up paddle, among others. The sports infrastructure of the training schools, such as volleyball nets and beach tennis courts, can be freely used outside scheduled class times. Competitions in footvolley, beach volleyball, and similar sports are also frequently held, contributing to the dynamic atmosphere along the beachfront (Memorial Niterói).

One of the main attractions of Icaraí Beach is the Itapuca Rock, located precisely at the boundary with Flechas Beach. Also known as the "Indian Stone," this rock formation was officially designated as a natural monument of Niterói in 1985 by the State Institute of Cultural Heritage (IBGE). Formed through a natural erosion process, its name derives from the Tupi-Guarani language, meaning "Pierced Stone" or "Laughing Stone." According to local Indigenous tradition, the rock is said to harbor the spirits of the young couple Jurema and Cauby, who were killed for living a forbidden love story. Moved by their fate, Jacy—considered by the people as the representation of the moon—allegedly interceded with Tupã, requesting that their souls be allowed to remain united at that place, thus eternalizing their love story (Memorial Niterói).

2.2 In Situ Analysis

The measurement of radiation levels In Situ was carried out using the portable radiation detector Atomtex model AT1125A, equipped with a sodium iodide scintillator detector (NaI(Tl)) and an integrated Geiger-Müller detector. This configuration allows for high-sensitivity measurements provided by the scintillator, as well as a wide range of ambient dose equivalent measurements enabled by the Geiger-Müller tube, making it suitable for various applications.

Thus, the device enables area monitoring and radiometry in locations that contain or may contain sources of radioactivity emitting X-rays and gamma radiation. It is recommended for applications where radiation fields of varying levels may be encountered, ranging from low levels close to background radiation to areas with radioactive sources with high radiation emission. It operates within a measurement range of 30 nSv/h to 100 μSv/h or 10 nSv to 10 Sv, with energy ranges from 50 keV to 3 MeV. The device also includes internal memory and allows data transfer to a PC via RS-232 or USB connection.

2.3 Monitoring

In the months preceding the field stage, a digital mapping of the study area was carried out using the Google Earth platform (Figure 1). This step had as its main objective the definition of the analysis points, as well as the planning of logistical and environmental aspects relevant to the execution of the fieldwork. During this preliminary period were established the environmental parameters to be monitored in the month prior to and during the fieldwork.

The in situ monitoring was standardized based on previously defined areas of interest, particularly two storm drains located at points 1 and 11 on the map (Figure 1). The sampling points were distributed along the sand strip, with distances ranging from 50 to 200 meters. The points closest to the sewer drains are spaced 50 meters apart, while those located in the central region of the beach are spaced 200 meters apart.

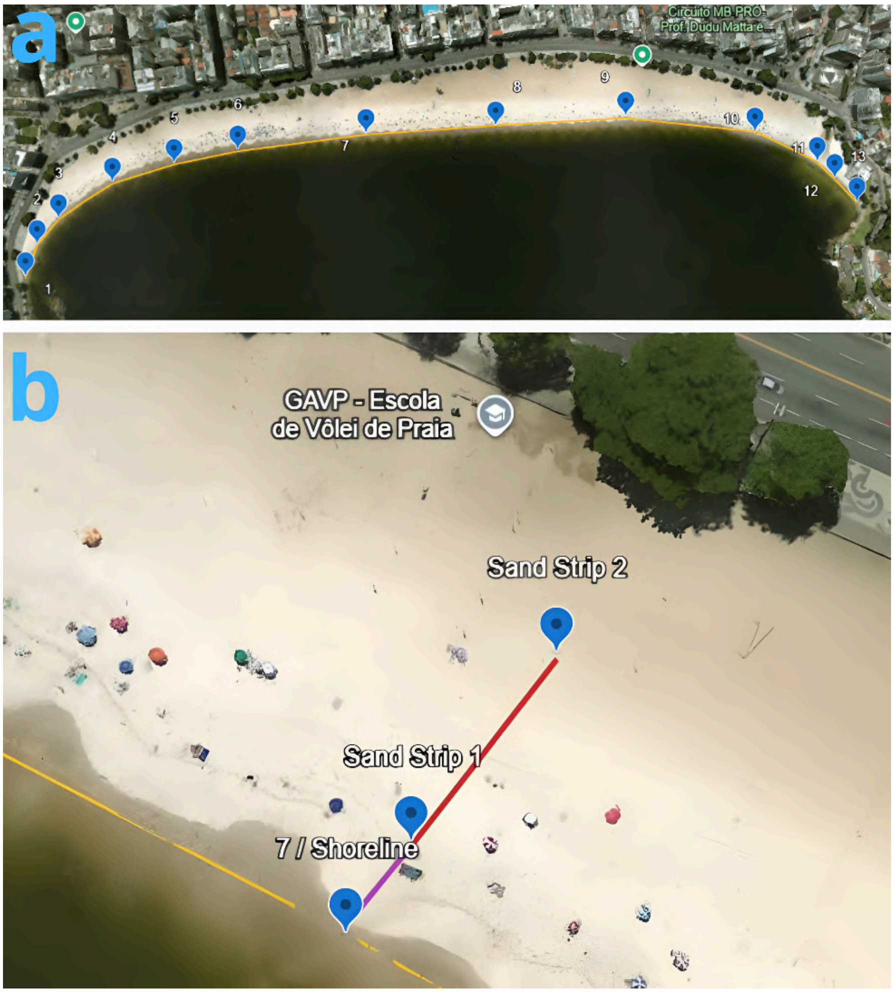

Figure 1. Icaraí Beach showing the location of sampling sites (image from Google Earth). (a) Distribution of the 13 sampling points along the shoreline. (b) Sections along the sand strip.

Each analysis point was subdivided into up to three distinct sampling sections, designated as: (1) Shoreline, (2) Sand Strip 1 , and (3) Sand Strip 2, which vary between 10 to 30 meters apart, except for the points located near the beginning and end of the beach, due to the limited width of the sand strip. This division allowed for the collection of more representative data regarding the beach's environmental dynamics and the potential variations in radionuclide concentrations across different zones of use.

3 Results and Discussion

The results are presented in Table 1. The recorded ambient equivalent dose rate values ranged from 20 nSv/h to 35 nSv/h, with estimated annual effective doses between 0.20440 and 0.28908 mSv/year. Sampling points 01 and 13 showed the highest ambient dose rates, exceeding 30 nSv/h with the Estimated Mean Annual Effective Dose (mSv/year) of 0.28908, 0.28032, 0.26864, and 0.28908 mSv/year, respectively. Points 5 and 6 showed the lowest ambient dose rates, ranging between 21 nSv/h and 28 nSv/h, and both presented an Estimated Mean Annual Effective Dose (mSv/year) of 0.20440.

Table 1 : Ambient Equivalent Dose Rate and Estimated Mean Annual Effective Dose at Sampling Locations.

Sampling Point	Shoreline (nSv/h)	Sand Strip 1 (nSv/h)	Sand Strip 2 (nSv/h)	Estimated Mean Annual Effective Dose (mSv/year)
Point 01	33 ±12%	-	-	0.28908
Point 02	34 ±16%	30 ±12%	-	0.28032
Point 03	24 ±12%	29 ±16%	29 ±20%	0.23944
Point 04	23 ±20%	27 ±16%	26 ±14%	0.22192
Point 05	21 ±12%	23 ±12%	26 ±20%	0.20440
Point 06	21 ±16%	21 ±16%	28 ±16%	0.20440
Point 07	20 ±16%	20 ±13%	33 ±12%	0.21316
Point 08	22 ±20%	21 ±20%	30 ±15%	0.21316
Point 09	27 ±20%	27 ±20%	28 ±10%	0.23944
Point 10	23 ±20%	24 ±15%	28 ±20%	0.21900
Point 11	26 ±20%	31 ±20%	35 ±20%	0.26864
Point 12	24 ±20%	24 ±12%	-	0.21024
Point 13	33 ±20%	-	-	0.28908

**The estimated mean annual effective dose was calculated considering continuous exposure over 8760 hours per year (24 h/day × 365 days), applying an outdoor occupancy factor of 0.2, as recommended by*

*UNSCEAR (2000).**

These points are located in areas with narrower sand strips or greater accumulation of heavy minerals. Notably, points 01 and 11 are located in the vicinity of stormwater outfalls and sewage discharge areas, which may influence the radiometric levels observed, which may indicate a potential influence of waste disposal on the increase in natural radiation levels. Additionally, points 02 and 13 are spatially close to points 01 and 11 and may therefore be affected by the dispersion of waste material along the beach. In contrast, points 05 and 06 exhibited the lowest values when compared to the others, suggesting a lower potential for external exposure in those regions.

In 1990, the International Commission on Radiological Protection (ICRP) established an annual dose limit of 1 mSv/year for the general public. The values observed in this study remain below this regulatory threshold, indicating that long-term exposure does not pose a significant radiological risk to individuals frequenting the beach. However, according to the UNSCEAR (2000) report, the global average outdoor exposure is approximately 0.07 mSv/year, meaning that the values recorded between 0.20440 and 0.28908 mSv/year are considerably higher than the global average, providing insights into the influence of local geology on the radiation levels observed. Nevertheless, these levels are not sufficiently elevated to represent any significant radiological hazard to the local population or occasional visitors.

4 Conclusion

The study revealed that ambient equivalent dose rates along the sand strip of Icaraí Beach exhibited significant variations in natural radiation levels throughout its extent. The recorded values ranged from 20 to 35 nSv/h, resulting in estimated annual effective doses between 0.20440 and 0.28908 mSv/year. Although all results remain below the annual exposure limit of 1 mSv/year established by the International Commission on Radiological Protection (ICRP, 1990), the measured doses exceed the global average outdoor exposure level of 0.07 mSv/year, as reported by UNSCEAR (2000).

The highest dose rates were observed near stormwater outfalls, suggesting a possible influence from deposition processes and accumulation of materials that may contain elevated concentrations of natural radionuclides. Conversely, the lowest dose rates were found in areas with minimal anthropogenic interference and wider sand strips, indicating a lower potential for external gamma radiation exposure.

While the observed values do not pose a significant radiological risk to public health for individuals frequenting the beach, the findings highlight the importance of continuous monitoring in urban areas with potential accumulation of radionuclides, particularly in locations subject to intense and sustained public use.

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Influence of detector geometry on small-field radiotherapy simulations

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Abstract
Radiotherapy is widely used in cancer treatment due to ionizing radiation's ability to damage biological tissue by interacting with cells. However, since radiation can affect both tumors and healthy cells, techniques that minimize unnecessary exposure are essential. One such technique involves the use of small fields, defined as the irradiated areas with dimensions smaller than 4x4 cm^2, often used in stereotactic treatments, particularly in regions such as the central nervous system, lungs, and vertebrae. Despite its usefulness, small-field dosimetry presents challenges due to the non-homogeneous radiation distribution, as the lateral range of secondary particles can exceed the field limits. The absorbed dose estimation can be performed experimentally, using appropriate detectors, or computationally via Monte Carlo simulations. In this study, the EGSnrc code was used to evaluate how the detector geometry influences the absorbed dose determination in small-field radiotherapy. Constructive Solid Geometry and MESH Geometry objects were considered. This study considered the percentage depth dose setup of a 6 MV photon beam, positioned 100 cm from the water phantom surface, and 10x10 cm^2 and 3x3 cm^2 field sizes. Detectors with 0.016 cm^3 volume were used, varying their geometric modeling with Constructive Solid Geometry (CSG) and MESH Geometry. Recent findings published by the MCMEG research group at the Federal University of Minas Gerais indicate that the use of CSG geometries presents a favorable relationship between result accuracy and computational time, and MESH geometries require significantly greater computational effort to achieve comparable uncertainty levels. Here, we present how this influence becomes evident in the context of small-field dosimetry.

Keywords: Radiotherapy; Small Field; PDD; Tetrahedral-mesh

1 Introduction

Radiotherapy is a medical treatment modality that uses ionizing radiation to manage various health conditions, particularly cancer. The primary goal of radiotherapy is to damage or destroy diseased cells in order to prevent the uncontrolled proliferation of malignant tissue. The interaction of high-energy photons with tissue generally occurs not only at the surface but also at greater depths, since these photons can penetrate soft tissue. As the radiation beam travels through the attenuating medium, it may be absorbed or scattered by the material. The probability of these interactions depends on the photon energy, the atomic number of the constituent atoms, and the density of the medium (Podgorsak, 2005).

In modern radiotherapy, ionizing radiation is produced by a linear accelerator (LINAC), which generates high-energy photons or electrons in the MeV range (Khan, 2014). These systems may be equipped with multileaf collimators, which shape and restrict the radiation field to conform it to the target surface, thereby minimizing exposure to healthy tissues. One particular configuration of the collimator permits the production of a small radiation field, generally defined as any field with an area equal to or smaller than 4x4 cm^2 at the surface of the patient or phantom (Parwaie et al., 2018).Small fields are particularly useful in stereotactic treatments, such as intracranial radiosurgery and extracranial procedures involving the lungs or vertebrae. These techniques allow for the precise delivery of high radiation doses to small targets while minimizing exposure to surrounding healthy tissues (Silva, 2022).

However, small fields present specific dosimetric challenges. Chand (2020) highlights that in such fields, the lateral dimensions are often smaller than the range of secondary charged particles generated by the primary beam. This results in out-of-field dose contributions and a non-uniform dose distribution, especially near the edges of the field. As Silva et al. (2016) explain, this phenomenon prevents a direct correlation between collision kerma and absorbed dose, complicating both measurement and simulation processes. Therefore, accurately characterizing absorbed dose deposition in small field configurations is essential to ensure the safety and effectiveness of the treatment planning in modern radiotherapy.

The accuracy of dosimetry in radiotherapy is strongly influenced by the choice of the detector according to the type of incident radiation. In small fields, secondary electrons generated by radiation interactions may have path lengths greater than the field dimensions. Consequently, a portion of the energy they deposit can occur outside the defined radiation field (in the penumbra region). This raises concerns about the accuracy of dose estimates obtained using detectors placed within the field. **Figure 1** illustrates the penumbra region in (a) the conventional field and (b) the small field. It shows how the penumbra becomes relatively larger in small fields.

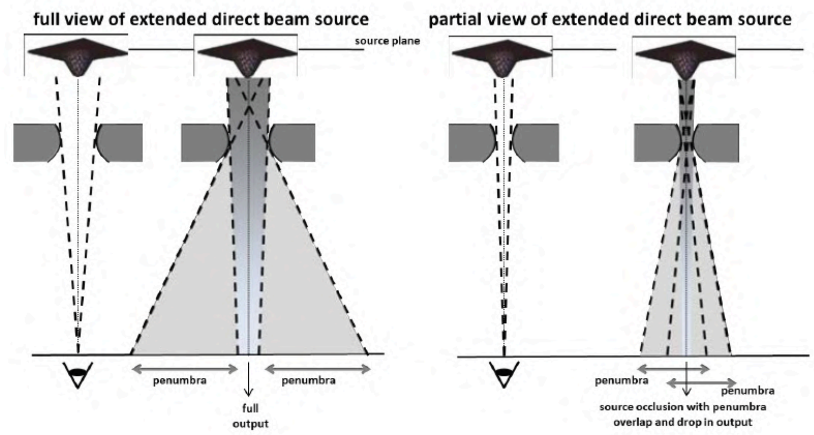

Figure 1: Schematic representation of source occlusion. On the left, the detector has a full view of the extended source, resulting in full output. On the right, part of the source is blocked, leading to penumbra overlap and reduced output due to partial occlusion (IAEA, 2017).

In radiotherapy, the Percentage Depth Dose (PDD) is fundamental for radiation beam characterization and treatment planning purposes (Podgorsak, 2005). It quantify how the absorbed dose delivered by a photon or electron beam varies with the depth in a water phantom and is defined as the percentage ratio between the dose (D) at a specific depth and the dose at a reference point, commonly used as the maximum dose (D_{max}) of the distribution (Silva, 2022; Scaff, 1997). To measure the PDD curve of a 6 MV LINAC square field, an ionization chamber is placed along the central beam axis in a 30x30x30 cm^2 water phantom with a Source-to-Surface Distance (SSD) set to 100 cm. It is known that parameters like energy of the radiation, field size, depth and SSD can affect the PDD values (Silva, 2022). **Figure 2** shows a schematic representation of the experiment.

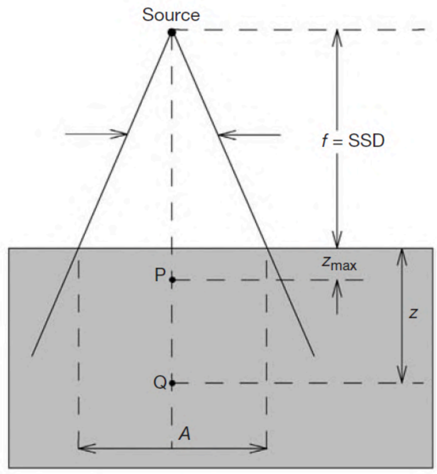

Figure 2: Geometry for PDD measurement. Point Q lies at depth z on the central beam axis, with point P at the depth of maximum dose Z_{max}. Field size A is defined at the phantom surface (Podgorsak, 2005).

In clinical dosimetry protocols, $PDD_{20,10}$ is used as a beam quality index (Silva, 2022), since it can be an indirect method to characterize the LINAC's spectrum (Fonseca et al. 2017). It is defined as the ratio between the percentage depth dose at 20 cm (PDD_{20}) and at 10 cm (PDD_{10}) depth (Equation 1) (Paixão and Fonseca, 2023).

$$PDD_{20,10} = \frac{PDD_{20}}{PDD_{10}} \tag{1}$$

In this context, Monte Carlo simulations can model the stochastic nature of particle interactions with matter. These simulations rely on random number generation and probabilistic models to solve complex physical problems, making them especially suitable for estimating the absorbed radiation dose in a given material based on scattering and absorption phenomena (Silva, 2016; Paula, 2014). Thus, Monte Carlo methods are widely used in fields such as radiotherapy, medical imaging, and radiation protection (Fonseca et al., 2017). One of the most established Monte Carlo codes in medical physics is EGSnrc (Fonseca et al., 2017), developed by the National Research Council of Canada, which allows accurate simulation of multiple scattering, boundary crossings, and energy deposition for particles with energies from a few keV up to hundreds of GeV (Kawrakow et al., 2000).

In EGSnrc, geometry can be created as Constructive Solid Geometry (CSG) that uses mathematically defined geometrical primitives such as planes, spheres, cylinders, and boxes. Operations, manually defined by the user, like union, intersection, and envelope, are required to represent more complex structures. On the other hand, tetrahedral-mesh geometry (MESH) is a significant advancement in anatomical modeling for Monte Carlo simulations. This type of geometry is modeled in external 3D software and imported into EGSnrc. They consist of collections of tetrahedra capable of representing complex anatomical structures (Takuya et al., 2015; Wang, 2023) with smaller computational time compared to voxel-based phantoms of similar geometrical complexity (Wang, 2023). It can be interpreted that there exists an international growing interest in working with anatomical realism, since, according to Lee (2017), ICRP adopted voxel phantoms based on tomographic images, followed by the usage of NURBS and polygonal meshes in 2007. Due to their flexibility, modeling speed, and anatomical detail, tetrahedral-mesh phantoms have become essential tools in radiation protection and medical imaging applications (Lee, 2017; Wang, 2023).

The difference in geometric representation is illustrated in **Figure 3**, which compares a voxelized cube with a tetrahedral-mesh cube (Takuya et al., 2017). **Figure 4** demonstrates the implications of these structural differences by showing a 2D dose distribution in a water phantom modeled using both (a) voxel and (b) tetrahedral elements. The comparison shows better conformity with the tetrahedral elements in the shape of the field.

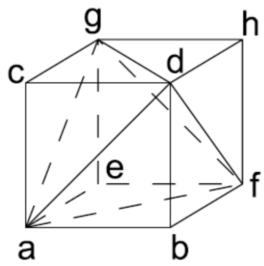

Figure 3: Tetrahedral-mesh of a cube geometry. Takuya *et al* (2017).

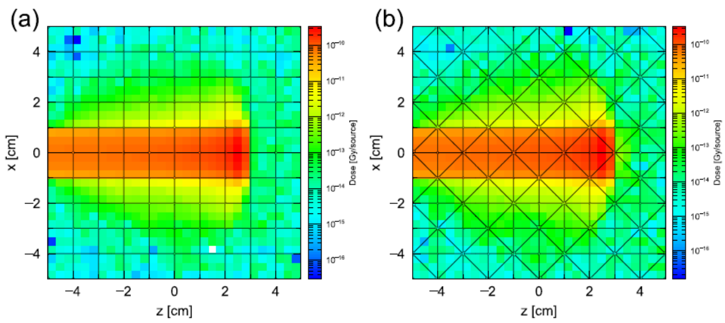

Figure 4: Dose distribution of proton beam in water phantom using (a) voxel and (b) tetrahedral mesh geometry(Takuya et al., 2017).

This work aims to investigate the influence of detector geometry (box, cylinder, ellipsoid, elliptical cylinder, and sphere) and modeling approach (CSG and MESH) on absorbed dose estimation in small-field radiotherapy using Monte Carlo simulations. PDD curves for 6 MV photon beams with 3x3 cm^2 and 10x10 cm^2 field sizes were evaluated. Moreover, the $PDD_{20,10}$ beam parameter was calculated to estimate the accuracy of the data. All the simulated data was compared with an experimental result.

2 Methodology

In this study, computational simulations were performed using the EGSnrc Monte Carlo code to investigate the influence of detector geometry modeling on the determination of absorbed dose in small-field radiotherapy.

Following a methodology similar to that employed by (Gomes and Fonseca, 2022), a 6 MV photon beam spectrum of a linac was considered, with field sizes of 10×10 cm^2 and 3×3 cm^2, incident perpendicularly on the surface of a 30×30×30 cm^3 water phantom inscribed in a 500×500×500 cm^3 universe made of dry air near sea level. The source-to-surface distance (SSD) was set to 100 cm, and the radiation detectors were distributed along the central axis of the beam. With this data, the percentage depth dose (PDD) was evaluated.

The detector was modeled with a sensitive volume of 0.016 cm^3, using two different geometric approaches. The first one was Constructive Solid Geometry (CSG), created from combinations of basic geometric primitives available in EGSnrc. The second was MESH Geometry, implemented through a 3D model composed of tetrahedral elements created using the free software Blender 4.0 for modeling, Paraview 5.12.0 for creating a .ply file for the model, and Gmsh 4.12.2 to create the tetrahedra in an

.msh extension file. **Figure 5** illustrates the five detector geometries simulated in this study: (a) cube, (b) cylinder, (c) ellipsoid, (d) elliptical cylinder, and (e) sphere.

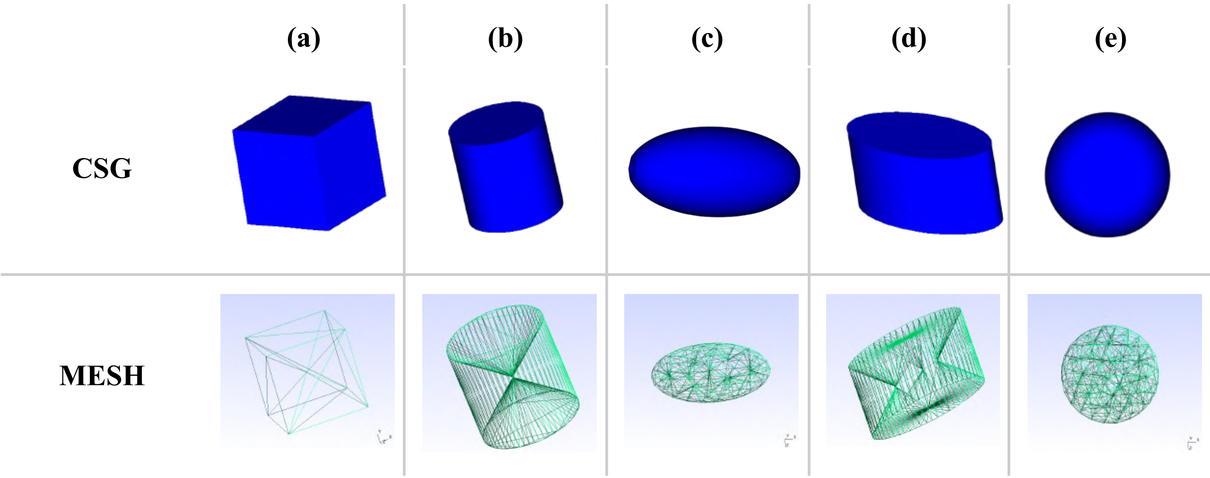

Figure 5: Detector geometries simulated in this study for CSG and MESH (a) cube, (b) cylinder, (c) ellipsoid, (d) elliptical cylinder and (e) sphere.

Simulations were carried out with 10^9 histories to maintain statistical uncertainties below 2%, except for the ellipsoid CSG case (3×3 cm²), which was run with 10^8 histories due to a code issue that prevented parallel processing at higher values of histories. The goal was to compare the resulting PDD curves and the value of $PDD_{20,10}$.

The focus was on how the detector geometry influences the dose distribution in small fields and discussing the trade-off between result accuracy and the computational cost associated with each modeling strategy.

Furthermore, the simulation results were compared with experimental data recently published in the master's dissertation of Silva (2022), entitled *Computational Dosimetry of Central Axis Parameters in Small Fields*. The experimental PDD data presented in that work were used as a reference to assess the accuracy of the simulated results.

3 Results and Discussion

Table 1 and **Table 2** present the results of the CSG and MESH simulations, respectively, for the small field case. Each geometry's $PDD_{20,10}$ value is shown along with its associated relative uncertainty and the percentage difference from the experimental reference value.

Table 1: CSG geometry $PDD_{20,10}$ results for a 3x3 cm² field.

Geometry	$PDD_{20,10}$	Relative uncertainty (%)	Difference (%)
Box	0.521 ± 0.003	0.67	-5.36
Cylinder	0.515 ± 0.005	0.97	-6.50
Ellipsoid	0.520 ± 0.011	2.10	-5.47
Elliptical Cylinder	0.565 ± 0.008	1.34	2.69
Sphere	0.522 ± 0.004	0.68	-5.22

Table 2: MESH geometry $PDD_{20,10}$ results for a 3x3 cm² field.

Geometry	$PDD_{20,10}$	Relative uncertainty (%)	Difference (%)
Box	0.518 ± 0.003	0.67	-5.93
Cylinder	0.517 ± 0.004	0.68	-6.05
Ellipsoid	0.519 ± 0.004	0.68	-5.78
Elliptical Cylinder	0.524 ± 0.005	0.87	-4.72
Sphere	0.525 ± 0.004	0.72	-4.65

Figure 6 bar plot shows the percent differences between simulated and experimental $PDD_{20,10}$ values in the case of a 3x3 cm^2 field size. All geometries underestimated the experimental values, except the elliptical cylinder CSG (2.69%). The average percentage difference is -3.97% for CSG geometries and -5.42% for MESH geometries.

Tables 3 and **4** present, respectively, the results of the CSG and MESH simulations for the conventional size field case. Again, each geometry's $PDD_{20,10}$ value is shown along its associated relative uncertainty and the percentage difference from the experimental reference value. The average percentage difference is -2.56% for CSG and -4.74% for MESH geometries.

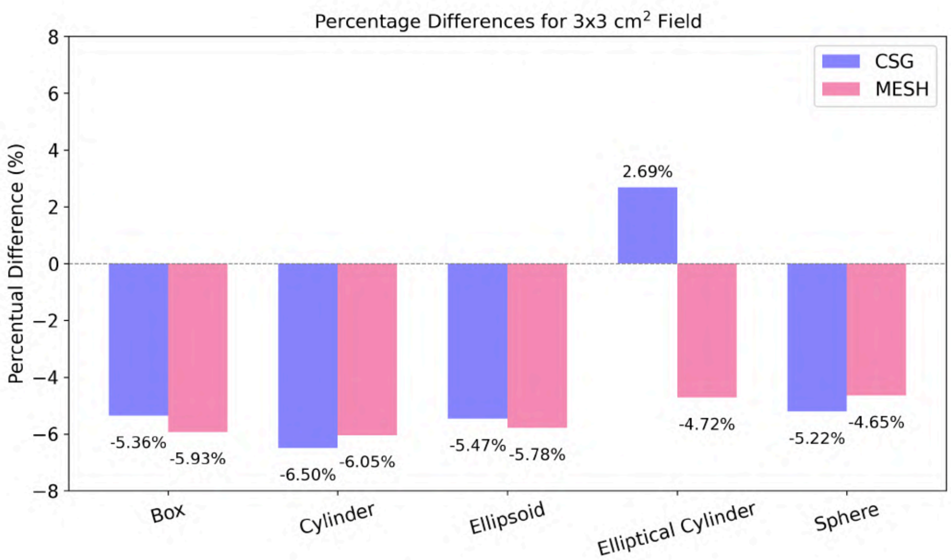

Figure 6: The percent difference of $PDD_{20,10}$ value from CSG and MESH simulation data compared to the experimental value in the case of a small field.

Table 3: CSG geometry $PDD_{20,10}$ results for a 10x10 cm² field.

Geometry	$PDD_{20,10}$	Relative uncertainty (%)	Difference (%)
Box	0.548 ± 0.011	2.05	-6.65
Cylinder	0.549 ± 0.015	2.81	-6.54
Ellipsoid	0.571 ± 0.012	2.02	-2.69
Elliptical Cylinder	0.618 ± 0.024	3.95	5.31
Sphere	0.574 ± 0.012	2.09	-2.25

Table 4: MESH geometry $PDD_{20,10}$ results for a 10x10 cm² field.

Geometry	$PDD_{20,10}$	Relative uncertainty (%)	Difference (%)
Box	0.541 ± 0.011	2.05	-7.76
Cylinder	0.559 ± 0.012	2.08	-4.79
Ellipsoid	0.569 ± 0.012	2.06	-3.03
Elliptical Cylinder	0.559 ± 0.005	0.83	-4.79
Sphere	0.567 ± 0.012	2.19	-3.33

The bar plot of the percent differences between simulated and experimental $PDD_{20,10}$ values in the case of a 10x10 cm^2 field size (**Figure 7)** shows that only elliptical cylinders overestimated the value of $PDD_{20,10}$ (5.31%). The average difference is less than 5% (-2.56% for CSG and -4.74% for MESH).

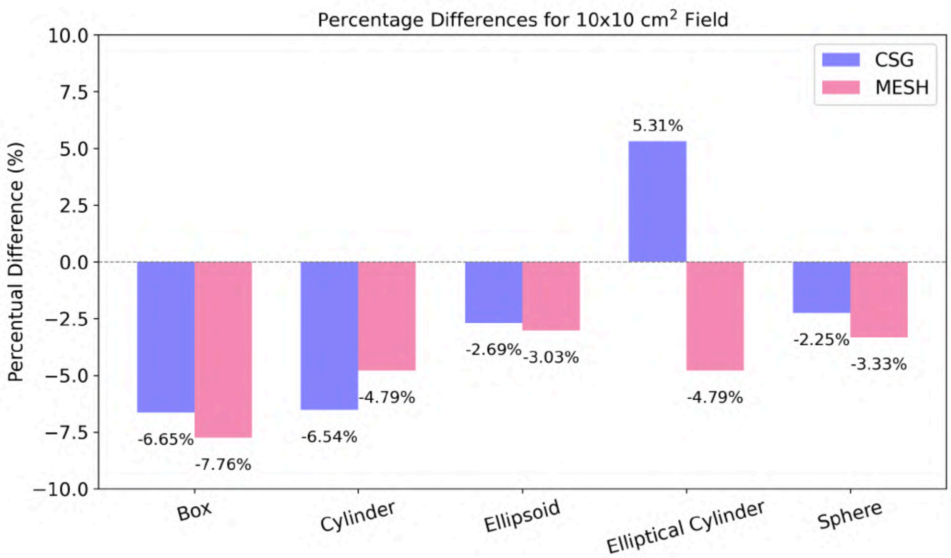

Figure 7: The percent difference of $PDD_{20,10}$ from CSG and MESH simulation data compared to the experimental data in the case of the conventional field.

The analysis of the relative uncertainties associated with the $PDD_{20,10}$ values (Tables 1 to 4 and **Figure 8**) reveals that MESH simulations resulted in lower uncertainties compared to the CSG with cylinder, elliptical cylinder, and sphere models for a 3x3 cm^2 field size.

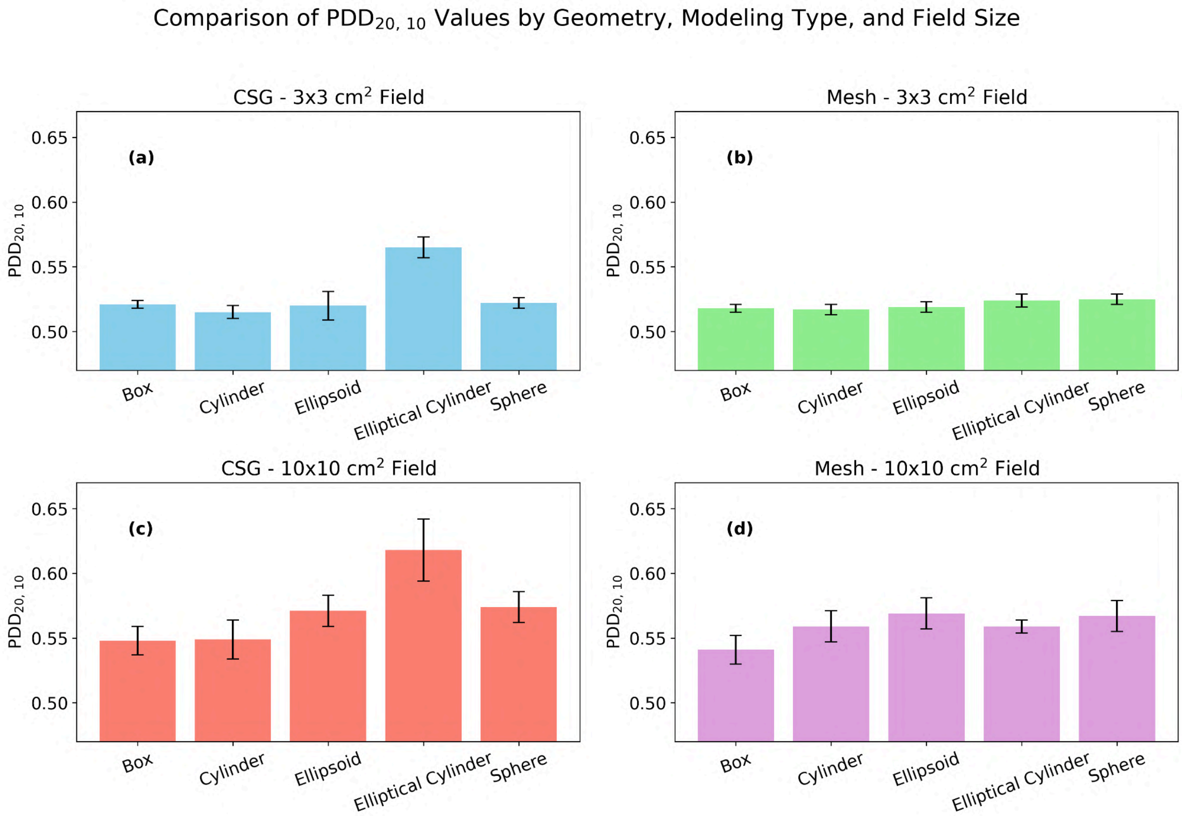

Figure 8: $PDD_{20,10}$ values and their respective relative uncertainties.

In the simulations performed with CSG for a 3x3 cm^2 field, the two cylindrical detector models consistently overestimated the experimental PDD values, particularly after the dose peak. In contrast, the Box, Ellipsoid, and Sphere geometries showed a tendency to underestimate the absorbed dose in the same depth interval, as shown in **Figure 9 (left)**. For the case of a 10x10 cm^2 field, the PDD curves exhibited more noticeable fluctuations compared to the small field, as presented in the PDD curves in **Figure 9 (right)**. However, the same behavior was observed for overestimating the dose in cylindrical shapes and underestimating it in the remaining geometries. The greater instability observed in this setup may be associated with a lower number of photon interactions within the small detector volume relative to the larger irradiated field. To mitigate these oscillations in conventional field measurements, one alternative is to increase the number of histories simulated and/or increase the detector's sensitive volume, thereby enhancing photon fluence and reducing statistical noise.

In the MESH geometry simulations for small fields, presented in **Figure 10 (left)**, the PDD curves showed significantly less statistical fluctuation compared to the CSG data. This improved stability is possibly a consequence of the finer spatial resolution of the tetrahedral mesh, which may present a better representation of the particle interactions in the interior of the detector. Despite this improvement, the MESH simulations still exhibited more noticeable oscillations in the case of the conventional field configuration, as easily observed in **Figure 10**, similar to what happened to the CSG simulation previously mentioned.

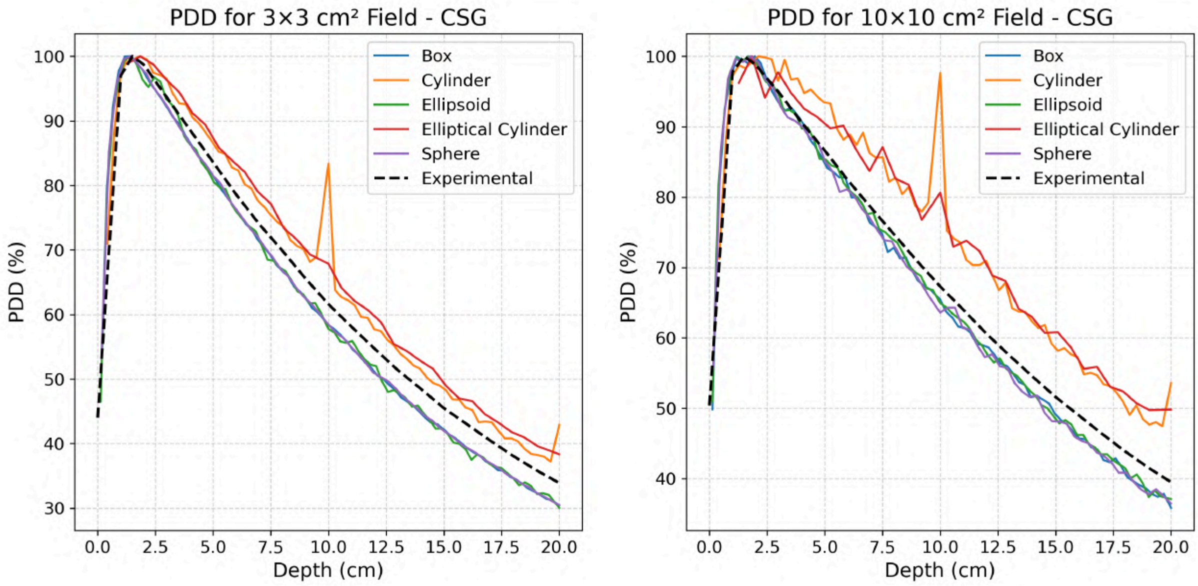

Figure 9: PDD curves for different CSG models compared to experimental data with small fields (on the left) and conventional fields (on the right).

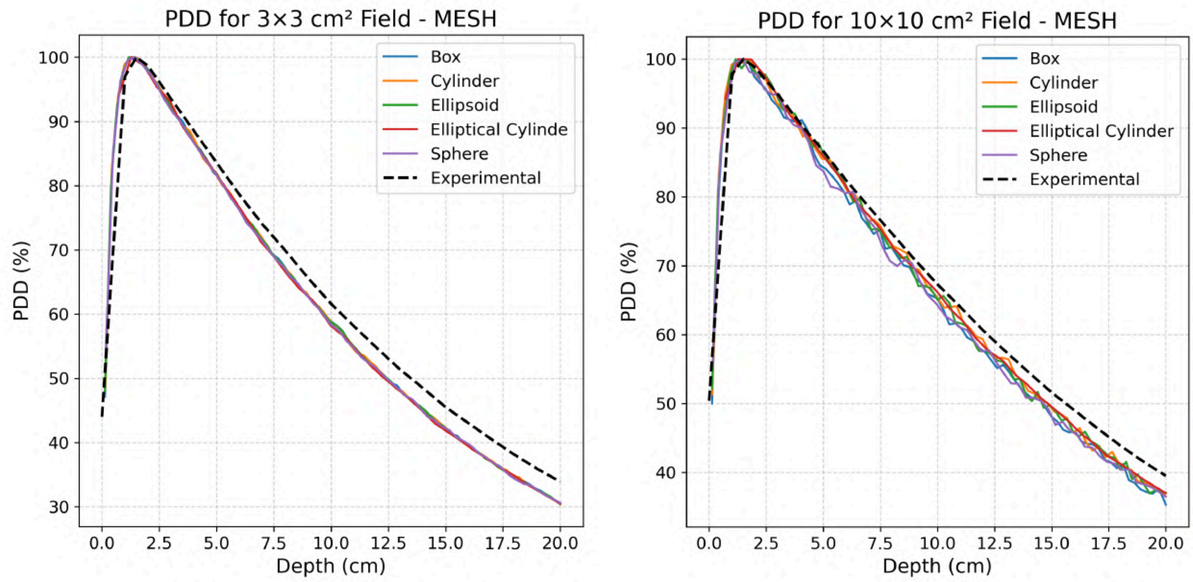

Figure 10: PDD curves for different MESH models compared to experimental data with small fields (on the left) and conventional fields (on the right).

Overall, a direct comparison between the CSG and MESH simulations highlights the superior agreement of the MESH geometries with the experimental data. The tendency of PDD underestimation observed in the simulations may be partially attributed to a slight displacement of the location of the maximum dose between the simulated and the experimental setups. According to Silva (2022), the maximum depth dose D_{max} for a 6 MV photon beam for the experiment is approximately 1.5 cm for a 3x3 cm^2 field and 1.5 cm for a 10x10 cm^2 field. For the simulations, the average value of D_{max} over all geometries was 1.40 cm for 3x3 cm^2 and 1.37 cm for 10x10 cm^2.

The MESH geometry allows for a more accurate representation of complex shapes of detectors and is easier to implement using 3D modeling software when compared to CSG modeling of complex geometries. This can be advantageous in simulations requiring detailed anatomical or structural fidelity.

In terms of computational performance, each CSG model required approximately 8,000.0 seconds on a single CPU and 230.0 seconds using parallel processing on 35 CPUs. For the MESH model, the total time of each simulation was about 10,000.0 seconds on a single CPU and 290.0 seconds using 35 CPUs.

The results of this study demonstrated that all simulations were able to reproduce the expected characteristics of the photon beam interaction in water, including the build-up regions and exponential decay in dose with depth. However, the modeling approach and detector geometry presented noticeable differences in results. It was possible to observe a more stable PDD curve with general agreement with experimental data **(Figure 10)**, despite the underestimation of dose. Larger fields presented greater statistical uncertainty for this scenario. Additionally, detector geometries with elongated shapes in the beam direction showed increased dose detection in the case of CSG simulations. In contrast, more compact geometries like the cube, sphere, and ellipsoid tended to produce results that behaved closer to the experimental. The MESH geometry is capable of modeling more complex geometries.

4 Conclusions

This study demonstrated that both the shape of the detector and the chosen modeling approach significantly influence the accuracy of the small field radiotherapy simulations. While CSG and MESH geometries were both able to reproduce the general behavior of the PDD curves, MESH models showed better agreement with experimental data, especially in the 3x3 cm^2 field. Elongated geometries, such as cylinders, tend to overestimate the dose in CSG simulations, while MESH models produce more stable results. The increase of oscillations in the 10x10 cm^2 field suggests that the small detector volume and/or the number of histories was not optimal for larger fields. Overall, the results show the importance of carefully selecting both detector shape and modeling method to ensure accurate Monte Carlo dosimetry.

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Influence of Photon Emission Spectra on Air Kerma Rate Constants for Radionuclides: A Monte Carlo EGSnrc Study

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Abstract

The air kerma rate constant is a critical parameter of utmost importance in several applications, including instrument calibration, radiation shielding design, and absorbed dose estimation in brachytherapy (Stabin, 2008).Traditionally, these values are determined using deterministic methods (Wasserman and Groenewald (1988) and Ninkovic and Adrovic (2005)). The feasibility of estimating air kerma rate constants through Monte Carlo simulations via EGSnrc code was recently demonstrated (Paixao and Fonseca (2024)). The influence of the photon spectrum database on the results of MC simulations ranged between 0% and 1%. In general, the DECDATA photon spectrum database provided results with better agreement with reference values. The results demonstrate good agreement with reference data. The results of this study will contribute to evaluating the impact of photon spectra on the air Kerma rate constant.

Keywords: EGSnrc; Monte-Carlo; Kerma Rate.

1 Introduction

The air kerma rate constant is a fundamental radiological quantity in medical physics, widely applied in dosimetry and radiation protection. It represents the rate at which kinetic energy from ionizing photons is transferred to a unit mass of air. According to the International Commission on Radiation Units and Measurements ICRU Report 33 (1980), the reference air kerma rate (RAKR) is recommended for specifying the strength of gamma emitting sources, being defined as the kerma rate in air at a standard reference distance, typically 1 m from the source. For a given radionuclide, the air kerma rate constant (Γ_δ) can be expressed as:

$$\Gamma_\delta = (K/t) \cdot (d^2/A) \tag{1}$$

where K/t is the air kerma rate, d is the distance from the point source, and A is the source activity. The subscript δ denotes the energy cut-off applied in the photon spectrum for calculation, which in this case is 20 keV. This threshold is commonly adopted in the literature and reflects practical constraints in clinical applications (Ninkovic and Adrovic, 2005).

A study conducted by Paixao and Fonseca (2024) estimated the air kerma constant rate, using photon spectra provided by RADAR in their Monte Carlo simulations. They calculated the relative percentage differences of their results with other authors, validating the methodology used. The reference values are from Wasserman and Groenewald (1988), Ninkovic and Adrovic (2005), and Smith and Stabin (2012), who estimated values of the air kerma rate or used it to estimate other associated quantities such as the exposure rate in relation to the kerma rate constant.

Wasserman and Groenewald (1988) had selected 30 radionuclides widely used in nuclear medicine. The authors used the exposure rate constant to obtain the values of the air kerma rate constant by applying analytical methods. In the work of Ninkovic and Adrovic (2005), Wasserman's data were expanded, bringing new data of the air kerma rate constant for 29 of the 30 radionuclides. In addition, they calculated data for 6 more radionuclides that were not previously used. Furthermore, this work was developed from experiments on two radionuclides that Ninkovic previously analyzed, where inconstancies were found (Ninkovic, 1992); (Ninkovic, 1993).

In the work of Smith and Stabin (2012), the exposure rate constant is provided for more than 1100 radionuclides. In all these works, the yield, the energy cutoff, and photon spectra were decisive in obtaining more conservative or more optimistic data, with spectrum energies above 15 or 20 keV being observed. In addition, Bremsstrahlung was neglected because its contribution in these cases was negligible.

This study aims to evaluate the influence of spectral data selection, comparing the RADARDecay (clinical) (RADAR, 2025) and DECDATA (metrological) (ICRP, 2008) on air kerma rate constants using the EGSnrc Monte Carlo code (Kawrakow et al., 2009), for 4 different radionuclides (C-11, F-18, Ga-68, and Ir-192). The results were compared to the data from the literature and mainly from the results obtained throughout analytical methods of Wasserman and Groenewald (1988) and Ninkovic and Adrovic (2005), and seek to provide evidence-based guidance for their appropriate use in clinical and experimental dosimetry.

2 Materials and methods

2.1 Monte Carlo EGSnrc code

The use of the EGSnrc code (Kawrakow et al., 2011) is due to its capacity and ease of modeling various situations in the area of medical physics. In it, the transport parameters for electrons and photons with a cutoff energy for particle transport of 1 keV were set for the simulations, and the remaining parameters were kept default. In the simulations, 10^9 particles were used in each run to improve statistical uncertainties associated with the MC method (Kawrakow et al., 2009). A computer with Intel (R) Core (R) processors i5-2400 x 4 with 8.0 GB of RAM and OS Ubuntu 22.04.4 LTS installed was used in the modelling and simulations. The simulation time for each input file was 240 seconds on average.

2.2 Computer Experiments

For calculating the air kerma rate constant, the EGSnrc cavity code was used. Four different radionuclides with low and high energy gamma spectra, C-11, F-18, Ga-68, and Ir-192, were used in the simulations. The model proposed by Paixao and Fonseca (2022) in the Monte Carlo and Microdosimetry Experimental Group (MCMEG) (2025) book *Computational Experiments Of Radiation Physics (English Edition) - Experiment Number 4 - Dosimetric Quantities* was used. The geometry modeled was a universe of air with density of $\rho=1.2048x10^-3$ g/cm³ and with 220 cm edges in size. The detector is a disc of 25 cm in radius, and it was placed at 100 cm from a punctual and isotropic source. The detector scores the photon fluence that reaches it. Figure 1 shows the image of the exercise provided by Paixao and Fonseca (2022).

For the radionuclides F-18 and C-11 the source was set as a monoenergetic emission. For the Ga-68 and Ir-192 sources, the spectra provided by RADARDecay (clinical) (RADAR, 2025) and, DECDATA (metrological) (ICRP, 2008) were used.

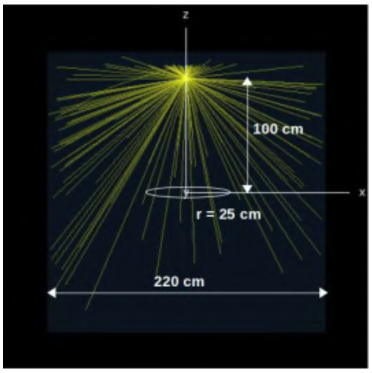

Figure 1: Model used in the simulations (Paixao and Fonseca, 2024).

After the simulations for the standard experimental distance of d = 100 cm, the simulations were performed for different positions between the target and the source, varying the position every 50 cm and then correcting the simulated air kerma with correction factors such as the sum of the yields for energies above 20 keV and the correction by the square of the distance as shown in the Equations 2 and 3 below:

$$\Gamma_\delta = K \cdot Y \cdot d^2 \tag{2}$$

$$Y = \sum Y_i \tag{3}$$

where the Γ_δ is the air kerma rate constant for that given distance, Y is the sum of all yields associated with energies above 20 keV in the photonic spectrum, and d is the distance between the target and the source.

2.3 Comparative Spectral Analysis

In this study, the RADARDecay (clinical) and DECDATA (metrological) spectra are analyzed to understand how different spectral databases influence dosimetric outcomes, particularly in estimating the air kerma rate constant. These spectra are widely adopted in the literature and in established calculation platforms, such as OLINDA software (RADAR/MIRDOSE and OLINDA/EXM, 2025) for RADARDecay, and the ICRP and other libraries for DECDATA, making it essential to assess the reliability and sensitivity of the calculations concerning the input data. Moreover, previous studies, such as those by Paixao and Fonseca (2024), have already demonstrated that the resulting values can vary depending on the spectrum used, underscoring the relevance of this choice as an object of investigation.

Differences between radioactive emission spectra arise from multiple factors, each of which can significantly influence dosimetric results. A key factor is the energy limit applied during spectral tabulation. For example, the MIRD (2025) dataset adopts a 10 keV cut-off, potentially excluding low energy photons that may still contribute meaningfully to dose calculations.

The method used to group spectral lines and perform yield summations also contributes to variability, ultimately impacting the total photon yield. These differences alter the weighted average photon energy of the spectrum and, consequently, affect the estimation of the air kerma rate constant, as demonstrated in Figures 2 and 3. In Monte Carlo simulations, these spectral variations are particularly relevant, as small shifts in energy distribution can modify photon interactions with matter, thereby altering the final dosimetric results.

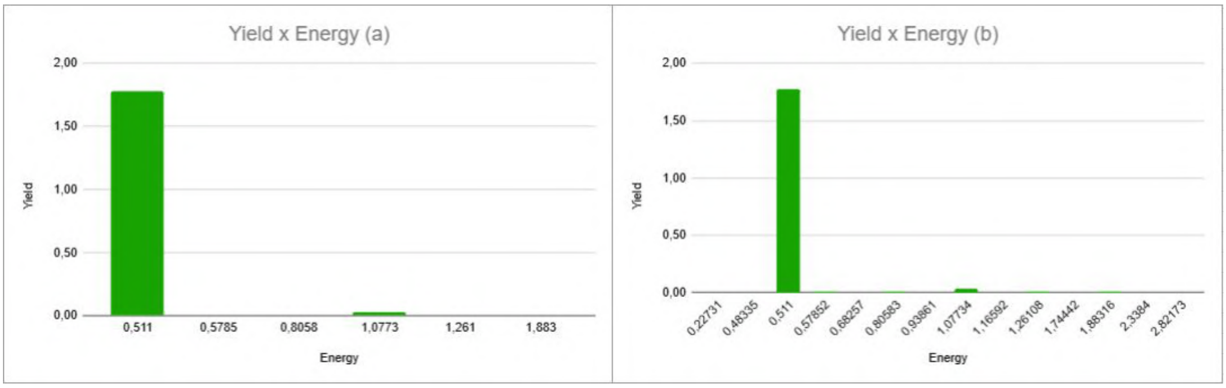

Figure 2: (a)Ga-68 RADARDecay spectra. (b)Ga-68 DECDATA spectra

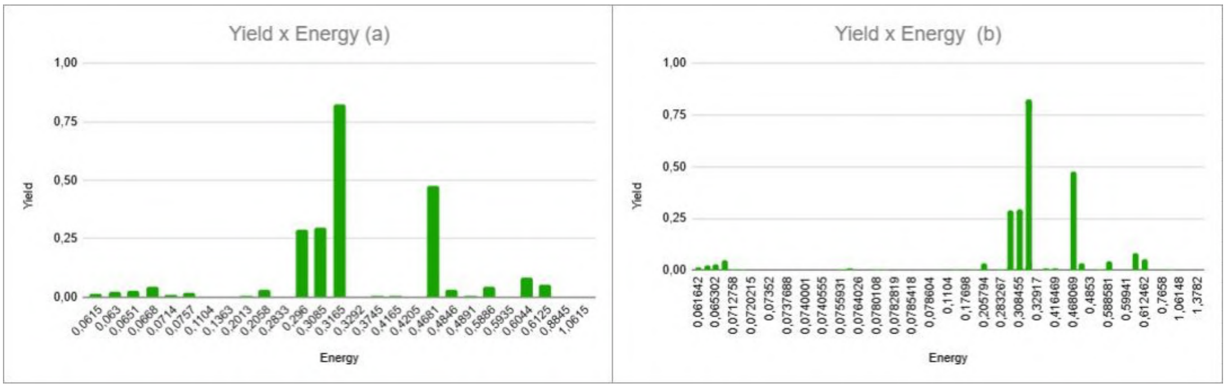

Figure 3: (a)Ir-192 RADARDecay spectra. (b)Ir-192 DECDATA spectra

After obtaining the air kerma rate constant values for the different positions, the study advanced to the next stage, which evaluated the impact of spectral selection on the results. This evaluation was performed by calculating the relative percentage difference at each point, aiming to determine whether these variations remained approximately constant or exhibited significant discrepancies. Point-to-point comparisons, as well as comparisons with reference data, were carried out by computing the percentage difference ($\Delta\%$) according to Equation 4.

$$\Delta\% = \left(1 - \left(\frac{q_0}{q}\right)\right) \cdot 100\% \tag{4}$$

where q_0 is the reference quantity value and q is the quantity calculated in this work.

3 Results

The air kerma was simulated for the radionuclides C-11, F-18, Ga-68, and Ir-192 using two different types of emission spectra available in the literature: RADARDecay and DECDATA. The results are presented in the following tables and figures. The air kerma rate constant was calculated at various source-to-detector distances. The relative computational uncertainty provided by EGSnrc was below 0.3% for all simulations. Figures 4 and 5 show the air kerma rate constants values calculated from EGSnrc simulations for RADARDecay and DECDATA spectra. The relative percentage differences were evaluated at all distances using Equation 4. For C-11, the results were 38.18 ± 5.03% using RADARDecay and 38.19 ± 5.03% using DECDATA. For F-18, both spectra yielded 37.02 ± 4.88%. The mean value for Ga-68 was 35.39 ± 4.84% with RADARDecay and 35.42 ± 4.89% with DECDATA. For Ir-192, the results were 30.05 ± 4.14% (RADARDecay) and 30.07 ± 4.23% (DECDATA).

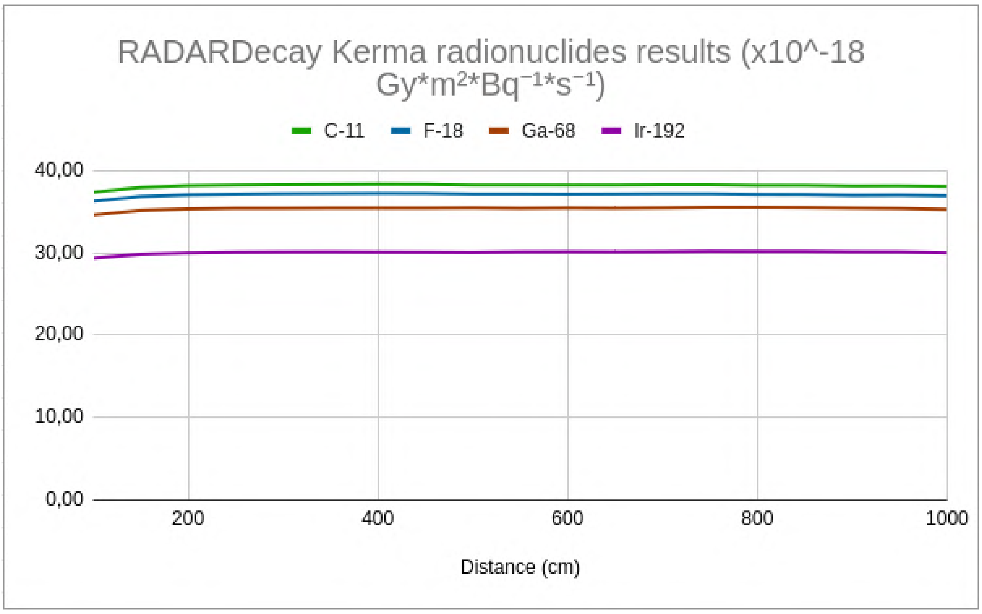

Figure 4: RADARDecay Kerma results for different radionuclides.

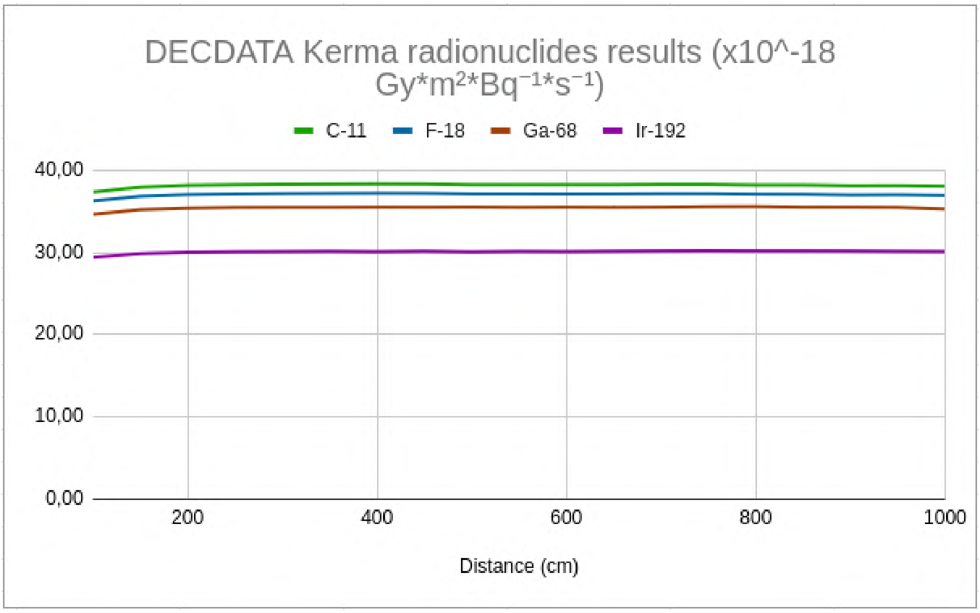

Figure 5: DECDATA Kerma results for different radionuclides.

The relative percentage differences between the simulated values and the reference data were analyzed. For C-11, F-18, and Ga-68, all differences remained below 4%, indicating good agreement. For C-11, the average differences observed were -1.46% for Wasserman, -1.34% for Ninkovic, 2.12% for Paixão, and -1.13% for MIRD. In the case of F-18, the average deviations were -1.27% for Wasserman, slightly above 2% for both Ninkovic and Paixão, and -1.14% for MIRD. For Ga-68, the mean deviations obtained using RADARDecay were 0.54% (Wasserman), -1.25% (Ninkovic), and -1.14% (MIRD). Using the DECDATA spectrum, the results were similar: 0.58%, -1.18%, and -1.07%, respectively. For Ir-192, larger differences were observed compared to the other radionuclides, although all remained below 8%. The largest deviations were associated with the MIRD reference, with mean differences of -5.43% for RADARDecay and -5.34% for DECDATA. The average values for Wasserman and Ninkovic were -1.23% and -0.84% using RADARDecay, and -1.14% and -0.75% using DECDATA, respectively.

In addition to the statistical analysis, the method of weighted average energy was employed to investigate how spectral characteristics may contribute to deviations from the reference values. This approach aimed to improve understanding of the influence of spectral shape and energy distribution on the final air kerma rate estimates.

Table 1 presents the results for the weighted average energy differences for Ga-68, with spectral deviations expressed in percentage values.

Table 1: Ga-68 RADARDecay and DECDATA weighted average energy spectra results.

		Wasserman	Ninkovic	MIRD
	$(\times 10^{-18}\ Gy * m^2 * Bq^{-1} * s^{-1})$	$(\Delta\%)$	$(\Delta\%)$	$(\Delta\%)$
Γ_{20} **RADARDecay**	34,72	-1,38	-3,12	-3,01
Γ_{20} **DECDATA**	34,74	-1,33	-3,06	-2,95

Table 2 shows the differences in the Ir-192 weighted average energy spectra.

Table 2: Ir-192 RADARDecay and DECDATA weighted average energy spectra results.

		Wasserman	Ninkovic	MIRD
	$(\times 10^{-18}\ Gy * m^2 * Bq^{-1} * s^{-1})$	$(\Delta\%)$	$(\Delta\%)$	$(\Delta\%)$
Γ_{20} **RADARDecay**	29,50	-3,03	-2,65	-7,15
Γ_{20} **DECDATA**	29,50	-3,02	-2,64	-7,14

4 Discussions

The comparative analysis between the simulated air kerma rate values and literature data revealed good agreement for most of the radionuclides studied, with relative percentage differences below 4% for C-11, F-18, and Ga-68. These findings suggest that the spectral data adopted from both RADARDecay and DECDATA provide consistent and reliable estimates for these positron-emitting radionuclides. Compared to the reference data provided by Wasserman and Groenewald (1988) and Ninkovic and Adrovic (2005), the simulations showed close agreement, with deviations typically around or below 1.5%. The results are also in line with those reported by Paixao and Fonseca (2024), although slightly higher differences around 2% were observed in some cases, likely due to variations in emission data and spectral preprocessing methods adopted in that work.

In contrast, comparisons with MIRD data revealed larger discrepancies, particularly for Ir-192, where deviations reached approximately -5.4%. This can be partially attributed to methodological differences in the MIRD approach, such as the use of a 10 keV energy cutoff, which excludes low energy photons from the calculations, and potential differences in the photon mass energy-absorption coefficients applied in dose conversion. To better understand these variations, the weighted average photon energy was calculated for each radionuclide, enabling the identification of subtle differences in spectral shape and energy distribution. This analysis showed that certain spectral libraries yield average energies more aligned with specific references. For instance, for Ir-192 and the DECDATA spectrum, the energy-weighted average resulted in estimates of the air kerma constant with up to 1% difference from reference values, as Wasserman, Ninkovic, and MIRD. This indicates that the choice of spectral data may influence the level of agreement with the reference used. Additionally, a statistical analysis across different source to detector distances was performed to assess the robustness of the results. The near identical mean values and standard deviations obtained using RADARDecay and DECDATA for each radionuclide confirm that distance-related uncertainties were well controlled. This reinforces the reliability of the simulations and emphasizes that the primary source of variation among the datasets lies in the spectral composition itself.

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5 Conclusion

This work evaluated the air kerma rate for four radionuclides (C-11, F-18, Ga-68, and Ir-192) through Monte Carlo simulations using photon spectra from RADARDecay and DECDATA. The results were compared to reference data from Wasserman et al., Ninkovic et al., Paixão et al., and MIRD. For C-11, F-18, and Ga-68, relative percentage differences remained below 4% across all references and spectra, indicating high consistency and confirming the reliability of both spectral datasets. For Ir-192, larger deviations were observed, particularly in comparison with MIRD, likely influenced by methodological factors such as energy cutoffs and spectral data processing. The results demonstrated that spectral composition, particularly the inclusion or exclusion of low energy lines, significantly impacts the calculated Γ_{20} values by affecting both the energy weighted mean and the total photon yield. These findings are consistent with those reported by Paixao and Fonseca (2024) and help explain discrepancies with analytical models such as those of Wasserman and Groenewald (1988),Ninkovic and Adrovic (2005), and MIRD (2025). The application of a statistical analysis across different source to detector distances reinforced the internal consistency of the simulation framework. In parallel, the analysis of weighted average photon energies provided insights into how spectral shape affects agreement with each reference dataset, especially for complex photon emitters such as Ga-68 and Ir-192. Considering the results, the DECDATA spectrum is more aligned with the datasets of Wasserman and Ninkovic, both of which are based on empirical measurements and are widely accepted in the field of medical physics. On the other hand, RADARDecay yields values more closely matching those reported by Paixão, whose methodology incorporates recent decay data and refined energy weighting techniques. In conclusion, both RADARDecay and DECDATA are valid for air kerma rate estimation. The choice between them should be guided by the specific context and reference framework adopted: DECDATA may be preferable for consistency with traditional dosimetric literature, while RADARDecay may be more suitable for studies aligned with recent computational methodologies.

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Mapping of natural radionuclides U, Th and K in the city of Belo Horizonte/Minas Gerais

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Abstract

The main radionuclides that make up natural radioactivity are uranium, thorium and potassium, and their decay products. Their distribution in soils depends on the radioactive content of the rocks from which they originated. Therefore, radiometric studies aim to determine areas with a high concentration of radionuclides and the possible risk associated with exposure to ionizing radiation. The objective of this work is to map the natural radionuclides U, Th and K through scintillator detectors in external environments in the city of Belo Horizonte. The area selected for study was Belo Horizonte/MG, which present geological characteristics suggesting the existence of areas with high background, due to their geological basis in Archean rocks of the granitic-gneissic complex and metasedimentary sequences of the great Precambrian unit of the Iron Quadrangle of Minas Gerais. Researchers from the Natural Radioactivity Laboratory of the Center for the Development of Nuclear Technology (LRN/CDTN) have carried out gamma scans of the city over the years, using the RS-230 spectrometer, coupled to a Global Positioning System (GPS). The coordinates were used in geoprocessing, through the ArcGIS® program, quantitatively and qualitatively identifying the radionuclides 238U, 232Th and 40K. The scan was carried out at 12,055 points in the city, with a maximum value of 2,924 counts per second (cps), an average of 179 cps and a median of 156 cps. The region of Belo Horizonte that presented the highest CPS value was the Venda Nova region, followed by the Pampulha, Northeast and Center-South regions. Considering the same distribution of total gamma count rates in Belo Horizonte divided into very low (3-100), low (100-150), medium (150-200), high (200-400) and very high (400-3000), presented by, the quantitative analysis of the points studied showed that 14.17% of them are in the very low range of counts per second; 33.68 in the low range; 25.17 in the middle range; 24.61 in the high range; and 2.37 in the very high range. However, the qualitative visualization on a map of the data could indicate hotspots, which are not grouped by region values but by specific locations in the city. In addition, it is important to note that the gamma sweep is performed approximately 1 m from the ground. The results indicate that, although most of Belo Horizonte has low to medium levels of gamma radiation, the presence of points with high concentrations reinforces the importance of continuous monitoring and spatial analysis to identify possible areas of radiological risk.

Keywords: Natural radionuclides; Scintillating detectors; Gamma radiation.

1 Introduction

Natural radioactivity is an intrinsic characteristic of the terrestrial environment, resulting mainly from the presence of radioactive elements such as uranium (U), thorium (Th) and potassium (K). These elements, in their most common isotopic forms (U-238, Th-232 and K-40), are the main responsible for natural gamma radiation, making up most of the human exposure to radiation of natural origin (UNSCEAR, 2020).

The distribution of these radionuclides is closely linked to the mineralogical composition and geological history of the rocks that make up the Earth's crust. Igneous rocks, such as granites and syenites, tend to have relatively high concentrations of uranium and thorium, while sedimentary rocks and derived soils can have significant potassium contents due to the presence of minerals such as feldspars and micas (Tzortzis et al., 2003; IAEA, 2014).

The severity of the biological effects caused by gamma radiation depends directly on the absorbed dose, the intensity of the exposure and the sensitivity of the tissues affected. From an environmental point of view, this form of radiation can pose a risk to both humans and ecosystems, especially in places contaminated by radioactive waste or by industrial activities that use radioactive materials. Prolonged exposure or exposure to high doses can cause severe deterministic effects, such as acute radiation syndrome, while lower but continuous or frequent levels increase the likelihood of stochastic effects, such as the emergence of cancer (Tauhata, 2013). These potential impacts underscore the importance of environmental monitoring and the adoption of mitigation measures to reduce the risks associated with ionizing radiation.

From an environmental and public health perspective, the presence of anomalous concentrations of natural radionuclides may pose a risk, since chronic exposure to ionizing radiation is associated with cumulative biological effects, including a higher likelihood of certain cancers. In this context, international organizations such as the International Atomic Energy Agency (IAEA) and the United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) recommend that radiometric mapping studies be carried out systematically, especially in inhabited regions or those subject to urbanization and mining processes (IAEA, 2014; UNSCEAR, 2020).

In Brazil, the diversity of geological contexts, as well as their territorial expansion, results in great variation in the levels of natural radioactivity. Studies show that geological formations composed of crystalline terrain, Archean shields, and folding bands present significant anomalies, which can generate exposures higher than the reference limits established for public individuals (Silva, 2024).

In this scenario, the application of portable scintillator detectors coupled to global positioning systems (GPS), associated with geoprocessing techniques, has proven to be an effective methodology for regional and urban mapping (Tzortzis, 2003). These studies allow the identification of areas with greater radiological potential, subsidizing public policies for environmental monitoring and protection of the population's health.

Although there are advances in the national and international literature, there are still significant gaps in the detailed knowledge of the spatial distribution of U, Th, and K in many Brazilian cities. Recent studies reinforce the importance of denser and more up-to-date investigations, especially in urban centers where the combination of favorable geology and intense urbanization can increase the risk of exposure of the population (Silva, 2024).

In view of the possibility of occurrence of high concentrations of natural radionuclides (U, Th and K) in urban areas, this study aims to study the mapping of the spatial distribution of ambient gamma radiation in the city of Belo Horizonte, using portable scintillator detectors of the RS-230 type. In this context, the research seeks to fill gaps that still exist on the radiometric characterization in external areas of the capital of Minas Gerais, contributing to the knowledge about the variability of natural radiation levels. For this, a gamma scan made at various points in the city was studied, conducted by researchers from the Natural

Radioactivity Laboratory of the Center for the Development of Nuclear Technology (LRN/CDTN), recording the counts per second (cps) obtained *in situ*. The collected data were georeferenced and organized on a map in the ArcGIS® environment, allowing the identification of distribution patterns and possible radiological hotspots. In addition, it is intended to discuss the implications of the results obtained for public health, urban planning and environmental management, emphasizing the importance of continuous monitoring strategies and mitigation of potential risks associated with exposure to ionizing radiation in urbanized areas.

2 Materials and Methods

2.1 Field of study

The area analyzed corresponds to the city of Belo Horizonte, capital of Minas Gerais, considered the sixth largest capital in the country. In 2022, the population estimate was 2,315,560 inhabitants, with a demographic density of 6,988.18 people per square kilometer (IBGE, 2025). From a geological point of view, about 70% of the urban area and the metropolitan region is based on Archean rocks of the granitic-gneissic complex (Belo Horizonte Complex) and on metasedimentary sequences belonging to the extensive Precambrian unit of the Iron Quadrangle, indicating the potential presence of zones with high levels of background radiation (Takahashi, 2022).

The municipality is part of the extensive São Francisco craton (Almeida, 1984), limited to the south by the Iron Quadrangle (QF). The geology of the QF is marked by three major main lithological sets: metamorphic complexes composed of Archean crystalline rocks; the archean sequence of the *greenstone belt* type represented by the Rio das Velhas Supergroup; and the paleoproterozoic metasedimentary sequence of the Minas Supergroup (Azevedo, 2007).

The Minas Supergroup is formed predominantly by metasedimentary rocks and is divided into subgroups called Sabará, Piracicaba, Itabira and Caraça. The Sabará Group represents the thickest unit of this supergroup, composed largely of intensely weathered schists and phyllites, conglomerates and metagrauvacas, rocks that can concentrate clay minerals and phyllosilicates, favoring the retention of potassium and traces of uranium in certain contexts (Corrêa, 2019). The Piracicaba Group is mainly home to metasedimentary rocks, such as phyllites, quartzites and metamorphized carbonates, which may also contain potassium-bearing minerals and associations with phosphate minerals, which in some cases are associated with thorium. The Itabira Group, on the other hand, is subdivided into two formations: Gandarela and Cauê, the latter being notable for containing banded iron formations (BIF), characterized by alternating layers of quartz and iron oxides, such as hematite and magnetite, which can occur together with accessory minerals that concentrated uranium during ancient geological processes (Carmo, 2023).

Therefore, the choice of Belo Horizonte as the study area is justified by the combination of geological and historical factors, which potentiate the occurrence of variations in natural radiation levels. The predominant presence of Archean rocks from the Belo Horizonte Complex and metasedimentary sequences from the Iron Quadrangle indicates a substrate favorable to the concentration of natural radionuclides, such as uranium, thorium and potassium.

2.2 Gamma screening

For the radiometric survey, over the last few years, gamma scans have been carried out at different points in the city of Belo Horizonte by researchers from the Natural Radioactivity Laboratory of the Center for the Development of Nuclear Technology (LRN/CDTN). For this, a portable spectrometer model RS-230 (RadiationSolutions, Canada) was used, a self-portable equipment sensitive to gamma radiation, coupled to a Global Positioning System (GPS) for precise recording of the geographic coordinates of each sampled point. The location information obtained was processed in a geoprocessing environment using the ArcGIS® software, allowing the quantitative and qualitative analysis of the presence and distribution of radionuclides ^{238}U, ^{232}Th e ^{40}K. From the data collected, the count-per-second (CPS) values were recorded, corresponding to the number of gamma radiation decay events detected per second, which makes it possible to estimate the intensity of natural radioactivity on the monitored surface.

3 Results and Discussions

Based on the data obtained through the gamma scan, which analyzed 12,055, the distribution of the total gamma count rates in Belo Horizonte was divided into: very low (3-100 CPS); low (100-150 CPS); medium (150-200 CPS); high (200-400 CPS), and very high (400-3000 CPS) (Duarte, 2017). To facilitate the visualization and spatial analysis of the results obtained, a map was made in the ArcGIS® software, which enabled the georeferenced representation of the measurements performed. This strategy allowed the recognition of patterns in the spatial distribution of radioactivity, favoring a more consistent evaluation of natural influences in the urban area of Belo Horizonte.

Based on the total gamma radiation count values, a heat map was prepared using the interpolation tool available in ArcGIS®. The interpolation was performed using the inverse distance weighting (IDW) method, which estimates values on a raster surface from the sampled points, assigning greater weight to the closest points (ESRI, 2024). The parameter used as an interpolation attribute was the gamma count in CPS (counts per second), and these values are considered as weights in the modeling of the spatial distribution of gamma radiation, as illustrated in Figure 1.

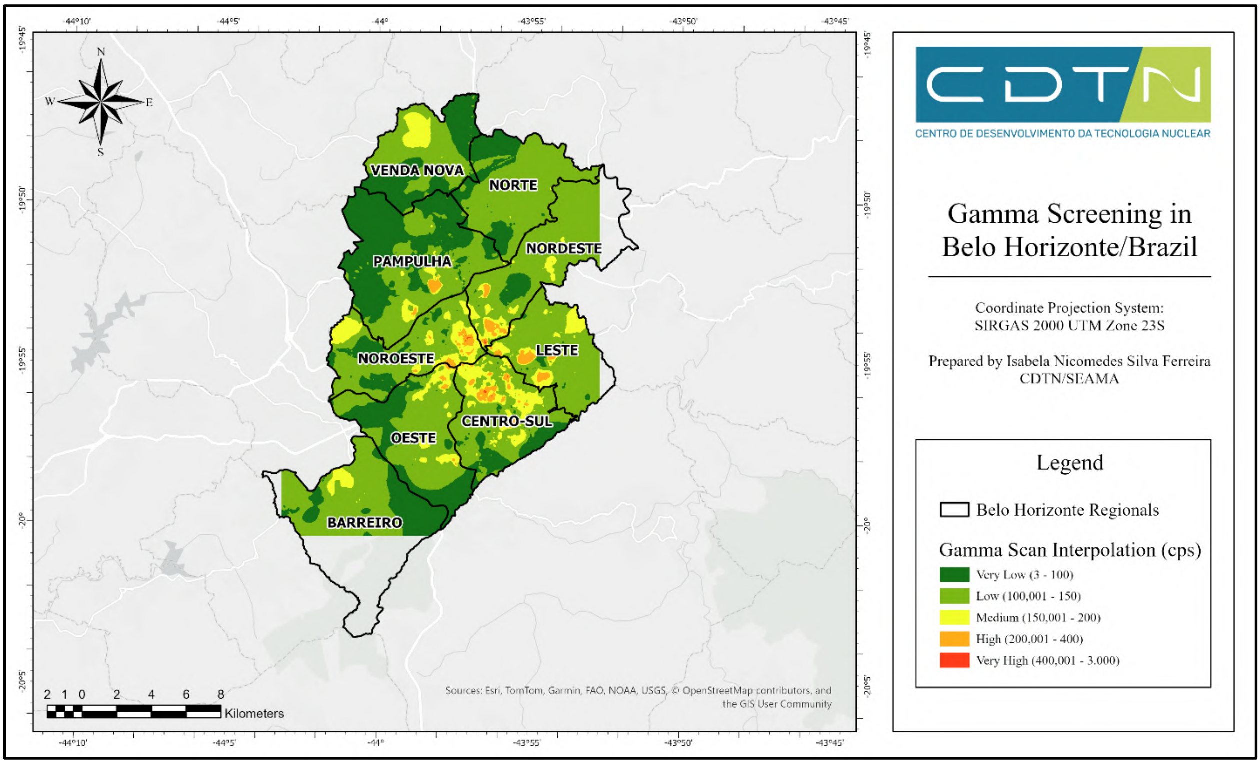

Figure 1: Gamma Screening in Belo Horizonte/Brazil.

The analysis of the data obtained by means of gamma scanning allowed the identification of the spatial variation of counts per second (CPS) of gamma radiation in the urban area of Belo Horizonte. It was observed that most of the sampled points presented SPC values classified between the low and medium ranges, comprising, respectively, intervals of 100 to 150 CPS and 150 to 200 CPS. These intervals together represent more than half of the measurements, indicating that, in general, the levels of gamma radiation in the city remain within standards compatible with those expected for urban environments located in Archean crystalline terrain.

On the other hand, it was possible to identify relevant percentages of points classified in the high (200–400 CPS) and very high (400–3,000 CPS) ranges, even if in a smaller proportion. The presence of these higher values indicates the occurrence of punctual radiometric anomalies, which may be related both to local variations in lithology, such as granitic outcrops or gneisses with higher levels of uranium and thorium, and to possible anthropogenic influences, such as areas of soil movement or reuse of rock materials in civil construction.

The spatial analysis evidenced in the heat map, constructed from the IDW interpolation, reinforces this heterogeneity. Specific regions with the highest concentration of SCP are noted, highlighting, according to the consolidated data, the regions of Venda Nova, Pampulha, Northeast and Center-South, which had the highest recorded counts. This distribution does not occur homogeneously across entire neighborhoods, but rather in isolated points or spots, which demonstrates the importance of detailed approaches at the local scale.

The identification of these *radiometric hotspots* makes it possible to evaluate priority areas for more detailed monitoring in the future, in addition to providing subsidies for complementary studies that can investigate the geological or anthropogenic origin of these anomalies. It is also important to consider that the measurements were taken approximately 1 meter from the ground, reflecting the external gamma radiation that can contribute to the exposure dose of individuals in open areas.

During the period of planning and construction of the capital, at the end of the nineteenth century, the demand for resistant and easily accessible materials favored the opening of quarries located close to the urban core. The rocks extracted from these quarries were widely used in the construction of historic buildings, paving streets and implementing infrastructure works, such as retaining walls and bridges. The availability of mineral resources directly contributed to the rapid growth of the city throughout the twentieth century (Ruchkys de Azevedo, 2007).

The quarries were fundamental for the development of Belo Horizonte, ensuring the supply of basic materials for civil construction since its foundation. Scattered in different parts of the city and its metropolitan region, these extraction areas mainly supplied gneisses, quartzites and granites, rocks that, due to their mineralogical characteristics, can contain significant levels of uranium, thorium and potassium. This intense use of local materials contributes to explain, in part, the occurrence of radiometric hotspots identified in the present study, since extraction residues, remaining blocks or even structures built with these rocks can influence the variation of gamma radiation at certain points in the urban area.

In the environmental and urban context of Belo Horizonte, these results indicate that, although most of the city has low to moderate levels of gamma radiation, the existence of points with high values reinforces the relevance of continuous monitoring, especially in regions of urban expansion or in areas where the use of local rock materials is frequent. These data can support territorial planning policies, risk prevention and good environmental management practices aimed at protecting the population against unnecessary exposure to natural ionizing radiation.

4 Conclusions

The results obtained in this study confirm that the city of Belo Horizonte presents, in its majority, levels of ambient gamma radiation classified between the low and medium ranges, compatible with the values expected for urban regions located in terrains of crystalline geological base. On the other hand, the identification of specific points with higher counts, classified in the high and very high ranges, demonstrates the occurrence of localized radiometric anomalies that may be associated both with natural geological characteristics, such as the presence of granitic and gneissic rocks, and with anthropogenic factors related to the history of use of rock materials in civil construction.

The elaboration of the thematic map and the spatial analysis by interpolation allowed a detailed visualization of the distribution patterns of gamma radiation, highlighting potential areas of attention for future investigations. Such information is essential to support environmental management actions, urban planning and continuous radiological monitoring, aiming to prevent risks associated with prolonged exposure of the population to natural ionizing radiation.

In addition, the history of quarrying and the extensive use of local rock materials in civil construction can directly influence the distribution of gamma radiation on the urban surface. Considering the high population density of the capital of Minas Gerais and its intense urban expansion, radiometric mapping of the region becomes essential to support environmental monitoring, territorial management and public health protection actions.

Thus, this work reinforces the importance of expanding radiometric surveys at the urban scale, especially in regions with a history of intense occupation and geological characteristics favorable to the concentration of natural radionuclides. It is recommended that complementary studies be carried out, covering more detailed analyses at critical points, in addition to extending the mapping to areas not yet contemplated, in order to complete the coverage of the entire urban area. As a future perspective, the need to estimate the doses of exposure of the public individual is also highlighted, deepening the assessment of health risks. Finally, it is noteworthy that this study meets the recommendations of the International Atomic Energy Agency (IAEA) for the monitoring of environmental radiation, and Belo Horizonte is one of the few Brazilian cities that have this type of systematic survey.

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Modeling and validation of a Whole Body Counter for *in vivo* measurements using CAD and MCNP6

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Abstract

Whole body counter (WBC) systems are of significant relevance for internal dosimetry, as this type of system is used to evaluate occupationally exposed individuals (OEIs) and the general public, in case of accidents (IAEA, 1996; Raabe, 1994). Due to factors such as limited space and economic feasibility, some systems feature shielding in the form of beds or small cabins made of materials like steel and lead (IAEA, 1996; Rahman, 2008; Paiva, 2017). Therefore, these aspects must be properly considered when proposing changes to the geometry or materials to improve these systems. Simulations using algorithms based on the Monte Carlo Method are widely employed in studies of this type of system, both for modifications and for calibration purposes (Paiva, 2016; Paiva, 2017; Ferreira, 2023). Through simulations, a proposed modification can be studied, evaluated, and refined before being implemented in the system. Thus, simulations can be valuable tools for studying and planning changes in the shielding and geometry of a WBC. Given this context, the objective of this study was to model, simulate, and validate a whole-body counter used for in vivo monitoring, aiming to improve the quality of internal dosimetry measurements performed by this system. To this end, computational simulations were carried out using MCNP6 (Monte Carlo N-Particle Transport Code), with the WBC's geometric model created in AutoCAD software, converted and imported into MCNP6 using the Visual Editor-Vised (Schwarz, 2005). In parallel, photon spectrometry readings were performed to provide the input spectra for the simulation. The simulated results were validated by comparing them with experimental data from previous studies on detector shielding, in which the model showed a variation of less than 10% compared to experimental data, and this variation was considered acceptable for validating the model proposed in this study.

Keywords: Internal Dosimetry; Whole Body Counter; MCNP6; AutoCAD.

1 Introduction

In vivo internal monitoring involves the use of highly sensitive detectors, such as NaI(Tl) scintillators or HPGe semiconductors, in order to detect ionizing radiation emitted by radionuclides present within the body of the monitored individual. These detectors can be employed in Whole Body Counter (WBC) systems, which are typically designed in bed or chair geometries to appropriately accommodate the monitored person (Raabe, 1994; IAEA, 1996; Knoll, 2010; ICRP, 2015).

Designing effective shielding for a WBC is not a trivial task, as the use of high atomic number (Z) materials alone can enhance system performance but may introduce other types of spectral noise. Ideally, the inclusion of copper and cadmium layers between the lead shielding and the detector helps to reduce noise originating from characteristic X-rays and backscattered photons (IAEA, 1996). However, these are expensive metals, and their incorporation into the shielding is not always straightforward. Therefore, it is advantageous to have tools that support the study and planning of shielding modifications in a WBC system, such as computational simulations, for which a validated model is essential.

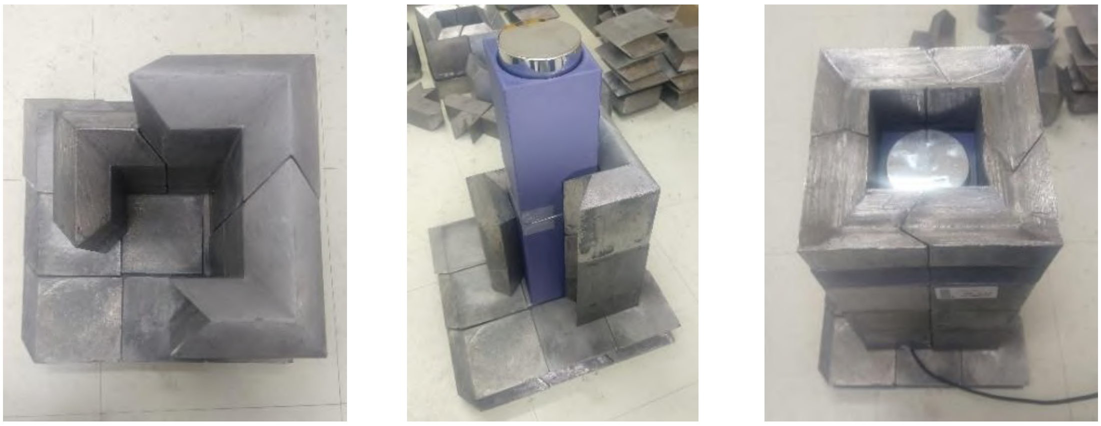
Simulations using Monte Carlo-based algorithms are widely applied in studies of this type of system, both for evaluating modifications and for calibration purposes (Paiva, 2016; Paiva, 2017; Ferreira, 2023). Through simulations, a proposed modification can be studied, assessed, and refined before being implemented in the actual system.

This work presents the modeling and validation of a whole-body counter system composed of a shadow-shield type shielded bed and an 8”×4” NaI(Tl) detector. A case study involving a change in the detector’s positioning was carried out. The study was conducted using CAD tools integrated with a Monte Carlo code and validated with experimental photon spectrometry data.

2 Materials and methods

2.1 Photon background spectra measurements

Measurements of the photon background spectrum in the monitoring room were carried out to define the source term for computational simulations using the MCNP6 code. Acquisitions were performed with a Canberra 3”×3” NaI(Tl) detector shielded by 5 cm thick lead bricks, as shown in Figure 1 (a). To ensure proper positioning and protection of the detector, a support structure was produced using 3D printing in ABS material (lilac-colored structure), as shown in Figure 1 (b). Only the front face of the detector was exposed. A slight collimation effect was achieved since the detector face was recessed approximately 5 cm from the front edge of the assembly, as observed in Figure 1 (c). This setup was oriented toward the six surfaces of the monitoring room where the WBC is located: floor, ceiling, and lateral walls. For all directions, the acquisition time used was one hour (3,600 seconds).



(a) (b) (c)

Figure 1: Detector assembly, support, and shielding. (a) Shielding structure. (b) Detector support. (c) Collimation setup.

The background photon spectra from the six directions of the monitoring room were corrected for energy drift, energy-dependent detection efficiency, and incomplete energy deposition effects (Antunes, 2021). An average spectrum was generated from the six measurements to provide a simplified representation of the photon field within the monitoring room. This simplified spectrum was then used in computational simulations to mimic the background photon emission in the monitoring environment.

2.2 WBC geometry modeling

Information used for the 3D CAD modeling of the geometry applied in this study was obtained through measurements and photographic records of the WBC. The modeling was performed using AutoCAD®. However, conventional CAD modeling does not meet certain constraints required for conversion to MCNP6 format. Therefore, files intended for use with MCNP6 must be modeled in a manner compatible with importation and conversion using Vised (Schwarz *et al.*, 2005; Pacheco, Mendes, Meira-Belo, 2023). The conversion process enabled geometric information imported from the CAD model, such as surfaces and cells. Definitions necessary for problem setup, including source specification, output requests (tallies), and the number of particle histories (NPS), were completed directly within the input text file.

The elemental composition (by mass fraction) and density of materials used in the simulations were obtained from the literature (Detwiler *et al*., 2021) and are presented in Table 1.

Table 1: Elemental composition of materials used on WBC modeling (Detwiler *et al*., 2021).

Element	Dry air	Lead	Carbon steel (1045)
C	0.000124	-	0.005
N	0.755268	-	-
O	0.231781	-	-
Air	0.012827	-	-
Fe	-	-	0.995
Pb	-	1.00	-
P ($g.cm^{-3}$)	0.001205	11.35	7.87

Standard photon (mcplib84) and electron (el03) transport libraries in MCNP6 were used. Similarly, the default cutoff energy for photon and electron transport was maintained at 1 keV for both. The number of particle histories (NPS) varied from 1.0E+08 to 5.0E+09, depending on the simulated case. Initially, both photons and electrons (mode p e) were transported in the simulations. However, after preliminary testing, it was confirmed that for the type of problem under analysis, transporting only photons (mode p) yielded equivalent results with significantly shorter simulation times compared to cases where both photons and electrons were transported.

2.3 Photon background simulation and obtention of FM factor

A spherical source with photon emissions directed inward from the shell toward the center of the sphere was proposed to simulate the photon background field in the monitoring room. A spherical surface with a radius of 200 cm was positioned such that its center coincided with the center of the 8”×4” NaI(Tl) detector crystal. The particle emission direction was defined using the ALE mode: random emissions within an angular distribution from 0° to 180° relative to the normal vector of the surface, with probabilities distributed over a hemispherical surface from the point of emission, according to the cosine law p(cos)=cos, -1≤ cos ≤1, where p(cos) is the probability associated with a given emission angle.

The source spectrum was implemented as a histogram (SI# H) with 1024 energy bins, each with a width (ΔE) of 3 keV. The energy bins and their respective emission probabilities for photons incidents on the WBC were derived from the simplified photon field spectrum inside the monitoring room, as described in the previous section.

The simulation output request (tally) was defined as the number of photon pulses in the crystal of the 8”×4” NaI(Tl) detector per photon emitted from the source (Tally F8:p). A total of 1024 energy bins were specified to obtain a photon count spectrum as a function of energy (e8 0 1e-5 0.003 1022i 3.072). Each bin had a width (ΔE) of 3 keV, corresponding to the energy calibration applied to the WBC’s NaI(Tl) detector. Consequently, the simulated spectrum energy range spanned from 3 keV to 3072 keV. Simulations were carried out on a DELL computer with an Intel® Core™ i7-4790 3.6 GHz CPU and 16 GB of RAM. Simulation time ranged from 9 hours to 10 days, depending on the case.

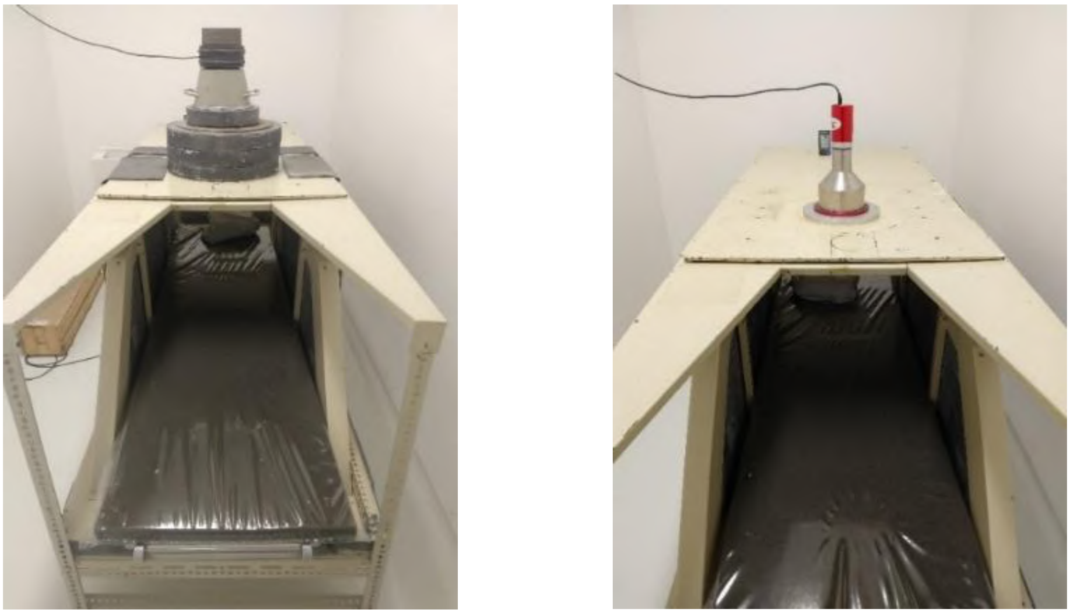
Based on these simulations, it was determined the number of photons required to be emitted by the source in order to match the total number of counts in the simulated spectrum with the reference experimental spectrum, acquired over 3,600 s using the 8”×4” NaI(Tl) detector in its current shielding configuration (Figure 2(a)). A multiplication factor (FM) was thus obtained, representing the ratio of photons to be emitted in the source. This factor was applied in all subsequent simulations to generate photon spectra corresponding to 3600-second acquisition periods, using their respective source definitions. Equation (1) was used to define the count scaling factor, considering the detector’s current shielding configuration.

$$FM = \frac{\sum TC_E}{\sum C_S} \tag{1}$$

Where *FM* is the multiplication factor used to scale the number of photons to be emitted, TC_E is the total counts in the experimental spectrum and C_S is the total counts in the simulated spectrum.

2.4 Validation of WBC modeling

To validate the WBC model, the FM factor obtained using the current shielding configuration (Figure 2 (a)) and background (BG) source modeling was tested. Simulated results were compared with an experimental acquisition performed under a different shielding configuration than the current one. In this study, the chosen configuration involved the removal of the upper lead shielding of the detector, as shown in Figure 2 (b). This shielding setup had been used previously, and experimental background spectra data were already available. A 3,600 second acquisition was conducted with the WBC using the 8”×4” NaI(Tl) detector in this unshielded configuration.



(a) (b)

Figure 2: Shielding configurations for validation. (a) Current configuration, initially modeled and used in the determination of the multiplication factor. (b) Configuration without the detector's upper lead shielding, used for validation purposes.

Figure 3 shows the background count spectra for the WBC detector with and without the upper shielding.

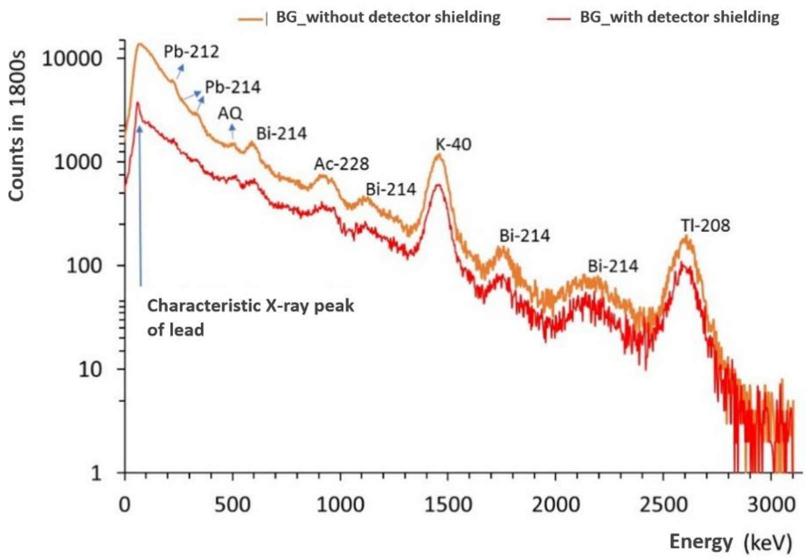


Figure 3: Background spectrum with and without detector shielding.

The WBC modeling with and without detector shielding can be seen in Figure 4. The multiplication factor obtained in the previous step was applied to the simulation results in the unshielded configuration. The simulated spectra were generated using the spherical surface source applied to the model without upper shielding (Figure 4 (c e d)).

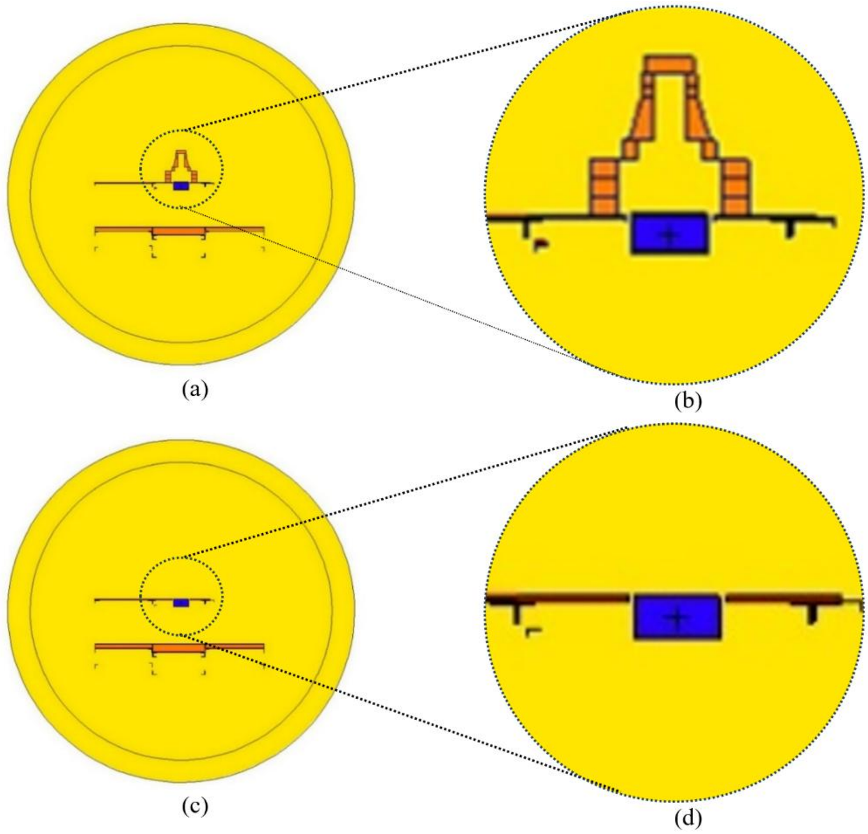


Figure 4: Comparison of the WBC model with and without shielding. (a) Front view with shielding. (b) Detailed view with shielding. (c) Front view without shielding. (d) Detailed view without shielding.

The resulting spectra were compared with the experimental spectrum acquired without the detector's upper shielding. The comparison was carried out both qualitatively, through visual assessment of the spectral shape, and quantitatively, through calculations of the mean energy of the spectrum, total count summation, and channel-by-channel percent difference. The mean energy of each spectrum was calculated using Equation (2).

$$\bar{E} = \sum_{3\ keV}^{3072\ keV} \frac{C_C}{TC} \times E_C \tag{2}$$

Where $\bar{E}$ is the mean energy of the evaluated spectrum, C_C is the total counts in the channel, TC is the total counts in the spectrum, and E_C is the energy of the corresponding channel.

3 Results and discussion

3.1 Photon background spectra measurements

For the photon background (BG) spectrum in the monitoring room, the mean emission probability across the measured spectra was adopted as a representative approximation for the BG. The spectrum representing the average of the photon background spectra from the monitoring room is shown in Figure 5.

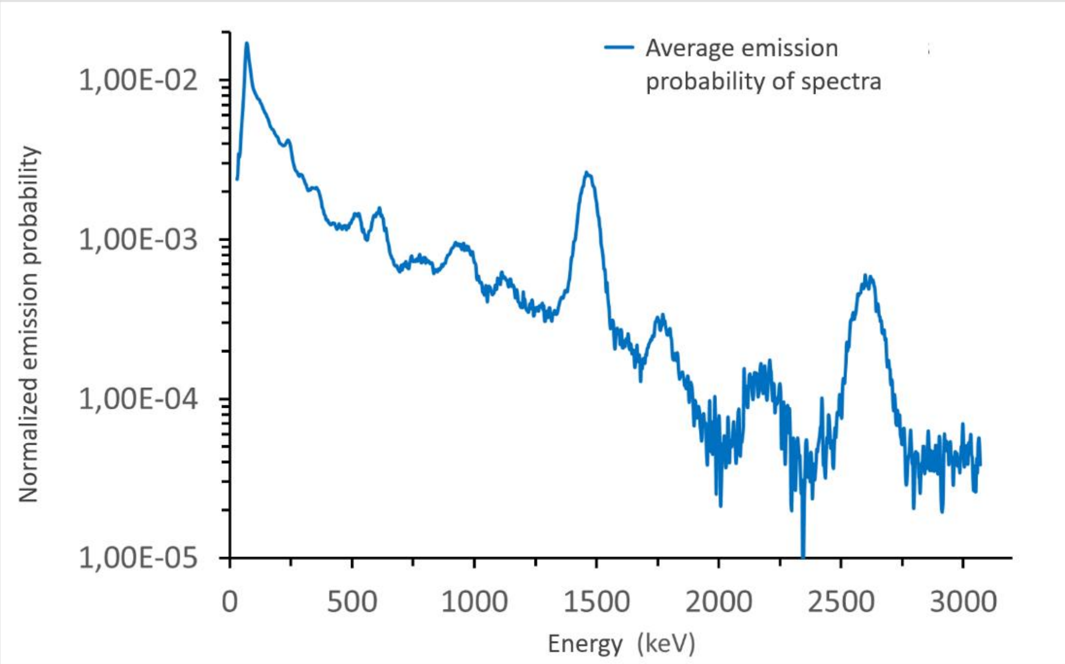
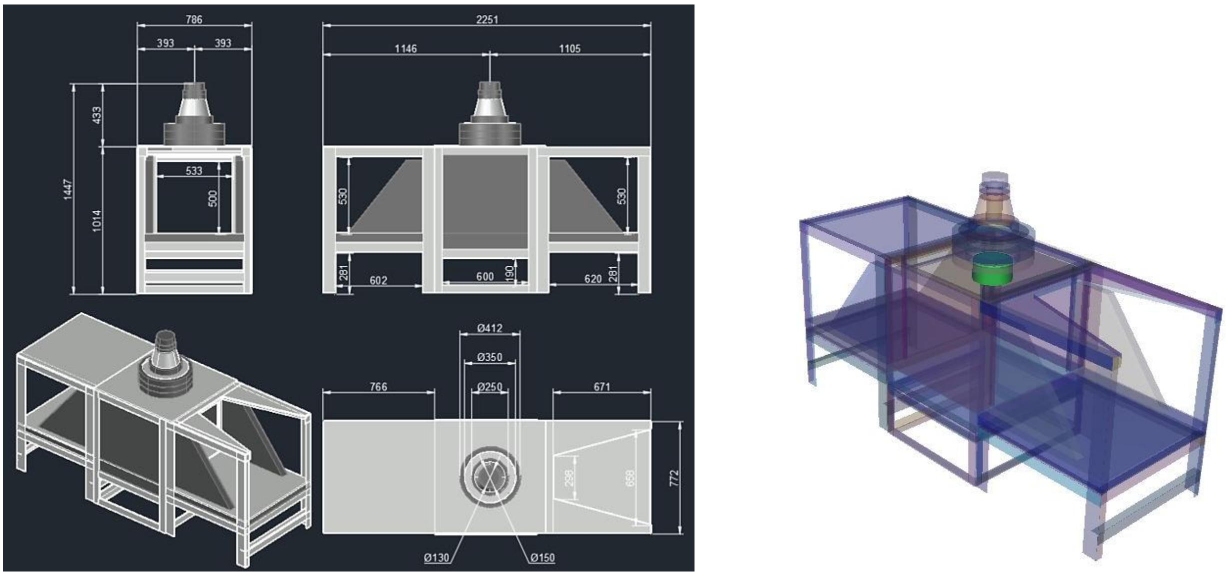


Figure 5: Mean photon emission probability from background spectra.

An undesired effect of the spectral correction process is an increase in noise level, i.e., increased variance in channel counts. This effect was particularly noticeable in the high-energy region of the spectrum. Nevertheless, the spectra remain visually very similar.

3.2 WBC geometry modeling

The CAD model of the WBC, based on physical measurements and photographic documentation, is shown in Figure 6. Relevant dimensions are included in Figure 6 (a). The CAD model was converted into the input file format for MCNP. The 3D representation of the WBC model in MCNP format, as generated using Vised®, is shown in Figure 6 (b). This geometric model was used in all simulations performed in this study.



(a) (b)

Figure 6: WBC geometry modeling results. (a) CAD model showing the main WBC dimensions in mm. (b) WBC model used on simulation.

3.3 Photon background simulation and obtention of FM factor

The simulated BG output spectrum is shown in Figure 7. The number of photons that had to be emitted from the source, based on the multiplication factor, so that the resulting simulated spectrum matched the total number of counts observed in the reference experimental spectrum (Figure 8) was 1.123E+10 photons. The BG curve simulated using this multiplication factor visually shows a strong resemblance to the measured BG spectrum, as demonstrated by the overlap of the two curves

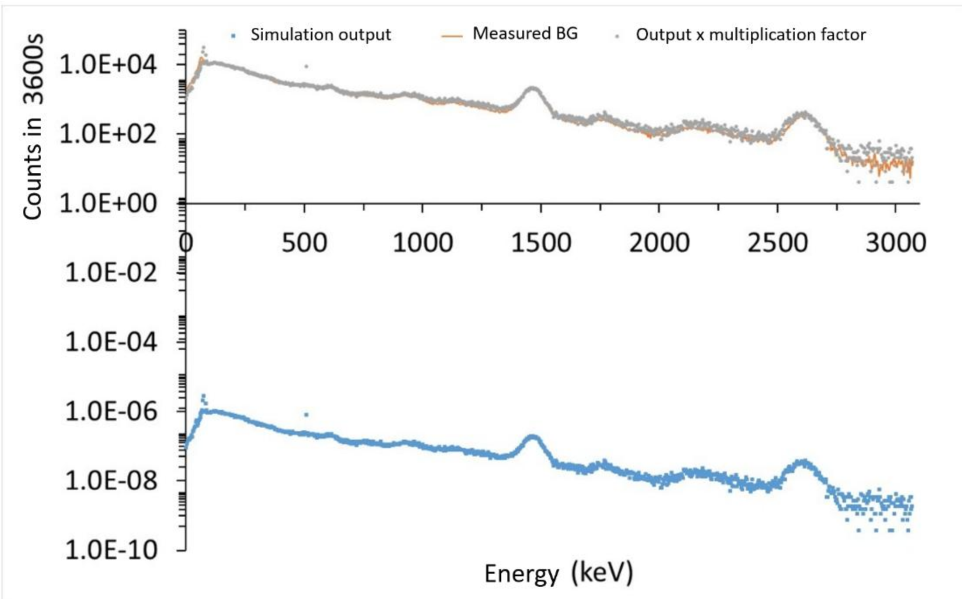


Figure 7: Comparison between the measured and simulated BG spectra.

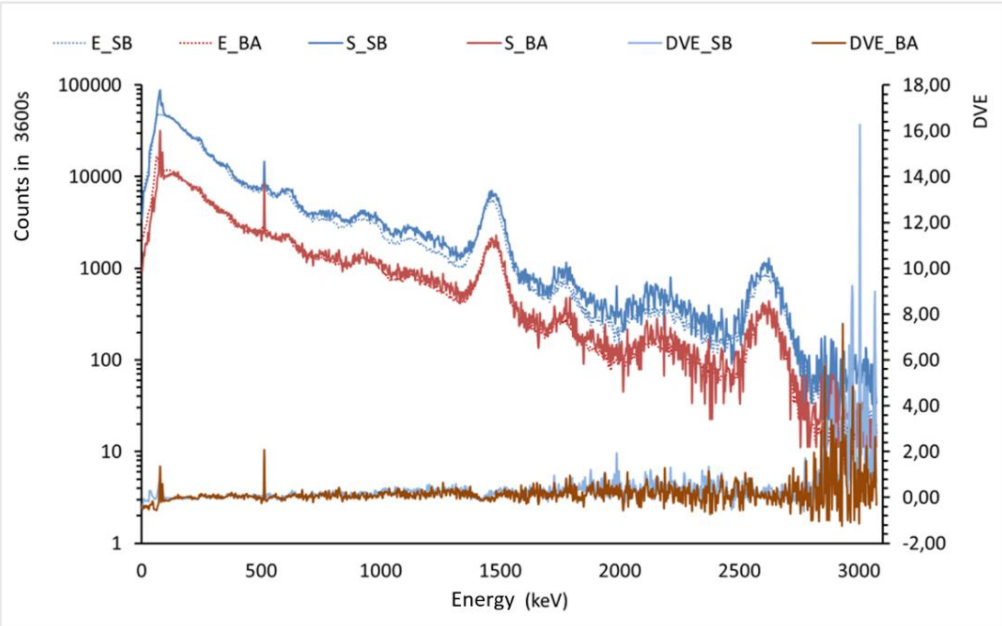
As observed, the simulation results closely matched the experimental spectrum in terms of behavior and spectral shape, key aspects for the validation of the study.

3.4 Validation of WBC modeling

The experimental spectrum acquired by the 8”×4” NaI(Tl) WBC detector without the upper shielding is shown in Figure 3. As expected, when compared to the spectrum obtained under the current shielding configuration, a substantial increase in count rates was observed across almost all channels. In the absence of the upper shielding, photon counts increased by an average of 260% in the low-energy range (50 to 300 keV); by 173% at the 1461 keV gamma line from K-40; and by 105% at the 2,614 keV gamma line from Tl-208. These results underscore the importance of the upper detector shielding, particularly in attenuating low-energy photons. Additionally, Table 2 shows a 225% increase in total counts for the spectrum without shielding compared to the current shielding configuration. The mean energy of the spectrum acquired without the upper shielding was lower than that of the spectrum obtained under the current shielding configuration. This occurs because the upper shielding is more effective at attenuating low-energy photons, thereby increasing the mean energy of the detected beam.

The simulated spectra were compared to the experimental spectra, as shown in Figure 8. The number of histories followed was NPS = 3.0E+09. Even so, relative errors greater than 10% were observed in the higher energy range, especially above the Tl-208 peak at 2,614 keV. This resulted in high variance in the estimated counts over 3600 seconds in this energy region. Visually, however, the simulated spectra were similar to the experimental ones. Quantitative results also showed close agreement, as seen in Table 2.

The total count summation of the simulated spectrum without shielding was 9.4% higher than that of the experimental spectrum. The differences in mean energy between simulated and experimental spectra were: 7.5% for the simulated spectrum with current shielding, and 6.5% for the simulation without shielding. It is also worth noting that the ratio between the mean energies of the simulated spectra (with and without shielding) was only 0.8% higher than the corresponding ratio observed in the experimental spectra.



Legend: E_SB–Experimental without shielding; E_BA–Experimental with current shielding; S_SB–Simulation without shielding; S_BA–Simulation with shielding; DVE_SB–Difference between experimental and simulated values without shielding; DVE_BA–Difference between experimental and simulated values with current shielding.

Figure 8: Comparison between experimental and corrected simulated spectra.

The channel-by-channel difference curve shows increased variation in the higher energy range. The simulated spectrum exhibits stable behavior, as evidenced by the relatively small difference in total counts between simulation and experiment. This is attributed to the use of the ALE random emission configuration, which closely resembles the physical behavior of environmental background radiation.

Additionally, in the difference curve, two prominent peaks can be observed in the low-energy region of the simulated spectra: one at 72 keV, corresponding to characteristic X-ray emissions from lead, and another at 511 keV, related to electron–positron pair annihilation.

Table 2: Comparison of experimental and simulated spectrum parameters.

Evaluated Parameters	**Experimental Data**		**Simulation Data**	
	BA	**SB**	**BA**	**SB**
Σ of counts	1514003	4926649	1514003	5391426
DVE Σ of counts	-	-	0.0%	9.4%
E_{Med} (keV)	472	415	508	442
DVE E_{Med}	-	-	7.5%	6.5%
BAE_{MED} /SBE_{MED}	1.14		1.15	
DVE BAE_{MED} /SBE_{MED}	-		0.8%	

Legend: SB – Without shielding; BA – Current shielding configuration; DVE – Percentage difference between experimental and simulated values, calculated as [DVE = (Simulado-Experimental) / Experimental].

For the shielding validation study, variations around 10% were considered acceptable. This variation is based on studies of uncertainty in internal dosimetry that have variations close to or greater than this value (Castellani *et al.*, 2013). Based on the comparison of the evaluated parameters shown in Table 2, the simulated model demonstrated values and behavior closely aligned with the experimental results. In particular, the comparison of total counts, used as a proportionality parameter between simulation and experimental outcomes, showed a difference of less than 10%, which falls within the acceptable range for this study. Therefore, the results validate the proposed model as a satisfactory representation of the physical WBC system.

4 Conclusions

The improvement of the shielding system of the WBC at Laboratory of Internal Dosimetry (LDI /CDTN) represents a key step toward enhancing the quality of internal dosimetry monitoring performed at CDTN. This directly benefits a large number of occupationally exposed individuals (OEIs), given the high volume of monitoring procedures carried out by LDI, and also extends to the general public, as this system may eventually be used in radiological emergency scenarios as well.

The measurements performed allowed for a proper characterization of the background radiation in the monitoring room, as the spectra obtained showed expected behavior and were consistent with previous measurements carried out at LDI. This enabled the use of a spherical source, based on the average emission probability across the measured spectra, as an adequate approximation of the background radiation field.

The proposed model yielded results that closely matched the experimental data, thus validating the model as a reliable representation of the physical WBC system. As a result, the validated model can be employed in future simulations and improvement proposals for the WBC. This will allow for computational verification of design changes and refinements before implementation, including modifications in shielding configuration, detector positioning, and even the location of the CDNT monitoring room itself.

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Monte Carlo simulation of a HPGe detector

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Abstract

This study presents a methodology for modeling and optimizing full-energy peak efficiency (FEPE) values using the Particle and Heavy Ion Transport Code System (PHITS), a general-purpose Monte Carlo code. A coaxial HPGe detector (CANBERRA GC5019) was modeled based on manufacturer specifications, and simulations were performed to calculate FEPE for several gamma emitters radionuclides (Am-141, Co-57, Cs-137, Co-60) at different source-to-detector distances. The initial simulation results overestimated FEPE compared to experimental data. To improve the agreement, critical geometrical parameters were systematically adjusted, including the window-to-crystal distance, upper and lower dead layer thicknesses.

The optimized configuration reduced the mean deviation between simulated and experimental efficiencies to below 4%, demonstrating the sensitivity of FEPE to small changes in detector geometry. The final model reliably reproduced experimental FEPE values across a range of energies (59.5 to 1332.5 keV) and measurement geometries. This work confirms that combining accurate experimental data with Monte Carlo simulations and geometrical calibration allows for robust FEPE prediction, especially in applications where direct calibration is limited or infeasible.

Keywords: Monte Carlo; PHITS; HPGe; Simulation;

1 Introduction

Gamma spectrometry underlies several nuclear techniques and radiological protection procedures. To determine the gamma-ray spectrum, the most efficient type of gamma detector is a semiconductor made of high-purity germanium (HPGe), due to its high resolution (YU et al., 2022).

In general, gamma detectors operate in the energy range of 0.01–3 MeV, and their response results from interactions between photons and the detector material: photoelectric effect, Compton scattering, and pair production. In all these processes, the energy of the gamma ray is transferred to electrons in the detector material. These electrons generate an electrical pulse, which is amplified by an electronic system, converted into a digital signal, and processed by a multichannel analyzer, which organizes the events according to their energy to form the gamma spectrum (MOLNÁR, 2004).

In practical gamma spectrometry, knowing the full-energy peak efficiency (FEPE) of the detection system relative to a radioactive sample is essential, as this parameter relates the activity of a radionuclide in the sample to the number of counts recorded in the corresponding peak of the spectrum, allowing for quantitative analysis of detected radionuclides.

To determine FEPE experimentally, gamma-ray sources with one or more known-energy peaks and known activity are often used. In practice, this determination can be difficult for various reasons:

- Absence of sources at certain energy levels
- Sources with limited or unsuitable shapes and dimensions
- Sources that emit cascade gamma rays require correction due to the detector's response time

Due to the nature of these experimental difficulties, particle transport computational methods are often used to estimate FEPE in various geometries and isotopes. In this work, the FEPE of an HPGe detector was calculated using the **Particle and Heavy Ion Transport Code System (PHITS)** (Sato et al., 2013), a FORTRAN code that simulates the transport and collision of various types of particles across a wide energy range.

Previous studies (JARGALSAIKHAN et al., 2025; PENABEÏ et al., 2024; SANTO et al., 2012; AGARWAL et al., 2011) have shown that the FEPE values obtained through simulation and validated experimentally for an HPGe detector with a gamma source can exhibit significant deviations when the simulations rely solely on the manufacturer’s specified dimensions. These discrepancies arise from the Monte Carlo method’s high sensitivity to the exact geometrical details and material properties that govern radiation interactions. In addition, the dead layer thickness—the superficial germanium region that does not contribute to detectable signals—cannot be directly measured experimentally and will therefore be estimated computationally in this study.

Given this incompatibility between experiment and simulation, one possible approach is to adjust certain detector parameters so that experimental measurements, within a selected range of energies and source-detector distances, can be satisfactorily reproduced through computational simulation.

The purpose of this study is to present a methodology for calculating the FEPE of an HPGe detector using the PHITS code.

2 Materials and methods

The full-energy peak efficiencies between point sources and a coaxial HPGe detector, model CANBERRA GC5019, were experimentally determined in a periodic calibration procedure for the detector's use in the neutron activation analysis laboratory at CDTN.

The sources were centered relative to the detector axis, and measurements were taken at distances sources-detector of 2, 5, 10, and 20 cm. These data were used to compare with simulation results and adjust the model to ensure the reliability of computational results.

Four point sources containing the radioisotopes Am-141, Co-57, Cs-137, and Co-60 were used. This set of radioisotopes has gamma-ray emission peaks at 59.5, 122.1, 136.5, 661.7, 1173.2, and 1332.5 keV.

FEPE ε can be calculated using the equation:

$$\varepsilon(E) = \frac{N}{A(t) \cdot P \cdot T}$$

Where:

- N: number of counts in the peak corresponding to energy E;
- P: probability of gamma-ray emission at energy E;

- A(t): source activity at the time of measurement;
- T: measurement time.

The activity at measurement time was calculated using the radioactive decay law:

$$A(t) = A_0 e^{-\lambda t}$$

Monte Carlo simulation

PHITS version 3.320 was used to model the HPGe detector based on the dimensions provided by the manufacturer, as shown in Table 1.

Table 1. Dimensions of the CANBERRA GC5019 HPGe detector provided by the manufacturer

Index	Detector Parameter	Value (mm)
A	Crystal diameter	69.5
B	Crystal length	52.5
C	Hole diameter	8.0
D	Hole depth	37.0
E	Window-crystal distance	5.0
F	Support thickness	3.2
G	Detector window thickness	1.5
H	Mylar and Kapton window thickness	4.3
I	Aluminum support thickness	0.76
J	Upper dead layer thickness	0.5
K	Radial dead layer thickness	0.5
L	Lower dead layer thickness	0.0

Figure 1 illustrates the detector's internal structure according to the indices in Table 1 (a), and its modeling in the PHITS code (b).

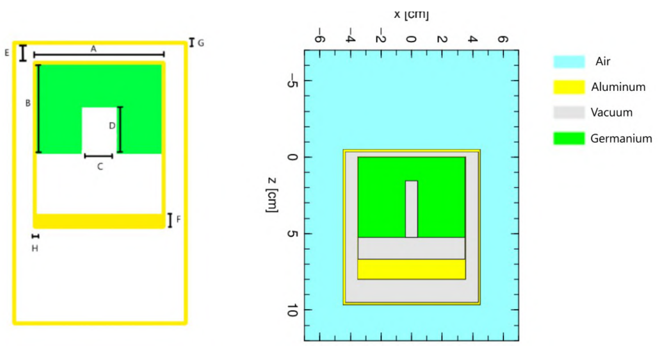

Figure 1.

To determine the system's detection efficiency using PHITS, the T-Deposit tally was selected. PHITS does not provide efficiency values directly; instead, the full-energy peak efficiency (FEPE) was derived from the energy deposited in the detector by photon interactions. The energy bin width was set to 2.4 keV, consistent with the experimental setup. Each photon absorption event was recorded in the corresponding energy bin, representing the energy deposited by electrons in the material during that event. The T-Deposit tally was configured to count the number of particles absorbed by the crystal at a specific energy per emitted particle from the source, providing a direct calculation of the FEPE relative to the detection system.

Simulations were run with approximately 10^8 photon trajectories, ensuring a statistical error of less than 1% for all results. The operating system installed on the computer used for the simulations was Windows 10. The CPU was an AMD Ryzen 5 5600GT with a clock frequency of 3.6 GHz (4.6 GHz Turbo) and 6 cores / 12 threads.

Dead Layer and Calibration

For computational modeling purposes, the effect of the dead layer in the HPGe detector was introduced using four parameters, referred to in this study as: upper dead layer, lower dead layer, internal dead layer, and radial dead layer. These values were adjusted individually by reducing the sensitive crystal's size in the respective region and adding an inactive germanium layer whose photons interactions would not be considered. Different combinations of dead layer parameters were used to evaluate and compare the efficiency results for the point sources.

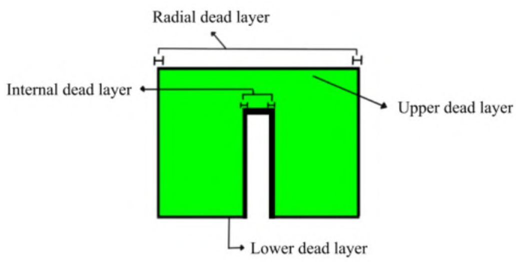

Figure 2. Illustration of the dead layer thicknesses used in this study.

Initially, simulations were conducted using the manufacturer's original dimensions (Table 1). The experimental data were acquired in 2021 from measurements of calibrated point sources, conducted as a periodic quality control procedure for the CDTN neutron activation laboratory. These data are presented in Table 2 in the following section. All experimental values were acquired with an uncertainty of less than or equal to 1%. Therefore, the simulation was configured to yield results with at most the same level of uncertainty.

3 Results and discussion

The reference parameter for result comparison was defined as the ratio between simulated and experimental efficiency values (PHITS/Exp), where a value of 1 represents the ideal case of perfect agreement. The adequacy of each simulation setup after modifying a parameter was measured by the average deviation of FEPE values across different energies and distances. Given that the uncertainty of each value is 1%, the uncertainty of the PHITS/Exp ratio is estimated at around 1.4%.

The first comparison below (Table 2) presents the absolute FEPE values and their ratios.

Table 2. Simulation results using the manufacturer's data

Energy (radionuclide)	Efficiencies calculated by simulation and experimentally								
	h = 2 cm			h = 10 cm			h = 20 cm		
	Phits (x 10^2)	Exp (x 10^2)	Ratio Phits/ exp	Phits (x 10^2)	Exp (x 10^2)	Ratio Phits/ exp	Phits (x 10^2)	Exp (x 10^2)	Ratio Phits/ exp
59.5 (Am-141)	8.79	4.20	2.09	1.26	0.78	1.61	0.37	0.25	1.46
122.1 (Co-57)	1.26	7.69	1.78	1.89	1.41	1.34	0.53	0.46	1.15
136.5 (Co-57)	0.37	7.46	1.78	1.81	1.37	1.32	0.54	0.45	1.19
661.7 (Cs-137)	0.53	1.98	2.02	0.61	0.41	1.48	0.20	0.14	1.38
1173.2(Co-60)	0.54	1.19	2.13	0.39	0.26	1.48	0.12	0.09	1.31
1332.5(Co-60)	0.20	1.19	1.92	0.35	0.24	1.49	0.11	0.08	1.30
Average deviation %			**98.8**			**45.5**			**30.0**

Energy in keV; *h* is the distance between the source and the detector window.

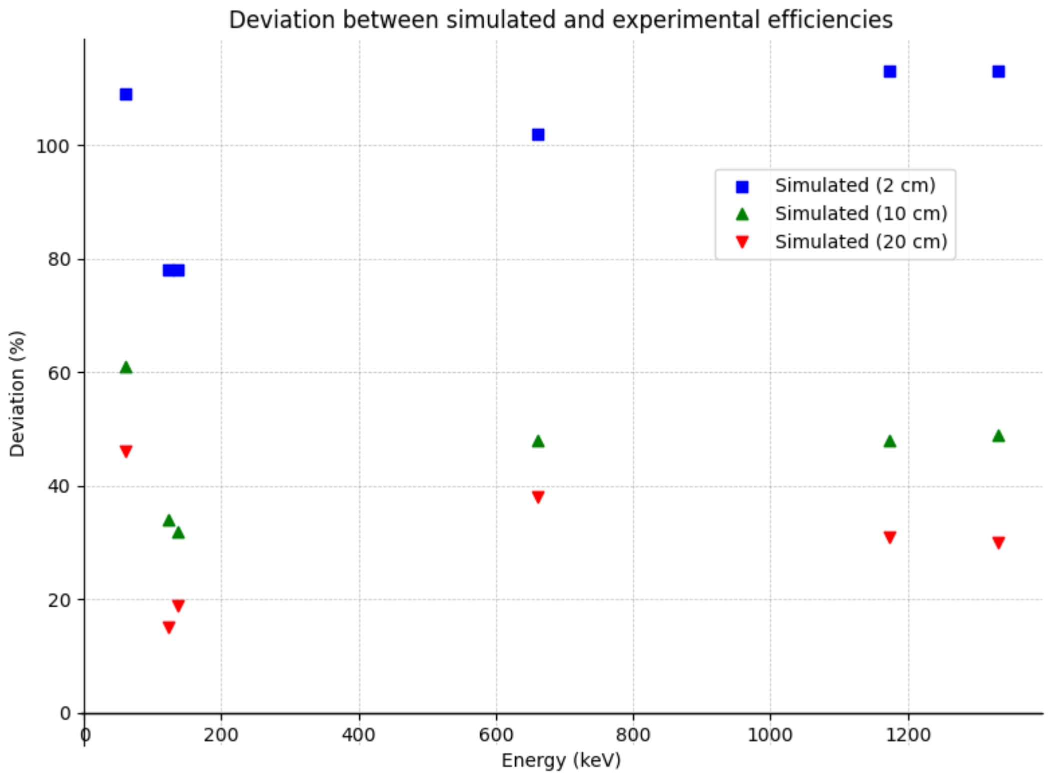

Figure 3. Graphical representation of the comparison between experimental values and simulation using the manufacturer's data

The simulated efficiencies for all source energies and source-detector distances were higher than the experimental ones. The first attempt to reduce this discrepancy involved decreasing the detector's radius, as this parameter most significantly affects the crystal volume. It was found that reducing the radius lowers all FEPE values as expected, it does not eliminate the discrepancy in average deviations between different distances. Thus, this parameter was not modified.

The next step was to examine the dependence of PHITS/Exp ratios on the window-crystal distance. Table 3 shows the comparison.

Table 3. Comparison of efficiencies varying the window-to-crystal distance

Energy (radionuclide)	Increase of the window-to-crystal distance				
	0 mm	15 mm	16 mm	**16,5 mm**	17 mm
	Phits/Exp	Phits/Exp	Phits/Exp	Phits/Exp	Phits/Exp
h = 2 cm					
59.5 (Am-141)	2.03	1.30	1.26	1.24	1.27
122.1 (Co-57)	1.78	1.06	1.03	1.02	0.99
136.5 (Co-57)	1.78	1.06	1.02	1.00	1.24
661.7 (Cs-137)	2.02	1.20	1.19	1.15	1.15
1173.2 (Co-60)	1.75	1.27	1.23	1.21	1.20
1332.5 (Co-60)	1.58	1.15	1.11	1.10	1.08
Average deviation %	**82.1**	**17.2**	**14.1**	**12.1**	**15.8**
h = 20 cm					
59.5 (Am-141)	1.46	1.20	1.23	1.23	1.26
122.1 (Co-57)	1.15	1.01	1.04	1.00	1.00
136.5 (Co-57)	1.19	1.01	1.00	0.99	1.01
661.7 (Cs-137)	1.38	1.14	1.14	1.14	1.21
1173.2 (Co-60)	1.31	1.15	1.15	1.14	1.17
1332.5 (Co-60)	1.30	1.15	1.15	1.14	1.16
Average deviation %	**25.2**	**11.0**	**11.6**	**11.1**	**13.6**

Energy in keV; h is the distance between the source and the detector window.

Increasing the window-crystal distance reduced all simulated efficiency values, especially at smaller source-detector distances. An ideal increase of **16.5 mm** was identified, where the ratios most closely approached unity across various source heights. Further increases beyond this point caused the ratios to diverge again.

From now on, the increment value will be fixed in future simulation configurations, and the same applies to the next parameters to be corrected.

To adjust the FEPE efficiency values at lower energies, the most relevant parameter is the upper dead layer thickness.

Table 4. Comparison of FEP efficiency varying the thickness of the upper dead layer

Energy (Radionuclide)	Increase in the thickness of the upper dead layer		
	0.00 mm	0.10 mm	**0.15 mm**
	Phits/Exp	Phits/Exp	Phits/Exp
	h = 2 cm		
59.5 (Am-141)	1.24	1.10	1.04
122.1 (Co-57)	1.02	0.99	0.98
136.5 (Co-57)	1.00	0.98	0.98
661.7 (Cs-137)	1.15	1.13	1.14
1173.2 (Co-60)	1.21	1.21	1.20
1332.5 (Co-60)	1.22	1.20	1.20
Average deviation %	**12.1**	**11.0**	**8.5**
	h = 20 cm		
59.5 (Am-141)	1.23	1.07	1.01
122.1 (Co-57)	1.00	0.98	0.97
136.5 (Co-57)	0.99	0.98	0.98
661.7 (Cs-137)	1.14	1.13	1.12
1173.2 (Co-60)	1.14	1.13	1.12
1332.5 (Co-60)	1.14	1.13	1.12
Average deviation %	**11.1**	**8.6**	**7.1**

Energy in keV; h is the distance between the source and the detector window.

The optimal increase found for the upper dead layer was **0.15 mm**. To improve results at higher energies, the lower dead layer was increased in steps of 0.1 mm. Table 5 shows these results.

Table 5. Comparison of FEP efficiency varying the thickness of the lower dead layer

Energy (Radionuclide)	Increase in the thickness of the lower dead layer			
	0.0 mm	0.4 mm	0.5 mm	**0.6 mm***
	Phits/Exp	Phits/Exp	Phits/Exp	Phits/Exp
	h = 2 cm			
59.5 (Am-141)	1.04	1.04	1.04	1.04
122.1 (Co-57)	0.98	0.97	0.97	0.98
136.5 (Co-57)	0.98	0.98	0.98	0.98
661.7 (Cs-137)	1.14	1.09	1.08	0.98
1173.2(Co-60)	1.20	1.15	1.14	1.12
1332.5(Co-60)	1.20	1.15	1.13	1.12
Average deviation %	**8.5**	**4.3**	**4.1**	**5.6**
	h = 10 cm			
59.5 (Am-141)	1.21	1.02	1.02	1.02
122.1 (Co-57)	1.02	0.98	0.96	0.96
136.5 (Co-57)	0.99	0.98	0.98	0.98
661.7 (Cs-137)	1.15	1.07	1.06	1.03
1173.2(Co-60)	1.15	1.08	1.06	1.05
1332.5(Co-60)	1.16	1.10	1.08	1.05
Average deviation %	**11.13**	**5.11**	**4.74**	**3.60**

Tab. 5 cont.				
Energy (Radionuclide)	Increase in the thickness of the lower dead layer			
	0.0 mm	0.4 mm	0.5 mm	**0.6 mm***
	Phits/Exp	Phits/Exp	Phits/Exp	Phits/Exp
h = 20 cm				
59.5 (Am-141)	1.01	1.00	1.00	1.01
122.1 (Co-57)	0.97	0.97	0.97	0.96
136.5 (Co-57)	0.98	0.98	0.98	0.98
661.7 (Cs-137)	1.12	1.06	1.05	1.02
1173.2(Co-60)	1.12	1.05	1.03	1.02
1332.5(Co-60)	1.12	1.05	1.03	1.02
Average deviation %	**7.11**	**3.55**	**2.69**	**2.27**

Energy in keV; h is the distance between the source and the detector window.

*column with the best parameters overall found

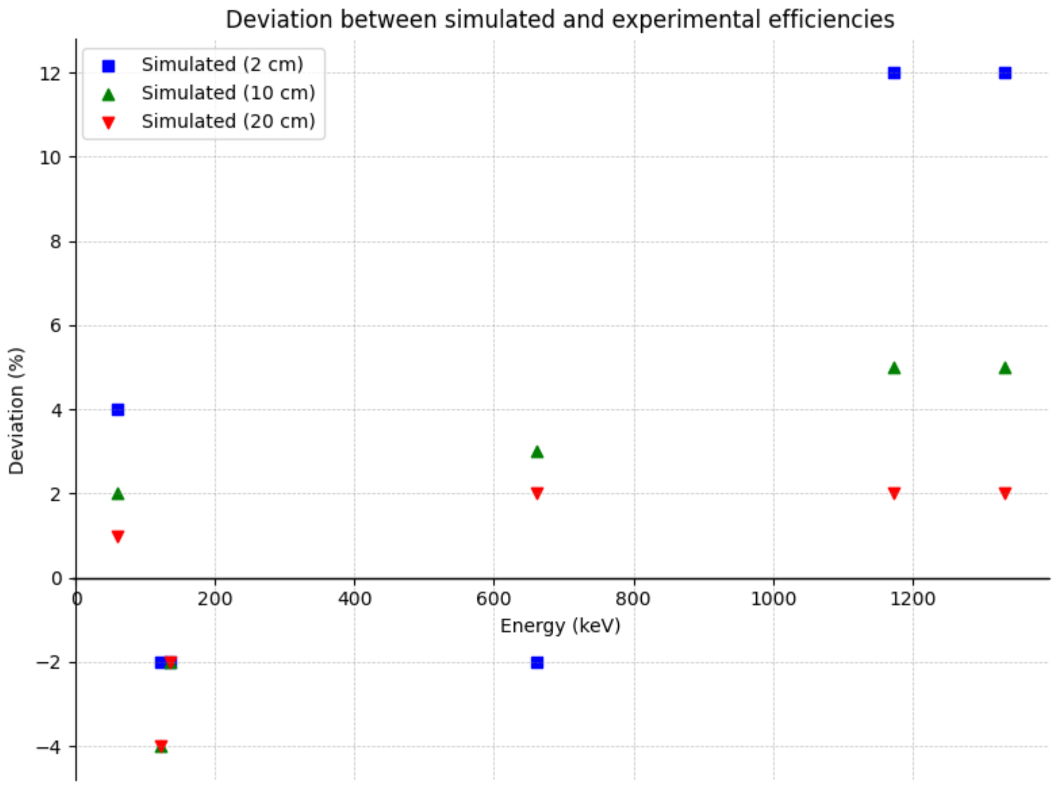

Figure 4. Graphical representation of the comparison between experimental values and the simulation after adjustments

The ideal thickness found was **0.6 mm**, reducing deviations in this energy range from 8,9% to 3,8%. In this optimal configuration, the deviations ranged from 1% to 12%. However, the two data points exhibiting 12% deviations—corresponding to the higher energies (1173.2 and 1332.5 keV) measured at the closest distance (2 cm)—can be considered outliers. Excluding these points, the deviation range narrows to 1–5%, which is consistent with the results reported by Penabeï et al. (2024), who measured point sources at different heights and found relative differences between experimental and simulated efficiencies ranging from −3.74% to 4.17%, and by Jargalsaikhan et al. (2025), who reported deviations of around 5% under similar experimental conditions.

With these adjustments, satisfactory agreement was obtained between experimental and simulated efficiency results for the GC5019 coaxial detector. This agreement was observed with point sources at energy between 59.5 and 1332.5 keV. The distances ranged from 2 to 20 cm from the detector window. The largest deviations were consistently observed when the sample was positioned closest to the detector, this trend that was similarly reported by Jargalsaikhan et al. (2025).

The table below summarizes the adjusted parameters, corresponding tables, and the optimized values:

Table 6. optimized detector parameters

Parameter	Table	Optimized Value (mm)
Window-crystal distance	4	21.5
Upper dead layer	5	0.65
Lower dead layer	6	0.6

4 Conclusion

A high-purity germanium (HPGe) detector was modeled in PHITS in order to simulate its full-energy peak efficiency (FEPE) for point sources of different energies and at different source-to-detector distances.

Using the dimensions provided by the manufacturer in the simulation results in significantly higher FEPE values compared to those observed experimentally. This work presents a methodology to correct key parameters that influence the FEPE value.

In the optimized configuration (highlighted column in Table 6), the average deviation between experimental and simulated values was 3.8%, with individual deviations ranging from 1% to 12%. The highest-energy values exhibited larger deviations at a height of 2 cm; however, this trend was not observed at 10 cm and 20 cm. Excluding the two data points corresponding to the higher energies (1173.2 and 1332.5 keV) measured at the closest distance (2 cm), the deviation range narrows to 1–5%. This provides a range of energies (59.5 to 1332.5 keV) and positions (2 to 20 cm in height—excluding positions below 10 cm for energies above 661.7 keV) within which the computational efficiency results can be considered reliable.

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Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms

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Abstract

Characterizing depth-dose distribution is essential for evaluate radiation behavior in tissue-equivalent materials, particularly in mammography, where the irradiated thicknesses is small and attenuation strongly depends on material composition. Monte Carlo simulations provide a precise way to model photon transport and analyze deposition at different depths (Lee 2024; Massera et al., 2024). Among the main tools available for this type of study, Geant4 stands out for its flexibility in simulating complex geometries and custom materials, and it has been widely validated in the biomedical field (Franco et al., 2020; Arce et al., 2025). In parallel, 3D printing has enabled the fabrication of customized homogeneous phantoms using materials such as ABS and PETG, which exhibit attenuation properties and densities close to soft tissue (Ivanov et al., 2018; Mettivier et al., 2022), supporting experimental dose evaluation using embedded dosimeters at various depths (Van et al., 2023). In this study, a 35 kV X-ray tube setup was simulated using the Geant4 code to estimate depth dose in realistic mammography conditions. Homogeneous phantoms were modeled with 3D-printed ABS (0.899 g/cm^3) and PETG (1.169 g/cm^3), each containing five layers with four thermoluminescent dosimeters ($CaSO_4$:Dy) per layer. The simulations were carried out using four Geant4 physics lists (G4EmStandardPhysics_option4, G4EmPenelopePhysics, G4EmLivermorePhysics, and G4EmStandardPhysics) to compare the influence of electromagnetic interaction modeling on dose results. Results showed excellent agreement between simulation and experiment. In particular, G4EmStandardPhysics_option4 reproduced the experimental mean dose with exact agreement. The Penelope and Livermore lists performed better in reconstructing depth dose profiles, particularly in deeper layers, with deviations below 2%. ABS demonstrated greater tissue equivalence (mean error: 0.5%) than PETG (1.2%), supporting its suitability as a breast tissue substitute in phantoms. These results confirm the robustness of Geant4 for mammography dosimetry simulation and highlight the importance of selecting the appropriate physics list according to the clinical goal—overall dose accuracy (StandardPhysics_option4) or spatial resolution (Penelope/Livermore).

Keywords: *Geant4; Mammography dosimetry; 3D phantoms; Physics lists; ABS/PETG.*

1 Introduction

The characterization of depth-dose distribution is essential for evaluating the behavior of radiation in tissue-equivalent materials, especially in mammography applications, where the irradiated thicknesses are small and attenuation is highly dependent on material composition. Monte Carlo simulations offer a precise approach for modeling photon propagation, enabling detailed analysis of energy deposition at different depths (Lee, 2024; Massera et al., 2024).

Among the main Monte Carlo codes available, Geant4 stands out due to its flexibility in simulating complex geometries and defining custom materials. This toolkit has been widely used in medical imaging research, including studies in mammography and dedicated breast computed tomography (breast CT), demonstrating consistent results when compared to experimental data (Franco et al., 2020; Sarno et al., 2018). Furthermore, the use of Geant4 in virtual clinical trials has shown its ability to accurately predict spatial dose distributions, even in scenarios with realistic structural and material variations, as demonstrated by Franco et al. (2020). Recent studies have also conducted detailed benchmarks to validate the accuracy and performance of Geant4 in biomedical applications, confirming its robustness for virtual clinical simulations and precise prediction of spatial dose distribution in complex anatomical configurations (Arce et al., 2025).

Several studies have explored the use of Geant4 in specific mammography applications, demonstrating its effectiveness in estimating key parameters such as dose distribution in glandular and adipose tissues, effective thickness, and detector response. Due to its ability to incorporate detailed anatomical geometries and realistic X-ray spectra, Geant4 has been employed in simulations aimed at optimizing clinical protocols, evaluating different anode/filter combinations, and investigating the influence of technical parameters on image quality and absorbed dose (Massera et al., 2024; Sarno et al., 2018). Additionally, its compatibility with custom breast models enables individualized simulations with improved dosimetric accuracy, supporting the development of dose reduction strategies without compromising diagnostic accuracy (Franco et al., 2020; Lee, 2024).

In parallel, 3D printing has been employed in the fabrication of customized homogeneous phantoms, enabling the construction of blocks using materials such as ABS and PETG, which exhibit densities and attenuation properties similar to those of soft tissues (Mettivier et al., 2022; Varallo et al., 2022; Ivanov et al., 2018). These phantoms enable the experimental evaluation of dose at various depths through internally placed dosimeters, providing a reliable basis for the validation of computational simulations (Van et al., 2023; Lee, 2024).

This study aims to compare the depth dose simulated using Geant4 with experimental data obtained from 3D-printed homogeneous phantoms made of ABS and PETG. This approach allows investigation of the influence of material composition and thickness on dose distribution, as well as validation of Geant4's capability to reproduce experimental results under controlled conditions.

2 Materials and Methods

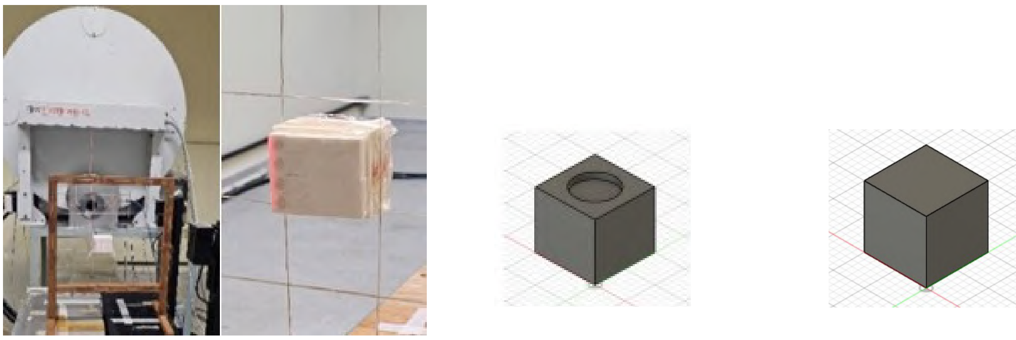
To evaluate the depth-dose distribution in different material configurations, modular and homogeneous phantom were fabricated using by 3D printing with ABS and PETG polymers. These phantom models were irradiated with X-ray beams under controlled conditions, and dosimetry was performed using $CaSO_4$:Dy thermoluminescent detectors (TLDs).

2.1 Experimental Irradiation of Phantom

To experimentally assess the depth-dose distribution under conditions representative of mammography exams, two homogeneous phantom were fabricated using commercial polymers: acrylonitrile butadiene styrene (ABS), with a density of 0.899 g/cm³, and polyethylene terephthalate glycol (PETG), with a density of 1.169 g/cm³. Both were manufactured via FDM (Fused Deposition Modeling) 3D printing, adopting a modular structure to accommodate the detectors and vary the material thickness along the beam path.

Each phantom consisted of a containment box (5 × 4 × 4 cm³) printed from the same material (ABS or PETG), responsible for holding and aligning the internal blocks. The box interior contained five layers, each with four cubic blocks (1 × 1 × 1 cm³), totaling 20 possible detector positions. Each block intended for dosimeter insertion contained a cylindrical hole (6.5 mm in diameter × 1.1 mm deep), Figure 1(b), to house a $CaSO_4$:Dy thermoluminescent dosimeter (TLD). Additional solid blocks (1 × 1 × 1 cm³) were used to build up the thickness between layers, as shown in Figure 1 (c).

Irradiations were performed separately for the ABS and PETG phantoms using an X-ray tube operating at 35 kV and 10 mA. The air kerma rate was experimentally measured at 1.48 mGy/min, and the exposure time was 3.38 minutes, resulting in an entrance dose of approximately 5 mGy. The TLDs were pre-calibrated and positioned at five distinct depths within the phantom, corresponding to progressive attenuation layers. All measurements were carried out in duplicate to ensure reproducibility and accuracy.



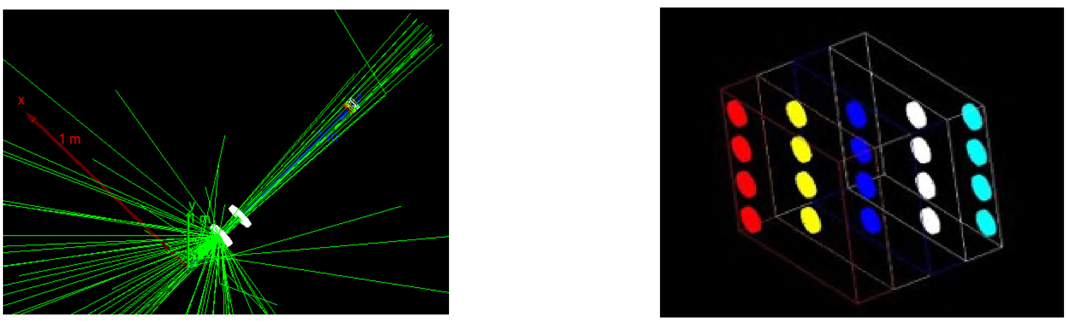
(a) (b) (c)

Figure 1: Positioning of the Phantom during irradiation (a) and 3D printed models for making the Phantom (b) and (c).

2.2 Computational Simulation with Geant4

The same geometric configuration used in the experimental irradiations was modeled in the Geant4 toolkit (version 11.3.0), compiled with multithreading support to optimize performance. The real dimensions of the blocks, their compositions, and the detector distribution were accurately reproduced. The ABS and PETG phantoms were simulated separately, each containing five layers with four sensitive volumes associated with the TLDs, totaling 20 detectors per simulation.

The radiation source was defined using a realistic 35 kV X-ray spectrum generated via SpekCalc, incorporating filtration equivalent to the experimental setup. The source was positioned to irradiate perpendicularly to the center of the phantom's front face. Four electromagnetic physics lists available in Geant4 were employed: G4EmStandardPhysics_option4, G4EmPenelopePhysics, G4EmLivermorePhysics, and G4EmStandardPhysics, with the aim of comparing different interaction models with matter.



(a) (b)

Figure 2: (a) Virtual experimental setup showing the photon beam emitted from the source and incident on the phantom. (b) Positioning of the thermoluminescent dosimeters (TLDs) within the 3D-printed phantom.

A total of 10^8 events were simulated for each condition (ABS or PETG), and the energy deposited in each sensitive volume was recorded and converted into absorbed dose, considering the mass of the detector (approximately 50 mg of $CaSO_4$:Dy per TLD). Since the number of photons simulated does not match the actual photon fluence from the experiment (approximately 10^{15} photons per irradiation), a normalization factor was applied, as described by Tucciariello et al. (2020). This factor was based on the ratio between the average experimental dose and the average simulated dose in each layer, enabling direct comparison with the experimental results.

Figure 2 presents a visualization of the virtual phantom, highlighting the layers and detector positions. In panel (a), the virtual geometry is shown, and panel (b) displays the complete experimental setup. The detectors in layer 1 are marked in red, layer 2 in yellow, layer 3 in dark blue, layer 4 in white, and layer 5 in light blue. Notably, the layer 5 detectors are positioned toward the rear of the phantom to measure the exit dose.

3 Results

3.1 Global Agreement Between Simulation and Experiment

Table 1 presents the comparison between experimental and simulated dose values for each thermoluminescent dosimeter (TLD) across the five layers of the homogeneous ABS and PETG phantoms. The data demonstrate a high level of agreement between experimental measurements and simulated values, particularly when using the G4EmStandardPhysics_option4 physics list, which reproduced the experimental mean dose within the experimental uncertainty for both ABS and PETG.

The other physics lists also showed satisfactory performance in reproducing the overall absorbed dose, with relative deviations below 1.23%. This highlighting the reliability of the Geant4 code for global dose estimation in mammographic applications.

Table 1: Comparison between simulated and experimental mean doses by layer for ABS and PETG.

Layer	**Detectors**	**Material**	**DoseExp (mGy)**	**Option4 (mGy)**	**Penelope (mGy)**	**Livermore (mGy)**	**Standard (mGy)**
1[a]	TLD11 – TLD14	ABS	2.93	2.90	2.93	2.95	2.90
		PETG	2.88	2.83	2.85	2.88	2.80
2[a]	TLD21 – TLD24	ABS	2.33	2.35	2.33	2.38	2.35
		PETG	1.98	2.08	2.08	2.10	2.03
3[a]	TLD31 – TLD34	ABS	1.65	1.58	1.58	1.58	1.60
		PETG	1.30	1.18	1.20	1.20	1.15
4[a]	TLD41 – TLD44	ABS	1.10	1.18	1.15	1.18	1.15
		PETG	0.75	0.78	0.73	0.73	0.78
5[a]	TLD51 – TLD54	ABS	0.73	0.73	0.73	0.75	0.70
		PETG	0.45	0.45	0.43	0.43	0.45
Mean	—	ABS	1.74	1.74	1.74	1.75	1.72

Layer	Detectors	Material	DoseExp (mGy)	Option4 (mGy)	Penelope (mGy)	Livermore (mGy)	Standard (mGy)
		PETG	1.46	1.46	1.46	1.46	1.44
Relative Error	—	ABS	—	0.00%	0.10%	0.69%	0.88%
		PETG	—	0.00%	0.18%	0.67%	1.23%

3.2 Layer-by-Layer Performance

The layer-wise analysis revealed important trends regarding the performance of the different electromagnetic physics lists. In the superficial layers (1st and 2nd), where the photoelectric effect is dominant, the Penelope and Livermore models showed good agreement with experimental data, with typical deviations below 1%. In the deeper layers (3rd to 5th), where multiple scattering becomes the prevailing interaction, the Livermore list performed slightly better than Penelope, especially in the PETG phantom.

However, G4EmStandardPhysics_option4 demonstrated uniform accuracy across all layers, with deviations consistently below 0.7%. This consistent behavior reinforces its ability to model both dominant low-energy effects and photon propagation in deeper regions of the phantom.

It is important to highlight that although the tests were conducted separately using homogeneous phantoms, the ultimate goal is to validate the use of ABS and PETG in combination to build heterogeneous breast phantoms that mimic real anatomical structures. The results obtained in this study indicate that both materials are promising for such applications, provided that the physics list is selected according to the analysis objective—either global dose estimation or detailed spatial profiling in depth.

4 Conclusions

This study assessed the accuracy of the Geant4 Monte Carlo toolkit in simulating depth-dose distributions in homogeneous 3D-printed phantoms made from ABS and PETG, under clinically relevant mammography conditions (35 kV).

The G4EmStandardPhysics_option4 physics list showed perfect agreement with experimental mean dose values for both ABS and PETG, supporting its use for global dose estimation.

The Penelope and Livermore models demonstrated superior performance in modeling depth-dependent dose variations, with errors < 2% in the deeper layers, making them suitable for applications focused on spatial resolution and detailed profiling.

Among the tested materials, ABS exhibited better superior tissue-equivalence, with a lower average error and density closer to that of real breast glandular tissue, making it the preferred material for representing glandular regions in future heterogeneous breast phantoms.

Although this study evaluated homogeneous materials separately, the results strongly support the combined use of ABS and PETG in constructing more anatomically realistic breast models, paving the way for more accurate clinical simulations using Geant4.

The next step of this project involves the design and irradiation of a heterogeneous ABS–PETG phantom, simulating the anatomical distribution of adipose (ABS) and glandular (PETG) tissues. This will aim to validate the combined performance of these materials and extend the application of Geant4 in personalized virtual clinical trials.

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^{99m}Tc Radiolabeled Papain Nanoparticle: *in silico* Dosimetry Evaluation using DM_BRA phantom and the experimental biodistribution data

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Abstract

Green nanotechnology has gained prominence in developing biocompatible materials for therapeutic and diagnostic applications in oncology and nuclear medicine. Ferreira et al. (2024b) presented an innovative system based on papain nanoparticle (P-NP) synthesized by gamma irradiation and radiolabeled with technetium-99m (^{99m}Tc) for molecular imaging using SPECT. Papain, a protease extracted from *Carica papaya*, exhibits anticancer, antibacterial, and antioxidant properties, making it an excellent nanocarrier for contrast agents and radionuclides. The formulation of papain nanoparticle enables the degradation of the tumor extracellular matrix, improving the penetration and distribution of diagnostic agents within the tumor microenvironment. Herein, the MCMEG research group conducted an *in silico* dosimetry calculation using the MCNP6.2.1 Monte Carlo code to estimate the absorbed dose in the organs and tissues of the voxelized phantom DM_BRA mouse. The calculations were performed using the biodistribution data obtained in the experimental process performed in laboratory using an injected activity of the ^{99m}Tc radionuclide in the mouse. The results indicate that ^{99m}Tc-P-NPs exhibit high tumor affinity, with a favorable distribution profile for breast cancer imaging.

This statement is evident when comparing the uptake of organs that act in physiological elimination of the healthy model with the 4T1 tumor model, since organs that act in the elimination of nanoparticles, removing them from the bloodstream through opsonization, followed by phagocytosis, such as the liver and spleen, showed percentage differences between the healthy model and the 4T1 tumor of 323% and 204%, and even higher values in breast cancer models (440% and 618%) respectively, which indicates greater retention of nanoparticles in animal models that have tumor volume. The similarity between experimental data and Monte Carlo calculations highlights the reliability of the simulation method for estimating radiation biodistribution in healthy mice. This comparison underscores the importance of validating computational models with experimental data to ensure accurate predictions of organ-specific dose distributions.

Keywords: Green Nanotechnology; Nanopapain, ^{99m}Tc; Dosimetry; MCNP6; MIRD.

1 Introduction

Papain is a plant-derived proteolytic enzyme extracted from the fruit *Carica papaya*. It is widely recognized for its ability to promote cutaneous absorption, facilitating drug penetration and enhancing therapeutic efficacy Fazolin et al. (2020). Papain-based pharmaceuticals exhibit diverse bioactive properties, including debriding, anti-inflammatory, anti-infectious, and anti-diabetic effects. Notably, papain also demonstrates antitumor activity, with promising potential as a drug carrier for cholangiocarcinoma treatment Varca et al. (2016); Müller et al. (2016). Additionally, it exerts potent anti-angiogenic effects, making it a candidate for both the prevention and treatment of angiogenesis-related diseases Mohr and Desser (2013). Papain nanoparticle synthesis can be achieved through various methods, such as complexation in aqueous solutions, electron beam irradiation, or gamma photon irradiation Varca et al. (2014a,b, 2016). Preserving enzymatic stability during synthesis is crucial, as bioactivity is a key factor for biological applications. Factors such as pH, temperature, and salt concentration can compromise stability. Moreover, the inclusion of pharmaceutical excipients to enhance solubility and viscosity may inadvertently promote microbial growth and further destabilize the enzyme Loyd V. Allen and Ansel (2013).

A study conducted by Ferreira et al. (2024b) demonstrated the potential of papain nanoparticles (P-NPs) as nanocarriers for radionuclides, targeting tumor tissues as diagnostic agents. Papain nanospheres were synthesized via irradiation-induced crosslinking and subsequently radiolabeled with technetium-99m ($^{99\mathrm{m}}$Tc). These nanoparticles were characterized through both *in vitro* and *in vivo* assays to explore their feasibility as novel nanoradiopharmaceuticals for breast cancer diagnosis using single-photon emission computed tomography (SPECT). Herein this study aims to further investigate the computational dosimetry of ^{99m}Tc-P-NPs using the Monte Carlo method.

The MCMEG research group Team (2025) has been actively engaged in modeling and calculating absorbed doses resulting from administered activity in nuclear medicine, using adult human and animal voxel phantoms Paixão et al. (2019); Gomes (2024); Ferreira et al. (2023); Gomes et al. (2024). In 2017, the DM_BRA mouse phantom was also developed and validated as a novel 3D voxel-based mouse model for preclinical and human first-estimation radiopharmaceutical dosimetry, using segmented images from the Digimouse CT database and validated through Monte Carlo simulations using MCNPx code Mendes et al. (2017), across photon energies from 0.015 to 4 MeV, showing strong correlation with reference organ masses and Specific Absorbed Fraction (SAF) data.

In this study, MCMEG, conducted *in silico* dosimetry calculations using the Monte Carlo N-Particle code (MCNP6.2.1) Werner et al. (2017). The objective was to estimate the dose distribution resulting from the injected activity of papain nanoparticles radiolabeled with ^{99m}Tc, based on computational simulations using the DM_BRA mouse voxel phantom, aiming to approximate the computational model to experimental conditions. Time-integrated activity coefficients (TIACs) for each organ were determined based on biodistribution studies from *in vivo* experiments with ^{99m}Tc-labeled papain nanoparticles Ferreira et al. (2024a). This approach enabled the calculation of the absorbed dose per unit of injected activity for both healthy and tumor-bearing mice, providing comprehensive dosimetric studies.

2 Materials and methods

2.1 Experimental Apparatus

In a previous study, the biodistribution to each organ was determined *in vivo* experiments with ^{99m}Tc-labeled papain nanoparticles Ferreira et al. (2024a). Biodistribution studies were performed in Healthy Balb/c Mice, 4T1 Tumor-Bearing Mice, and Spontaneous Breast Cancer Mice (MMTV-PyMT). The organ uptake was expressed as a percentage of the injected activity per gram of tissue (% IA/g) or organ. The authors suggested that the biodistribution results indicate that the uptake of ^{99m}Tc-P-NPs in the spleen and liver is evident, but expected, considering that the nanoparticles can be removed from the bloodstream through opsonization followed by phagocytosis. The results indicate that the ^{99m}Tc-P-NPs were excreted mainly through the renal-urinary system, with secondary clearance occurring through the hepatic system, and accumulation of the nanoparticle in the tumor decreased over time Ferreira et al. (2024a).

2.2 Monte Carlo Dose Calculation

The Monte Carlo code used for this computational modeling was MCNP6.2.1, a general-purpose three-dimensional simulation tool capable of transporting 37 different particle types for applications such as criticality, shielding, dosimetry, and detector response Goorley et al. (2016). In this study, the Monte Carlo transport parameters included a 1 keV cut-off energy for photons and electrons, XCOM photon and Compton cross-sections, NIST bremsstrahlung cross-sections, SIMPLE bound Compton scattering, and Rayleigh scattering.

A computational scenario was modeled to estimate organ-specific dose deposition in the DM_BRA mouse voxel phantom using experimental biodistribution data of ^{99m}Tc-labeled papain nanoparticle in healthy and tumor-bearing Balb/C mice. The radionuclide emissions considered were beta particles, electrons, and photons, with each type separately configured in the source card. Absorbed dose calculations incorporated the individual yield of each emission type, based on DECDATA software (Table 1).

The simulations were performed on a cluster with 48 Intel(R) Xeon(R) Gold 5412U processors, 128GB of RAM, and Ubuntu 22.04.4 LTS. A total of 38 tasks were used with 5×10^6 particles to maintain statistical uncertainties below 3%, resulting in a computational time of approximately 3 minutes. The +F6 tally was applied to compute track-length energy deposition, converting simulation results to mGy/particle using a factor o 1.602×10^{-10} joules/MeV.

Table 1: Value of the total yield (Y) of the decay spectrum of the emitted particles(ICRP (2008)).

	Radiation yield			
Radionuclide	γ	e^-	β^+	β^-
^{99m}Tc	6.4670	5.5166	–	3.70E-05

2.3 Computational Internal Dosimetry Calculations

Internal dosimetry was performed following the MIRD schema (Howell et al., 1999), which combines mathematical models with radionuclide biodistribution and the first equations were published on Pamphlet No. 1 (Loevinger and Berman, 1968). The absorbed dose in a target region is calculated using the time-integrated activity (Ã) and the absorbed dose rate per unit activity (S-value) (Snyder, 1975):

$$D = \tilde{A} \cdot S \tag{1}$$

To calculate the S-values, the radioactivity source spectrum provided by DECDATA software package and the DM_BRA mouse voxel phantom was used to create the input files for MCNP6.2 code. The S-values were calculated for the ^{99m}Tc source and the MC code scores the absorbed dose per number of emitted particles. The result can be multiplied by the yield (number of emitted particles per decay) of the radionuclide to calculate the S-value. Time-integrated activity coefficients (TIACs) for each organ were determined based on biodistribution studies from *in vivo* experiments with 99mTc-labeled papain nanoparticles provided by Ferreira et al. (2024a). Table 2 shows the TIAC values measured for healthy Balb/C mice and Balb/C mice with 4T1 tumor volume, TIAC/g values for each organ were also calculated.

With the TIAC values, it is possible to calculate the dose deposited per injected activity (mGy/MBq) per organ, taking into account the total yield of each type of particle by the product of the dose obtained through Monte Carlo computer simulation. This expression can then be represented by (2):

$$D = \tilde{a} \cdot \sum Y \cdot D^{MCNP} \cdot CF \tag{2}$$

Where $\tilde{a}$ is the time-integrated activity coefficient, D^{MCNP} is the dose obtained using the Monte Carlo code MCNP6.2.1, Y is the yield for each type of radiation from ^{99m}Tc, and CF the conversion factor to mGy/MBq.

Table 2: Time-integrated activity coefficient values for healthy mice, mice with 4T1 tumor volume, and mice with MMTv-PyMT for different organs.

Organs	**Healthy mice**		**Mice with tumor**		**Mice with MMTv-PyMT**	
	TIAC (s)	**TIAC (h)**	**TIAC (s)**	**TIAC (h)**	**TIAC (s)**	**TIAC (h)**
Blood	—	—	—	—	—	—
Heart	61.54	0.0171	50.05	0.0139	30.60	0.0085
Lung	89.33	0.0248	259.85	0.0722	74.32	0.0206
Spleen	253.77	0.0705	830.90	0.2308	2039.89	0.5666
Thyroid	—	—	—	—	—	—
Brain	5.15	0.0014	4.33	0.0012	2.42	0.0007
Pancreas	6.83	0.0019	6.11	0.0017	5.69	0.0016
Liver	3163.06	0.8786	14305.78	3.9738	18346.35	5.0962
Stomach	222.51	0.0618	189.11	0.0525	94.61	0.0263
Kidneys	4226.60	1.1741	2839.14	0.7887	947.39	0.2632
Bladder	487.33	0.1354	216.09	0.06	788.20	0.2189
Muscle	1647.71	0.4577	996.92	0.2769	732.92	0.2036
Bone	3118.69	0.8663	2394.21	0.6651	2331.20	0.6476
Spine	—	—	—	—	—	—
Bone Marrow	5.16	0.0014	7.23	0.002	2.90	0.0008
Intestines	894.75	0.25	647.81	0.18	413.23	0.1148

2.4 Animal Phantom

Computational animal phantoms are widely used for dosimetry calculation (Laazouzi et al., 2024; Dogdas et al., 2007; H., 2018; Keenan et al., 2010). They can be used in different MC codes to provide dosimetric data for different radionuclides. In this study, the voxel phantoms named DM_BRA mouse Mendes et al. (2017) was used.

The DM_BRA was modeled in a matrix size of $152 \times 396 \times 42$, with voxels of dimensions $0.25 \times 0.25 \times 0.5\,\mathrm{mm}^3$. It comprises 22 segmented regions (organs) and has 25.7 g in mass. In the DM_BRA, each organ was segmented as a uniform region. The distinction between wall and content for intestine, stomach, heart and bladder segmentations was not taken into account. In addition, the bones of the DM_BRA were segmented as a single structure, called the skeleton. No distinction was made between cortical and trabecular (spongiosa) bone, and the DM_BRA phantom model shows the segmentation of organs as homogeneous tissues, with few internal details. All information about organs compositions and densities were provided by the developers. Tissue composition and density values were taken from ICRP(2009) (Mendes et al., 2017).

Figure 1: The DM_BRA voxel phantom images from AMIDE® software. Coronal (A) Sagittal (B) and transversal (C) slices. Whole body (D) and skeleton (E) 3D rendering, from Mendes et al. (2017)

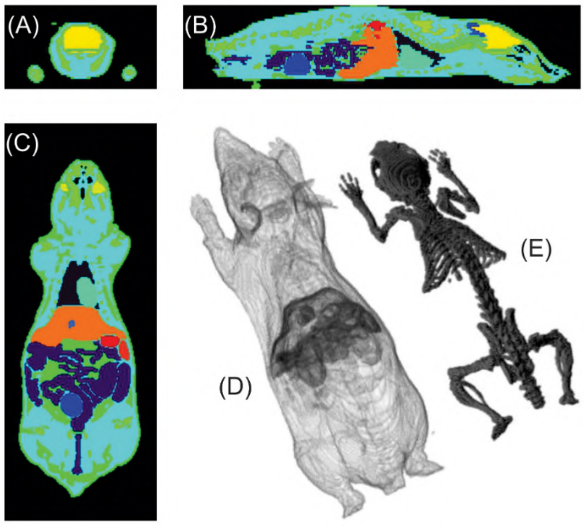

Table 3 presents the average organ masses obtained from the experimental process for healthy Balb/C mice, 4T1 tumor-bearing mice, MMTV-PyMT mice, and the DM_BRA voxel phantom.

Table 3: Comparison of organ masses between Healthy Balb/c Mice, 4T1 Tumor-Bearing Mice, and Spontaneous Breast Cancer Mice (MMTV-PyMT), and the DM BRA phantom.

Mass per organ (g)				
Organs	Healthy Mice	4T1 tumor Bearing Mice	MMTv-PyMT	DM_BRA
Adip. Tissue	—	—	—	2.379
Adrenals	—	—	—	0.005
Blood	0.2451	0.0888	0.2168	—
Brain	0.4013	0.3624	0.4083	0.35
Brown Fat	—	—	—	0.142
Eye	—	—	—	0.006
Gall Bladder	—	—	—	0.009
Hard. Gland	—	—	—	0.032
Heart	0.0697	0.0724	0.0875	0.215
Intestine	0.7411	0.7714	0.8529	2.311
Kidney	0.2172	0.2259	0.2204	0.482
Liver	0.8748	1.0338	0.9936	1.987
Lung	0.116	0.1305	0.1323	0.167
Muscle/Soft	0.0679	0.0755	0.0795	12.254
Pancreas	0.119	0.1247	0.1349	0.047
Skeleton	0.0102	0.036	0.0214	2.311
Skin	—	—	—	2.228
Spinal Cord	0.0517	0.0502	0.0725	0.099
Spine	1.6039	1.1307	1.6451	—
Spleen	0.0734	0.1658	0.0847	0.14
Stomach	0.1033	0.1272	0.1290	0.245
Testes	—	—	—	0.159
Thyroid	0.117	0.0293	0.1352	—
Tumor MMTv-PyMT	—	—	0.1274	—
4T1 Tumor	—	0.1484	—	—
U. Bladder	0.0893	0.0683	0.1089	0.238
Total Body	18.35	20.95	22.58	25.7

Some segmented organs in the DM_BRA phantom, such as the thyroid and testicles, were not included in the simulations due to the lack of biodistribution data for these structures. Organs such as the liver, spleen, and kidneys showed percentage differences in their masses when compared to the healthy experimental mouse model and the DM_BRA phantom, of 127%, 91%, and 122%, respectively. These differences are mainly attributed to differences in the body mass of the animals in the experimental model, which were 18.35 g for the healthy model, 20.95 g for the 4T1 tumor model, and 22.58 g for the MMTV-PyMT model, compared to the computational phantom, which has a total body mass of 25.7 g.

The absorbed dose per injected activity data calculated with MCNP6.2 in this work were compared to the data available in the experimental results provided by Ferreira et al. (2024a). The comparisons were performed by calculating the percentage difference (Δ%) through Equation 3:

$$\Delta = \left(1 - \left(\frac{q_0}{q}\right)\right) \cdot 100\% \tag{3}$$

where q_0 is the reference quantity value and q is the quantity calculated in this work.

3 Results

Table 4 provides the total radiation dose in mGy/MBq estimated for the organs of the healthy Balb/C mice, 4T1 tumor-bearing mice, MMTV-PyMT mice using the MCNP6.2.1 code. Figure 2 shows the estimated doses for the different Balb/C mice models.

Table 4: Total dose estimated for different organs of healthy Balb/C mice, 4T1 tumor-bearing mice, MMTV-PyMT mice with the MCNP6.2.1 code, and percentage differences in relation to healthy mice.

	Healthy mice		Mice with 4T1 tumor			Mice with MMTv-PyMT		
Organs	Dose(mGy/MBq)	SD (mGy/MBq)	Dose(mGy/MBq)	SD (mGy/MBq)	Δ% (4T1)	Dose(mGy/MBq)	SD (mGy/MBq)	Δ% (MMTv)
Lungs	1.68	0.0211	4.64	0.0447	176,28	2.24	0.0288	33,50
Intestines	1.53	0.0052	1.59	0.0061	3,63	1.43	0.0056	-6,90
Bladder	5.74	0.0354	2.76	0.0301	-51,91	9.20	0.0606	60,38
Muscle	0.71	0.0014	0.74	0.0017	4,74	0.74	0.0017	5,55
Heart	1.11	0.0146	1.54	0.0193	39,36	1.37	0.0175	24,00
Bone Marrow	0.53	0.0123	0.67	0.0172	25,55	0.57	0.0150	6,57
Pancreas	1.68	0.0323	2.12	0.0441	26,56	2.24	0.0451	33,48
Bone	3.73	0.0089	3.05	0.0100	-18,24	3.02	0.0107	-19,04
Brain	0.30	0.0048	0.29	0.0059	-5,51	0.27	0.0058	-10,79
Kidney	23.36	0.0497	16.49	0.0531	-29,41	6.35	0.0333	-72,81
Liver	4.79	0.0109	20.28	0.0287	323,55	25.89	0.0345	440,75
Spleen	5.24	0.0432	15.92	0.0971	204,04	37.60	0.1622	618,20
Stomach	3.07	0.0239	3.75	0.0303	22,25	3.30	0.0266	7,53

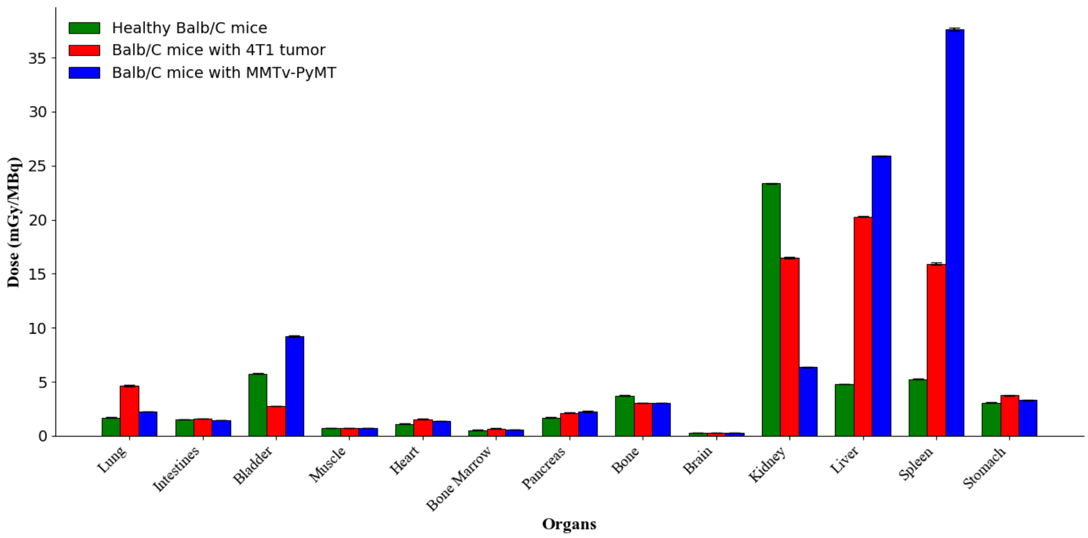

Figure 2: Total dose estimated for different organs of healthy Balb/C mice, 4T1 tumor-bearing mice, MMTV-PyMT mice with the MCNP6.2.1 code.

The spleen and liver consistently received the highest doses across all groups, with tumor-bearing mice exhibiting significantly higher doses compared to healthy mice. Notably, the MMTV-PyMT model group showed the highest radiation doses in these organs, suggesting a greater impact of this tumor type on biodistribution and radiation absorption. The kidneys also showed elevated doses in tumor-bearing mice, particularly in the MMTV-PyMT group. Conversely, organs such as the brain, pancreas, bone marrow, heart, and bladder consistently received minimal doses across all groups. These findings indicate that tumor presence and type substantially influence radiation dose distribution, with metabolically active organs such as the spleen and liver being most affected.

Figures 3, 4, 5 compare the doses (mGy/MBq) and the percentage of injected dose per gram for each organ in healthy mice, as calculated using experimental measurements and Monte Carlo simulations with MCNP6.2 for the different Balb/C mice models.

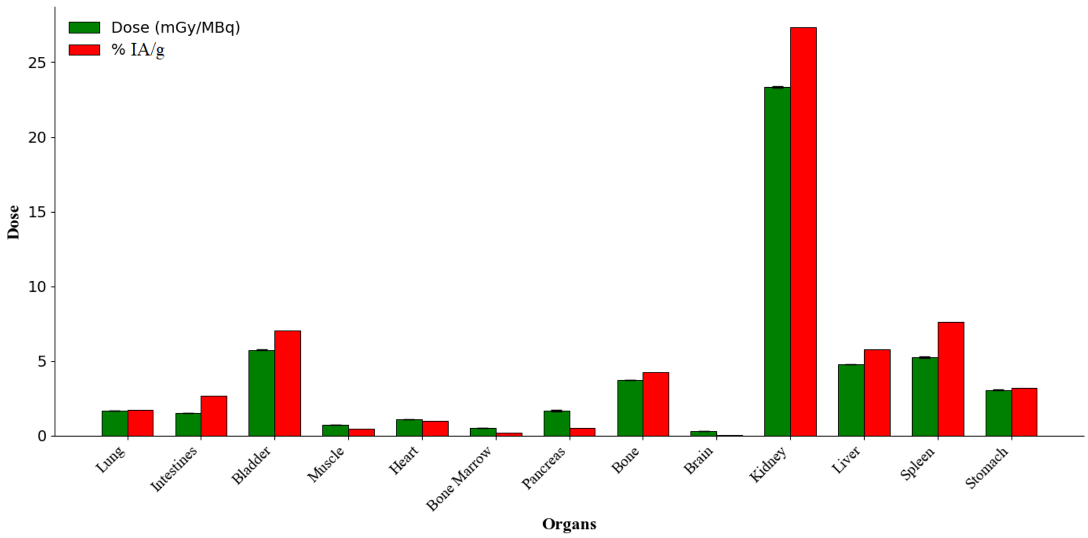

Figure 3: % Injected activity per gram for each organ in relation to the doses (mGy/MBq) obtained using the monte carlo method with MCNP6.2 for healthy mice.

Figure 3 presents the dosimetric and biodistribution results for healthy mice. The kidneys exhibited the highest values in both absorbed dose and percentage of injected activity per gram (%IA/g), suggesting that the primary route of excretion for the radiolabeled papain nanoparticles is renal. The liver and spleen displayed moderate uptake and dose levels, consistent with expected accumulation in organs associated with the reticuloendothelial system (RES). Additionally, the bladder showed a relatively high %IA/g compared to dose, which aligns with rapid clearance through the urinary tract. Minimal accumulation in muscle, heart, and brain indicates limited nonspecific uptake in these tissues.

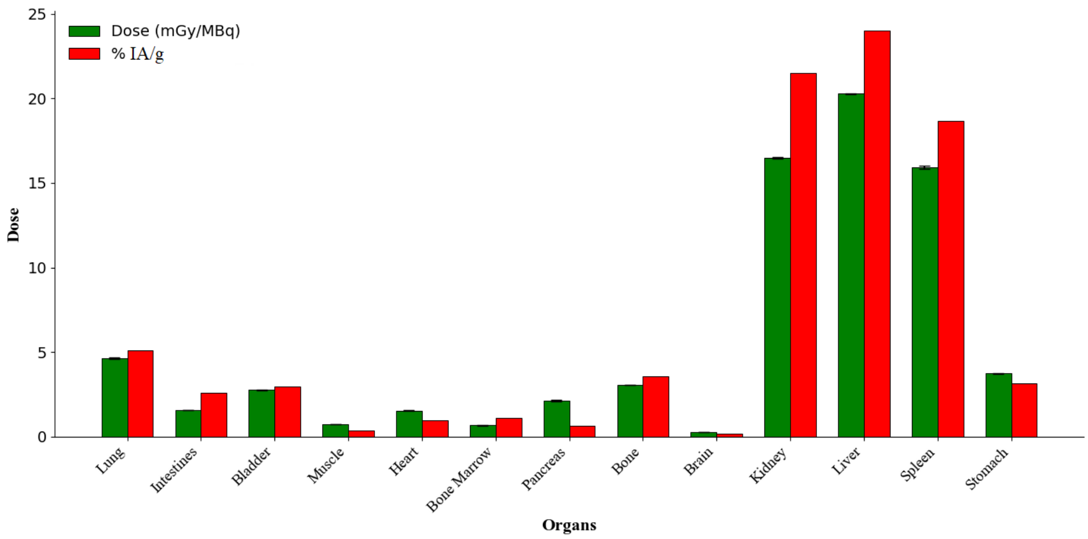

Figure 4: % Injected activity per gram for each organ in relation to the doses (mGy/MBq) obtained using the monte carlo method with MCNP6.2 for 4T1 tumor-bearing mice.

In Figure 4, corresponding to the 4T1 tumor-bearing mice, both uptake and dose in the liver and spleen increased in comparison to the healthy group. This suggests an enhancement in RES activity, potentially driven by systemic effects of the tumor. Conversely, the kidneys demonstrated a slight reduction in both absorbed dose and %IA/g, indicating a possible alteration in renal clearance dynamics. Slight increases in pancreas and bone values may reflect tumor-related metabolic or immunological changes. Bladder and intestinal data remained moderately consistent, although a slight reduction in bladder dose was observed.

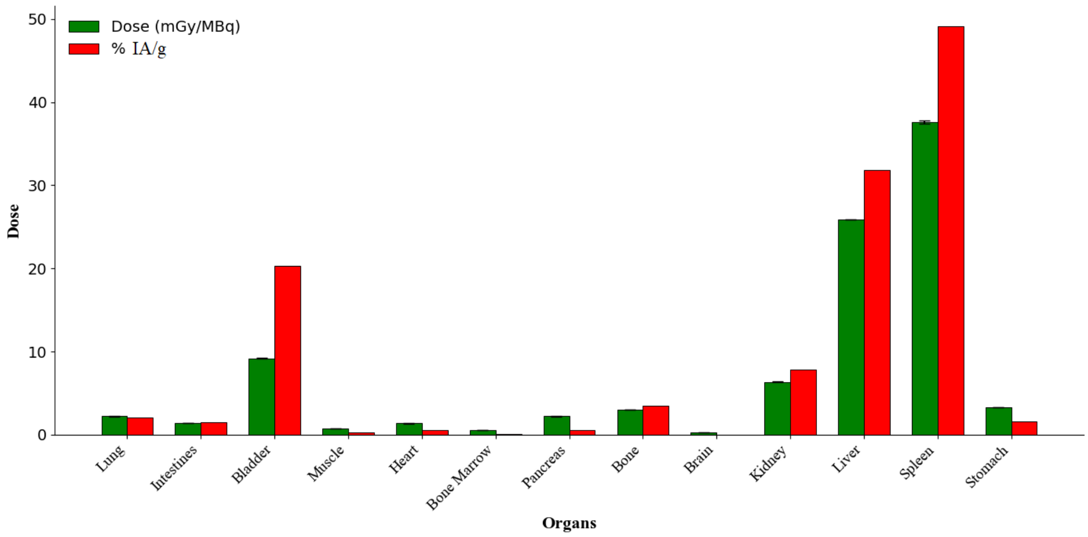

Figure 5: % Injected activity per gram for each organ in relation to the doses (mGy/MBq) obtained using the monte carlo method with MCNP6.2 for MMTv-PyMT model.

Figure 5 illustrates results for the MMTV-PyMT tumor model, which showed a marked shift in biodistribution. The spleen emerged as the organ with the highest nanoparticle accumulation in both metrics, likely due to intensified immune or macrophage activity associated with tumor burden. The liver and bladder also showed increased uptake, with the liver dose surpassing values seen in the 4T1 model. Notably, %IA/g in muscle rose sharply, while the corresponding dose remained moderate, possibly indicating localized inflammatory responses or tumor-induced changes in vascular permeability. Kidney dose further decreased, reinforcing the notion of altered excretory dynamics in the presence of systemic tumor load.

Overall, the presence of tumors substantially influences the biodistribution profile of papain nanoparticles, particularly in the liver, spleen, and muscle. Among the models studied, the MMTV-PyMT group displayed the most significant alterations, suggesting a more profound systemic impact on nanoparticle pharmacokinetics. While absorbed dose and %IA/g generally correlate, their non-linear relationship highlights the importance of considering both tissue uptake and organ mass in internal dosimetry assessments.

4 Conclusions

The aim of this study was to estimate the dose deposited per injected activity (mGy/MBq) of technetium-99m (^{99m}Tc)-labeled papain nanoparticles, based on previous biodistribution studies, through computer simulations using MCNP6.2.1—a Monte Carlo-based code. A voxelized mouse phantom, named DM_BRA, was employed to compare the simulation results with those obtained from the experimental *in vivo* biodistribution study of the papain nanoparticles. The results obtained through *in silico* dosimetry indicate that the physiological uptake background of papain nanoparticles is primarily associated with organs such as the liver and spleen, while the excretion pathway involves the urinary system, as evidenced by increased doses in the kidneys and bladder. When comparing healthy mice to tumor-bearing and MMTV-PyMT tumor groups, it is evident that the presence of tumors influences the biodistribution of papain nanoparticles, and consequently, the absorbed dose in specific organs or tissues. These findings highlight the sensitivity of papain nanoparticles to physiological alterations associated with malignant neoplasms.

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Natural Radionuclide Activity and Radiological Risk Assessment in Brazilian Coffee Samples

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Abstract

Seventeen samples of distinct brands of Brazilian coffee were purchased from popular markets of the Southeast region of Brazil and were analyzed by gamma spectrometry through the High Purity Germanium (HPGe) detector to measure the activity concentrations of the nuclides ^{40}K, ^{137}Cs, ^{226}Ra and ^{228}Ra. The samples presented substantial values of ^{40}K activity concentration, ranging from 502 to 623 Bq kg^{-1}. In all samples, the activity concentration of ^{137}Cs stayed below the detection limit of 0.2 Bq kg^{-1}. The concentrations of ^{226}Ra and ^{228}Ra varied from nondetectable to 2.6 Bq kg^{-1} for ^{226}Ra and from nondetectable to 7.8 Bq kg^{-1} for ^{228}Ra. Based on these values, we estimated the radiological risk to the health of the consumer and of the farm and factory workers, including external hazard indices and post-infusion internal dose estimates. The radium equivalent and the annual effective dose rate were also determined. All results presented values below the limit recommended by UNSCEAR (2000) 290 μSv.year^{-1} and by ICRP (2012) 1 mSv.year^{-1}. The values obtained for the lifetime cancer risk are considered negligible, since they are below the intervention limit (10^{-4}). Therefore, the product poses no risk to the population.

Keywords: HPGe; Activity Concentration; Dosimetry;

1 Introduction

Brazil is widely recognized as the world's largest coffee producer, responsible for approximately 36.8% of global production in 2024, a position maintained for over a century due to favorable climatic conditions, extensive agricultural areas, the application of modern technologies, and a well-established tradition in cultivation, particularly of Arabica coffee (IBGE, 2024). This economic and historical relevance makes it essential to ensure not only the quality of the product but also its safety for consumption. From a radiological perspective, this is crucial to safeguard the Brazilian population and maintain the reliability of Brazilian coffee in international markets.

The ingestion of radionuclides is a relevant source of exposure to ionizing radiation; therefore, understanding the abundance of radionuclides and radioactivity levels in food is fundamental to assessing population exposure (Paiva, 2022). The evaluation of radiation doses received by the population is a subject recognized by international organizations, emphasizing the importance and need for food assessments (Garcez, 2019; Cruz da Silva, 2020; Silva, 2022). In this context, studies that verify the natural concentration of radionuclides in agricultural products, including coffee, play a strategic role.

Terrestrial gamma radiation is primarily associated with radionuclides from the ^{238}U and ^{232}Th decay series, as well as the naturally occurring isotope ^{40}K present in the Earth's crust. Additionally, due to the radiological accident that occurred in the state of Goiás in 1987, it is also important to assess the presence of the radionuclide ^{137}Cs in agricultural samples. Gamma radiation emitted by these natural and artificial radionuclides can be present in agricultural products, representing a potential pathway for internal exposure in humans (Singh, 2009; Filgueiras, 2021).

The objective of this study is to quantify the levels of natural radioactivity of ^{226}Ra, ^{228}Ra, and ^{40}K, as well as the presence of the radionuclide ^{137}Cs, in coffee samples produced in Brazil. Furthermore, it aims to evaluate the annual effective dose associated with coffee consumption and to estimate the resulting cancer risk, comparing the obtained results with international reference values. The data generated in this work provide baseline values on the natural radioactivity present in coffee, supporting risk assessments related to its consumption and occupational exposure of workers handling large quantities of coffee beans in warehouses or processing facilities.

2 Materials and methods

The coffee powder samples were prepared under clean conditions to avoid any external contamination. After opening the packaging, each sample was transferred into low-background polypropylene containers, appropriate for gamma spectrometric analysis. Once sealed, the containers were stored for at least 45 days in a temperature-stable environment, maintained between −10 °C and −12 °C, to prevent density changes due to microorganism growth. This approach ensures the original geometry of the sample holder is preserved, as well as the establishment of secular equilibrium between the radionuclides of the natural decay series.

The efficiency calibration of the detection system was carried out using the LabSOCS simulation software, which models the detector setup and sample geometry through the Geometry Composer tool. This process accounts for photon attenuation within the material and corrects self-absorption effects using Monte Carlo based algorithms (Bronson, 2003).

The gamma measurements were performed using a High-Purity Germanium detector (HPGe), model GC3020 (Mirion/Canberra), with a 30% relative efficiency. The detector features a coaxial crystal (62 mm in diameter and 40 mm in height) connected to a low noise 2002C RC preamplifier. The assembly is cooled by liquid nitrogen via a 7500SL cryostat and stored in a 30-liter dewar. The system operated under a bias voltage of 4500 V, as specified by the manufacturer.

Radiation shielding included a 10 cm thick low background lead housing, lined internally with 1 mm of tin and 1.6 mm of copper, and externally protected by 9.5 mm of carbon steel. The inner chamber of the shielding unit measured 27.9 cm in diameter and 40.6 cm in depth.

Data acquisition was conducted using a DSA 1000 digital multichannel analyzer (Mirion/Canberra), operating over an energy range from 30 keV to 2 MeV, with 8192 channels. The system was fully controlled by the Genie 2000 Gamma Analysis software, which provided spectrum acquisition, calibration, peak fitting, and radionuclide identification.

The gamma spectrometry analyses were carried out at the Environmental Analysis and Computer Simulation Laboratory (LAASC), at the Federal University of Rio de Janeiro (UFRJ). To validate the detection system and verify the reliability of the results, a comparison was performed at the Laboratory of Gamma Spectrometry (LGO), operated by the Nuclear Energy Commission of Brazil (CNEN). The LGO is traceable to the Brazilian Laboratory of Metrology of Ionizing Radiations (LNMRI/CNEN), which in turn is recognized by the Bureau International des Poids et Mesures (BIPM). The LGO has demonstrated consistent performance in intercomparison exercises, such as those organized by the MAPEP program (Mixed Analyte Performance Evaluation Program) of the United States Department of Energy (USDOE).

The spectra obtained were used to quantify the activity concentrations of naturally occurring radionuclides by evaluating characteristic gamma transitions: the 1460.8 keV photopeak was used for potassium-40 (^{40}K) with an emission probability ($P\gamma$) of 10.7%, the 609.3 keV energy for radium-226 (^{226}Ra) via the decay of ^{214}Bi with $P\gamma$ of 44.26%, and the 911.1 keV energy for radium-228 (^{228}Ra) via ^{228}Ac with $P\gamma$ of 25.8%. Each sample was measured for a counting time of 18,000 seconds. The activity concentration (AC, in $Bq \cdot kg^{-1}$) was calculated according to the following expression (IAEA, 1989):

$$AC = \frac{N}{\varepsilon \cdot t \cdot P\gamma \cdot m} \tag{1}$$

where:

- N is the net peak area;
- ε is the detection efficiency;
- m is the sample mass (kg);
- t is the counting duration (s);
- Pγ is the gamma emission probability.

The combined uncertainty associated with the result was estimated by standard error propagation, incorporating contributions from each variable of the equation (1):

$$\delta AC^2 = (\frac{1}{\varepsilon \cdot t \cdot P\gamma \cdot m})^2 \cdot (\delta N^2) + (\frac{N}{\varepsilon^2 \cdot t \cdot P\gamma \cdot m})^2 \cdot (\delta \varepsilon^2) + \left(\frac{N}{\varepsilon \cdot t \cdot P\gamma^2 \cdot m}\right)^2 \cdot (\delta P\gamma^2) \quad (2)$$

To determine the activity, it is essential to establish the limit at which the detector can provide reliable data. For this purpose, the minimum detectable activity (MDA) was calculated using the following equation.

$$MDA = \frac{2.71 + 466 \cdot \sigma}{\varepsilon \cdot t \cdot P\gamma \cdot m} \quad (3)$$

where σ is the standard deviation of the background signal in the energy window of interest.

2.1 Dose due ingestion estimation

To estimate the annual effective dose from ingestion, the ingestion coefficient factor (Cf) recommended by ICRP 72 (1996), as updated by ICRP 119 (2012), was applied. Additionally, the activity concentration data ($Bq \cdot kg^{-1}$) obtained from the samples for each radionuclide and the statistical data on household coffee consumption ($kg \cdot year^{-1}$), provided by the Brazilian Institute of Geography and Statistics (IBGE, 2024), were used. The effective ingestion dose for each analyzed sample was then calculated using Equation (ICRP, 2012).

$$ED = Cf \cdot U \cdot Ac \quad (4)$$

where: *ED* is the annual effective dose ($\mu Sv.y^{-1}$).
Cf is the dose coefficient published by ICRP 119 ($\mu Sv.Bq^{-1}$)
U is the amount of coffee consumed in one year ($kg.y^{-1}$)
Ac is the activity concentration in the coffee ($Bq.kg^{-1}$)

The coefficient factors (Cf) from ICRP 119 (2012) are 0.1920 $\mu Sv \cdot Bq^{-1}$ for ^{226}Ra, 1.14 $\mu Sv \cdot Bq^{-1}$ for ^{228}Ra, and 0.042 $\mu Sv \cdot Bq^{-1}$ for ^{40}K. Since no data on radionuclide transfer to coffee infusion were found, a conservative assumption of complete transfer was adopted, consistent with previous studies on cancer risk and dose estimation for tea and coffee infusions (Garcez, 2019; Cruz da Silva, 2020).

The calculation of annual effective dose in human organs or tissues is performed using tissue weighting factors (Wf). These factors vary according to organ sensitivity, thyroid (0.04), lungs (0.12), and brain (0.01) as recommended by ICRP (2012).

$$AED = ED \cdot Wf \quad (5)$$

2.2 Cancer risk

Cancer risk estimation can be performed either based on effective dose or on the activity of the ingested radionuclide. In this study, the calculation was based on ingestion dose, following USEPA (1999), which recommends a risk factor (RF) of 0.05 Sv^{-1} for all cancer types in adults. This means that for each sievert of effective dose, an estimated 5% increase in cancer mortality risk is expected for the exposed individual.

$$Rc = ED \cdot RF \quad (6)$$

Cancer risk evaluation is interpreted according to two thresholds: results equal to or lower than 10^{-6} are considered insignificant, whereas values greater than 10^{-4} are sufficient to warrant interventions and further investigations.

3 Results and discussion

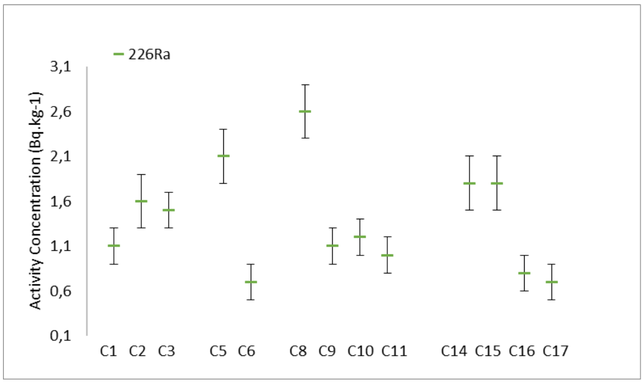
The activity concentration results show little variation for each radionuclide, with ^{40}K accounting for approximately 99.38% of the total contribution. ^{40}K presented an average value of 562 Bq·kg^{-1}, while ^{228}Ra and ^{226}Ra averaged 2.4 Bq·kg^{-1} and 1.1 Bq·kg^{-1}, respectively. No significant variation was observed among the results for each radionuclide, with maximum measurement ranges (difference between maximum and minimum values) of 121 Bq·kg^{-1} for ^{40}K, 7.8 Bq·kg^{-1} for ^{228}Ra, and 2.6 Bq·kg^{-1} for ^{226}Ra. Among the three radionuclides, only ^{228}Ra and ^{226}Ra showed results below the detection limit.

Table 1: Activity concentrations of ^{40}K, ^{226}Ra, ^{228}Ra, and ^{137}Cs in coffee samples (Bq.kg^{-1}).

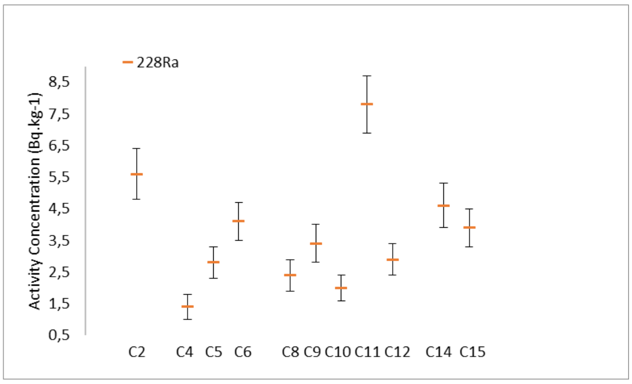
Sample	**^{226}Ra**	**^{228}Ra**	**^{40}K**	**^{137}Cs**
C1	1.1 ± 0.2	BDL	502.1 ± 24,0	BDL
C2	1.6 ± 0.3	5.6 ± 0.8	557.4 ± 26.2	BDL
C3	1.5 ± 0.2	BDL	623.2 ± 28.5	BDL
C4	BDL	1.4 ± 0.4	590.1 ± 27.3	BDL
C5	2.1 ± 0.3	2.8 ± 0.5	593.6 ± 27.0	BDL
C6	0.7 ± 0.2	4.1 ± 0.6	544.8 ± 24.9	BDL
C7	BDL	BDL	561.8 ± 25.9	BDL
C8	2.6 ± 0.3	2.4 ± 0.5	571.2 ± 25.1	BDL
C9	1.1 ± 0.2	3.4 ± 0.6	530.3 ± 25.1	BDL
C10	1.2 ± 0.2	2.0 ± 0.4	597.7 ± 27.9	BDL
C11	1.0 ± 0.2	7.8 ± 0.9	572.8 ± 29.9	BDL
C12	BDL	2.9 ± 0.5	542.1 ± 25.7	BDL
C13	BDL	BDL	591.3 ± 27.5	BDL
C14	1.8 ± 0.3	4.6 ± 0.7	539.3 ± 26.0	BDL
C15	1.8 ± 0.3	3.9 ± 0.6	567.0 ± 26.4	BDL
C16	0.8 ± 0.2	BDL	551.2 ± 25.4	BDL
C17	0.7 ± 0.2	BDL	533.7 ± 24.4	BDL

BDL, below detection limit.

The coffee samples were purchased in two Brazilian states, Rio de Janeiro and Minas Gerais, both located in the southeastern region of the country. Minas Gerais is historically one of the largest coffee producers in Brazil. The selection aimed to diversify the brands, attempting to cover as much of the territory as possible. Figures 1 and 2 show the distribution of activity concentrations by sample. Analysis of these figures reveals the reproducibility of results for samples C14 and C15, which may indicate similarities in soil composition at the cultivation sites or the origin of the coffee beans.



a)



(b)

Figure 1: (a) Distribution graph of ^{226}Ra Activity concentration. (b) Distribution graph of ^{228}Ra Activity concentration.

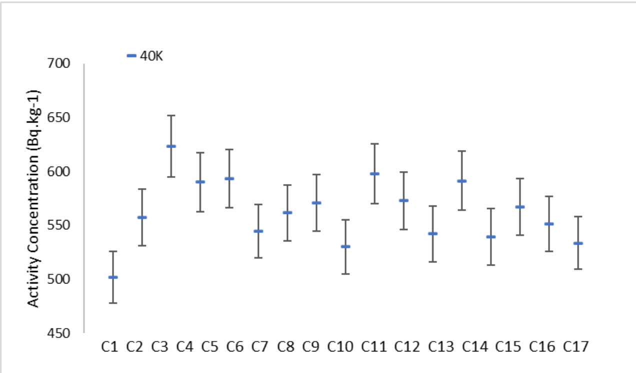


Figure 2: Distribution graph of ^{40}K Activity concentration.

3.1 Effective dose due to ingetion

The distribution of the annual effective dose due to ingestion of coffee samples (Table 2) shows that values ranged from 0.107 to 0.166 mSv·year^{-1}, with an average of 0.133 mSv·year^{-1}. The main contributor to this dose was ^{40}K, representing 88.9% of the total effective dose (Table 3), followed by ^{228}Ra (10.3%) and ^{226}Ra (0.8%). This predominance of ^{40}K is consistent with its known abundance in agricultural products, reflecting its essential role in plant metabolism and natural occurrence in soils.

When compared with international studies (Table 4), the mean annual effective dose obtained here falls between the value reported by Alharbi (2017) (0.241 mSv·year^{-1}) and that reported by Al-Alawy (2020) (0.077 mSv·year^{-1}), and is higher than the value found by Khandaker (2020) (0.022 mSv·year^{-1}). These differences are expected, as radionuclide levels in agricultural products are highly dependent on local environmental conditions and cultivation practices.

An additional observation from Table 2 is the variability associated with ^{228}Ra, which showed detectable levels in several samples (up to 0.045 mSv·year^{-1} in C11). While the overall contribution of this radionuclide to total dose is small, its presence may indicate localized soil characteristics or specific fertilization practices influencing radium uptake by coffee plants.

Another point worth noting is the distribution of doses by organ (thyroid, lungs, and brain), all of which remained well below 0.02 mSv·year^{-1}. This pattern reinforces that even for organs with higher sensitivity to internal exposure, the resulting doses from coffee ingestion are radiologically insignificant. Moreover, the cancer risk values, consistently on the order of 10^{-6}, fall within the range considered negligible.

Table 2: Effective dose due to ingestion and cancer risk.

Sample	^{40}K (mSv.y^{-1})	^{226}Ra (mSv.y^{-1})	^{228}Ra (mSv.y^{-1})	ED (mSv.y^{-1}) ^{40}K, ^{226}Ra and ^{228}Ra	Thyroid (mSv.y^{-1})	Lungs (mSv.y^{-1})	Brain (mSv.y^{-1})	Cancer Risk
C1	0.106	0.001	0.000	0.107	0.004	0.013	0.001	10^{-06}
C2	0.117	0.002	0.032	0.151	0.006	0.018	0.002	10^{-06}
C3	0.131	0.001	0.000	0.133	0.005	0.016	0.001	10^{-06}
C4	0.124	0.000	0.008	0.132	0.005	0.016	0.001	10^{-06}
C5	0.125	0.002	0.016	0.143	0.006	0.017	0.001	10^{-06}
C6	0.115	0.001	0.023	0.139	0.006	0.017	0.001	10^{-06}
C7	0.118	0.000	0.000	0.118	0.005	0.014	0.001	10^{-06}
C8	0.120	0.003	0.014	0.136	0.005	0.016	0.001	10^{-06}
C9	0.112	0.001	0.019	0.132	0.005	0.016	0.001	10^{-06}
C10	0.126	0.001	0.011	0.138	0.006	0.017	0.001	10^{-06}
C11	0.121	0.001	0.045	0.166	0.007	0.020	0.002	10^{-06}
C12	0.114	0.000	0.017	0.131	0.005	0.016	0.001	10^{-06}
C13	0.124	0.000	0.000	0.124	0.005	0.015	0.001	10^{-06}
C14	0.113	0.002	0.026	0.141	0.006	0.017	0.001	10^{-06}
C15	0.119	0.002	0.022	0.143	0.006	0.017	0.001	10^{-06}
C16	0.116	0.001	0.000	0.117	0.005	0.014	0.001	10^{-06}
C17	0.112	0.001	0.000	0.113	0.005	0.014	0.001	10^{-06}

Table 3: Distribuition of the Effetive dose in brazilian coffee.

Nuclideos	Contribuição	Média (mSv.y^{-1})	Máximo (mSv.y^{-1})	Mínimo (mSv.y^{-1})
^{40}K	88.9%	0.118	0.131	0.106
^{226}Ra	0.8%	0.001	0.025	0.000
^{228}Ra	10.3%	0.013	0.045	0.000

Table 4: Comparison the annual effective dose in coffee powder.

	Average of annual effective dose (mSv.y^{-1})
Present study	0.133
Khandaker (2020)	0.022
Alharbi (2017)	0.241
Al-Alawy (2020)	0.077

4 Conclusions

High-resolution gamma spectrometry analysis allowed the determination of activity concentrations of ^{40}K, ^{226}Ra, and ^{228}Ra in coffee samples available on the Brazilian market. The detection limits achieved were 7 Bq·kg^{-1} for ^{40}K, 0.5 Bq·kg^{-1} for ^{226}Ra, and 0.7 Bq·kg^{-1} for ^{228}Ra. In several samples, the activity concentrations of ^{226}Ra and ^{228}Ra were below these limits, indicating a naturally low presence of these radionuclides in the product. The artificial radionuclide ^{137}Cs was not detected in any sample, even with a detection limit of 0.2 Bq·kg^{-1}, demonstrating the low likelihood of artificial radioactive contamination in the analyzed coffee.

The estimated annual effective dose from coffee consumption in Brazil showed a maximum value of 166 μSv·year^{-1} and an average of 133 μSv·year^{-1}, both below the UNSCEAR reference value (290 μSv·year^{-1}) and the ICRP recommended limit (1 mSv·year^{-1}). The mean cancer risk was calculated as 6.67×10^{-6} (0.000667%), significantly lower than the UNSCEAR reference risk (4.38×10^{-5}·year^{-1}).

When compared with studies from other countries, the values obtained in this work lie between those reported by Alharbi (2017) (0.241 mSv·year^{-1}) and Al-Alawy (2020) (0.077 mSv·year^{-1}), and are higher than those reported by Khandaker (2020) (0.022 mSv·year^{-1}). Such differences may be associated with natural variations in soil composition, climatic conditions, and regional agricultural practices, which are known to influence radionuclide concentrations in agricultural products.

Based on these findings and considering the average coffee consumption reported by IBGE (2024), it can be concluded that coffee ingestion does not pose a significant radiological risk to public health in Brazil, contributing minimally to internal exposure from naturally occurring radionuclides. These results provide one of the first national assessments of radiological risk from coffee consumption and serve as a reference for future studies on radioactivity in widely consumed food products in Brazil.

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New Low-Cost Process of PIN Diode Fabrication for Radiation Detection

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Abstract

This work reports the fabrication and electrical characterization of silicon PIN radiation detectors using a low-cost microfabrication route based on standard wet etching and in-house Spin-On Glass (SOG) dopants. Devices were built on n-type FZ wafers (20–160 kΩ·cm) without substrate thinning (400 µm), employing 1.5 µm thermal SiO_2 for masking/passivation and Al metallization. Two sizes were evaluated (3AWA16: 2×2 mm; 3AY3: 8×8 mm). I–V curves were measured in the dark and under pulsed X-rays (Spectro 70X), with irradiation applied during the sweep (t = 2.5 s). Low reverse leakage (nA) and clear rectification were observed; under irradiation, transient spikes appeared only under reverse bias (3AWA16: 150–600 nA at ~12–23 V; 3AY3: up to ~1.3 µA at ~17–22 V), while forward traces remained unchanged. C–V measurements (1 MHz; −10 to +10 V) are reported strictly as capacitance: for 3AWA16, C_{max} 2.16×10^{-11}F $\rightarrow$ 2.57×10^{-11}F and C_{min} 3.43×10^{-12}F $\rightarrow$ 3.44×10^{-12}F. For 3AY3, C_{max} 7.30×10^{-9}F $\rightarrow$ 7.17×10^{-19}F and C_{min} 3.87×10^{-11}F $\rightarrow$ 3.85×10^{-11}F. These results demonstrate a simple, reproducible approach to low-cost PIN detectors responsive to low-dose X-rays, supporting applications in radiation monitoring, nuclear instrumentation, and optoelectronics.

Keywords: Diode PIN; X-Ray Detection; Low-Cost process; Sensors.

1 Introduction

Silicon-based radiation detectors are fundamental components in high-energy physics experiments, playing a key role in tracking systems and the detection of charged particles in accelerators such as the Large Hadron Collider (LHC) at CERN (Rossi and Bruning, 2012). The continuous evolution of these experiments especially considering the upcoming upgrades under the High Luminosity LHC (HL-LHC) program imposes stringent requirements on sensor technology: high radiation tolerance, precise timing resolution, and the ability to operate under high fluence rates, with nominal values approaching 10^{34} cm^{-2} s^{-1}. Under such extreme conditions, device performance progressively degrades due to the interaction of ionizing and non-ionizing radiation with the silicon crystal. Radiation-induced damage leads to several detrimental effects that directly affect sensor operation: (a) an increase in full depletion voltage, resulting from changes in the effective doping concentration; (b) a rise in leakage current, due to the creation of generation and recombination centers; and (c) a decrease in charge collection efficiency, caused by carrier trapping (Lutz, 2007; Abdel et al., 2013). Among these effects, the increase in depletion voltage following substrate type inversion is particularly critical, demanding technological strategies that can mitigate such impacts throughout the detector's operational lifetime. Various sensor architectures have been developed to meet the demands of modern experiments. LGADs (Low Gain Avalanche

Detectors), for instance, offer internal signal amplification with timing resolutions in the tens of picoseconds range, making them suitable for particle timing applications. However, their structure requires multiple doping steps with strict control over concentration profiles, and they suffer from gain degradation at high fluence levels. HV-MOS (High Voltage CMOS) sensors, on the other hand, integrate electronics and sensing elements on the same substrate, offering good spatial resolution and scalability, although their performance becomes limited under very high radiation fluences. In this context, silicon PIN diodes comprising an intrinsic region sandwiched between P-type and N-type doped layers remain a robust and widely adopted solution, both in particle physics experiments and in medical and space applications (Ronchin et al., 2004; Srithanachai and Niemcharoen, 2011; Valtonen et al., 2016). Their simple structure enables reliable operation in full depletion mode, providing high sensitivity and good charge collection linearity. They also exhibit remarkable radiation hardness, especially when designed with reduced thickness. In contrast to more complex sensor architectures, PIN diodes stand out for their simplicity of fabrication, involving fewer critical steps and greater compatibility with standard microelectronics processes. Reducing the active thickness of the detector (i.e., the intrinsic region) is one of the most effective strategies to mitigate radiation effects. Ultra-thin sensors (with thicknesses in the range of 50–100 μm) can operate at significantly lower depletion voltages and delay the onset of type inversion, even in substrates with relatively low initial resistivity (Mandić et al., 2010). However, fabricating such thin devices presents technological challenges: conventional mechanical thinning is not viable for achieving these dimensions with sufficient precision and structural integrity, particularly in fragile substrates. In this regard, MEMS-based micromachining techniques have shown great promise especially anisotropic wet etching with potassium hydroxide (KOH), which offers excellent etching uniformity and high selectivity toward silicon oxides and silicon nitride. In the present work, we propose and evaluate a simplified fabrication process for PIN diodes based on high-resistivity Float Zone (FZ) silicon wafers, where doping is achieved using home-building Spin-On Glass (SOG) sources for both boron (P-type) and phosphorus (N-type) diffusion. This approach significantly reduces fabrication cost, offers greater control over doping profiles, and eliminates the need for commercial pre-doped wafers. Although the process includes localized wet etching capabilities, the devices characterized in this study intentionally retained the full 400 μm thickness of the intrinsic region, allowing for analysis of the junction quality and detector response without mechanical or chemical thinning. The results obtained from the fabrication and electrical characterization of the PIN diodes are presented and discussed, highlighting the potential of this approach as a simple, effective, and low-cost alternative for applications in high-radiation environments, particularly where prototyping flexibility and resource efficiency are key.

2 Materials and Methods

2.1 Operating Principle of PIN Diode Detectors

The sensor structure is based on a conventional PIN diode architecture, which consists of three distinct regions with controlled doping profiles: a shallow P-type surface layer, typically doped with boron; a relatively thick, intrinsic (undoped or lightly doped) region, formed by a high-resistivity monocrystalline silicon substrate; and a deep N-type layer, commonly doped with phosphorus. This configuration establishes a P–I–N junction, where the intrinsic region plays a critical role in defining the devices electrical and sensing properties. Under reverse bias conditions, the intrinsic layer becomes fully depleted of free carriers, creating a wide depletion region that extends across the entire thickness of the intrinsic silicon. This extended depletion width is fundamental to the operation of the detector, as it enables the generation of a strong and uniform electric field across the bulk of the device. When ionizing radiation or photons interact with the detector, they produce electron-hole pairs within the depletion region. These charge carriers are then rapidly separated and collected by the internal electric field electrons drift toward the N-type contact, and holes toward the P-type contact resulting in a measurable electrical signal. The use of high-resistivity silicon (typically $> 20\ \mathrm{k\Omega \cdot cm}$) as the substrate material is essential to maximize the depletion width at relatively low bias voltages, improving charge collection efficiency and minimizing leakage currents. Moreover, the absence of heavy doping in the intrinsic region reduces the probability of carrier recombination, further enhancing the signal-to-noise ratio of the device. Figure 1 presents a simplified schematic of the PIN diode structure. This structure is widely employed in particle and photon detection systems due to its high sensitivity, fast response time, and relatively simple fabrication process when compared to more complex sensor technologies such as LGADs or monolithic CMOS detectors.

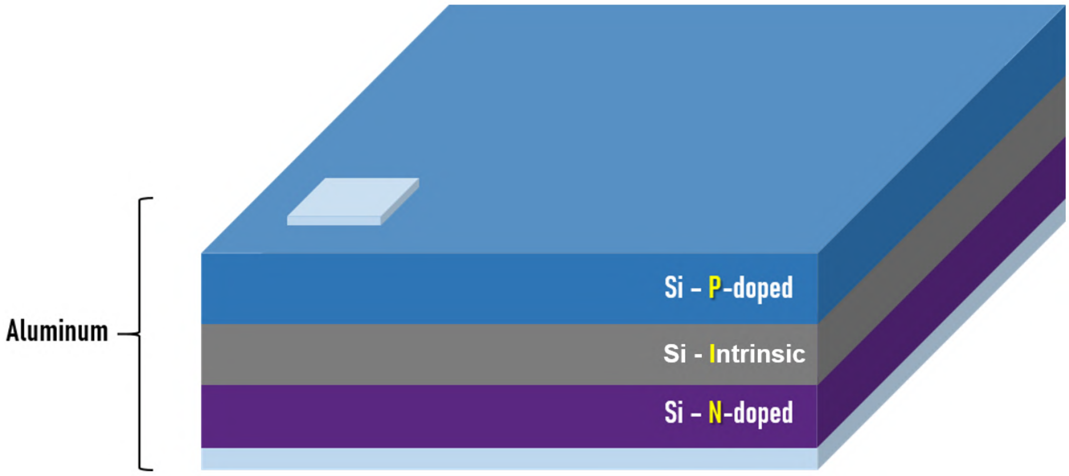

Figure 1: Simple diagram of PIN diode structure with electrical contacts.

2.2 Fabrication

PIN diodes were fabricated using high-resistivity silicon wafers produced by the Float Zone (FZ) method, oriented in the <100> direction, n-type, phosphorus-doped, with resistivity values ranging from 20 kΩ·cm to 160 kΩ·cm, corresponding to doping concentrations between 4.2×10^{10} cm^{-3} and 2.2×10^{11} cm^{-3}. These substrates are particularly suitable for radiation sensor applications due to their low defect density and high minority carrier lifetime. Prior to processing, the wafers underwent a standard chemical cleaning sequence, consisting of:

- RCA-1 (NH_4OH:H_2O_2:H_2O, 1:1:5 at 75 °C for 10 min), to eliminate organic and metallic contaminants;
- RCA-2 (HCl:H_2O_2:H_2O, 1:1:4 at 75 °C for 10 min), focused on the removal of alkali ions and cations like $A1^{3+}$, Fe^{3+} and Mg^{2+}.

A 1.5 µm-thick thermal silicon dioxide (SiO_2) layer was grown on both wafer surfaces, serving as both an etch mask and a passivation layer. Photolithography was first applied to the rough (unpolished) side to open windows in the oxide layer. These exposed areas were then subjected to anisotropic wet etching in a 10% w/w KOH solution at 80 °C, enabling localized thinning of the substrate and the formation of the intrinsic region. The etch time was adjusted according to the desired thickness. Subsequently, aligned photolithography was performed on the polished side to open oxide windows aligned with the thinned regions on the opposite side. P-type (boron), in polished side, and N-type (phosphorus), in unpolished side, doping were carried out via simultaneous thermal diffusion at 950 °C for 30 minutes, using Spin-On Glass (SOG) sources prepared in-house. The SOG dopants were deposited by spin coating and thermally cured. The in-lab preparation of SOG dopants is a key advantage of this process, offering flexibility, control over concentration, and significant cost reduction, especially when compared to commercial pre-doped wafers or advanced deposition systems. After junction formation, aluminum metallization was performed by thermal evaporation. On the polished side, the metal contacts were patterned by photolithography to define the active regions of the devices. On the rough side, aluminum was deposited uniformly, serving as the common back contact. The developed process combines accessible microfabrication steps, the use of laboratory-prepared SOG dopants, anisotropic wet etching, and standard lithographic and metallization techniques, resulting in a robust, low-cost, and reproducible approach for the fabrication of PIN diodes for radiation detection, suitable for nuclear instrumentation, optoelectronics, and photodetection applications. The final result of the fabrication process is illustrated in Fig. 2.

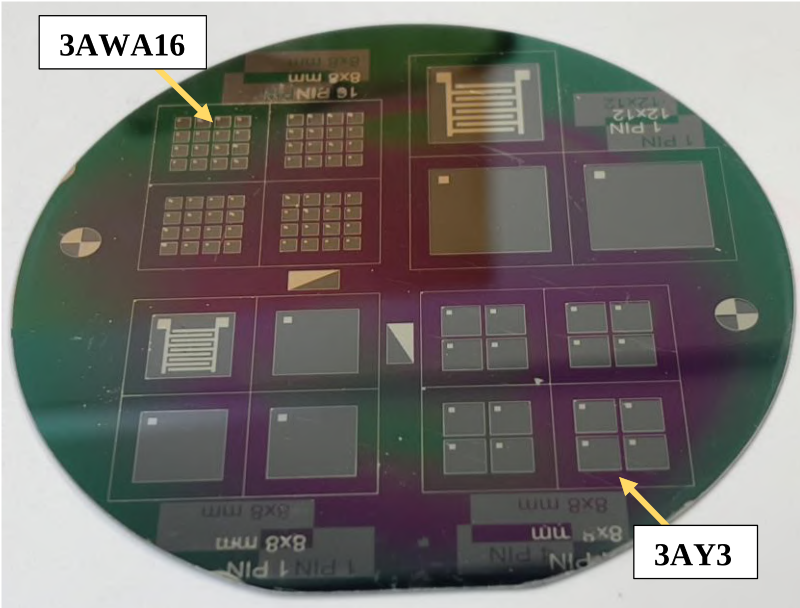

Figure 2: Final result of the PIN diode fabrication on the silicon wafer.

2.3 Electrical Characterization

Electrical characterizations were carried out at room temperature using a Keysight B2912B precision source/measure unit, connected to a Wentworth Laboratories probe station. All measurements were performed in the dark to eliminate photo-induced currents and with identical cabling and grounding across dark and irradiated runs. A voltage sweep ranging from –10 V to +30 V was applied to evaluate the diode response under both reverse and forward bias conditions. Although the most significant current variations in PIN diodes occur under forward bias due to exponential carrier injection, the region of interest for radiation detection is reverse bias, where the depletion region extends across the intrinsic layer and enables efficient collection of photogenerated carriers. Therefore, particular attention was given to the reverse regime, where leakage currents, depletion behavior, and sensitivity to irradiation are most critical. During testing, the devices exhibited low reverse leakage currents (in the nanoampere range) and clear rectifying behavior, with exponential current increase under forward bias—responses that are characteristic of well-fabricated PIN structures. These results point to good junction formation, low interface defect density, and effective development of the intrinsic depletion region (Sze and Ng, 2006). Additionally, to evaluate the devices under ionizing radiation, an X-ray source (Spectro 70X, Dabi Atlante) was employed, operated in pulsed mode. The samples were irradiated during the I–V sweep itself, with no delay between exposure and measurement; the total exposure time was t=2.5 s. Instrument parameters (voltage step and SMU integration time) were kept constant across dark and irradiated acquisitions, ensuring comparability. To complement the device analysis, Capacitance–Voltage (C–V) measurements were performed at room temperature using an Agilent E4980A – Precision LCR Meter. The DC bias was swept from –10 V to +10 V in 100 mV steps, while the small-signal capacitance was recorded at a 1 MHz test frequency.

3 Results

Electrical characterization of the devices under dark conditions was performed within the voltage range of –10 V to +30 V. Measurements were carried out at room temperature using a Keysight B2912B precision source/measure unit, connected to a Wentworth Laboratories probe station. The analyzed samples, identified as 3AWA16 and 3AY3, did not undergo intrinsic layer thinning, thus retaining the original substrate thickness of approximately 400 µm. The active areas of the devices were 4 mm²

(2 × 2 mm) for sample 3AWA16 and 64 mm² (8 × 8 mm) for sample 3AY3. Figure 3 shows the current–voltage (I–V) curves obtained for both samples under dark conditions. Both devices exhibited clear rectifying behavior, with low leakage current under reverse bias remaining in the nanoampere range confirming the formation of a high-quality p–i–n junction with good passivation and low defect density. These reverse characteristics are particularly relevant, as this is the operational regime for radiation detection. Sample 3AWA16 showed extremely low leakage current across the full reverse range, while in forward bias, the current increased exponentially, reaching approximately 100 mA at -10 V. Although the forward regime confirms diode functionality, the low reverse current is the primary indicator of the junction's integrity for sensor applications. Its small active area contributes to minimizing lateral leakage and edge effects, which is crucial in structures without mesa isolation or guard rings. Sample 3AY3, with a significantly larger area, also showed diode-like behavior, maintaining reverse leakage at a similar level despite the larger junction. In forward bias, the current increased more gradually, also reaching approximately 100 mA at -10 V. The reduced slope in the direct regime can be attributed to higher series resistance, increased junction capacitance, and possible non-uniformities effects that scale with area. Importantly, the large perimeter-to-area ratio increases susceptibility to recombination and lateral leakage, potentially impacting reverse operation in high-fluence conditions. Overall, both devices operate reliably under reverse bias, exhibiting low leakage and robust depletion behavior, even without thinning the intrinsic region validating their feasibility as radiation detectors.

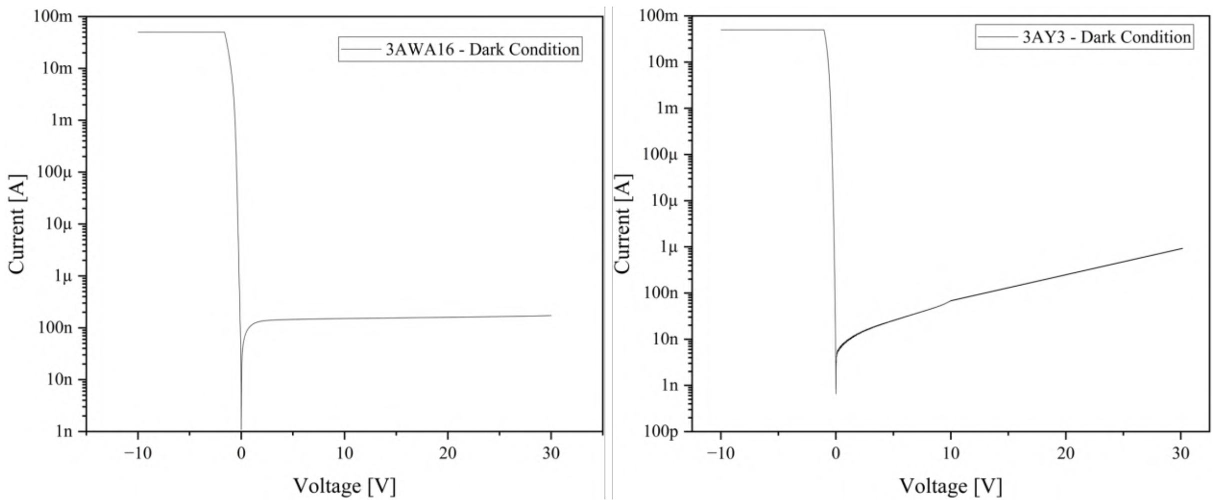

Figure 3: Current–voltage (I–V) characteristics of PIN diode samples 3AWA16 and 3AY3 under dark conditions. Both devices exhibit typical diode-like rectifying behavior, with low current under reverse bias and dominant conduction under forward bias.

Samples 3AWA16 and 3AY3 were exposed to X-rays using a Spectro 70X source (Dabi Atlante), at a dose rate of 0.58 µSv/h, to assess their sensitivity under low-level ionizing radiation. Figure 4 shows the I–V curves under this condition. Both devices maintained their PIN diode characteristics, with minimal degradation in leakage behavior. However, distinct current oscillations emerged under reverse bias in both samples absent in the dark measurements. These transient current peaks are attributed to the generation of electron–hole pairs in the intrinsic region caused by X-ray photon absorption. The internal electric field separates and collects these carriers, producing visible spikes in current. In sample 3AWA16, oscillations occurred between 3 V and 7 V, with amplitudes from 150 nA to 600 nA. In sample 3AY3, the peaks appeared in the narrower range of 4 V to 6 V, but reached higher values, up to 1.3 µA, due to the larger active volume enhancing carrier generation. Importantly, forward current remained essentially unchanged after exposure, indicating that the radiation dose was not sufficient to induce permanent damage or increase generation–recombination activity. These observations confirm that both devices are capable of detecting low-dose ionizing radiation via carrier excitation in the depletion region.

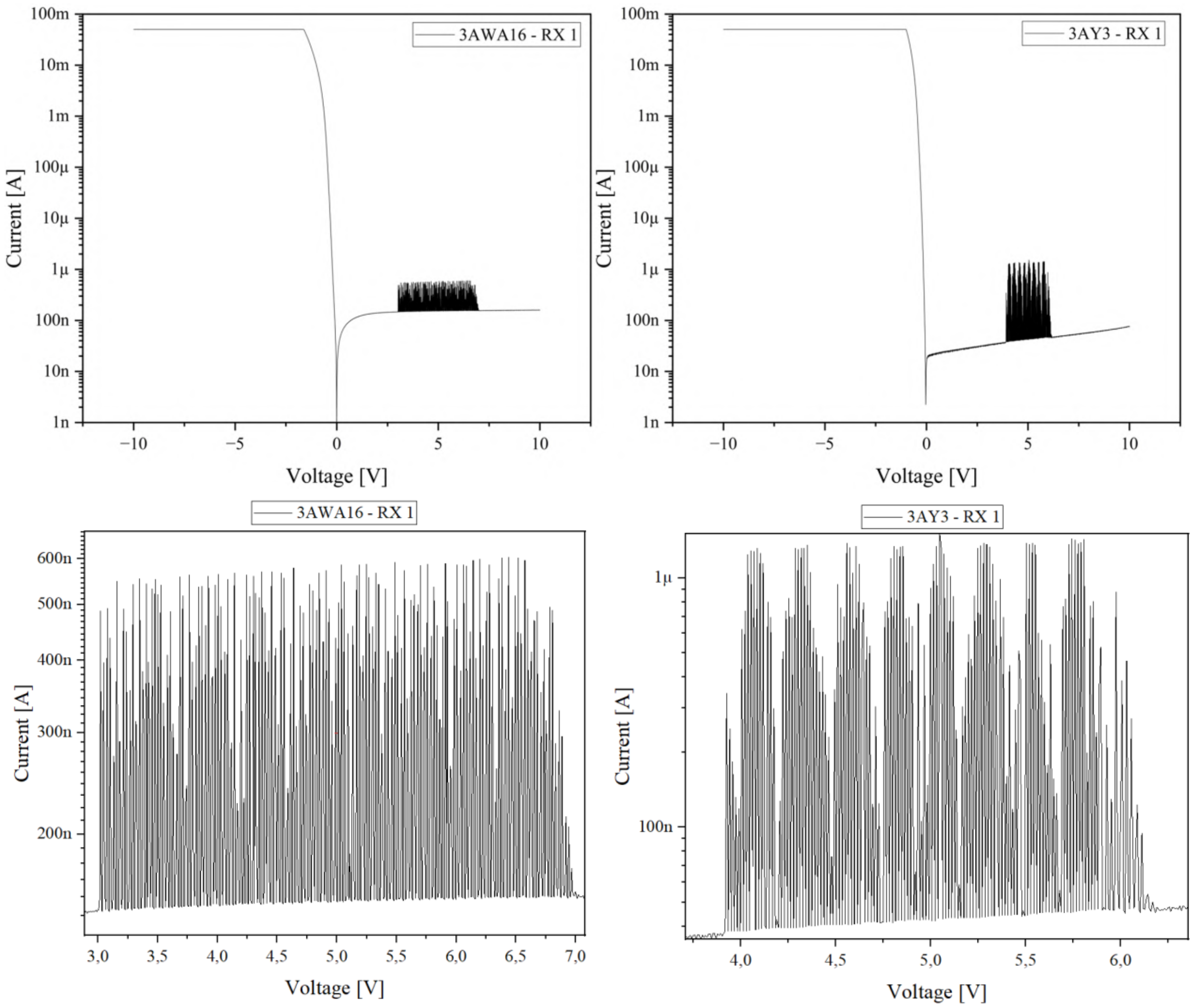

Figure 4: Current–voltage (I–V) characteristics of PIN diode samples 3AWA16 and 3AY3 under exposure to ionizing radiation (0.58 µSv/h). Under reverse bias, both devices exhibit oscillatory current peaks arising from X-ray–induced carrier generation in the intrinsic region.

Extended-range measurements (–10 V to +30 V) under irradiation were also performed, as shown in Figure 5. This approach allows broader exploration of the reverse bias regime, where the electric field plays a critical role in enhancing carrier collection. Both samples preserved their diode characteristics with stable forward conduction and low reverse leakage. In the reverse region, however, the oscillatory photocurrent response intensified:

- For sample 3AWA16, peaks were observed from +12 V to +23 V, with amplitudes ranging between 150 nA and 600 nA;
- For sample 3AY3, oscillations occurred between +17 V and +22 V, with magnitudes reaching 1.3 µA.

The shift of detection peaks to higher reverse voltages suggests stronger carrier drift and enhanced charge separation due to the intensified electric field. The number and intensity of pulses also increased, indicating consistent photon interactions. Once again, the larger-area device exhibited greater sensitivity, confirming the correlation between active area and radiation-induced signal. These results reinforce the viability of thick, unthinned PIN diodes as reliable low-dose radiation sensors. The oscillatory behavior in reverse bias, coupled with forward regime stability, demonstrates operational robustness and sensitivity to ionizing events key for detector applications in environments such as medical diagnostics, aerospace, or environmental monitoring.

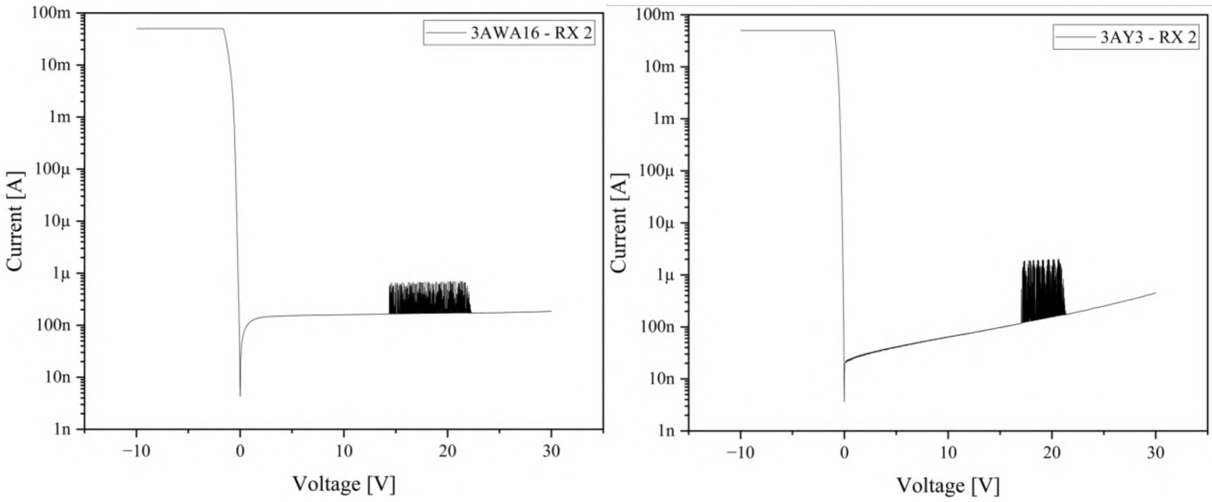

Figure 5: Current–voltage (I–V) characteristics of samples 3AWA16 and 3AY3 under X-ray exposure (0.58 µSv/h) with an extended bias range (–10 V to +30 V). Oscillatory peaks in forward bias indicate photogenerated carrier detection in the intrinsic region of the PIN structure.

Under dark conditions, the I–V curves show rectifying behavior with reverse leakage in the nanoampere range and smooth traces with no spikes. Under X-ray exposure**,** transient current bursts appear exclusively under reverse bias and remain absent under forward bias. Their amplitudes and bias windows are consistent with photogeneration and drift collection in the depletion region of the PIN diode and with the pulsed emission of the source: for device 3AWA16, 150–600 nA between ~12–23 V; for 3AY3, up to 1.3 µA between ~17–22 V. These values far exceed the local baseline fluctuation in the dark and increase with (i) active area and (ii) reverse-bias magnitude, both expected when pair generation volume and drift field strengthen. Moreover, the invariance of forward-bias curves before/after exposure indicates no permanent damage and argues against instrument instabilities. Taken together (a**)** temporal coincidence with the exposure window, (b) restriction to reverse bias, (c) scaling with area/field, and (d) absence in dark runs these controls are sufficient to rule out spurious electrical oscillations and to attribute the spikes to the collection of X-ray-induced carriers.Capacitance–Voltage (C–V) measurements were performed at room temperature using an Agilent E4980A LCR (1 MHz test frequency), sweeping the DC bias from –10 V to +10 V in 100 mV steps. Considering the extremes along the sweep, sample 3AWA16 exhibited C_{max} of $2.16 \text{ X } 10^{-11}$F in the dark and $2.57 \text{ X } 10^{-11}$F under X-rays, while C_{min} changed from $3.43 \text{ X } 10^{-12}$F to $3.44 \text{ X } 10^{-12}$F. For sample 3AY3, C_{max} measured $7.30 \text{ X } 10^{-9}$F in the dark and $7.17 \text{ X } 10^{-9}$F under X-rays. C_{min} changed from $3.87 \text{ X } 10^{-11}$F to $3.85 \text{ X } 10^{-11}$F. Results are reported strictly in terms of capacitance (maximum and minimum values along the sweep), with no depletion estimates.

4 Conclusions

This work demonstrated the fabrication and operation of silicon PIN radiation detectors using a simplified, low-cost process based on lab-prepared SOG doping and wet chemical etching. Even without thinning the intrinsic region, the devices exhibited low leakage and strong rectification. In I–V measurements, under pulsed X-ray irradiation applied during the sweep (t=2.5 s), a reverse-bias response appeared as transient current spikes: for 3AWA16, amplitudes of 150–600 nA at ∼12–23 V; for 3AY3, spikes up to **∼**1.3 µA at ∼17–22 V. Forward-bias traces remained unchanged. The absence of spikes in the dark, confinement to reverse bias, and temporal coincidence with the exposure window rule out spurious oscillations as the signal origin. Across devices, the smaller-area device (3AWA16) showed higher current stability and more defined spikes, whereas the larger-area device (3AY3) exhibited series-resistance and perimeter influences. C–V (1 MHz) measurements are reported strictly as capacitance, these results complement the irradiated I–V outcomes and will support future comparisons with the literature, while this paper reports capacitances directly without conversion to depletion parameters. The proposed fabrication route using SOG doping and conventional processing reinforces the feasibility of developing radiation-sensitive devices without reliance on expensive tools or commercial dopant sources, enabling applications in academic, industrial, and clinical contexts. As future work, we recommend absolute dose/flux calibration, noise spectral analysis, a systematic thermal-stability campaign (I–V(T), C–V(T), cycling and Arrhenius), and extending C–V to C–F.

Acknowledgements

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Occupational exposure assessment in ROLL procedures with Tc-99m: A Monte Carlo simulation approach

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Abstract

The radioguided occult lesion localization (ROLL) procedure is used in breast-conserving surgeries, particurlaly for non-palpable lesions. Despite the use of low activities of ^{99m}Tc, the proximity between the healthcare professional and the patient can result in significant radiation exposure to radiosensitive organs. This study used Monte Carlo simulations with the MCNPX 2.7.0 code to estimate the equivalent and effective dose rates of both a healthcare professional and a patient during the procedure. Computational anthropomorphic phantoms (FASH3) representing the two individuals were employed, and three protection scenarios were analyzed: no shielding, with a 0.5 mmPb lead apron, and with a protective screen. Equivalent dose rates were evaluated in various organs, emphasizing those with higher radiosensitivity. The results showed that using the lead apron significantly reduced the equivalent dose rate to the professional's thoracic region. The effective dose rate dropped from 3.4 µSv/h (without protection) to 0.83 µSv/h. The protective screen also provided effective shielding, but it was less efficient than the apron. The most exposed organs for the patient were the breast (535 µSv/h), liver (133 µSv/h), and lungs (96 µSv/h). In conclusion that the use of personal protective equipment is essential even in low-dose procedures. This reinforces the importance of applying the ALARA principle in clinical nuclear medicine practice.

Keywords: ROLL; Dosimetry; Occupational exposure; Virtual anthropomorphic phantom; Monte Carlo simulation.

1. Introduction

Breast cancer remains one of the leading causes of mortality among women in Brazil and worldwide. Thanks to mammography, ultrasonography, and other advanced imaging techniques, the desease is increasingly being detected in its early stages. To remove non-palpable lesions, techniques such as Radioguided Occult Lesion Localization (ROLL) have gained prominence for their precision and convenience in breast-conserving surgery, confirming their established use in modern clinical practice (Sarlos et al., 2009; Akerman et al., 2009; Ocal et al., 2011).

In the ROLL procedure, a radiopharmaceutical is injected directly into the breast lesion to guide resection using an intraoperative gamma probe (ICRP, 2015). Although low radiation levels result from this procedure, the proximity of the medical team to the patient can result in significant exposure to the professional's lens, thyroid, and breast. Since 2011, the ICRP has lowered the annual dose limit for the lens of the eye from 150 mSv to 20 mSv/year, with a five-year average and a single-year maximum of 50 mSv (ICRP, 2012).

In these contexts, an accurate assessment of occupational dose is essential to support radiation protection strategies for professionals exposed to ionizing radiation. The strrategies include the use of shielding, optimization of exposure time, and distancing onself from the radiaton field. However, experimental studies with direct dosimetry are not always feasible due to variability in clinical scenarios. Therefore, Monte Carlo - based simulations have become a robust and widely accepted tool in the scientific community. These simulation allow estimation of dose deposition in organs and tissues with high precision by modeling realistic situations using computational anthropomorphic phantoms and clinical geometries (Roser et al., 2019; Park et al., 2021; Poorbaygi et al., 2023).

This work aims to contribute to the literature on occupational safety in nuclear medicine by estimating, via Monte Carlo simulations using MCNPX 2.7.0 (Pelowitz, 2011), the doses absorbed by a healthcare professional positioned alongside the patient during the ROLL procedure. The study employing realistic computational anthropomorphic phantoms (Cassola *et al*., 2010) and parameters consistent with clinical practice. Analysis of the results highlights reflection on the importance of radiation protection even in procedures considered low-dose, and contributes to the application of the ALARA principle in hospital routines.

2. Materials and Methods

2.1 Radiation transport code MCNPX 2.7.0

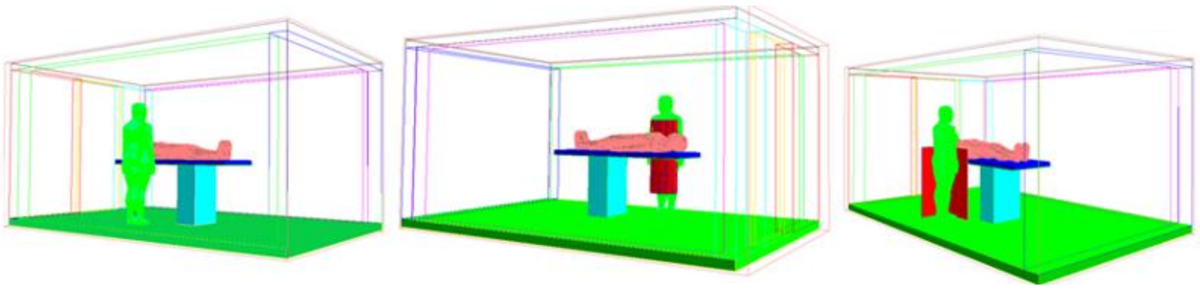
Monte Carlo simulations were performed using the particle-transport code MCNPX 2.7.0 (Pelowitz, 2011). This code is widely used in fields such as nuclear reactor physics, radiation protection, nuclear medicine, and dosimetry research. MCNPX 2.7.0 enables the simulation of the behavior of particle - such as energetic neutrons, photons, electrons, and protons within different media and complex geometries. This makes Monte Carlo simulations a powerful tool for modeling environments and conditions where radiation transport is a significant factor.

2.2 Computational anthropomorphic phantom FASH3

In this study, two female computational anthropomorphic phantoms, known as FASH3, were employed. A phantom was built in a standing posture to represent the physician performing the ROLL procedure. Another phantom was built in supine position to represent the patient with breast cancer. These phantoms were constructed based on the adult female anthropometric properties from the ICRP 89 reference model (ICRP, 2003). The standing phantom was used to represent the phisician and the the patient was represented by a phantom in the supine position. Both anthropomorphic phantoms have a height of 1.63 m and a body mass of 60 kg each. The FASH3 phantom features a voxel matrix of 221 rows, 128 columns, and 677 slices, with cubic voxels measuring 2.4 mm on each side.

2.3 Configuration of protective equipment

Three distinct configurations were simulated to evaluate the impact of the personal protective equipment (PPE) used by physicians during the ROLL procedures on the radiation dose they were exposed in different protection scenarios. In the first scenario, the physician performed the procedure without wearing a lead apron or shielding screen. This scenario provides a baseline measurement of direct radiation exposure for the professional (Figure 1A). In the second scenario, the physician wore a lead apron with an equivalent thickness of 0.5 mm Pb, which covered the torso and other radiosensitive body regions; this configuration enabled assessment of the apron's effectiveness in reducing direct exposure as well as the residual dose received by the professional (Figure 1B). In the third scenario, a 0.5 mm thick lead shielding screen was placed between the physician and the radiation source. The physician were not use an apron.This was done to evaluate the amount of radiation attenuated by the shield during the procedure (Figure 1C).



(A) (B) (C)

Figure 1: Exposure scenarios: (A) No shielding between the physician and patient; (B) the physician wearing a lead apron (B); (C) the physician is behind a lead shielding screen (C).

The evaluation of these scenarios enabled a comparative analysis of the dose rates to which the physician is exposed in each situation. The analyses highlighted the importance of protective equipment in the clinical environments and contributes to the implementation of practices that minimize occupational risks associated with radiation exposure during ROLL procedures.

As showed in Figure 1, both the physician and the patient are in a procedure room filled with atmospheric air.The surgical room dimensions are 4.5 m in length, 3.0 m in width, and 2.8 m in height, and its walls are made of 20 cm-thick concrete. The patient was positioned on a surgical table with a metal base and a polyurethane mattress measuring 70 cm in width, 190 cm in length, and 5 cm in thickness. The physician was positioned 20 cm from the patient's thoracic surface. All chemical and physical characteristics of the materials used were obtained from the literature (McConn et al., 2011).

2.4 Definition of the ^{99m}Tc source

This study employed the decay scheme of ^{99m}Tc generated by the DECDATA software, based on ICRP Publication 107 (ICRP, 2008). The scheme includes two main photon energies: 0.0186 MeV and 0.1405 MeV. Their respective emission yields are 0.077 and 0.879 photons per disintegration. Based on this information, a photon point source was modeled and positioned within the patient's right breast.

2.5 Dosimetric Calculation

To quantify the equivalent dose rate in selected organs and tissues, as well as the dose distribution throughout the body of the computational anthropomorphic phantoms, Equation 1 was used:

$$\dot{H}\left(\frac{\mu Sv}{h}\right) = \sum_{R} 1.602 \times 10^{-10} \times A \times \frac{F6}{N} \times k \times w_R \times t \times 10^{6} \qquad (1)$$

where 1.602×10^{-10} J/MeV is the conversion factor from MeV/g/source-particle to Gy/source-particle; A is the total activity of the source (74 MBq); F6 is the the average energy deposited in the volume of interest; N is the total number of voxels in the selected organ or tissue; k represents the total number of photons per decay emitted by the source (k = 0.96); w_R is the radiation weighting factor, which is equal to 1 for photons and electrons, and *t* is the exposure time (3.6×10^{3} s).

2.6. Calculation of effective dose rate

The effective dose rate was calculated based by multiplying the equivalent dose rate of each organ and tissue and its tissue weighting factor (w_T), as described in Equation 2 (ICRP, 2007):

$$\dot{E}\left(\frac{\mu Sv}{h}\right) = \sum_{T} \dot{H} \times \, w_T \qquad (2)$$

Monte Carlo simulations were performed on a 32-core PC with 128 GB of RAM, with a computational time of approximately 12 hours to simulate one 10^9 particle histories, ensuring that errors were reduced to less than 1% in all irradiation scenarios studied.

3. Results and Discussion

3.1 Assessment of occupational exposure

Table 1 presents the values of equivalent and effective dose rates, expressed in µSv/h, based on the use or non-use of protective equipment, for the physician during a ROLL procedure using the ^{99m}Tc radioisotope inside the patient's right breast. The physician's exposure was evaluated in three scenarios: without lead apron, with lead apron, and with a lead shielding screen.

Table 1: Equivalent and effective dose rates (µSv/h) for the physician's main radiosensitive organs and tissues, based on the use or non-use of lead protective equipment. Relative errors (%) are shown in parentheses.

Organ	Without apron	With apron	With lead shield
Red bone marrow	2.0 (0.2)	9.0×10^{-1}(0.1)	1.6 (0.2)
Colon	3.7 (0.2)	8.2×10^{-1} (0.3)	1.5 (0.2)
Lungs	3.2 (0.2)	7.1×10^{-1} (0.3)	3.1 (0.2)
Stomach	5.6 (0.2)	1.3 (0.4)	4.9 (0.2)
Breast	7.1 (0.2)	1.6 (0.3)	7.1 (0.2)
Remainder tissue	1.3×10^{-1} (0.1)	3.0×10^{-2} (0.1)	1.3×10^{-1} (0.2)
Gonads	2.0 (1.0)	4.2×10^{-1} (2.2)	4.8×10^{-1} (0.2)
Bladder	1.8 (0.7)	3.7×10^{-1} (1.5)	3.9×10^{-1} (0.2)
Esophagus	2.7 (0.4)	6.3×10^{-1} (0.9)	2.7 (0.2)
Liver	4.4 (0.2)	1.0 (0.3)	4.1 (0.2)
Thyroid	5.1 (0.5)	1.1 (1.1)	5.1 (0.2)
Bone surface	6.1×10^{-1} (0.1)	2.3×10^{-1} (0.1)	4.6×10^{-1} (0.2)
Brain	1.4 (0.3)	1.3 (0.3)	1.4 (0.2)
Salivary glands	1.3 (0.4)	1.2 (0.4)	1.3 (0.2)
Skin	2.1 (0.1)	6.7×10^{-1} (0.1)	1.5 (0.2)
Eye lens	2.6×10^{2} (0.1)	8.3×10^{1} (0.2)	1.9×10^{2} (0.2)
Effective dose rate	3.4 (0.3)	8.3×10^{-1} (0.5)	2.8 (0.2)

Analyzing the values in Table 1, it is observed that the most irradiated organs are the eye lenses, breasts, and stomach, especially in the unprotected condition. The eye lenses were the most irradiated organs, with an equivalent dose rate of 260 µSv/h without an apron. This is a critical situation, as the eye lenses are highly sensitive to radiation, and cumulative exposure increases the risk of cataracts. Using the shield, the equivalent dose rate drops to 190 µSv/h, and using the apron reduces this to 83 µSv/h.

When evaluating the exposure of the breasts and stomach, the breast receives a notable equivalent dose rate of 7.1 µSv/h in both unshielded and shielded setups. With the use of a protective apron, this value decreases significantly to 1.6 µSv/h. Similarly, the stomach is initially subjected to 5.6 µSv/h, which is reduced to 1.3 µSv/h when the apron is applied. These results highlight the effectiveness of protective aprons in minimizing radiation to radiosensitive organs.

The thyroid and gonads are other radiosensitive organs that also benefit from protection. Without an apron, the gonads is exposed to a equivalent dose rate of 2.0 µSv/h to 0.40 µSv/h with an apron and 0.48 µSv/h with an shield. The thyroid is exposed to a equivalent dose rate of 5.1 µSv/h in both the unprotected and shielded scenarios, which drops to 1.0 µSv/h with the apron.

The overall effective dose rate also reflects the importance of using protective equipment in ROLL procedures. Without an apron, the effective dose rate is 3.4 µSv/h; however using a apron reduces it to 0.83 µSv/h, which is a significant decrease. This reinforces the importance of using lead aprons and shields to protect the body from unwanted radiation and decrease the risk of long-term adverse effects. Thus, the use of protective equipment such as aprons and shields is essential to ensure the safety of the professional, minimizing exposure to radiation levels that could, over time, cause adverse health effects.

3.2 Assessment of patient exposure

Table 2 presents the equivalent and effective dos rates in µSv/h for the patient undergoing the ROLL procedure.

Table 2. Equivalent and effective dose rates (in µSv/h) for the patient's main organs. Relative errors (%) are shown in parentheses.

Organ	Equivalent dose rate (µSv/h)
Red bone marrow	$1.3{\times}10^1$ (0.1)
Colon	8.3 (0.1)
Lungs	$9.6{\times}10^1$ (0.1)
Stomach	$5.0{\times}10^1$ (0.1)
Breast	$5.4{\times}10^2$ (0.1)
Remainder tissue	2.3 (0.1)
Gonads	1.6 (1.1)
Bladder	1.1 (0.8)
Esophagus	$6.1{\times}10^1$ (0.1)
Liver	$1.3{\times}10^2$ (0.1)
Thyroid	$1.6{\times}10^1$ (0.3)
Bone surface	5.1 (0.1)
Brain	1.5 (0.3)
Salivary glands	3.1 (0.2)
Skin	$1.4{\times}10^1$ (0.1)
Eye lens	3.8 (0.7)
Effective dose rate	$9.3{\times}10^1$ (0.2)

According to Table 2, it is observed that during the ROLL procedure, higher equivalent dose rates are more pronounced in certain organs during the ROLL procedure. Notably, the breast, liver, lungs, esophagus, thyroid, and skin are among the most irradiated organs. The breast received the highest equivalent dose rate (540 µSv/h). Due to the breast's high radiosensitivity, this equivalent dose rate is particularly significant and underscoresthe importance of minimizing radiation exposure to reduce the risk of adverse effects. The liver, with an equivalent dose rate of 130 µSv/h, also ranks among the most irradiated organs. Due to the anatomical proximity of the region where the radiopharmaceutical is administered, the lung may receive a considerable dose of radiation. The lungs are exposed to an equivalent dose rate of 96 µSv/h, which make them one of the most exposed organs. Since the lungs are near the region of interest for ROLL in the breast, they also receive a significant dose. The esophagus, with a equivalent dose rate of 61 µSv/h, is also impacted by the procedure. Located near the breast, this organ is susceptible to radiation doses that may increase the risk of side effects, especially with repeated exposure. The thyroid and skin recorded equivalent dose rates of 16 µSv/h and 14 µSv/h, respectively. The thyroid is especially radiosensitive, so its exposure may pose long-term risks, such as an increased incidence of thyroid cancer. This justifies the need for additional protection whenever possible.

Finally, the overall effective dose rate for the patient is 93 µSv/h, reflecting the cumulative effects on various organs and tissues. These results highlight the importance of radiological protection measures for the patient, such as the potential use of dose-reduction techniques or protective barriers to minimize exposure to the most impacted organs, particularly in repetitive procedures.

3.3 Dose distribution in the physician's body

Figure 2 shows the absorbed dose distribution in the physician's body, as represented by the FASH3 anthropomorphic phantom, during a ROLL procedure in the following situations: (A) no apron, (B) with apron, and (C) with a lead shield.

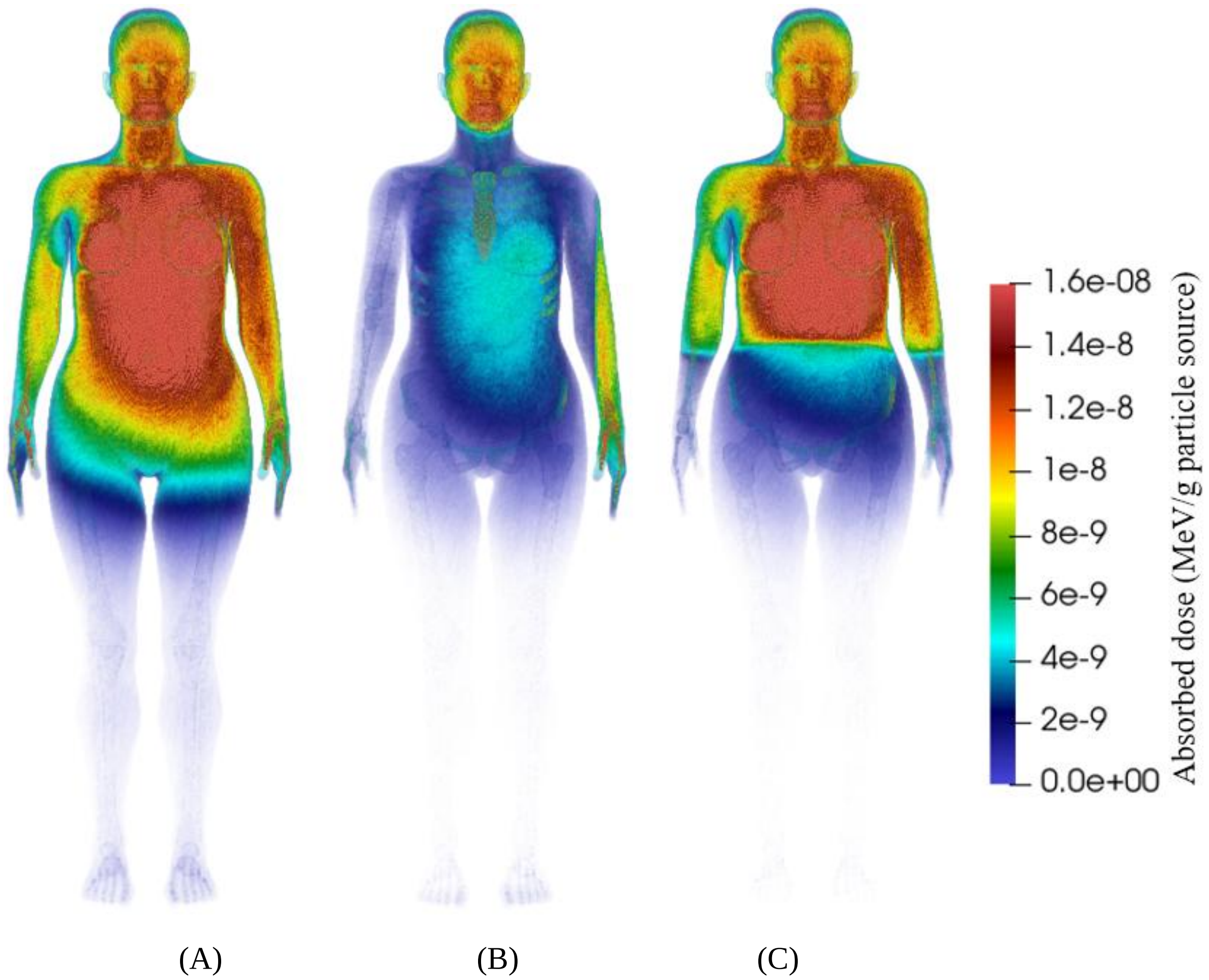


Figure 2: Distribution of absorbed dose in the physician's body.

Based on the information on the absorbed dose distribution in the physician's body, it is possible to highlight that in scenario (A), without a lead apron, there is a higher concentration of absorbed dose in the trunk region, especially in the chest area where the breast and nearby organs such as the liver and lungs are located. The absence of a lead apron results in greater exposure of these organs, leading to a more intense absorbed dose distribution in regions close to the radioisotope application site.

In scenario B, the use of apron significantly reduces the dose distribution in the thoracic region compared to scenario A. The apron considerably decreases patient exposure, particularly in the chest region, where radiosensitive organs such as breasts and liver are located. The dose reduction provided by the apron underscores its effectiveness in minimizing direct exposure to the region of interest and protecting internal organs.

Finally, in scenario C, with a lead shield, a reduction in dose is observed, though it is less pronounced in the chest region then with the apron. While the shield provides an additional protective barrier, it effectiveness is lower than that of the apron in directly protecting the patient's thoracic region. Nevetheless, it contributes to reducing overall exposure.

The combination of both protection, the apron and the lead shield, can be an effective strategy for minimizing a physician's exposure during procedures such as ROLL.

4. Conclusions

The results underscore the importance of using personal protective equipment. Lead aprons and shields to reduce the physician's occupational exposure during the ROLL procedure with the ^{99m}Tc radioisotope. The evaluation of equivalent dose rate values for different organs shows that, without protection, the physician is exposed to significant radiation levels, especially in radiosensitive organs like the eyes, breasts, and stomach. The use of the lead apron substantially reduces this exposure, particularly in the thoracic region, where crucial organs like the liver and breasts are located, decreasing the equivalent dose in these organs and, consequently, the risks of long-term adverse effects.

Additionally, the shield provides extra protection, albeit with less effective than the apron because its protection its is more generalized and less focused on specific regions. However, combining both the apron and the shield together is an effective strategy for minimizing the medical team's exposure. The combination creates a significant barrier against scattered radiation and enhances safety during ROLL procedures.

Regarding the patient's exposure, as expected and confirmed, the results show that organs near the radioisotope application site, such as breasts, liver, lungs, and esophagus, receive considerable radiation doses. The effective dose rate for the patient underscores the importance of adopting strategies that minimize exposure to radiosensitive organs, especially for repeated procedures on the same patients. In summary, its is essential to use lead aprons and shields is essential to ensure occupational safety and minimize the medical team's radiological exposure. Specific protection strategies for patients should also be considered to reduce risks associated with the procedure ROLL.

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Parametrization and Validation of a HPGe Detector by Monte Carlo Simulation

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Abstract

Accurate gamma-ray detector characterization is essential for reliable radiometric measurements, particularly in challenging environments like nuclear waste characterization, where complex geometries hinder direct experimental calibration. A detailed computational model for the FALCON 5000® High Purity Germanium (HPGe) detector has been established using the MCNP® Monte Carlo code. The objective was to create a precise model accurately representing the detector's physical response through incorporation of its detailed geometrical dimensions and material compositions. Validation involved Monte Carlo simulations predicting detector efficiency, specifically modeling the detection of the main Cobalt-60 (^{60}Co) photopeak emissions from a calibrated source across various source-detector geometries. These obtained efficiencies was compared with independently acquired experimental data, demonstrating agreement typically within 10%. This correlation confirms the accuracy and predictive capability of this parametrized computational model, providing a fundamental tool that advances the accurate prediction of detector performance and the interpretation of complex data for future applications of the spectrometry technique such as radioactive waste characterization.

***Keywords**: Gamma* Spectrometry; Detection Efficiency; Monte Carlo Simulation.

1 Introduction

The safe management of radioactive waste is one of the biggest challenges facing the nuclear industry today. An important step in this process is accurate radiological characterization, which requires the quantification of radionuclides present in large waste containment structures (Rahman et. al., 2024). Gamma spectrometry is the most suitable technique for this purpose, and High Purity Germanium (HPGe) detectors are the preferred choice due to their excellent energy resolution, allowing the identification and quantification of gamma emitters radionuclides in complex spectra (Abdelati et. al., 2018).

For field applications, such as in situ measurement of large waste structures, portable HPGe spectrometers, like the FALCON 5000®, offer advantages in the spectrometry technique. Their portability mainly due to its electric cooling system eliminate the need for cooling gases and liquids, facilitating operations in remote or hard-to-reach locations. However, the accuracy of quantitative analyses depends entirely on an intrinsic computational calibration of the equipment (Hafızoğlu, 2024). The experimental calibration procedure faces difficulties such as replicating complex measurement geometries (heterogeneous radionuclide matrices, non-uniform source distributions in complex structure geometries).

Monte Carlo simulation techniques emerge as a good tool for overcoming such obstacles. It consists of representing particle interactions through functions that represents the physics of the problem in a computer code. A validated computational model of the detector allows for the calculation of detection efficiency for any sample geometry (Thakur et. al., 2023). However, a common challenge in implementing these models is obtaining the boundary conditions necessary for modeling the internal components of the detector equipment, which often occurs due to inaccuracies in the detector parameters provided by the manufacturer. Initial results with nominal values for crystal dimensions, dead layer thickness and window thickness can lead to significant deviations, often exceeding 30% (Luís et. al., 2010).

The objective of this work is to establish a new and detailed computational model for the HPGe FALCON 5000® detector using the MCNP®6.2 Monte Carlo code, aiming to represent the detection efficiency of the equipment, which was achieved by evaluating its responses to variations in the previously mentioned geometric dimension parameters. The model's validation was performed by comparing the detection efficiencies predicted by the simulation, using a ^{60}Co point source, with experimentally obtained data, confirming the accuracy and predictive capability of the model for current and future applications.

2. Materials and Methods

2.1. Monte Carlo Modeling

Simulations were performed using the MCNP®6.2 code to model the HPGe Falcon 5000® detector. Figure 1 shows the initial geometric model, which was built based on the detector's nominal parameters provided by the manufacturer (Mirion Technologies (Canberra), Inc. (2017)). It consists of a cylindrical germanium crystal with a 30.4 mm radius and 30.9 mm length, represented by the green structure, and enclosed in a 2.0 mm thick aluminum casing with a 1.2 mm thick window, represented by the red structure in Figure 1.

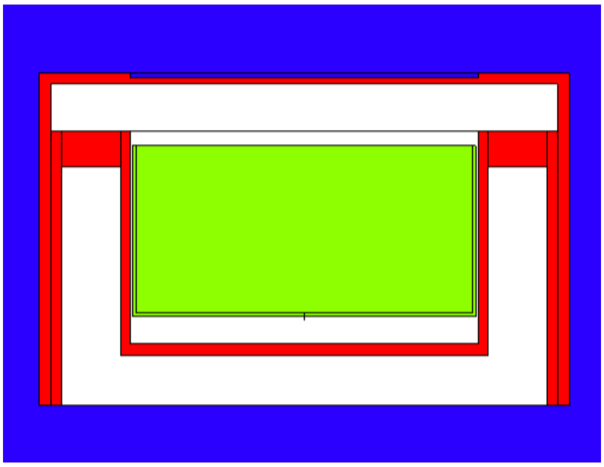

Figure 1 *- HPGe detector Falcon 5000® simulated geometry.*

An isotropic Cobalt-60 point source, with its two main gamma emissions (1173.2 keV and 1332.5 keV), was modelled at 30 cm distance from the detector window. The F8 Tally Card, specifically used to determine the pulse count by each energy bean within the detector volume, measured the absolute peak detection efficiency (C. J. Werner et.al., 2017).

For the detection efficiency analysis, each parameter was varied individually while keeping the others at their nominal values. The following variations were investigated:

Window Thickness: 1.2 mm (nominal), 1.4 mm, 1.6 mm, 1.8 mm and 2.0 mm.

Crystal Radius: 30.4 mm (nominal), 29.4 mm and 28.4 mm.

Crystal Length: 30.9 mm (nominal), 29.9 mm and 28.9 mm.

Front Dead Layer Thickness: 0.3 µm (nominal), 3 µm, 10 µm, 20 µm, 30 µm and 300 µm.

Lateral Dead Layer Thickness: 0.5 mm (nominal), 0.7 mm, 0.9 mm, 1.1 mm and 1.3 mm.

Back Dead Layer Thickness: 0.7 mm (nominal), 1.4 mm, 2.1 mm and 2.8 mm.

2.2. Experimental Validation

To validate the final computational model, experimental measurements were carried out using the Falcon 5000® detector and a ^{60}Co calibration source at the Instrument Calibration Center in IPEN. Measurements were taken with the source positioned at four distinct distances from the detector window: 1.5 m, 2.0 m, 3.0 m and 3.5 m. The acquired spectra was analyzed using InterSpec® software to determine the counting rate for each emission energy interval.

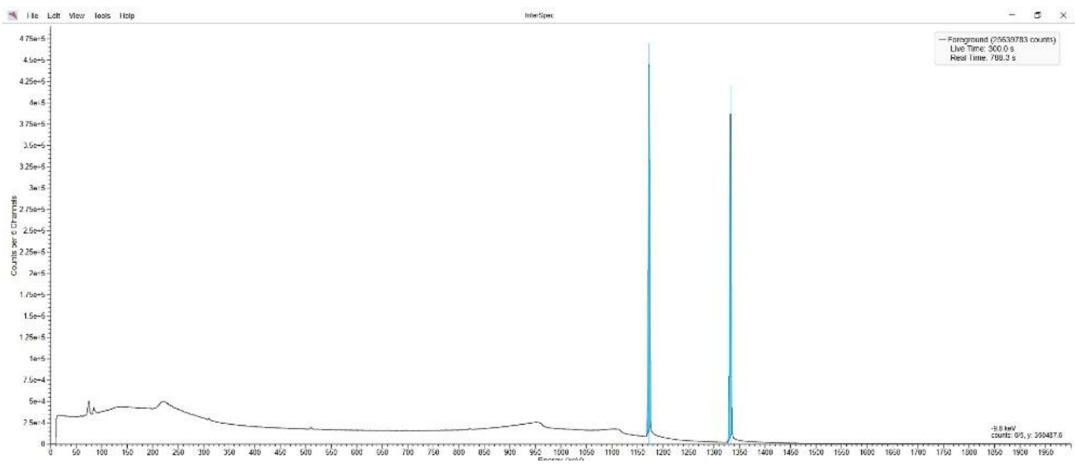

Figure 2 *- Energy spectrum pattern of ^{60}Co source measured at 200 cm distance from the detector.*

The experimental results followed the pattern shown in Figure 2 and were then compared with the results obtained from simulations that utilized both the nominal and optimized detector parameter sets.

3 Results and Discussion

The results show the influence of each parameter on the detection efficiency and the validation of the nominal model with the optimized geometry.

3.1. Parameters Analysis

The first step of the study involved establishing simulations with the nominal parameters provided by the manufacturer as a reference for detection efficiency, and subsequently analyze the variation of each parameter in the detector response.

3.1.1. Influence of Detector Window Thickness

The aluminum window at the front of the detector acts as an attenuator for incident photons. The effect of its thickness on detection efficiency was evaluated by varying it from 1.2 mm to 2.0 mm. For the high energy emission of ^{60}Co decay, the impact of this variation was minimal. The 2,0 mm window thickness case led to an increase in the population of lower-energy photons in the detector crystal, result of the interaction higher-energy photons in the window (Fig. 3) and their counts in the detector crystal.

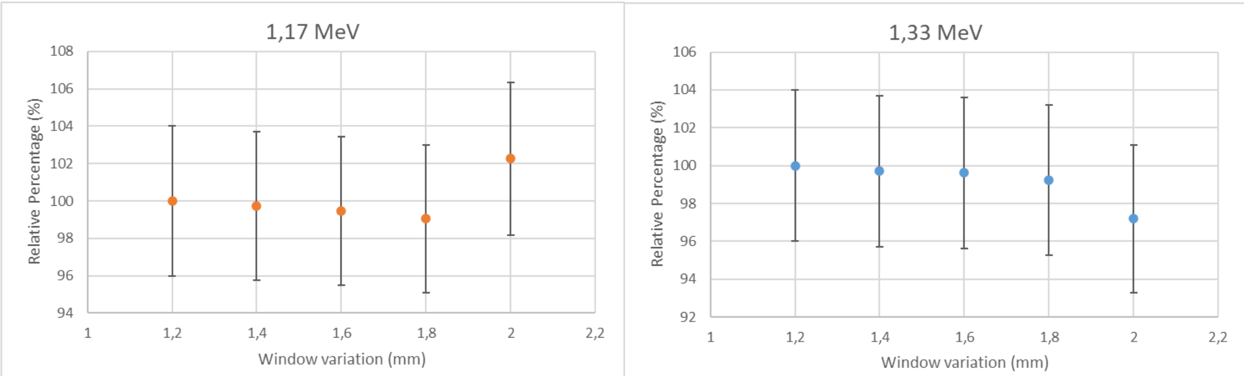

Figure 3 *- Effect of window thickness variation on the detection efficiency.* ***Left****: Low energy emission from* ^{60}Co *decay.* ***Right****: High energy emission from* ^{60}Co *decay.*

3.1.2. Influence of Crystal Dimensions (Radius and Length)

Varying the crystal radius (Fig. 4) and length (Fig. 5) affected the detector's efficiency for the energy interval analyzed. A reduction in both parameters leads to a decrease in the detector's sensitive volume, which impacts the photons emitted by ^{60}Co source. This happens because the mean free path of these photons in germanium is long, and a smaller crystal offers less interactions, thereby reducing the probability of an event in the energy ranges studied.

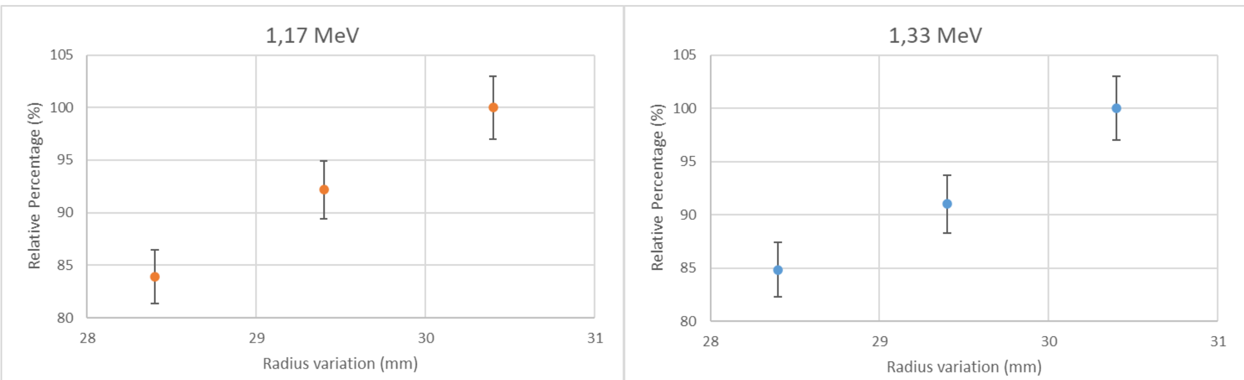

Figure 4 *– Effect of the detector crystal radius variation on the detection efficiency.* ***Left****: Low energy emission from* ^{60}Co *decay.* ***Right****: High energy emission from* ^{60}Co *decay.*

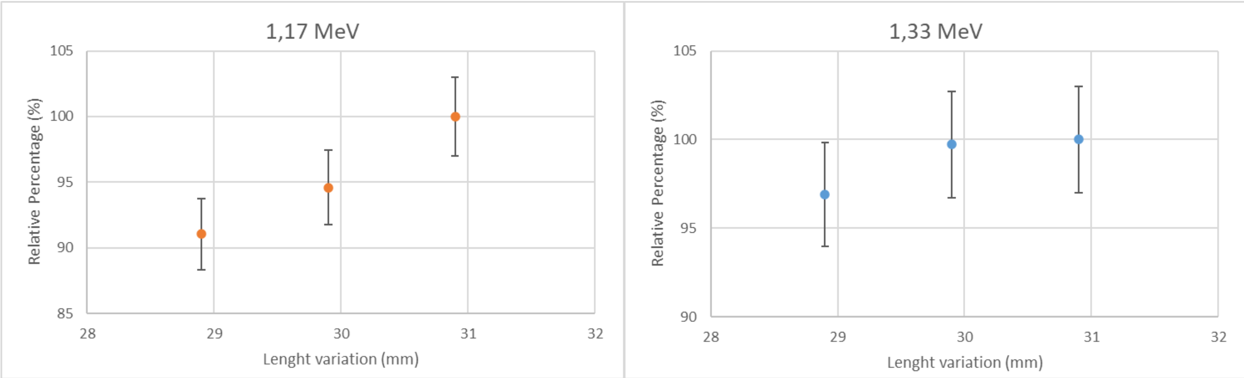

Figure 5 – *Effect of the detector crystal length on the detection efficiency.* ***Left****: Low energy emission from ^{60}Co decay.* ***Right****: High energy emission from ^{60}Co decay.*

3.1.3. Influence of Dead Layers

Similar to the window thickness, the variation impact of a HPGe dead layer on efficiency for ^{60}Co photopeaks was inconclusive. Figure 6 shows the variation in dead layer thickness altered the efficiency only in the thousand-fold variatiofwn (0.3 mm), the others parameters values maintained variations of less than 1%. Dead layers represent boundary regions of the detector crystal where generated charges may not be correctly accounted for. Therefore, their adjustment is relevant for calibration, but it is not a critical parameter for optimizing the detector for high-energy sources like ^{60}Co. The side and back dead layers, although distinct parameters, have a similar effect to altering the crystal dimensions for efficiency calculation.

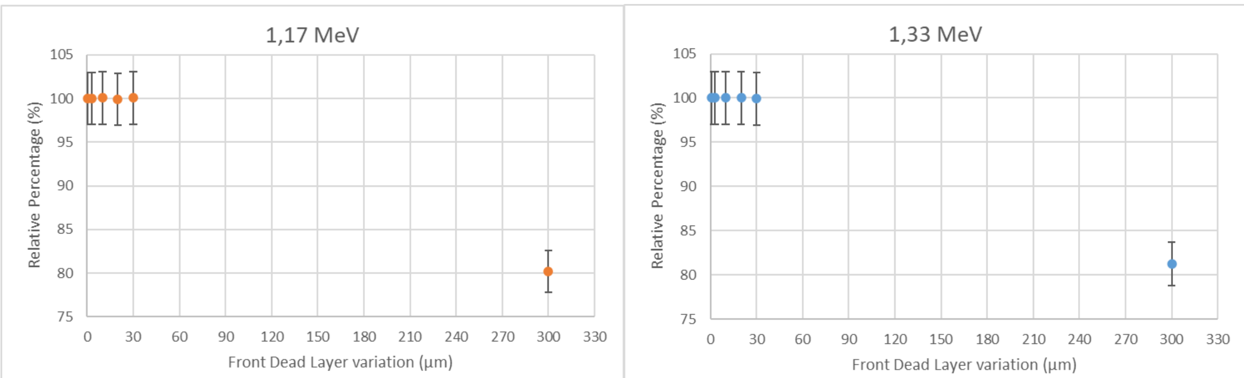

Figure 6 *- Analysis of the front dead layer thickness.* ***Left****: Low energy emission from ^{60}Co decay.* ***Right****: High energy emission from ^{60}Co decay.*

3.2 Model Optimization

Table 1 presents the combination of parameters that yielded the best results for the model optimization step. To achieve the maximum correspondence between simulated efficiencies and experimental measurements, the side dead layer was increased by 0.7 mm, and the back dead layer was increased by 2.1 mm. The front dead layer thickness was adjusted to 0.3 mm, and the window thickness was increased by 0.2 mm. It is worth noting that, for the analysis of the efficiency of each energy peak, increasing the side and back dead layers is equivalent to a decrease in the crystal dimensions (radius and length). An alternative approach would be, for example, to assume a crystal length smaller than the nominal value and apply a smaller or no increase to the back dead layer. However, for the detector model, it was decided to modify the dead layers instead of the crystal dimensions, as it is recognized that uncertainties in dead layer dimensions are greater than uncertainties in crystal dimensions.

Table 1 *- Detector model parameters optimization.*

Parameters	Nominal value	Optimized value
Window thickness (mm)	1,2	1,4
Crystal radius (mm)	30,4	30,4
Crystal length (mm)	30,9	30,9
Front dead layer thickness (µm)	0,3	300
Lateral dead layer thickness (mm)	0,5	1,3
Back dead layer thickness (mm)	0,7	2,8

3.3. Model Validation

Table 2 was generated by comparing the efficiencies derived from simulations, which utilized nominal parameter values, with those obtained from laboratory experiments. The efficiency was calculated according to the following equation:

$$\boldsymbol{Efc = \frac{p.MC}{\left(\frac{Exp}{t.A}\right)}}$$

Where:

Efc: Detection efficiency comparison;

p: Emission probability;

MC: Counts obtained from the Monte Carlo simulation;

Exp: Photopeak counts (Experimental data);

t: Detector live-time;

A: Source activity.

Table 2 – Comparison between Simulation and Experimental with nominal parameters.

SDD (cm)	Energy	
	1,17 MeV	1,33 MeV
150	1,47 (2)	1,28 (2)
200	1,38 (2)	1,28 (3)
300	1,33 (2)	1,34 (2)
350	1,35 (2)	1,35 (2)

Table 3 presents the comparison between the simulations and the experimental results, utilizing the optimized parameters cited in Table 1.

Table 3 - Comparison between simulation and Experimental with optimized parameters

SDD (cm)	Energy	
	1,17 MeV	1,33 MeV
150	1,23 (2)	1,06 (2)
200	1,14 (2)	1,06 (2)
300	1,12 (2)	1,10 (2)
350	1,13 (2)	1,11 (2)

The comparison of the values in Table 2 reveals a significant difference (approximately 30%) between the simulated and experimental results. In Table 3, however, there was a reduction in this discrepancy to approximately 10%. This further indicates that the optimized parameters lead to a better convergence with the experimental data.

4 Conclusions

A sensitivity analysis was performed for the geometric parameters of an HPGe Falcon 5000® detector using Monte Carlo simulations. The study demonstrated that, for the high energy emissions of ^{60}Co decay, the detection efficiency is influenced by the active dimensions of the crystal (radius and length, subtracting the dead layers), while the other parameter studied, the detector window thickness, has inconclusive results in the analysis of detection efficiency.

It was shown that adjusting the thicknesses of the side and back dead layers is an effective method for optimizing the active volume of the crystal in the model, leading to better agreement between simulations and experimental measurements. The resulting optimized model was successfully validated against experimental measurements at a variety of distances proving its statistical concordance.

It is concluded that the optimized MCNP model is a tool that can be used to validate detection efficiency in complex geometries, contributing to the interpretation and analysis of experimentally obtained results. Thus, it is possible to use the optimized model for the inverse problem in the characterization of radioactive waste in nuclear power plant equipment.

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ASSESSING PHOTON AND ELECTRON RANGES IN HEALTHCARE WASTE FOR RADIATION STERILIZATION

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Abstract

The use of ionizing radiation as an alternative method for the pathogenic inactivation of Healthcare Waste (HCW) has shown promise compared to conventional methods such as incineration, which are associated with significant environmental impacts. The technique of pathogenic inactivation by ionizing radiation is still under investigation, and preliminary results have shown promising potential for effective and environmentally safe treatment of healthcare waste. This study aims to determine the penetration ranges of photons and electrons in a representative HCW medium to identify the most suitable energies for maximizing microbiological inactivation efficiency through radiation. A mean density of 0.15 g/cm³ was adopted based on the findings of Neves (NEVES,2015), who analyzed HCW generated in hospitals in Belo Horizonte, Brazil. The chemical composition and elemental weight fractions of the HCW were based on data provided by Silva (SILVA et. al 2024), which analyzed typical healthcare waste materials for use in radiation simulation models. For photons, mass attenuation coefficients were obtained using the XCOM software developed by the National Institute of Standards and Technology (NIST), covering photon energies from 0.01 MeV to 20 MeV (NIST,XCOM 2024). For electrons, Mass Stopping Power and Continuous Slowing Down Approximation (CSDA) data were extracted from the ESTAR database, also from NIST (NIST,STAR 2024). The results of the mass attenuation coefficient, Mass Stopping Power and CSDA determined by XCOM and ESTAR, respectively, were used to infer the ranges of photons and electrons in HCW. The results allowed to estimate the mean range of the photons ($\overline{R_\gamma}$) between 40.95 cm (0.1 MeV) and 374.32 cm (20 MeV), i.e. an increase of 814 % over the energy range and the Practical Mean Range for electrons ($\overline{R_e}$) was between 0.06 cm (0.1 MeV) and 36.93 cm (20 MeV), a 62.75% increase in range in the HCW offering guidance for the selection of optimal energy levels that ensure adequate radiation penetration for complete pathogenic inactivation. The analysis of the results showed that the energy of 10 MeV for photons is quite appropriate for large volumes of HCW and the same energy of 10 MeV of electrons is suitable for irradiation of small volumes of HCW.The results reinforce the importance of energy optimization in the design of radiation-based HCW treatment systems to determine the appropriate energy to achieve adequate penetration according to the volume of HCW to be irradiated.

Keywords: Healthcare Waste Composition, Pathogenic Inactivation, Photon Range, Electron Range, XCOM, ESTAR.

Introduction

Infectious healthcare waste (HCW) contains a wide range of pathogenic microorganisms that must be effectively inactivated prior to disposal to prevent risks to public health and the environment. One promising approach is the use of ionizing radiation, which can effectively inactivate pathogens provided that the minimum sterilizing dose is delivered homogeneously throughout the volume of the waste package, as established by Tata ((TATA, 1995).

The development of radiation-based treatment methods for HCW represents a potential alternative to conventional techniques, such as incineration, which often lead to significant environmental emissions and secondary pollutants.

An important technical factor for the success of such radiation-based treatment systems is the choice of beam energy, which must be aligned with the penetration range of photons or electrons into the health waste material. This range is influenced by chemical composition and physical properties – especially density and the proportion of each element in the irradiated medium.

Thus, the objective of this study is to determine the photon and electron ranges in a representative medium of HCW using the mass attenuation coefficient, mass stopping power data and CSDA obtained from the XCOM and ESTAR software developed by NIST. By correlating the radiation range with the expected depth of waste to be irradiated, this work aims to provide technical guidance for selecting optimal beam energies that ensure homogeneous dose distribution, maximize microbiological inactivation, and minimize energy inefficiency or unnecessary exposure of healthcare waste.

Methodology

To estimate the medium penetration ranges of photons and electrons in healthcare waste (HCW), a representative medium was modeled based on its average elemental composition and density. These physical parameters are essential for calculating both the mass attenuation coefficients for photons and the mass stopping power for electrons, which determine how radiation propagates and is absorbed within the material.

Determination of Chemical Composition and Density of the HCW

The elemental composition and stoichiometric weight fractions of HCW were derived from the study by Silva (SILVA et al,2024), which analyzed typical materials found in infectious HCW. These data provided the elemental basis needed to perform photon and electron interaction simulations.

Table 1- Stoichiometric chemical proportion of the elements in the Total Waste Mixture (A1+A4+B+D).

Chemical element	Proportion of Element in HCW Mixture (A1+A4+B+D)
Carbon – C	52,526 %
Oxygen – O	25,349 %
Chlorine – Cl	9,851 %
Hydrogen – H	6,770 %
Sodium - Na	1,395 %
Calcium – Ca	0,528 %
Sulphur - S	0,400 %
Silicon - Si	0,155 %

Nitrogen - N	0,086 %
Fluorine - F	0,053 %
Iron - Fe	0,042 %
Boro – B	0,018 %
Potassium – K	0,018 %
Chromium - Cr	0,011 %
Aluminium – Al	0,006 %
Nickel - Ni	0,001 %
Titanium – Ti	0,001 %
Copper - Cu	0,001 %
Total Characterized	97,211 %
Miscellaneous elements in Unknown proportions	2,789 %

The mean density of the modeled HCW was set to 0.15 g/cm³, based on the results reported by Neves (NEVES,2015) This low density reflects the predominantly organic and porous nature of the healthcare waste.

Mean Range Estimation for Photon ($\overline{R_\gamma}$) by Interaction Modeling (XCOM)

Photon attenuation was calculated using the XCOM: Photon Cross Sections Database. XCOM was used to obtain mass attenuation coefficients (μ/ρ) across a broad range of photon energies of 0.1 to 20 MeV. These coefficients were combined with the density value to calculate the linear attenuation coefficients, and ultimately estimate radiation penetration depth (range) in the HCW medium.

The photon range was estimated from the linear attenuation coefficient (μ), calculated as:

$$\mu = \left(\frac{\mu}{\rho}\right) \times \rho \tag{1}$$

Where:

μ = linear attenuation coefficient (cm^{-1})

μ/ρ = mass attenuation coefficient (cm^2/g)

ρ = material density (g/cm^3)

Photons undergo exponential attenuation, and their mean range is intrinsically linked to the mass attenuation coefficient (μ/ρ). The intensity of a photon beam after traversing a thickness x is given by (Hubbell & Seltzer, 1995):

$$I = I_0 \times e^{-\mu . x} \tag{2}$$

The mean free path (λ) is defined as the inverse of the linear attenuation coefficient (μ) (Attix, 1986):

$$\lambda = \frac{1}{\mu} = \frac{1}{\left(\frac{\mu}{\rho}\right) \times \rho} \tag{3}$$

The mean range of photons ($\overline{R_\gamma}$) can be approximated as the mean free path, considering that most energy is deposited after multiple interactions:

$$(\overline{\boldsymbol{R_\gamma}}) \approx \boldsymbol{\lambda} = \frac{\mathbf{1}}{\left(\frac{\boldsymbol{\mu}}{\boldsymbol{\rho}}\right)\times\boldsymbol{\rho}} \tag{4}$$

Mean Range Estimation for Electron ($\overline{\boldsymbol{R_e}}$) by Interaction Modeling (ESTAR)

The CSDA range represents the total average distance traveled by an electron from the point of injection into a material until it loses all of its kinetic energy. It assumes a continuous and smooth energy loss, ignoring multiple scattering or stochastic fluctuations.

The CSDA range is defined as the integral of the inverse of the mass stopping power (S/ρ)(E) which describes the energy loss per unit path length and per unit mass density (ICRU,37):

$$CSDA = \int_0^{E_0} \frac{1}{(\frac{S}{\rho})(E)} dE \tag{5}$$

The ESTAR program (Electron STopping power And Range tables), developed by the NIST, performs this calculation. Its input requires:

The chemical composition of the material, specified by stoichiometric formula and mass fractions;

The density of the medium in g/cm3.

Based on these inputs, ESTAR uses cross-section data and theoretical models to compute:

1. The total mass stopping power (S/ρ) as a function of energy;
2. The CSDA range, via numerical integration over tabulated (S/ρ) values up to the incident energy E_0
3. The physical range in centimeters, derived from the CSDA range divided by the material density:

$$R_{CSDA} = \frac{CSDA}{\rho}(cm) \tag{6}$$

This method provides a reliable and widely used estimation for electron transport in homogeneous media.

Due to multiple scattering, the actual range is shorter than R_{CSDA}. A common empirical correlation is the Practical Mean (Andreo et al., 2017):

$$\overline{R_e} \approx 0.6 \times R_{CSDA} \tag{7}$$

The Practical Mean Range is an empirical correction to the CSDA range (Continuous Slowing-Down Approximation) to account for electron multiple scattering in materials. It represents the average depth where most electrons deposit their energy before complete absorption.

Results

Range Estimation for Photon beams

Using the chemical elements and proportions of the HCW in Table (1), the density of 0.15 g/cm3 the XCOM provided the mass attenuation coefficients (μ/ρ) and through Equation (4) we were able to estimate the mean range of the photons for energies ranging from 0.1 to 20 MeV Table (2).

Table(2) - Mean Range Estimation for Photon ($\overline{R_\gamma}$) for beam energy of 0,1 to 20 MeV in HCW.

Photon Energy (MeV)	HCW Mass Attenuation Coefficient - μ/ρ (cm^2/g)	Mean Range Estimation for Photon - $\overline{R_\gamma}$ (cm)
1.000E-01	1.628E-01	40.95
1.500E-01	1.430E-01	46.62
2.000E-01	1.304E-01	51.12
3.000E-01	1.132E-01	58.89
4.000E-01	1.014E-01	65.75
5.000E-01	9.259E-02	72.00
6.000E-01	8.565E-02	77.84
8.000E-01	7.524E-02	88.61
1.000E+00	6.767E-02	98.52
1.022E+00	6.696E-02	99.56
1.250E+00	6.053E-02	110.14
1.500E+00	5.510E-02	120.99
2.000E+00	4.733E-02	140.85
2.044E+00	4.678E-02	142.51
3.000E+00	3.808E-02	175.07
4.000E+00	3.270E-02	203.87
5.000E+00	2.919E-02	228.39
6.000E+00	2.672E-02	249.50
7.000E+00	2.490E-02	267.74
8.000E+00	2.351E-02	283.57
9.000E+00	2.243E-02	297.22
1.000E+01	2.155E-02	309.36
1.100E+01	2.084E-02	319.90
1.200E+01	2.025E-02	329.22
1.300E+01	1.975E-02	337.55
1.400E+01	1.933E-02	344.89
1.500E+01	1.898E-02	351.25
1.600E+01	1.867E-02	357.08
1.800E+01	1.818E-02	366.70
2.000E+01	1.781E-02	374.32

Range Estimation for Electrons beam

Using the chemical elements and proportions of the HCW in Table (1), the density of 0.15 g/cm3 the ESTAR provided the mass stopping power (S/ρ) and through Equations(6), (7) we were able to estimate the mean range of the electrons for energies ranging from 0.1 to 20 MeV Table (3).

This value is not exact as in the case of photons, as it depends on several factors such as:

1. Lateral scattering (multiple collisions);
2. Angular distribution of emerging electrons;
3. Initial beam energy;
4. Irradiation geometry.

Table (3) shows the Mass Stopping Power and CSDA of healthcare waste for energies ranging from 0.1 to 20 MeVand the estimation of Practical Mean Range for electrons ($\overline{R_e}$).

Table(3) Practical Mean Range for electrons ($\overline{R}_e$) for HCW as a function of the energy of the electrons beam.

Electron Energy (MeV)	Mass Stopping Power form HCW (MeV cm^2/g)	CSDA (g/cm^2)	$\overline{R_e} \approx 0.6 \times R_{CSDA}$ (cm)
1.000E-01	3.926E+00	1.502E-02	0.06
1.250E-01	3.428E+00	2.186E-02	0.09
1.500E-01	3.091E+00	2.956E-02	0.12
1.750E-01	2.849E+00	3.800E-02	0.15
2.000E-01	2.668E+00	4.708E-02	0.19
2.500E-01	2.416E+00	6.684E-02	0.27
3.000E-01	2.251E+00	8.833E-02	0.35
3.500E-01	2.137E+00	1.112E-01	0.44
4.000E-01	2.055E+00	1.350E-01	0.54
4.500E-01	1.994E+00	1.598E-01	0.64
5.000E-01	1.948E+00	1.851E-01	0.74
5.500E-01	1.912E+00	2.111E-01	0.84
6.000E-01	1.884E+00	2.374E-01	0.95
7.000E-01	1.846E+00	2.911E-01	1.16
8.000E-01	1.822E+00	3.456E-01	1.38
9.000E-01	1.808E+00	4.007E-01	1.60
1.000E+00	1.800E+00	4.562E-01	1.82
1.250E+00	1.796E+00	5.954E-01	2.38
1.500E+00	1.805E+00	7.342E-01	2.94
1.750E+00	1.820E+00	8.721E-01	3.49
2.000E+00	1.836E+00	1.009E+00	4.04
2.500E+00	1.866E+00	1.279E+00	5.12
3.000E+00	1.894E+00	1.545E+00	6.18
3.500E+00	1.921E+00	1.807E+00	7.23
4.000E+00	1.946E+00	2.066E+00	8.26
4.500E+00	1.970E+00	2.321E+00	9.28
5.000E+00	1.993E+00	2.573E+00	10.29
5.500E+00	2.015E+00	2.823E+00	11.29
6.000E+00	2.036E+00	3.070E+00	12.28
7.000E+00	2.076E+00	3.556E+00	14.22
8.000E+00	2.113E+00	4.034E+00	16.14
9.000E+00	2.150E+00	4.503E+00	18.01
1.000E+01	2.185E+00	4.964E+00	19.86
1.250E+01	2.269E+00	6.087E+00	24.35
1.500E+01	2.349E+00	7.170E+00	28.68
1.750E+01	2.426E+00	8.217E+00	32.87
2.000E+01	2.501E+00	9.232E+00	36,93

Discussion

Comparative Analysis of Photon and Electron Range in HCW: Optimal Energy Selection for Different Volume Applications

The main results of the analysis of the data in Table (2) and (3) reveal significant differences in the use of photons and electrons at different energies.

For photons the mean range $\overline{R_\gamma}$ it was between 40.95 cm (0.1 MeV) and 374.32 cm (20 MeV), an increase of 814% over the energy range.

The conversion of electrons into photons in linear accelerators has an efficiency of only 16%, drastically reducing the useful dose rate in HCW. This factor, together with the choice of energy for adequate penetration, is fundamental for optimizing the HCW treatment process by ionizing radiation.

Based on this, we can see that for large volumes, the use of photons is recommended, due to their greater penetration, compensating for their low energy efficiency. Energy of 10 MeV photons with a range of about 309.4 cm thick allows to treat thicker volumes (e.g. HCW container), in these cases despite the loss of 84% of the conversion energy, depth is indispensable.

For electrons, the Practical Mean Range ($\overline{R_e}$) was between 0.06 cm (0.1 MeV) and 36.93 cm (20 MeV), an increase of 62.75% in range, but with absolute values much lower than photons.

In the case of small volumes, the electron beam is recommended, as it showed greater dose efficiency at depths ≤40. With a beam of 10 MeV, the range of electrons is about 19.9 cm, which covers typical HCW bads (15 – 24) cm thick. The great advantage of using electrons is that there is no loss in beam conversion, that is, 100% of the beam energy falls on the medium, greatly increasing the dose rate in the HCW.

Conclusion

The physicochemical characterization of healthcare waste (HCW) — including its density and elemental composition — proved to be a crucial step in accurately estimating the mean range of photons and practical range of electrons, using data from the XCOM and ESTAR databases provided by NIST. The chemical composition dominated by carbon, oxygen, chlorine, hydrogen and sodium, combined with the low material density (0.15 g/cm^3), directly affects attenuation coefficients and stopping powers, thus influencing the optimal energy selection for irradiation processes.

The comparative analysis revealed that photons exhibit significantly greater mean penetration depths than electrons, with an increase of up to 814% across the 0.1 to 20 MeV range. Conversely, electrons present much shorter practical ranges but offer superior dose delivery efficiency, as their energy is fully deposited in the medium with no losses from conversion.

Based on the data evaluated, it is concluded that an energy level of 10 MeV is optimal for both photons and electrons:

For small to moderate volumes (up to 30 cm thick), a 10 MeV electron beam is recommended, achieving a practical range of approximately 19.9 cm and a high dose rate, ideal for typical compacted HCW bags.

For large volumes (thickness ≥ 2 m), 10 MeV photons are more suitable, with an average penetration of approximately 309.4 cm. Despite the low conversion efficiency (~16%), the achieved depth justifies their use in applications such as containerized HCW treatment.

Choosing the correct energy based on the physical characteristics of the waste in conjunction with an appropriate dose rate enables effective microbiological inactivation while maintaining operational safety and technical feasibility, reinforcing the potential of radiation-based technologies as a sustainable alternative to conventional incineration.

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SMS Notification System for Potential Radiological Risk Scenarios

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Abstract

Nuclear Engineering, in general, deals with critical systems where operational safety is a fundamental requirement to ensure the secure use of its technology. In this context, multiple detection and monitoring mechanisms are implemented with the aim of preventing radiological risk scenarios [1]. This work proposes the development of a Control System capable of integrating ionizing radiation detectors with an SMS messaging API. The system's architecture allows a predefined dose rate threshold (setpoint) to be configured, such that, when exceeded, an automatic notification is sent to a group of previously registered users [2]. The solution was designed with a focus on flexibility and multipurpose application, making it suitable for use in a variety of contexts, such as nuclear power plants, waste storage facilities, specialized laboratories, and nuclear research centers. Furthermore, the system has the potential to serve as a communication link between responsible authorities and nearby communities. Expected outcomes include enhanced initial response capability, reduced exposure to ionizing radiation, and improved communication channels during critical situations. The system can also be integrated into broader emergency management platforms, contributing to coordinated radiological protection strategies.

Keywords: Control System; Radiation Detection; SMS Notification;

1 Introduction

Nuclear Engineering, in general, deals with critical systems in which operational safety is a fundamental requirement to ensure the safe use of its technology. In this context, multiple detection and monitoring mechanisms are implemented with the aim of preventing radiological risk situations [1,8].

However, even with these safety mechanisms, incidents involving nuclear materials and their by-products can occur. These events may be caused either by some type of threat or by unexpected occurrences characterized by the loss of control over radiation sources. This can directly or indirectly pose a danger to human life, health, or property [5].

Regarding their scope, in certain incidents, these hazards may have a limited reach, meaning they pose a threat only to individuals working directly with radiation sources or with the equipment containing them, as well as to their immediate working environment. However, under certain conditions, as a consequence of large-scale incidents, the local population and critical infrastructure may also be affected.

This work proposes the development of a Control System capable of integrating ionizing radiation detectors with an SMS messaging API, with the purpose of providing rapid and broad notifications. The system can serve as a tool to inform a group of technical users, alert the population of potential risks, and coordinate incident response teams. The proposal was designed with a focus on flexibility and multipurpose application, making it suitable for use in various contexts, such as nuclear power plants, waste storage facilities, specialized laboratories, and nuclear research centers. Furthermore, the system has the potential to serve as a communication link between the responsible authorities and nearby communities.

The system's architecture consists of a radiation detector connected to an electronic controller. This setup allows for the configuration of a predefined dose rate value as a trigger point (setpoint), so that, once exceeded, an automatic notification is sent to a group of previously registered users through an SMS system.

2 Materials and Methods

2.1 Materials

For this experiment, the embedded system consisted of a Raspberry Pi 3B controller and a BG51 radiation detector, as illustrated in Figure 1.

The choice of the Raspberry Pi is justified by its convenience as a controller: it has power and communication inputs that allow for the straightforward integration of the BG51 detector via GPIO (General Purpose Input and Output) pins, in addition to a built-in WiFi module, which enables the implementation of SMS messaging through a web application using Twilio.

As for the selected detector, its choice is supported by its low power consumption (25µA), making it ideal for applications in radiation-harsh environments, requiring little to no maintenance over extended periods of operation. Furthermore, the detector can detect gamma and beta radiation, perform energy discrimination, and offers a wide operational range between 70 keV and 2 MeV, with a dose rate measurement range spanning from 0.1 µSv/h to 100 mSv/h.

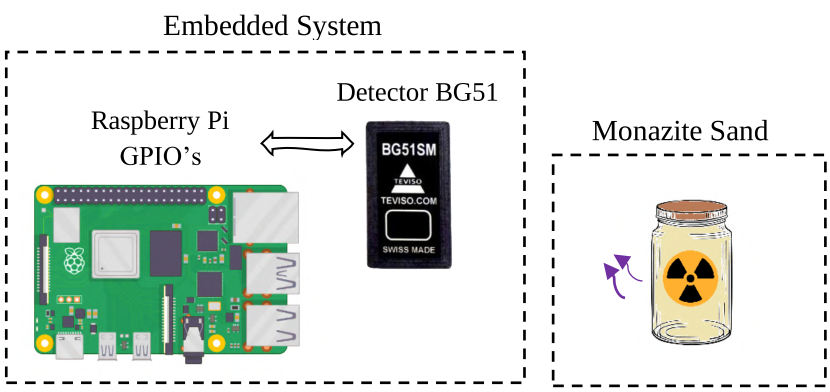

Figure 1 - Diagram of the Experiment

Source: The Authors (Own).

To test and validate the system, a sample of monazite sand — a naturally occurring radioactive material (NORM) — was used. This type of sand contains a concentration of thorium sufficient to trigger the detector and generate counts related to its dose rate. These dose rate values were used to define the radiation thresholds that would trigger SMS notifications.

2.2 Nuclear Instrumentation

Based on the graph in Figure 2 and the information from the BG51 datasheet, the detector's conversion factor was identified, establishing a relationship of 5 CPM for each 1 µSv/h of dose rate. To apply this factor, the dose calculation methodology adopted was based on the number of events per minute (CPM) recorded by the system [3].

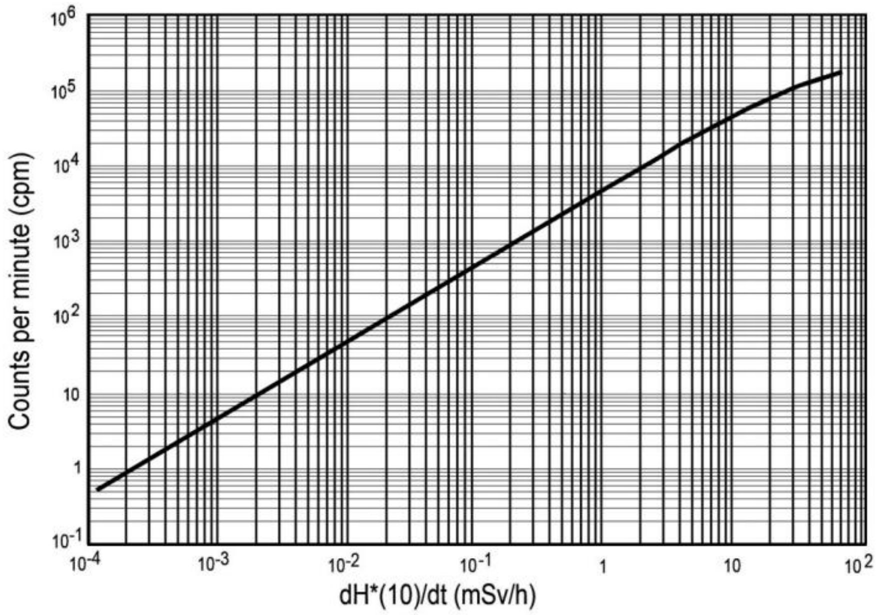

Figure 2 - Counts vs. Dose Rate for the BG51 Detector

Source: BG51 datasheet, Televiso [3].

To ensure the reliability of the proposed detection system, a comparison of dose rate values was carried out using, under the same conditions, the PRD/PRD-ER RadEye detector by Thermo Scientific, a NaI(Tl) scintillator. Figure 3 illustrates the conversion factor of counts as a function of the dose rate for the RadEye detector.

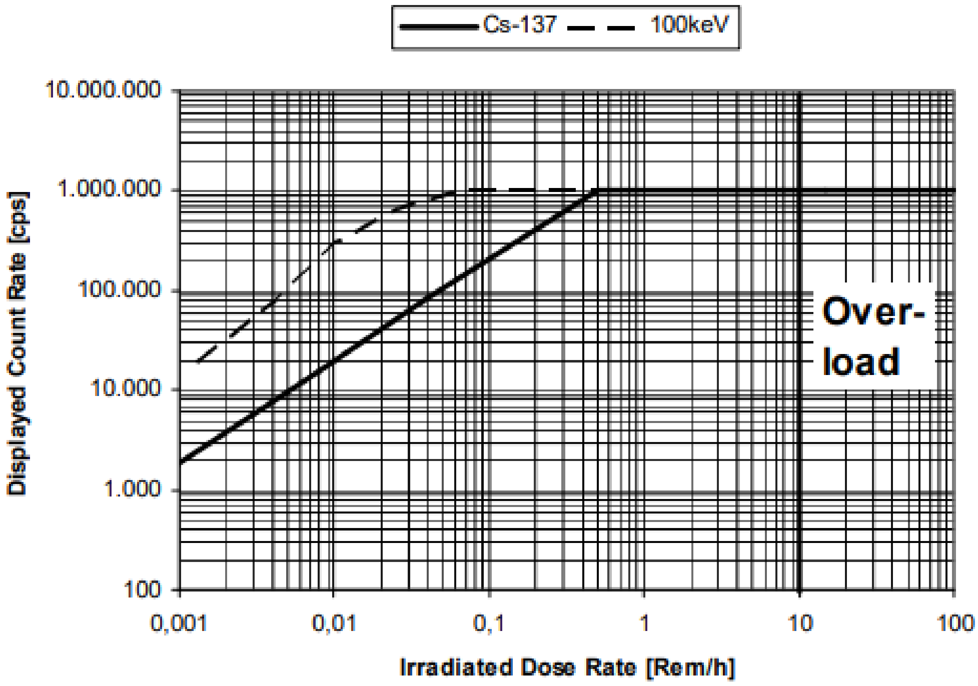

Figure 3 - Counts vs. Dose Rate for the RadEye Detector

Source: RadEye datasheet, Thermo Scientific [4].

For the RadEye detector, by analyzing its graph and the information provided in its datasheet, it was possible to determine the conversion factor, which establishes approximately 1.5 CPS/μR/h, equivalent to 9000 CPM for each 1 μSv/h (150 s^{-1}/μSv/h), considering 660 keV radiation photons (Cs-137)[4].

2.3 SMS Sending System Configuration

There are essentially two methods for implementing an SMS system when using an electronic controller such as the Raspberry Pi: via SIM card and via web service.

The SIM card method uses an SMS module capable of sending messages through a mobile network service. In this setup, a physical SIM card is required — registered and with available credit — as well as mobile signal coverage and service availability in the region. Once connected to the SMS module, the Raspberry Pi, through a script, utilizes the carrier's messaging service via the SIM card. In this method, the sender is the phone number associated with the SIM card used in the SMS module, and the number of messages sent is limited by the mobile carrier's service restrictions.

The other method is via a web service, such as Twilio. In this approach, the Raspberry Pi can be programmed with a dedicated script. This script allows defining the sender, recipients, and message body. Similar to the SIM card method, a trigger can also be programmed in the code to execute the SMS sending function. When triggered, the system makes a request to Twilio's server, which then generates and delivers the required message to the specified recipients. In this case, the only requirements are an internet connection and the availability of the web-based SMS service.

For this work, the software-based method via the Twilio service was chosen.

2.3.1 Twilio Configuration

The configuration and implementation of Twilio is straightforward. First, it is necessary to create an account on the platform. Through this process, you will associate your account with your **Twilio phone number**, which is a virtual number specifically reserved for your account. In addition, your account will be linked to an **Account SID** and an **Auth Token** for authentication.

The remaining configuration steps can be found in Twilio's online documentation at: https://www.twilio.com/docs/messaging/quickstart.

Figure 4 illustrates a sample code in the Python programming language for sending messages. In this code, you simply replace your account SID and authentication token in the fields **"TWILIO_ACCOUNT_SID"** and **"TWILIO_AUTH_TOKEN"**, respectively. Additionally, the field corresponding to the SMS message body can be edited to include a customized message. In our case, it will be a notification regarding the dose rate measured by the BG51 detector.

Send an SMS Using Twilio with Python

Python

```python
# Download the helper library from https://www.twilio.com/docs/python/install
import os
from twilio.rest import Client

# Find your Account SID and Auth Token at twilio.com/console
# and set the environment variables. See http://twil.io/secure
account_sid = os.environ["TWILIO_ACCOUNT_SID"]
auth_token = os.environ["TWILIO_AUTH_TOKEN"]
client = Client(account_sid, auth_token)

message = client.messages.create(
    body="Join Earth's mightiest heroes. Like Kevin Bacon.",
    from_="+15017122661",
    to="+15558675310",
)

print(message.body)
```

Figure 2 - Python Script for Sending SMS

Source: Twilio Documentation.

Once the system was implemented and configured, the jar of monazite sand was placed near the detector in order to increase the dose rate counts and thus trigger the SMS sending mechanism.

3 Results and Discussion

When the radioactive sample was placed near the detector, an increase in the dose rate was observed, as expected. This increase activated the function that executes the SMS sending script, which successfully sent a message to a previously selected phone number containing information about the radiation measured and processed by the embedded system.

Figure 5 shows the SMS message received via Twilio.

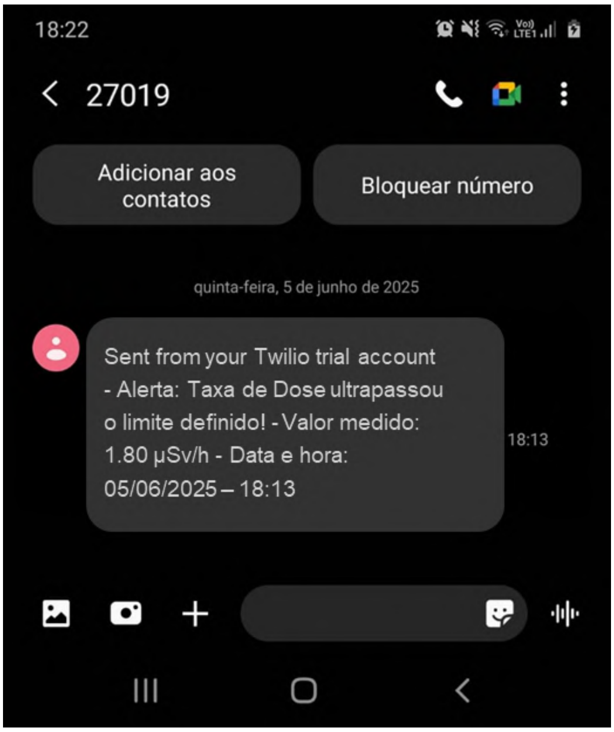

Figure 3 - SMS Screenshot

Source: The Authors

3.1 Feasibility and Scalability of the Twilio-Based Notification System

Although the message was successfully sent, it is still necessary to assess whether this notification method is viable and efficient under operational criteria. One of the most relevant operational criteria relates to Quality of Service (QoS) policies.

QoS policies determine how communication standards between different devices operate. In this study, we focus on the Reliability policy, which has two distinct profiles: Reliable and Best Effort.

The Reliable profile ensures that data transmission will occur, even if it requires multiple resend attempts. The Best Effort profile, on the other hand, attempts to transmit the data but does not guarantee delivery if there is any interruption along the path. In other words, information may be lost if the network is not robust enough or if, for any reason, the message fails to be successfully delivered.

These QoS profiles are crucial in our case, as we are working with the transmission of information that could be critical in an emergency scenario. Therefore, any failure in message delivery—regardless of the reason—represents a significant vulnerability to the system's ability to fulfill its intended purpose: to serve as a risk mitigation tool in potential radiological events.

3.2 Limitations of the Proposed Notification System

Beyond the Quality of Service policies, another important limitation is the mass messaging capability of the system.

Additionally, the system must ensure authenticity in the sending of messages to prevent malicious users—either intentionally or unintentionally—from generating panic by sending alerts to the general population.

4 Conclusions

The proposed SMS-based detection and notification system proved to be functional and efficient, achieving its intended goal of automatically notifying, via SMS, a previously selected user about the presence of radiation at levels exceeding the defined trigger threshold.

The application allows for customization of the message body. This enables the transmission of information regarding the radiological event according to the specific purpose of the application. Thus, in addition to fulfilling its alert function, details such as the measured dose rate, the date and time of the event, the location, and other relevant information can be communicated to professionals involved, response teams, or the local population with a greater degree of specificity.

Despite the success achieved, some limitations related to service quality policies and system availability must be considered, since a reliable system with high availability and the ability to consistently send messages is essential for this application.

In summary, the proposed system may represent a promising tool for rapid response strategies in radiological events, through either mass or targeted notifications, and can be employed in various contexts such as nuclear power plants, waste storage facilities, specialized laboratories, and nuclear research centers.

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Spectrophotometric Analysis of a 3D-Printed Photopolymerizable Resin Exposed to High-Dose Gamma Radiation

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Abstract

Radiation processing relies on precise dosimetry to ensure that materials receive the intended absorbed dose and undergo the expected modifications. Detectors must operate within the dose range of the application, with solid radiochromic materials standing out for their ability to change color when exposed to radiation. Recent studies suggest that certain resins can serve as radiochromic detectors, offering economic and operational advantages over traditionally used dosimeters. In this work, the response of a water-washable photopolymerizable resin, manufactured using a masked stereolithography three-dimensional printing process (MSLA 3D printing), was evaluated in the dose range from 100 to 500 kGy. This resin was selected because its formulation allows post-processing with water, simplifying handling and reducing the use of organic solvents during fabrication. All irradiations were carried out using a panoramic ^{60}Co irradiator at the Gamma Irradiation Laboratory (LIG/CDTN/CNEN). For this study, seven different wavelengths were analyzed by spectrophotometry. Thickness measurements of each printed piece were taken to normalize the absorbance values and minimize variations related to geometry. The wavelength corresponding to the most consistent response was selected, considering the associated uncertainties and response trends. The absorbance spectra obtained showed that the resin exhibits an asymptotic exponential response within the studied dose range and wavelength, considering the associated uncertainties with a 95% confidence level. Gamma radiation induced two new bands in the visible spectrum of the analyzed samples. These bands account for the color change of the samples, which became yellowish with the increased absorbed dose. Such optical changes indicate the formation of new color centers, which is characteristic of materials designed for high-dose dosimetry. Therefore, it was possible to confirm that the photopolymerizable resin has potential as a radiation detector, as expected, and can be applied in radiation dosimetry. Moreover, the dosimetric system developed, from material fabrication to signal reading, proved feasible for applications in radiation processing, highlighting the potential of additive manufacturing to produce detectors with tailored shapes and consistent properties.

Keywords: Dosimetry; radiochromic; additive manufacturing; resin.

1 Introduction

Radiation processing of materials, initiated in 1954 by the Ethicon Division of Johnson & Johnson, consists of applying ionizing radiation using commercial irradiators (IAEA, 2008). Various materials—such as food, medical supplies, blood bags and components, semi-precious stones, and polymers—are irradiated for commercial or research purposes. Each application has a typical dose range. For example, for the irradiation of blood components, the recommended dose range is 15 to 50 Gy, and for food irradiation, up to 10 kGy. Polymer modification may require from 10 kGy to about 200 kGy (McLaughlin et al., 1989), and treatment of gemstones to add commercial value may range from a few Gy to the MGy range.

Radiation dosimetry plays an essential role in the processing of materials by radiation. This is because it is necessary to ensure that the required dose, agreed upon by the client, is correctly administered, ensuring that the desired changes occur (Miller, 1986; International Irradiation Association, 2022). To achieve this, the facilities use dosimeters – devices responsible for correlating changes in some of their physicochemical properties with the dose absorbed by the detector (Attix, 2004). Among the most widely used options, solid radiochromic dosimeters stand out due to the visible change in color after irradiation, allowing direct qualitative analysis (Fowler, 1963).

These dosimeters are based on the radiochromic effect — a change or intensification in the color of certain materials when exposed to ionizing radiation. This change can be observed in films and polymers, such as acrylic (PMMA), with or without the addition of dyes, which are widely used commercially (Sisti Galante, Villavicencio and Campos, 2004; Louback and Meira-Belo, 2011), in addition to other materials such as resins (Hosni et al., 2013). Resins are natural or synthetic polymers formed by macromolecules composed of repeating units. Among their various types, epoxy resins stand out, and more recently, light-curing resins are widely used in additive manufacturing. Photopolymerization occurs due to the presence of a photoinitiator in the resin composition, which, when exposed to light, undergoes a chemical reaction that triggers the polymerization process (Stansbury and Idacavage, 2016; Machado et al., 2024).

Although patents protect the exact composition of light-curing resins, it is known that most of them contain acrylate or (meth)acrylate and/or epoxy monomers and oligomers (Layani, Wang and Magdassi, 2018; Li et al., 2018). Epoxy resins and methyl methacrylates are described in the literature as suitable materials for the manufacture of radiation detectors through color change (radiochromic effect) (Meleshevich, Strakovskaya and Kovalenko, 1987). Thus, given the similarity in chemical composition, it is reasonable to assume that photopolymerizable resins also present a radiochromic response correlatable with absorbed dose.

Several light-curing resins, such as those identified as standard, crystal, and water-washable resin, are commercially available. These resins vary in chemical composition, aiming to modify specific properties and thus adapt them to their intended purposes. Water-washable resin, in particular, stands out by eliminating the need for isopropyl alcohol in the post-printing wash step, allowing the residue removal process to be performed solely with water. This characteristic makes the process simpler, more economically viable, and environmentally sustainable. When comparing water-washable resin to regular resin, there are advantages and disadvantages. Although it has a higher price per kilogram, it saves on solvents in the long run. However, water-washable resin has limitations for specific applications that require greater mechanical performance than standard resin (Shahidi et al., 2024), even considering that the use of isopropyl alcohol in the washing step may affect the mechanical properties of the resins (Liu et al., 2024).

Water-washable resin offers significant dosimetry advantages, especially compared to the current PMMA. A detector made with water-washable resin can be integrated into a dosimeter and, consequently, into a dosimetric system that can be developed and operated entirely within the facility, from production to reading. PMMA detectors, in turn, are generally purchased from specialized suppliers, and the import cost tends to be higher than the investment required for an additive manufacturing system. Furthermore, detectors printed with water-washable resin offer greater geometric flexibility, enabling reproducible manufacturing of various shapes with controlled dimensions. Multiple detectors can be produced simultaneously from digital models (Peerzada et al., 2020). PMMA, in turn, is typically available in cut acrylic sheets, resulting in predominantly rectangular geometries and greater susceptibility to dimensional variations.

Therefore, this work aims to investigate water-washable resin as a raw material for manufacturing radiochromic detectors using additive manufacturing for application in radiation processing, and to develop and evaluate a dosimetric system based on these detectors, integrating fabrication, irradiation, and signal acquisition into a practical approach for high-dose applications.

2 Methodology

Initially, a batch of samples needed to be printed using water-washable resin supplied by 3D Lab. Printing was performed with a Creality Halot-Mage Pro MSLA LCD 3D printer installed at the Gamma Irradiation Laboratory (LIG/CDTN/CNEN). The parts were washed and cured using Creality's Wash & Cure UW-02 unit, which was also installed in the same laboratory.

The printed model was developed based on the dimensions of the Red Perspex 4034 commercial dosimeter, manufactured by Harwell. The design was created using the free TinkerCad software and subsequently exported in .stl format to Halot Box, a slicing software provided by the 3D printer manufacturer. For printing, standard procedures were followed as detailed in the equipment manual. The parameters used were a motor speed of 5 mm/s and an initial exposure time of 30 s with 1.6 s of exposure per layer. After printing the batch, a water wash was performed, followed by a five-minute cure in each main orientation.

The printed units were used as radiation detectors and irradiated at LIG with a Category II multipurpose panoramic irradiator, model IR-214, GammaBeam™, manufactured by MDS Nordion. The equipment houses a dry-stored Cobalt-60 source, whose maximum activity in 2023 was 60,000 Ci (MDS Nordion, 2018). During irradiation, the samples were positioned with limiters in an ABS sample holder to ensure electronic balance and positional stability. Irradiation was performed at a dose rate of 3672.48 Gy/h, previously determined with a Fricke dosimeter, and the applied doses were 100, 150, 200, 250, 350, and 500 kGy.

After irradiation, the samples were stored away from light for approximately 24 hours before spectrophotometric reading. Absorbance spectra were obtained in scanning mode using a Shimadzu Mini 1240 UV spectrophotometer installed at the Luminescent Dosimetry Laboratory (LDL/CDTN/CNEN). Furthermore, non-irradiated samples were selected as references (background). Therefore, before each new reading, the spectra of these samples were reacquired to make specific corrections for that set of measurements, minimizing potential instrumental variations.

The data were analyzed using specific absorbance, that is, absorbance normalized per unit length. The associated uncertainties were calculated for a 95.45% confidence interval by calculating the quadratic sum of the type A and type B uncertainties for the sources evaluated. Furthermore, the effective degrees of freedom were calculated using the Welch-Satterthwaite formula, and the coverage factor was consulted in Student's T-distribution table (International Organization For Standardization, 2008). By evaluating specific points in the spectrum, it was possible to define the wavelength used to analyze the resin.

It is worth noting that, at the coordinates chosen for irradiation, a study of the homogeneity of the radiation field was performed. To this end, 90 orange acrylic plates (4 mm high and 1 mm wide) were irradiated on a vertical support, arranged in 18 columns and five rows. Each plate was analyzed at a wavelength of 606 nm, as per Louback & Meira-Belo (2011), using a UV Mini 1240 spectrophotometer. Subsequently, the net specific absorbance was determined by subtracting the background. This procedure created a two-dimensional dose distribution map, which allowed identifying a region most suitable for sample irradiation.

3 Results and Discussion

Some of the printed and irradiated samples are illustrated in Figure 1. The color change is visible, indicating the occurrence of the radiochromic effect and attesting to the resins' functionality as gamma radiation detectors. Furthermore, the sample's transparency was maintained throughout the analyzed range, a property necessary for more accurate dosimetric analysis (Shinde et al., 2025).

Figure 1: Samples made of water-washable resin as a function of absorbed dose.

Another interesting factor arising from the manufacturing process is the homogeneity in the thickness of samples from the same batch. Interestingly, the thickness variation within a single batch of dosimeters was minimal, which contributes to more accurate results. (Kojima et al., 1992). The American Society for Testing and Materials recommends that absorbance be given as a function of thickness for PMMA-based dosimeters to reduce variations in absorbance (ASTM, 2019). Therefore, measuring the thickness of a sample is critical to the accuracy of the results. Thus, detectors manufactured from 3D-printed resin, similar to polymethyl methacrylate, have the advantage of batch-by-batch homogeneity, which reduces variations in absorbance. The variation coefficient for the samples' thickness was approximately 1%, indicating batch homogeneity.

The processed spectra obtained with the spectrophotometer are shown in Figure 2a, and Figure 2b shows the net contribution of gamma radiation to the spectrum. An increase in absorbance is observed in some areas of the spectrum as the absorbed dose increases, demonstrating the viability of the water-washable photopolymerizable resin used in this study as a gamma radiation detector. As expected, the obtained spectra reveal the emergence of new absorption bands, attributed to the interaction with gamma radiation, which induces the formation of new color centers (Prasad and Lal, 2022).

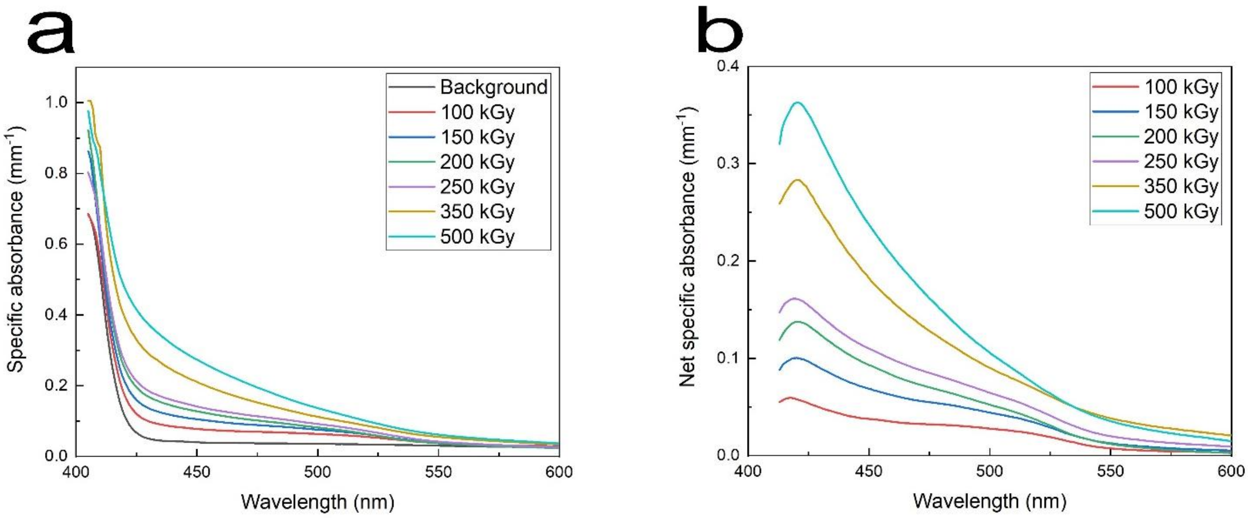

Figure 2: (a) Absorbance spectra total and (b) contribution of gamma radiation

Analysis of the net spectra reveals the presence of two leading bands: one centered at 420 nm, characterized by a distinct absorption peak, and another around 500 nm. This indicates that the complementary colors yellow, orange, and red are predominantly transmitted or reflected by the material, consistent with what was observed in Figure 1.

Seven wavelengths were chosen for analysis and initial calibration, each adjusted by an asymptotic exponential dosimetric curve, present in the literature (Meira-Belo, Rodrigues and Grynberg, 2013; Lundahl et al., 2014). Table 1 presents the equation parameters used in the adjustment and the R^2 found for each wavelength used.

Table 1: Results of asymptotic exponential fitting with the model: $y = A \cdot (1 - e^{-B \cdot x}) + C$.

Wavelength (nm)	Parameters			R^2
410	A	B	C	0.809
421	0.513	3.10E-3	-0.114	0.985
440	1.41	6.84E-4	-0.0405	0.988
444	1.03	7.29E-4	-0.0319	0.989
480	0.918	7.72E-4	-0.0299	0.995
499	0.305	1.51E-3	-0.0110	0.992
511	0.167	2.41E-3	-0.00806	0.983

Thus, the wavelength that best fitted the asymptotic exponential curve was 480 nm, whose curve and calculated uncertainties are shown in Figure 3. It is worth noting that the uncertainty determined by LIG for irradiation is 10% for the entire range used; however, this was not displayed in the graph to improve visualization.

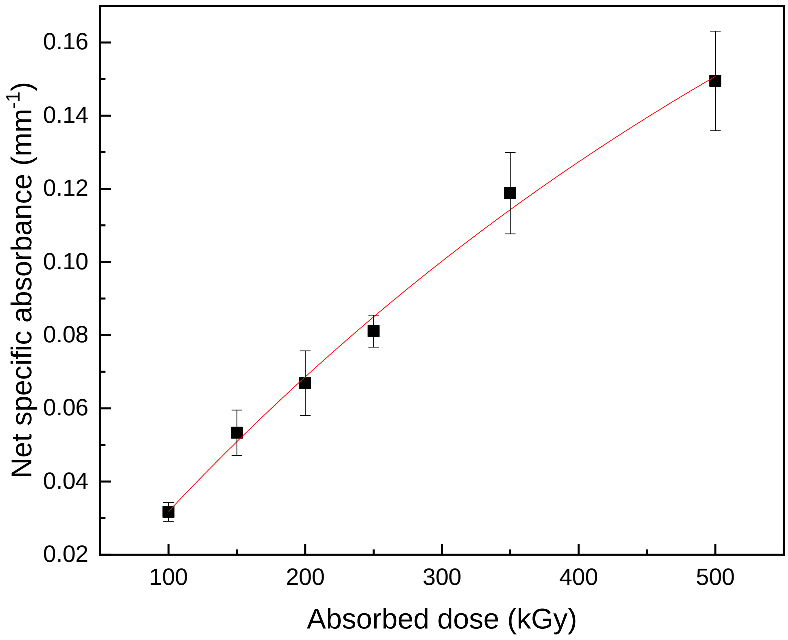

Figure 3: Net specific absorbance as a function of absorbed dose.

The adopted model adequately fits the experimental data, especially considering the uncertainties. Including expanded uncertainties in the analysis allows for a more realistic and applicable interpretation, facilitating the identification of graphical patterns. The wavelength that demonstrated the most consistent behavior is consistent with parallel studies related to signal fading, standing out for its greater stability. Although it does not present the most significant signal amplitude, as observed in the spectra, the 480 nm point was considered the most suitable for dosimetry application in radiation processing.

Daily irradiation in the laboratory continued as usual during this study; therefore, routine interruptions (dose fractionation) did not significantly affect the detector response. This behavior is essential for the detector in radiation processing dosimetry, as exposure to ionizing radiation is usually given in fractions (Lundahl et al., 2014).

4 Conclusions

A water-washable photopolymerizable resin was investigated as a potential detector for high doses of gamma radiation. The resin was printed as rectangular prisms, irradiated with a Cobalt-60 source, and its dosimetric properties were analyzed by UV-VIS spectrophotometry. The samples responded within the analyzed range (100 to 500 kGy), showing a visible color change and increased absorbance in the visible region of the spectrum. The radiation formed two prominent bands responsible for the observed coloration. After analyzing seven wavelengths with asymptotic exponential adjustments, it was determined that the appropriate wavelength for reading the resin is 480 nm. Therefore, it is concluded that the resin can serve as a raw material for a radiochromic detector with potential application in radiation processing, given its response range, comparable to that reported for acrylics and other resins in the literature. Furthermore, the sample fabrication process was simplified by using water as a solvent and the ability to rapidly print multiple samples with greater homogeneity in thickness. Thus, the developed dosimetric system can be replicated, encompassing sample printing, irradiation (using ABS supports), reading mode and wavelength, as well as analysis and calibration.

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MCNP simulation of the effects of prostatic edema in LDR brachytherapy with ^{125}I using the adapted ICRP 110 voxel male phantom

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Abstract

Prostate brachytherapy, an oncology treatment involving the insertion of radioactive seeds, is commonly associated with the development of edema, an inflammatory response that alters the volume of the prostate. This volumetric variation raises significant concerns about the delivery of the radiation dose to the target organ, and could compromise the effectiveness of the treatment due to the displacement of the sources over time. This study aims to quantify the impact of prostatic edema on dose distribution in brachytherapy. For this purpose, the ICRP 110 male phantom was used, whose original prostate volume was altered with variations of 10% to 50% to emulate different degrees of edema. A dedicated system generated the input files for the MCNP code of the ^{125}I seeds, calculating the positioning and relative distances for each level of edema. The results of the simulations showed a progressive reduction in the absorbed dose as the prostate volume increased. While the original volume of 16.5 cm^3 of the prostate received 100% of the planned dose (144 Gy), the volumes with edema of 10%, 20%, 30%, 40% and 50% received 99%, 97%, 94%, 91% and 87% of the dose, respectively. It can be concluded that prostate edema resulting in an increase of more than 20% of the original volume leads to clinically significant losses in the dose absorbed by the organ, which can negatively impact the efficacy of brachytherapy treatment. These findings underline the importance of considering edema when planning and monitoring prostate brachytherapy.

***Keywords**:* LDR brachytherapy; Iodine-125 seeds; Voxel phantom; MCNP.

1 Introduction

Prostate cancer is a serious public health concern in Brazil. According to INCA/RJ, this malignant neoplasm is responsible for 28.6% of cancer deaths among men in the country. In 2023, around 17,000 men died from the disease, an alarming average of 47 deaths per day (CAPES, 2023). In Brazil, prostate cancer is the second most common type of cancer among men, second only to non-melanoma skin cancer. Age is a crucial risk factor, with incidence and mortality increasing significantly after the age of 60. It is estimated that between 2023 and 2025, the country will register 71,730 new cases annually (INCA/2023).

It's important to note that not all prostate tumors behave in the same way. While some types can grow rapidly and metastasize, leading to death, most develop slowly (taking approximately 15 years to reach 1 cm^3), often without showing symptoms or posing a significant threat to men's health throughout their lives (INCA/2023).

The prostate, a gland exclusive to the male reproductive system, is located in the lower abdomen. Similar in shape and size to an apple, it is located just below the bladder and in front of the rectum, surrounding the initial portion of the urethra, the channel through which urine is eliminated. The prostate's main function is to produce part of the semen, the liquid that carries the sperm during ejaculation. Considered predominantly a cancer of old age, around 75% of global prostate cancer cases occur after the age of 65.

In low dose rate (LDR) brachytherapy treatment, which involves the insertion of multiple ^{125}I seeds into the prostate, it is common for edema to appear as an inflammatory response by the body. This edema, which represents an increase in the volume of the prostate due to the trauma of the procedure, is generally not considered during initial treatment planning. However, this volumetric change can lead to displacement of the radiation sources, significantly affecting the dose administered. Deviations in the prescribed dose can not only compromise the target volume, but also adversely impact surrounding healthy tissues and organs. Accurately quantifying the actual dose administered and these deviations in prostate brachytherapy is a challenge (Almanza, 2018).

To investigate the impact of prostate edema, this study used the ICRP 110 male voxel phantom, adapting the planes that delimit the voxel and those that make up the lattice structure. This modification made it possible to simulate variations of 10% to 50% in the original volume of the prostate, emulating edema.

A dedicated system, used in this work, was developed and improved to automate the necessary calculations. It prepared the scenario for the phantom by defining the volume variations and simulation parameters for the MCNP code. This system automatically generated the input files for the ^{125}I radioactive seeds (cell, surface and TR cards). For each percentage of edema simulated, the system individually calculated the positioning and distances between the seeds (taking into account the inter-seed effect) in correlation with the prostate edema in the ICRP 110 male phantom.

In all, six input files were built and simulated in the Monte Carlo MCNP code. The first file represented the phantom with the original prostate volume (16.5 cm^3) and 73 planned and distributed ^{125}I seeds. The other five files simulated the swollen prostate, with volumes varying from 10% to 50% above the original.

2 Techniques and Materials

2.1 LDR Prostate Brachytherapy

Low dose rate (LDR) prostate brachytherapy treatment is divided into three main stages: planning, implantation and post-implant evaluation. During planning, the aim is to create a dose distribution that is in line with dosimetric criteria. This phase involves outlining the prostate and surrounding structures, such as the urethra and anterior wall of the rectum, defining the volume to be treated and determining the practical details of the implant, including the number, activity and position of the seeds.

Planning can be carried out preoperatively or intraoperatively. Studies indicate that intraoperative planning generally results in more accurate dose delivery to the prostate, as well as reducing urinary and rectal toxicities associated with the procedure (Pé-Leve, 2018).

2.2 REX phantom (Voxel Model, ICRP 110)

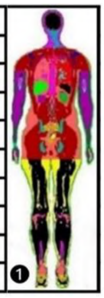
The anthropomorphic REX phantom, a male voxel model recommended by ICRP publication no. 110 (ICRP 110, 2009), represents a standard reference adult.

For the construction of this computational phantom, whole-body computed tomography images of a 38-year-old man with a height of 176 cm and a body mass of just under 70 kg were used (the ICRP Reference Standard Male is 176 cm and 73 kg). The individual, who had undergone whole-body irradiation due to leukemia, showed no visible signs of disease on the images. He was in a supine position with his arms parallel to his body.

The data set consisted of 220 slices of 256 x 256 pixels. The original voxel size was 8 mm high with an in-plane resolution of 2.08 mm, resulting in a voxel size of 34.6 mm³. In total, 122 structures were segmented (67 of which were bones or groups of bones), including many of the organs and tissues later identified in the characterization of the ICRP reference anatomical data (Cordeiro, 2013). The main characteristics of the reference adult male computational phantom (REX) are summarized in Table 1.

Table 1 - Technical data of the ICRP male phantom in voxels.
(* Additional skin slices from the head and sole of the foot.)

Properties	**REX**
Height (m)	1.76
Mass (Kg)	73.0
Total voxel number	1,946,375
Slice Thickness (mm)	8.0
In-plane Voxel Resolution (mm)	2.137
Voxel volume (mm^3)	36.54
Number of columns	254
Number of lines	127
Number of slices	220 (+2)*



❶ Image of the ICRP male voxel phantom visualized in the MORITZ graphic encoder and the main characteristics of the ICRP anthropomorphic voxel model.

2.3 Radioactive Isotope ^{125}I

In this work, the ^{125}I Amersham model 6711 (Figure 1) source was used as a reference. It is modeled as an inner silver cylinder containing an emitting film. The outer shell, made of titanium, has a diameter of 0.81 mm and a length of 4.5 mm, and is closed by two ellipsoids (Duggan et al, 2004).

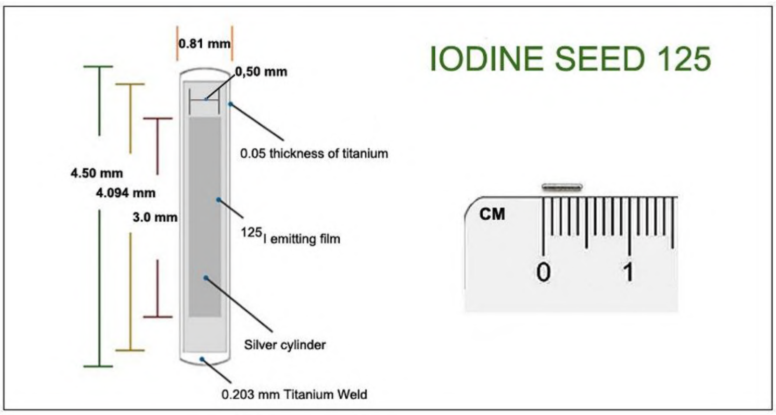


Figure 1: Geometric illustration of the ^{125}I Amershan Model 6711 seed.

2.4 System development

In this work, a dedicated system was used to generate the ^{125}I seed input files for the MCNP code, which was decisive in providing accurate technical data. This system was designed and developed specifically for prostate LDR brachytherapy work.

Programmed in Java (a high-level, object-oriented language), the system automates all the seed positioning calculations and generates the input files. It can take into account the predictability of edema or not. The seeds are inserted into the prostate of an ICRP 110 voxel male phantom, which is adapted by the system for each case study, with or without edema.

The system has the following options on its home screen (Figure 2):

- **IC Student Module**: Aimed at future work with undergraduate students, allowing them to register their data in the system.
- **Researcher Module**: Records the date, time and person responsible for each input file generated.
- **Calculation/Generation of INPUT MCNP**: Module for entering data and selecting parameters to create the input file. It also includes the feature of modifying voxel phantom planes for EDEMA predictability studies.
- **Reports**: Allows access to the complete history of the system's use in generating input files.
- **OUTPUT Export**: Extracts the results data from the MCNP output file.

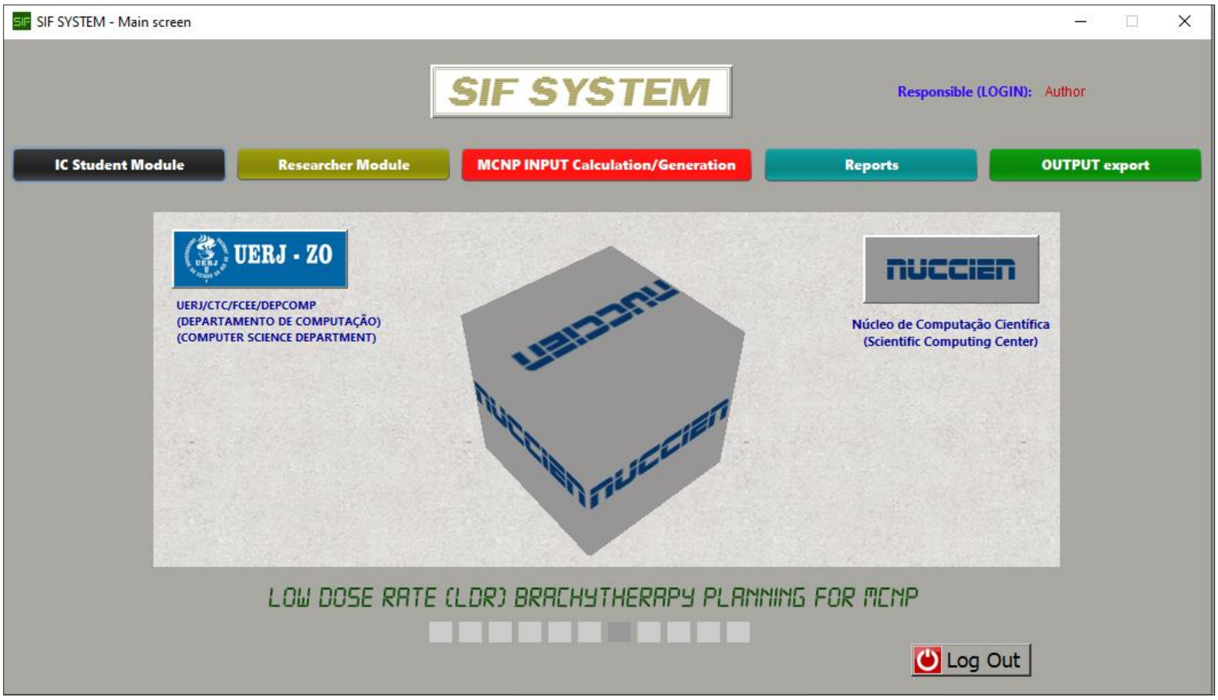


Figure 2: Overview of the system designed to generate input files.

In the computer environment, the system recreates the reference template (Figure 3a) used in the clinical process of inserting the ^{125}I seeds. Once the seeds have been inserted, it is possible to generate the input file with the “EDEMA predictability” feature. To do this, simply activate this feature and select the desired percentage. In this configuration, the system repositions the seeds and performs a recalculation based on the new prostate volume, according to the conditions illustrated in Figure 3b.

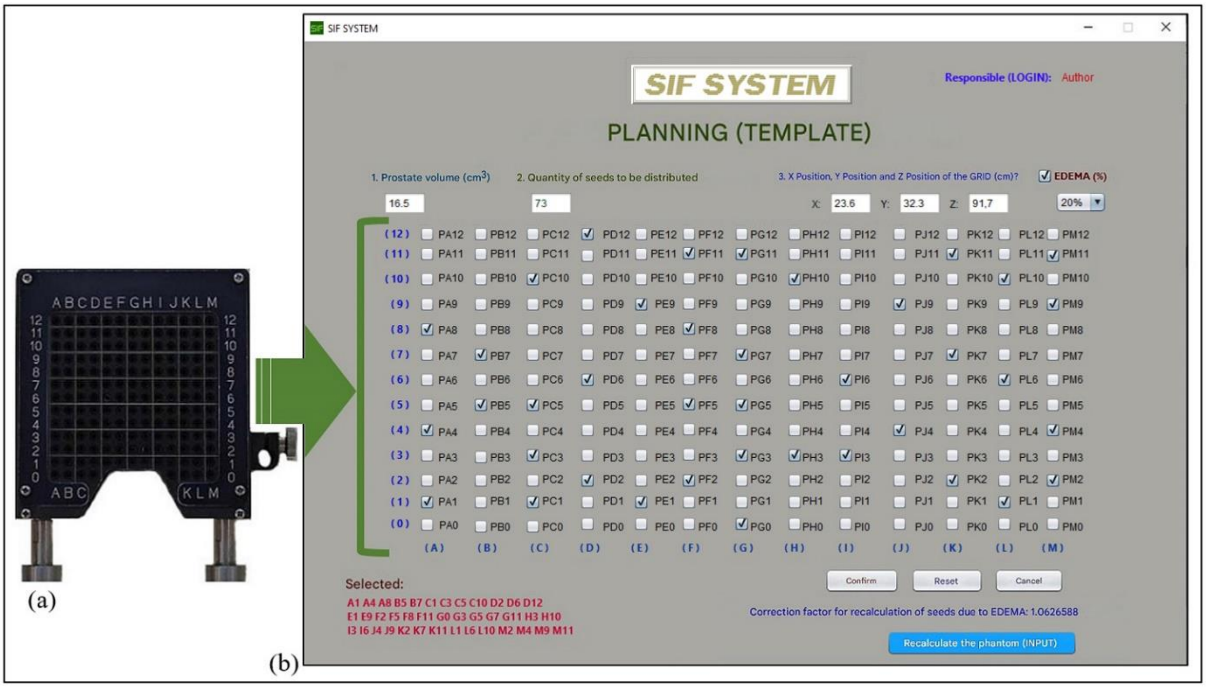


Figure 3: (a) The template used in the clinical procedure. (b) The computer system and the correlation with the template.

The “MCNP INPUT Calculation/Generation” option in the system interface is crucial for creating the input file for the MCNP code. The process begins by requesting the volume of the prostate in cm^3. Once this information has been filled in and confirmed, the system determines the total number of seeds for the treatment. Once the number of seeds has been defined, the model's location data must be entered in order to spatially position the seeds in the lattice structure (prostate geometry) of the phantom. The exact position is obtained using the Moritz Geometry Tool program (Riper, 2008), which allows 3D visualization of the geometry in a voxel structure. Next, the system asks the user if they want to enable edema predictability. If enabled, the user selects the desired percentage (10%, 20%, 30%, 40% or 50%). This percentage recalculates the total volume and, after selection, the system returns the correction factor for recalculating the seed distribution.

At this stage, the “Recalculate phantom (INPUT)” button is activated, allowing the user to load the desired voxel phantom. To study the impact of edema on prostate brachytherapy planning, it is essential that the working phantom is adapted to each proposed scenario. To simulate prostate inflammation, changes are needed to the voxel planes, the lattice planes and the rectangular parallelepiped that encompasses the phantom.

It is important to note that, in the computer simulation, the analysis will focus only on the prostate, using the definitions described in tally F8 of the MCNP input file. These changes to the plans are carried out automatically by the system: it tracks specific data extracted from the phantom input file and carries out all the necessary recalculations based on the percentage of edema selected.

The system's graphic interface has been carefully designed to mirror the template used in the procedures. Its visual structure is made up of columns identified by the letters A to M and rows numbered 0 to 12. When a user, for example, selects the "H10" position on the interface for inserting seeds, the system is designed to ask for two crucial pieces of data: the quantity of seeds to be inserted in that position and the individual depths of each one. Subsequently, the system describes the instruction cards (i.e. the guidelines) that are relevant to the seeds already positioned. This workflow is consistent and applies identically to all additional seeds.

When the process is complete, a new window appears, allowing you to fill in the final technical data (Figure 4a). This data includes the initial numbering of cells, surfaces and materials, which is essential for correct sequencing in the phantom when it is inserted. Once the "Generate Input File" button is pressed, the system asks the user to enter the "File name (8 characters)" where all the instructions will be stored (Figure 4b). After confirmation ("OK"), the user will be notified of the successful creation of the input file and its specific location within the computer folder (Figure 4c). Finally, the program automatically opens the file, allowing work to begin immediately on preparing the simulation in the MCNP code.

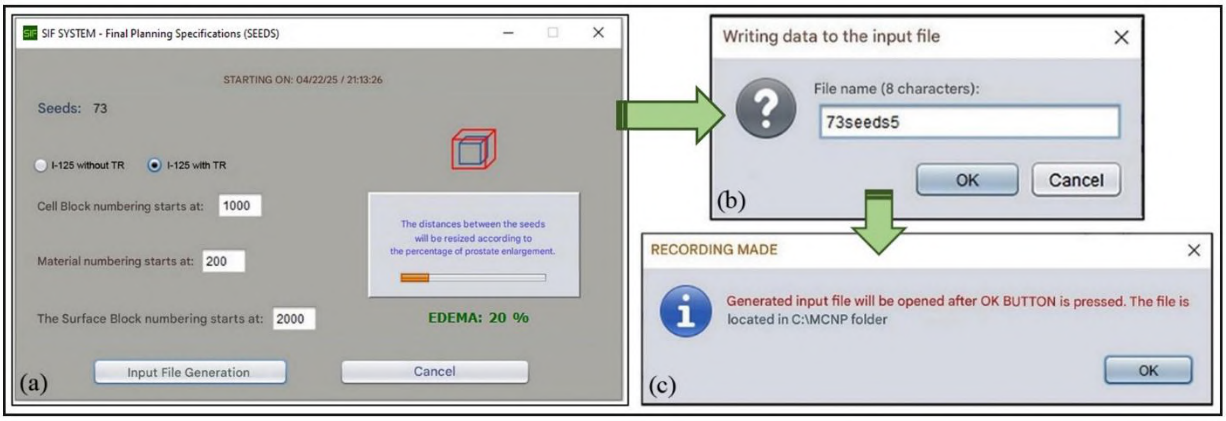
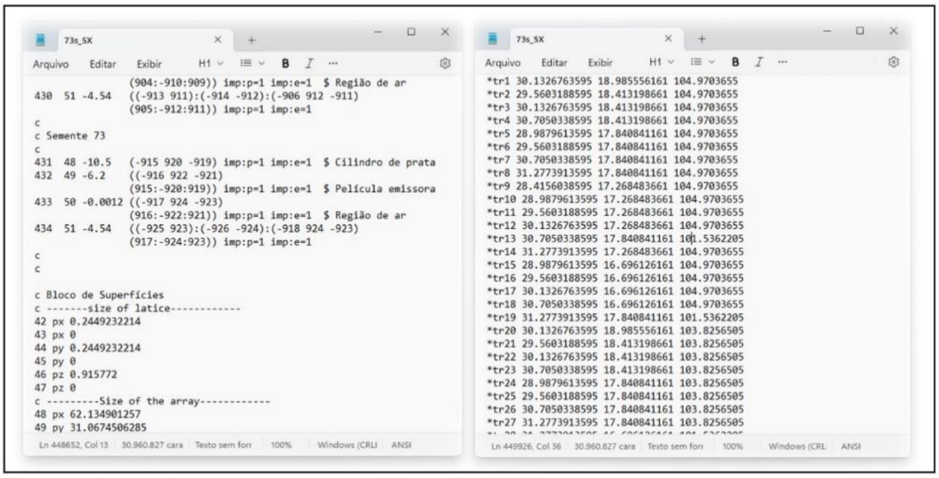


Figure 4 - Final sequence for generating the MCNP code input file for ^{125}I seeds.

3 Preparation and computer simulation

For the analysis proposed in this work, six input files were generated by a system developed specifically for this purpose. These files, destined for the MCNP code, contained the complete description of cells, surfaces, materials and TR cards for the spatial positioning of 73 ^{125}I seeds in the prostate of the ICRP 110 (REX) male voxel model, which has a volume of 16.5 cm^3. The first file, called "73s_X", included the REX phantom with the original prostate volume, containing the 73 seeds distributed as planned and following the Siemens template. The remaining files, "73s_1X", "73s_2X", "73s_3X", '73s_4X' and "73s_5X", represented the prostate with different levels of edema, ranging from 10% to 50%. In these cases, the distances between the 73 seeds were recalculated to follow the proposed edema in the X, Y and Z coordinates. Figure 5 shows sections of the input file "73s_5X" generated by the system. This file details the recalculation of the positions of the ^{125}I seeds, taking into account 50% prostatic edema.



```
73s_5X
Arquivo  Editar  Exibir
                (904:-910:909)) imp:p=1 imp:e=1  $ Região de ar
430  51 -4.54   ((-913 911):(-914 -912):(-906 912 -911)
                (905:-912:911)) imp:p=1 imp:e=1
c
c Semente 73
c
431  48 -10.5   (-915 920 -919) imp:p=1 imp:e=1  $ Cilindro de prata
432  49 -6.2    ((-916 922 -921)
                (915:-920:919)) imp:p=1 imp:e=1  $ Película emissora
433  50 -0.0012 ((-917 924 -923)
                (916:-922:921)) imp:p=1 imp:e=1  $ Região de ar
434  51 -4.54   ((-925 923):(-926 -924):(-918 924 -923)
                (917:-924:923)) imp:p=1 imp:e=1
c
c

c Bloco de Superfícies
c -------size of latice-----------
42 px 0.2449232214
43 px 0
44 py 0.2449232214
45 py 0
46 pz 0.915772
47 pz 0
c ---------Size of the array-----------
48 px 62.134901257
49 py 31.0674506285
Ln 448652, Col 13 | 30.960.827 cara | Texto sem forr | 100% | Windows (CRLI | ANSI
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73s_5X
Arquivo  Editar  Exibir
*tr1 30.1326763595 18.985556161 104.9703655
*tr2 29.5603188595 18.413198661 104.9703655
*tr3 30.1326763595 18.413198661 104.9703655
*tr4 30.7050338595 18.413198661 104.9703655
*tr5 28.9879613595 17.840841161 104.9703655
*tr6 29.5603188595 17.840841161 104.9703655
*tr7 30.7050338595 17.840841161 104.9703655
*tr8 31.2773913595 17.840841161 104.9703655
*tr9 28.4156038595 17.268483661 104.9703655
*tr10 28.9879613595 17.268483661 104.9703655
*tr11 29.5603188595 17.268483661 104.9703655
*tr12 30.1326763595 17.268483661 104.9703655
*tr13 30.7050338595 17.840841161 101.5362205
*tr14 31.2773913595 17.268483661 104.9703655
*tr15 28.9879613595 16.696126161 104.9703655
*tr16 29.5603188595 16.696126161 104.9703655
*tr17 30.1326763595 16.696126161 104.9703655
*tr18 30.7050338595 16.696126161 104.9703655
*tr19 31.2773913595 17.840841161 101.5362205
*tr20 30.1326763595 18.985556161 103.8256505
*tr21 29.5603188595 18.413198661 103.8256505
*tr22 30.1326763595 18.413198661 103.8256505
*tr23 30.7050338595 18.413198661 103.8256505
*tr24 28.9879613595 17.840841161 103.8256505
*tr25 29.5603188595 17.840841161 103.8256505
*tr26 30.7050338595 17.840841161 103.8256505
*tr27 31.2773913595 17.840841161 103.8256505
Ln 449926, Col 36 | 30.960.827 cara | Texto sem forr | 100% | Windows (CRL | ANSI
```

Figure 5: Parts of input file “73s_5X” from recalculation of seed positioning of ^{125}I in prostate with 50% Edema.

All the files were validated using the Moritz Geometry Tool graphic coder (RIPER, 2008). Figure 6 shows a partial view of the phantom, highlighting the anatomical location of some organs. Figure 7 illustrates the position and distribution of the radioactive seeds within the prostate volume, according to the modeled implantation plan. The simulation was carried out using the MCNP code (MCNPX, 2005), and each file required a simulation time of approximately 1,500,000,000 stories.

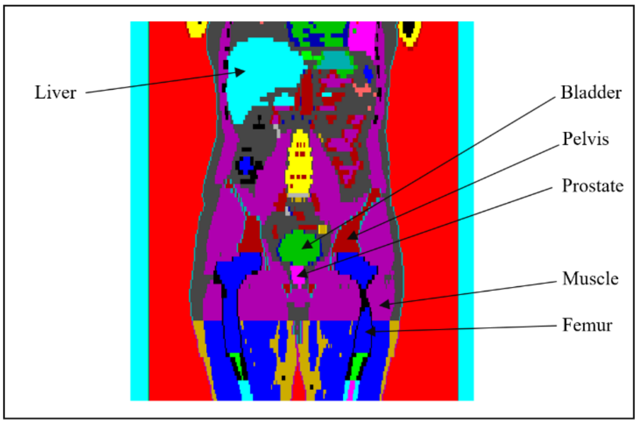

Figure 6: Visualization of the different structures present in the male voxel phantom (REX) of ICRP 110.

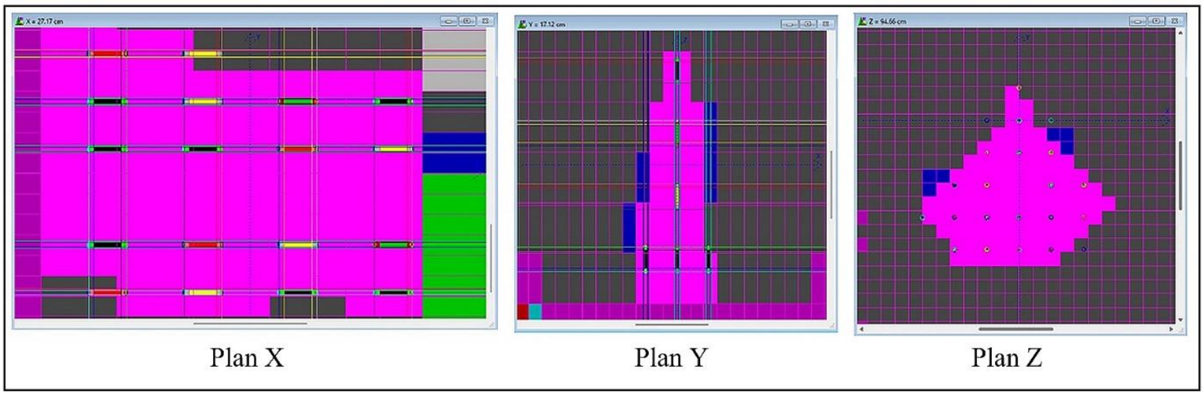

Figure 7: Example of using the Moritz Geometry Tool graphic encoder to check the input files of the MCNP code for the REX phantom with ^{125}I seeds.

4 Results

The results obtained from the computer simulation contribute to a deeper understanding of the dose distribution in low dose rate (LDR) brachytherapy scenarios for the treatment of prostate cancer. The feasibility and accuracy of these simulations were directly influenced by detailed knowledge of the prostate volume and the specific technical conditions of LDR brachytherapy planning, which include the implantation methodology and the spatial distribution of the radioactive seeds within the prostate tissue.

The primary analysis focused on assessing the impact of prostatic edema on dose delivery. A progressive lag in dose delivery to the target organ is observed, which correlates directly with the percentage of edema. The simulations covered variations in edema in increments of 10%, from 10% to 50% increase in prostate volume. Table 2 summarizes these findings, detailing how each increment in edematous volume results in a measurable change in the dose absorbed by the prostate. This lag is critical, as it can compromise the effectiveness of the treatment if it is not properly taken into account.

To visualize and quantify this dose-volume relationship, Dose-Volume Histograms (DVHs) were generated for each of the simulated edema phases (Figure 8). The DVH is a gold-standard graphic tool in radiotherapy that allows the radiation dose received by different tissue volumes to be correlated, whether it be the target organ or adjacent sensitive structures. Analysis of the DVHs reveals the changes in dose coverage and homogeneity of distribution as prostatic edema progresses, offering a clear representation of the challenges posed by this condition.

Prostatic edema is not a static condition. The medical literature establishes that each phase of edema has a specific duration, and this inflammation gradually decreases over time until the prostate returns to its original pre-treatment volume. This temporal dynamic of edema has profound implications for the planning and execution of LDR brachytherapy. During the initial planning phase, the treatment team must necessarily discuss and integrate these considerations about the temporal evolution of edema. The ability of computer simulation to provide additional data and hypothetical scenarios is a distinguishing feature of this work. By simulating dose behavior under different levels of edema, the clinical team acquires valuable information that, combined with statistical techniques and clinical expertise, allows for the determination of more resilient and effective treatment strategies, optimizing dose delivery to the tumor and minimizing dose to organs at risk, even in the face of prostate volume fluctuations caused by edema.

Table 2 - Dose Values and Gap Percentages for Conditions with and without Prostatic Edema.

EDEMA (%)	Dose to organ (Gy) ***Planned (144 Gy)***	Percentage of total Gy planned (%)
0	144.00	100
10	142.56	99
20	139.68	97
30	135.54	94
40	131.04	91
50	125.28	87

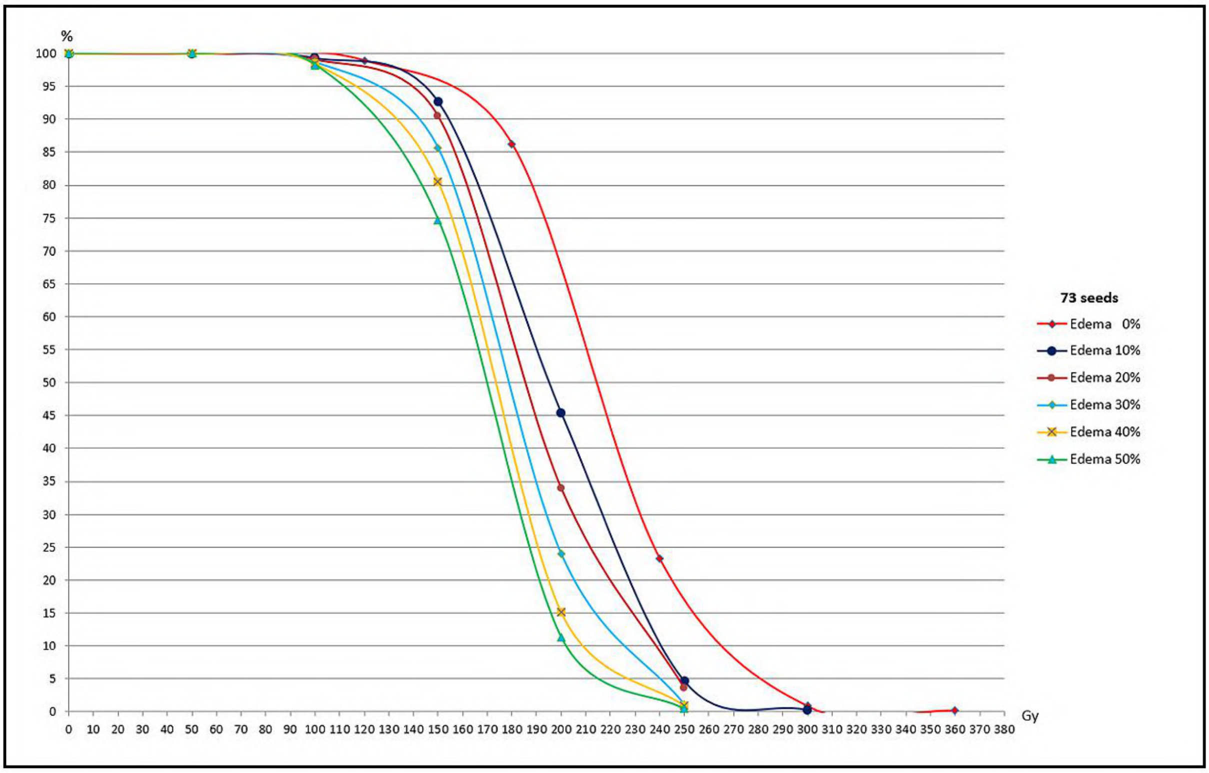

Figure 8: Dose-Volume Histograms (DVHs) of the results of the MCNP code input files in the EDEMA study.

5 Conclusions

The results obtained in this study highlight the need for greater accuracy in low dose rate (LDR) brachytherapy planning protocols for prostate cancer, especially in view of the inflammatory response caused by the implant - prostatic edema. Using the MCNPX code, computer simulations were carried out with the REX voxel phantom, making it possible to clearly demonstrate the impact of prostatic edema on the spatial distribution and effective delivery of the prescribed dose.

It was observed that the presence of edema, with volumetric variations of 30% to 50%, generates significant lags in the coverage of the tumor target, especially when this inflammatory process lasts for a period of up to 30 days. This scenario can seriously compromise therapeutic efficacy, considering the half-life of the radioactive ^{125}I seed, which is approximately 59.4 days. The prolongation of inflammation affects the geometry of the implant, altering the x, y and z distances between the seeds, and directly influences the inter-seed effect, which is decisive for the success of the treatment.

It is important to note that each patient has a unique anatomy, which requires a specific quantity and distribution of seeds inside the prostate. Therefore, it is essential to consider edema as a critical variable during the planning and post-implantation phase, with a view to guaranteeing the effective dose and preserving sensitive adjacent structures, such as the urethra and bladder. Failure to adequately monitor this factor can result in undesirable side effects and the need for auxiliary therapies to mitigate symptoms.

In this way, this work reinforces the relevance of integrating computer modeling, clinical knowledge and physiological monitoring in the context of prostate brachytherapy, with a view to optimizing therapeutic results and improving patients' quality of life.

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Volumetric dose evaluation using 6 MV linear accelerators for blood irradiation

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Abstract

Irradiation of blood and blood components is used to inactivate transfused lymphocytes with the main purpose of preventing transfusion associated graft-versus-host disease (TA-GVHD) in accordance with recommendations established by ANVISA (RDC n° 34) and CNEN (NN 6.16). Both Institutions defined dosimetry requirements to ensure proper control of blood irradiation to guarantee the safety and quality of the transfused product. The blood irradiator should deliver 25 Gy to the center of the irradiator unit, but should not exceed 50 Gy or be less than 15 Gy. The protocols show that the irradiation needs to be done by a blood irradiator enabling the irradiation of blood components in the absence of dedicated irradiators. On protocol says how irradiation should be performed, under what conditions or paparameters. This study proposes the application of Fricke dosimetry to blood irradiation using linear accelerator and the development of a protocol that complies with current regulations. To evaluate the volumetrically absorbed dose, it was used a phantom of polymethylmethacrylate (PMMA) with dimensions of 30.0 x 20.0 x 17.0 cm³ and density of 1.17 g.cm-³ with 25 plates. The phantom allows the irradiation of 40 bags of polyethylene bags with dimensions of 4.0 x 4.0 x 0.4 cm³, filled with 1.4 g of standard $FeSO_4$ solution and allocated in 8 plates, known as Fricke solution, each containing 5 cavities of 4.0 x 4.0 x 0.5 cm³. Irradiations were performed using a Varian Clinac 2100 C linear accelerator with 6 MV photon beam at the University Center for Cancer Control (CUCC) of Rio de Janeiro State University (UERJ), located in Rio de Janeiro, Brazil. A Treatment Planning System (TPS) was executed using a field size of 40.0 X 40.0 cm² covering the entire phantom, with a source-to-phantom central axis distance to 100 cm. Two parallel opposed (anterior–posterior) isocentric beams covering the phantom both length-wise and breadth-wise were placed and normalized to deliver 25 Gy at the center of the phantom. The Fricke dosimetry depends on the oxidation of ferrous ions (Fe2+) to ferric ions (Fe3+) caused by interaction of the ionizing radiation with the solution. The increased concentration of ferric ions is measured at a wavelength of 304 nm using a Cary 50 Bio UV-VIS spectrophotometer. The obtained results of mean absorbed dose were compared with the mean dose from the treatment planning system of the PTV (25.00 Gy). The mean dose delivered to the Fricke was 24.71 ± 0.06 Gy, with a relative error of -1.18% compared to the mean dose from PTV. Fricke dosimetry has shown to be a potential tool for quality control in blood component irradiation using linear accelerators, with type A uncertainty of 0.23%. The results comply with ICRU Report 62 and Anvisa recommendations, with uncertainties below 5%. This work was supported by the Universal CNPq Project, call CNPq/MCTI n° 10/2023, process n° 420745/2023-8.

Keywords: Linear accelerator; Fricke Dosimetry; Blood irradiation.

1 Introduction

Transfusion‑associated graft‑versus‑host disease (TA-GVHD) in immunodeficient patients causes a potential transfusion reaction and is a fatal complication in more than 90% of cases, occurring in patients who receive blood components, as a consequence of allogenic T-lymphocytes from immunocompetent donor or graft are transfused to recipient or host which usually immunocompromised, therefore cannot destroy foreign lymphocytes by the recipient’s immune system..

Transfused graft T- lymphocytes can cause human leukocyte antigen incompatibility or recognize the recipient’s cells as foreign antigens. They then initiate a rejection response by releasing cytokines such as interleukin-1 and tumor necrosis factor, which initiate the recipient’s inflammatory cells. The proliferation of T lymphocytes leads them to attack the recipient’s tissues and bone marrow (IAEA, 1997).

Blood component irradiation to inactivate lymphocytes is the only proven method of preventing TA-GVHD, because the effect of radiation breaks the DNA molecule of the T-lymphocyte, preventing its division and capacity to proliferate in the body, thereby rendering the T- lymphocytes inactive. Generally, the radiation sources used in this process are gamma radiation from 60Co or 137Cs sources or X-rays, employing linear accelerators and self-shielded irradiators to reduce the risk. According to Pelszynski et al. (1994), the effect of T-cell inactivation increases exponentially with increasing doses of gamma irradiation.

Furthermore, there is international interest in replacing gamma-ray irradiators for blood irradiation due to the high costs associated with governmental regulations pertaining to radioactive elements on site (Janatpour et al., 2005). In Brazil, irradiation is performed at blood centers, which are regulated by the National Health Surveillance Agency (Anvisa), the National Nuclear Energy Commission (CNEN), and accredited by the AABB (American Association of Blood Banks). With respect to regulatory documentation in Brazil, both entities consider dosimetry with well-established doses for the blood bag irradiation area to be fundamental and necessary.

The irradiation of blood components is regulated by Resolution RDC No. 34 (Anvisa, 2014), which establishes the Good Practices in the Blood Cycle. According in Article 4, item XXVIII: "Irradiated blood components are those subjected to irradiation procedures that ensure a defined minimum dose on the mid-plane of the irradiated unit". To ensure the safety of the blood irradiation process and to prevent TA-GVHD, Anvisa’s resolution in Article 61, § 1 stipulates that the exposure time must be set so that all blood and components should deliver 25 Gy to the center of the irradiator, with but should not exceed 50 Gy or be less than 15 Gy.

1.1 Fricke dosimetry applied to blood irradiation

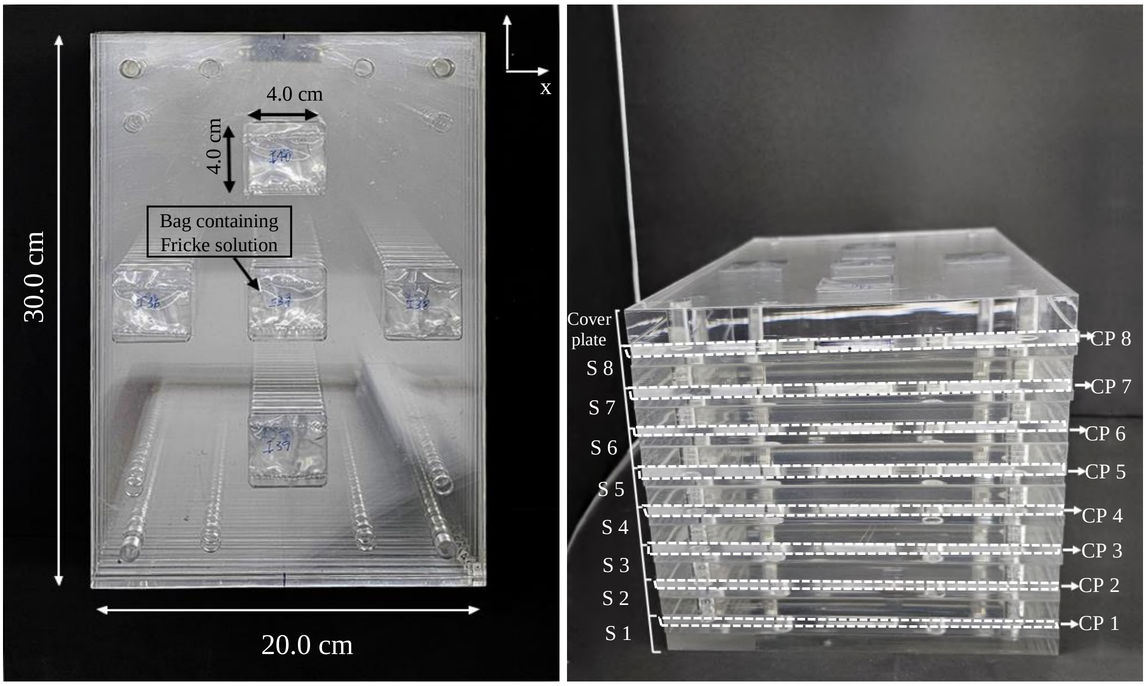
To supply the need for hemocenters, hospitals and clinics to perform annual dosimetry for the irradiation of blood components as required by ANVISA RDC No. 34 (2014), it is feasible to implement a dosimetric protocol to establish a primary standard for absorbed dose to water, applicable to blood irradiators or linear accelerators used in this process. Chemical dosimetry using a standard $FeSO_4$ solution, known as the Fricke solution, as a tool for the dosimetry and quality control of blood irradiators to reducing the likelihood of TA‑GVHD (Mantuano et al., 2018; Mantuano et al., 2020; Mantuano et al., 2021; Rosado et al., 2020).

The interaction of radiation with the Fricke solution results in the oxidation of ferrous ions (Fe^{2+}) to ferric ions (Fe^{3+}). The increase in ferric ion concentration is measured by optical absorption spectrophotometry at the maximum absorption wavelength of 304 nm (Fricke, 1927; Klassen et al., 1999). The Fricke dosimeter contains 96% water; therefore, the radiation attenuation in the solution is similar to that of water, making it tissue‑equivalent (Olszanski et al., 2002). In situations that do not require traceability to national standards, this system can be used for the absolute determination of the absorbed dose without calibration, since the chemical yield of radiation induced ferric ions is well characterized. Furthermore, it has demonstrated good performance as a reliable absorbed‑dose standard solution for blood irradiation and other applications, since Fricke dosimetry is classified by ISO as a Type I reference standard dosimetry system and is recommended for primary absolute dosimetry of the absorbed dose to water by TRS 398 (ISO, 2015; IAEA, 2000; Mantuano et al., 2018; Mantuano et al., 2020; Salata et al., 2015; De Almeida et al., 2014).

2 Methods

The blood irradiation protocols for linear accelerators described in the literature vary in dose distribution, in the types and sizes of containers used to hold the blood bags to be irradiated, and also in the number of bags per procedure and their arrangement within those containers.
To analyze the dose distribution volumetrically and develop dosimetry for volumetric dose evaluation using a Fricke solution, a polymethylmethacrylate (PMMA) phantom was designed and constructed for Silva (2023). The phantom measures 30.0 × 20.0 × 17.0 cm^3 (ρ = 1.17 $g \cdot cm^{-3}$) and is composed of 25 plates. The phantom consists of a cover plate and eight sets of three plates: two flat plates with thicknesses of 1.0 cm and 0.5 cm, respectively, and one plate with five 0.5 cm−thick cavities for positioning the bags containing Fricke solution (Figure 1).

The phantom enables the irradiation of 40 polyethylene bags (4.0 × 4.0 × 0.2 cm^3) filled with 1.4 g of Fricke solution, which are placed in eight plates with five cavities each, each cavity measuring 4.0 × 4.0 × 0.5 cm^3. For the irradiation of the bags containing Fricke solution in the linear accelerator, they were arranged in ascending order starting with the first cavity plate and numbered from 1 to 40. This sequence was maintained in all irradiations, and an additional three Fricke solution bags were used as control.



(a) (b)

Figure 1: (a) Top view of the blood-irradiation phantom for a linear accelerator, showing bags 36–40. (b) Side view of the phantom with the eight cavity plates (CP) outlined and numbered.

Bags 1 to 5 were placed in the first set (S 1), with bag 3 positioned at the center. Bags 6 to 10 were placed in the second set (S 2), with bag 7 at the center. This pattern was repeated successively until all 40 cavities were filled. The phantom's central plate belongs to set 4, with bag 17 located on the central axis.

A total dose plan of 25 Gy was delivered to the center of the phantom in order to ensure that no part of the component receives a dose less than 15 Gy or more than 50 Gy according to Anvisa (2014) recommendation.

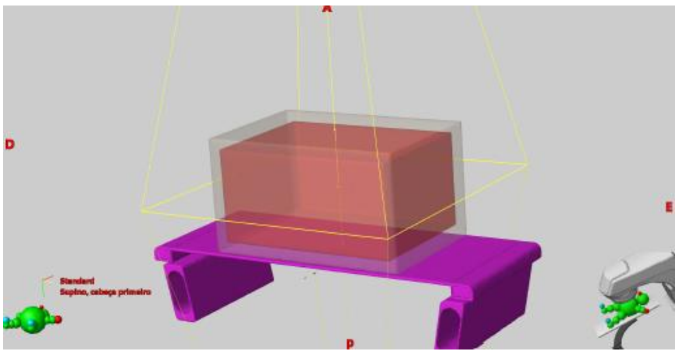


Figure 2: phantom arranged in Varian Eclipse® contouring application for dose-volumetric calculation.

Three irradiations using the same treatment plan were performed on the Varian Clinac 2100C linear accelerator, employing 6 MV photon beams at the Cancer Control University Center of the State University of Rio de Janeiro. The plan was delivered using 4fields of 641 monitor units (MU) each and 40,0 x 40,0 cm² field. The source−to−axis distance (SAD) of the phantom was set at 100 cm. For the dose distribution, two isocentric, parallel and opposing beams (anterior and posterior) were arranged to deliver a relative dose of 25 Gy at the center of the phantom (Figure 3).

Figure 3: The bags containing Fricke solution arranged in the phantom were positioned on the table of the linear accelerator.

2 Results

The mean absorbed dose in the Fricke solution is calculated using equations (1) and (2):

$$\Delta OD = (OD_i - OD_c) \cdot [1 + 0.0012.(25 - T_i)] \cdot [1 + 0.0069 \cdot (25 - T_r)] \tag{1}$$

$$D_F = \frac{\Delta \mathrm{OD}}{\mathrm{G(Fe^{3+}) \cdot L \cdot \rho \cdot \varepsilon}} \tag{2}$$

Where ∆OD in equation (1) is defined as the difference in optical density (OD), measured at 304 nm, considering the effect of temperature; OD_i and OD_c are the optical densities of the irradiated solution *i* and of the control solution, respectively; T_i is the temperature in °C of the Fricke solution during irradiation, and T_r is the temperature in °C of the Fricke solution during spectrophotometer reading.

In equation (2), $G(Fe^{3+})$ is the chemical yield of the solution for the radiation energy, established for each energy through the measurements performed; ε is the molar linear extinction coefficient of the ferric ions, equal to 2174 cm^2/mol at 304 nm; L is the optical path length of the cuvette; and ρ is the density of the Fricke solution at 25 ºC.

The readings of both the irradiated and control Fricke dosimeters were taken on the day following irradiation, using a Cary 50 Bio UV-VIS spectrophotometer and the Fricke parameters listed in Table 1. Therefore, it was possible to calculate the Fricke dose in Gy for each bag from the three irradiations performed, using the solution's absorbance at 304 nm obtained by spectrophotometry.

Table 1: Fricke parameters

Parameters	**Values with their units of measurement**
L	1 cm
$G(Fe_{+3})$	1.637 x 10^{-6}mol/J
ε	2174 cm^2/mol
ρ	1.022 g/cm^3

The 25 Gy irradiation at the center of the phantom yielded a mean dose of 24.71 ± 0.45 Gy with combined standard uncertainty of 1.82% (k = 1), which is 1.18 % lower than the expected dose of 25.0 Gy.

For statistical analysis, it is possible to evaluate the dose differentiation points relative to the three irradiations, their mean, and the Dose prescription from treatment planning system (Gy) based on Table 2.

In Irradiation A, although a maximum dose of 25.85 Gy was observed in Bag 29, corresponding to Set 6 of the phantom, the average dose for Bags 1 to 5 that positioned in the first plate set, was approximately 25.44 Gy, while the mean dose for Bags 26 to 30 was 24.85 Gy. Thus, there is a pattern of higher doses in sets 1 and 8 across all three irradiations. For the lowest doses, irradiations A, B, and C show values around the phantom's axis in sets 3, 4, and 5, respectively. Type A uncertainty was calculated as the standard deviation of the three measurements.

Table 2: Dose Statistics

Irradiation	$\boldsymbol{D_{max}}$ **(Gy)**	**Bag** $(\boldsymbol{D_{max}})$	$\boldsymbol{D_{min}}$ **(Gy)**	**Bag** $(\boldsymbol{D}_{\text{min}})$	$\boldsymbol{D_{cent,Bag\ 17}}$ **(Gy)**	$\boldsymbol{D_{mean}}$ **(Gy)**
A	25.85	29	23.15	12	24.01	24.80 ± 0.09
B	25.95	03	24.51	16	24.73	25.02 ± 0.06
C	25.26	36	23.54	23	23.98	24.30 ± 0.07
Mean (A, B e C)	25.44	-	23.87	-	24.24	24.71 ± 0.06
PTV	26.11	39	24.78	18	25.00	25.37 ± 0.06

In Figure 4, it can be observed that the dose distribution is uniform throughout most of the phantom, considering the 15 to 50 Gy range recommended by Anvisa Resolution RDC No. 34.

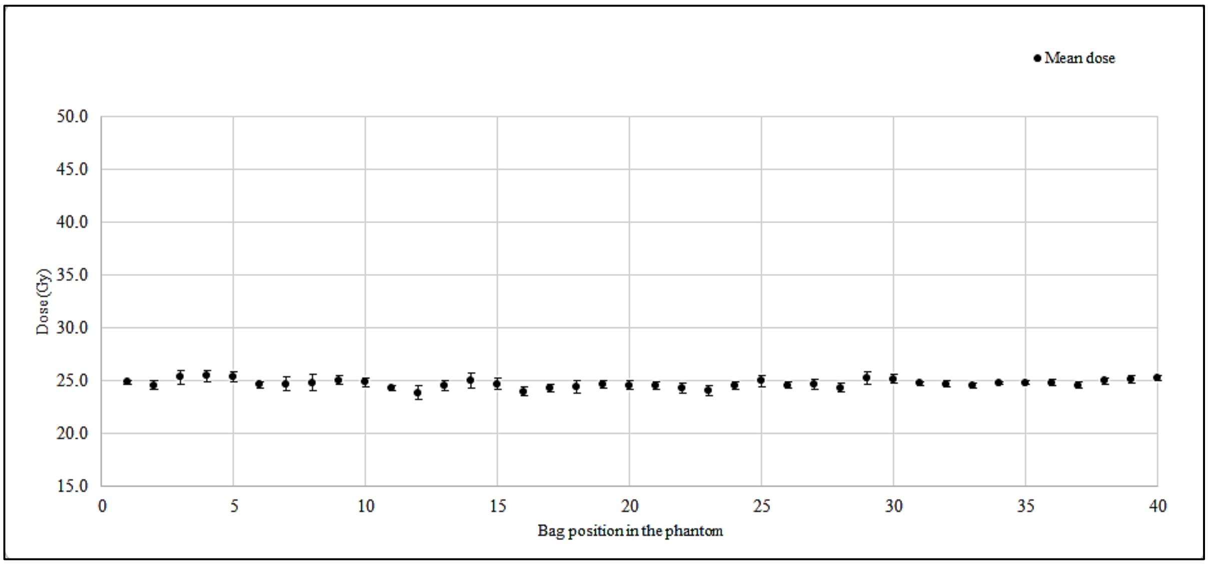


Figure 4: Mean dose distribution in the phantom

4 Conclusion

Volumetric evaluation of the dose distribution using Fricke dosimetry proved to be an effective tool for quality control in the irradiation of blood components with linear accelerators with mean dose of 24.71 ± 0.45 Gy (k=1). The results comply with ICRU Report 62 recommendations—showing uncertainties below 5%—and the dose distribution falls within the range stipulated by ANVISA's RDC No. 34, thereby meeting the annual dosimetry requirements for blood components at hemotherapy centers, hospitals, and clinics as specified by ANVISA and CNEN.

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Photons Response of an Albedo Monitor

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Abstract

The Institute of Radiological Protection and Dosimetry (IRD) operates a personal neutron monitoring service using an albedo-type monitor developed in-house[1]. This study aimed to evaluate the monitor's response to photons in terms of the operational quantity $H_p(10)$, as neutron fields from commercial sources are typically accompanied by a photon component. The monitor was modeled in Geant4 and irradiated following the ISO 4037-3 standard[2,3]. Results showed that the monitor exhibits greater sensitivity at low photon energies compared to high-energy fields. The dosimetric system includes two pairs of thermoluminescent dosimeters (TLDs) TLD 600 and TLD 700: one for detecting incident radiation and an albedo pair for detecting backscattered radiation. The simulations showed that the most appropriate algorithm for assessing the quantity Hp(10) due to photons considers only the TLD-700 detectors, as these measure the photon contribution generated by commercial neutron sources.

Keywords: Monte Carlo, Geant4, X-and gamma-rays; Dosimetry.

1 Introduction

Since the 1970s, numerous institutions worldwide have adopted albedo dosimeters for neutron dosimetry at energies up to 10 keV, by assessing the flux of neutrons backscattered from the human body through the albedo technique (Piesch, 1977).

The Institute of Radiological Protection and Dosimetry (IRD) of the Brazilian Nuclear Energy Commission (CNEN) operates a neutron individual dose assessment service using an AlNor-type albedo neutron monitor developed in-house (Martins, 2008). The monitor's sensitive elements consist of two pairs of Thermoluminescent Dosimeters (TLDs): TLD-600, which is sensitive to both neutrons and photons, and TLD-700, which is sensitive only to photons. Both TLDs exhibit similar photon sensitivity, given their nearly identical effective atomic numbers, 8.2 and 8.1, respectively (Freitas, 2018). The IRD monitor scheme is shown in Figure 1.

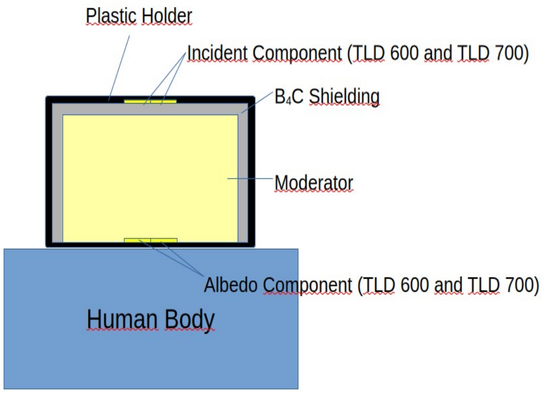


Figure 1: IRD albedo monitor's design.

Figure 1 shows both pairs of TLD, a B_4C cover to shield the albedo pair of incident neutrons and a moderating body to thermalize the albedo neutrons.

The IRD monitor was calibrated in neutron fields from various sources, such as ^{241}Am-Be, ^{252}Cf, and thermal neutrons, to assess the operational quantity Personal Dose Equivalent (Hp(10)), based on the signal difference between TLD-600 and TLD-700, considering the ratio of incident to albedo neutrons (Martins et al., 2010).

However, the sensitivity of thermoluminescent crystals to neutrons complicates the assessment of photon doses when these are comparable to neutron doses. Neutron fields from commercial sources are invariably accompanied by a photon component. In the case of the ^{241}Am-Be source, for instance, the ^{241}Am decays by alpha emission, followed by a gamma of approximately 60 keV energy. The alpha particle subsequently reacts with ^{9}Be, producing an excited ^{12}C nucleus and a neutron. The excited ^{12}C then decays by emitting a 4.4 MeV gamma photon. The predominant photon energy in this case is 60 keV (Piesch, 1977).

In the study conducted by Hoedlmoser et al., the TLD-700 measured a photon contribution from two neutron sources ^{241}AmBe and ^{252}Cf using active and passive detectors. The study verified a contribution of approximately 3% and 5% for the ^{252}Cf and Am-Be sources, respectively (Hoedlmoser et al., 2012). Measuring the photon contribution in neutron fields involves several factors, such as the detector's energy dependence and its non-negligible neutron sensitivity, as well as the presence of neutrons themselves, which poses challenges to photon measurement when their contribution is comparable to that of neutrons.

The IRD albedo neutron monitor was not calibrated to measure Hp(10) due to photons fields, including those generated by commercial neutrons sources. As a result, users of such monitors are often required to carry two different devices—one for neutrons and another exclusively for photons. For practical purposes, however, it would be ideal to use a single monitor capable of evaluating both photon and neutron doses.

Hence, the aim of the present study is to investigate, through Monte Carlo simulations, the feasibility of using this same albedo neutron monitor to assess photon doses for occupationally exposed workers using the toolkit Geant4.

2 Materials and Methods

To perform the simulations, Geant4 version 10.7.2 was installed on a notebook equipped with an Intel CORE i3 processor featuring four processing threads. A description of the monitor on the ISO 4037-3 slab phantom was implemented in the code and irradiated with a square plane parallel beam, with a 30 x 30 cm^2 cross section. Monitor, phantom and source are embedded in air.

The simulated radiation fields are monoenergetic and have mean energies corresponding to the N-20, N-30, N-40, N-60, N-80 and N-100 qualities recommended in ISO 4037-1, as well as the reference energy of ^{137}Cs. The source emitted 10^7 photons incident perpendicularly to the front face of the monitor. The dose in the monitor's sensitive elements was calculated as the ratio between the total energy deposited in the crystals by each particle and their masses.

Since the objective of this study is to evaluate the photon response of the IRD monitor in terms of the operational quantity Hp(10), this quantity was obtained by the product of the air kerma (K_{air}) at the measurement point and the conversion coefficient $h_{pK}(10)$ [Sv/Gy], which converts K_{air} into the operational quantity.

K_{air} was determined by replacing the IRD monitor with a chip (with the same dimensions as the crystals) filled with air and irradiating it with the defined radiation fields. The calculation of K_{air} is shown in Equation 1.

$$K_{air}=\sum^{n} E_i \cdot \phi_{E_i} \cdot \left(\frac{\mu_{tr}}{\rho}\right)_{E_i} \tag{1}$$

where E_i is the energy of the field, Φ is the fluence in the detector, calculated by the ratio between the sum of the particle path lengths within the detector and its volume, and $(\mu_{tr}/\rho)_{Ei}$ is the mass energy-transfer coefficient of air for each radiation field energy.

2.1 Monitor Calibration and Hp(10) Response

Initially, it was necessary to calibrate the IRD monitor in terms of Hp(10) for photon fields in order to subsequently evaluate its response. The calibration of the monitor was performed based on the combined sum of the absorbed doses in the crystals, as shown in equation 2.

$$H_m=\sum^{n} c_i \cdot D_i \tag{2}$$

where H_m is the operational quantity measured by the monitor, and c_i are the coefficients that convert the dose D_i in the i-th crystal of the monitor from gray to sievert. The coefficients were determined by applying the analytical approach of the least squares method (Equation 3) using a Python code.

$$c_i=\left(Doses^T \cdot Hp(10)\right)^{-1} Doses \cdot Hp(10) \tag{3}$$

In equation 3, Doses and Hp(10) are matrices in which the columns represent the crystals in the monitor and the rows correspond to the absorbed dose values for each simulated radiation field. These coefficients were applied in Equation 2 to ultimately obtain the personal equivalent dose on the monitor. The response was calculated as the ratio Hm/Hp(10).

The albedo monitor evaluates the operational quantity due to neutrons based on the signal difference between the TLD-600 and TLD-700 detectors. Since the TLD-700 is sensitive only to photons, the monitor was also calibrated in terms of Hp(10) considering only the TLD-700 crystals, applying the least squares method to determine the coefficients of the combined sum.

2.2 Monitor Energy Dependence

The energy dependence in Hp(10) (E.D.) of the monitor was verified from the ratio between the absorbed dose in the crystal and the Hp(10) obtained from air kerma, as shown in equation 4 (Cardoso, 2021).

$$E.D. = \frac{\left(\frac{D_{TLD}}{Hp(10)}\right)_E}{\left(\frac{D_{TLD}}{Hp(10)}\right)_{Cs}} \tag{4}$$

Where D_{TLD} is the dose in a single crystal of the monitor. The ideal energy dependence curve is flat over a given energy range, ensuring that the dosimeter does not overestimate or underestimate a quantity of interest, such as Hp(10) in the present study.

3 Results and Discussions

The values of radiation energy, Air Kerma (K_{air}), Hp(10) and the doses in each monitor TLD are shown in Table 1.

Table 1: Values of air kerma, absorbed dose in the monitor crystals, and personal equivalent dose.

Energy (keV)	**K_{air} (nGy)**	**Hp(10) (nSv)**	**D_{6i} (nGy)**	**D_{7i} (nGy)**	**D_{6a} (nGy)**	**D_{7a} (nGy)**
16.3	28.4 ± 1.2	9.0 ± 0.4	28 ± 2	27 ± 1	11 ± 1	10.1 ± 0.7
24.6	14.2 ± 0.5	11.6 ± 0.4	19 ± 1	17 ± 1	11.8 ± 0.5	11 ± 2
33.3	6.7 ± 0.2	8.1 ± 0.2	13 ± 1	10.1 ± 0.8	9 ± 1	9 ± 1
47.9	3.55 ± 0.06	5.9 ± 0.1	7.1 ± 0.1	7 ± 1	7 ± 2	6.8 ± 0.4
65.2	3.00 ± 0.09	5.7 ± 0.2	5.6 ± 0.6	5.4 ± 0.5	4.9 ± 0.6	4.9 ± 0.7
83.3	3.3 ± 0.1	6.2 ± 0.2	6 ± 1	6.3 ± 0.8	5.3 ± 0.4	4.9 ± 0.5
662	34 ± 1	41.0 ± 2	39 ± 8	34 ± 2	31 ± 7	35 ± 9

It is observed that the dose values between the crystals of the incident pair shown in Table 1 are consistent with each other and become comparable to the doses of the albedo pair starting from the energy of 33.3 keV.

Several algorithms can be used to evaluate the quantity Hp(10) due to photons. In this study, two algorithms were analyzed: in the first, all four crystals of the monitor were considered to provide (H_m), and in the second, only the TLD-700 crystals were taken into account (H_{m7}).

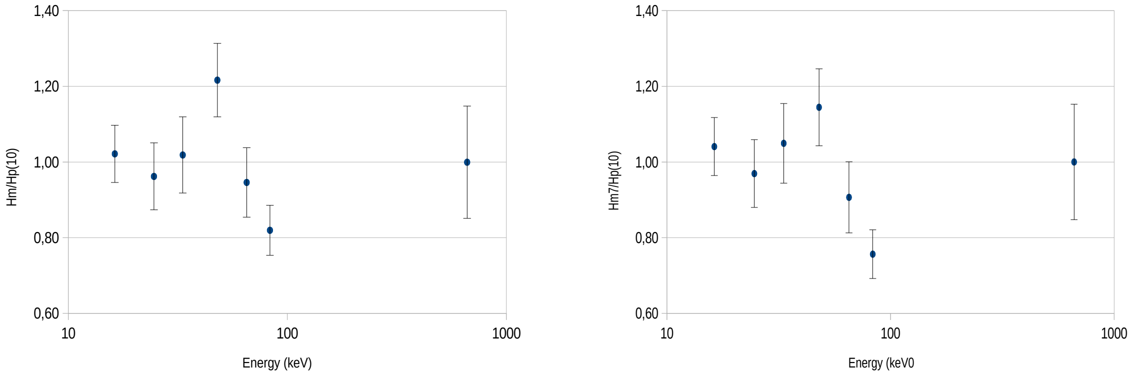
The choice of the second algorithm is based on the fact that the photon sensitivity of the TLD-600 and TLD-700 is the same. Neutron fields from commercial sources are always accompanied by a photon component. Therefore, the signal difference between these crystals is used to eliminate the photon contribution and assess only the neutron component. By calibrating the TLD-700, which is sensitive only to photons, in terms of Hp(10), it becomes possible to quantify the photon contribution.

The absorbed dose values in the crystals and the personal equivalent dose were applied in equation 2 to evaluate both combinations. The weighting coefficients for each monitor crystal are presented in table 2.

Table 2: Crystals weights coefficient for both algorithm.

Weighting factors for the four TLD (Sv/Gy)				**Weighting factors for TLD 700 (Sv/Gy)**	
c_1	c_2	c_3	c_4	c_1	c_2
-0.3065	0.2812	-0.8869	2.0125	-1.0181	2.0624

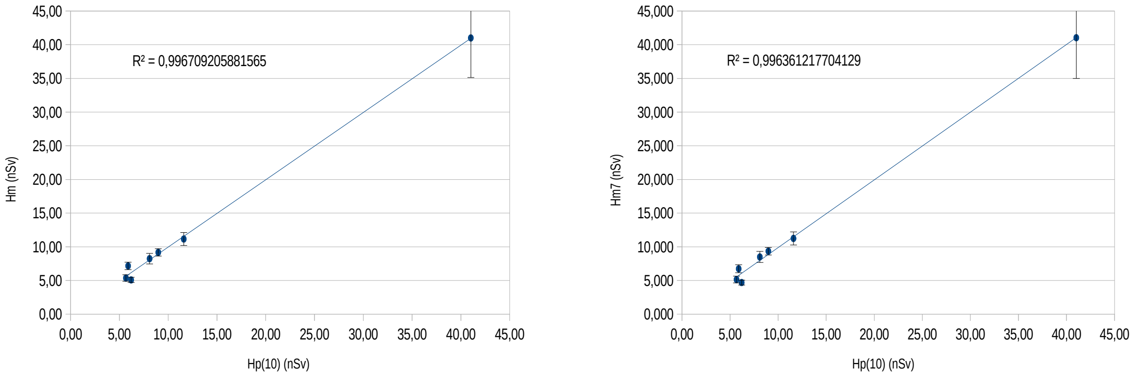
After applying the weighting coefficients and calculating the operational quantity using both combinations, the ratio H_m/Hp(10) for each beam energy was calculated and evaluated using the graph shown in Figure 2.



(a) (b)

Figure 2: Response of the IRD monitor considering both algorithm. (a) First combination: all four crystals. (b) Second combination: only the TLD-700 crystals.

Figure 2 shows that both the first algorithm, which considers all four crystals, and the second, which considers only the TLD-700 crystals, are equivalent, as there is no significant difference in the shape of the response curves. Both curves show a maximum deviation of approximately 20% from the reference value at the energies of 47.9 keV and 83.3 keV. The equivalence of each curve can be verified better in Figure 3, which presents H_m as a function of Hp(10) for both algorithms.



(a) (b)

Figure 3: Personal equivalent dose as a linear function of Hp(10). (a) First algorithm. (b) Second algorithm.

The linear fits for each algorithm are shown in Figures 3(a) and 3(b), along with the coefficients of determination, R^2 = 0.9967 and 0.9964, respectively. In both cases, the R^2 values were very close to unity, indicating that both algorithms are capable of providing a reliable estimate of Hp(10) due to photons for the radiation fields used. Thus, using the algorithm shown in equation 4, that considers only the TLD-700 crystals of the monitor, it is possible to verify and assess the photon contribution originating from neutron sources.

$$H_m = -1.0181 \cdot D_{7i} + 2.0624 \cdot D_{7a} \quad (4)$$

where D_{7i} is the absorbed dose or reading value in TLD 700 from incident pair and D_{7a} is the same for TLD 700 from albedo pair.

Figure 4 shows the energy dependence curve of the monitor obtained using equation 4.

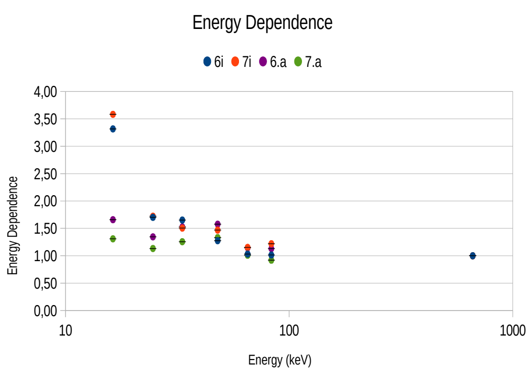


Figure 4: Energy dependence curve of the IRD monitor. The TLD-600 and TLD-700 crystals of the incident pair are presented as 6i and 7i, respectively, while the crystals of the albedo pair are presented as 6a and 7a.

The energy dependence curve indicates that, within the range of 33.3 to 83.3 keV, the monitor response overestimates the quantity by a factor of approximately 1.5, gradually converging to unity with increasing photon energy. For the incident pair, at the lowest energy evaluated, the quantity is overestimated by a factor of 3.5.

This behavior necessitates the application of complex correction factors, particularly at 16.4 keV. While some dosimeters employ suitable filtration to effectively flatten the energy dependence curve, the implementation of such filters in the IRD monitor could have implications for production costs and may require design modifications. Consequently, a thorough evaluation of this approach is essential in order to determine the most advantageous cost–benefit balance.

4 Conclusions

The response of the IRD albedo neutron monitor to photon fields was evaluated using the coefficient of determination (R^2) for two algorithms: the combined sum considering the readings from all four crystals, and the one considering only the TLD-700 readings. Both approaches are equally capable of providing a representative estimate of the personal equivalent dose due to photons for the radiation fields used in the simulations of this study, as shown in the graphs in Figure 3.

However, the operational quantity due to neutrons is assessed by subtracting the photon contribution generated by commercial neutron sources, that is, the signal from the TLD-700. Therefore, the most appropriate algorithm is the combined sum considering only the TLD-700 readings from the monitor.

The monitor exhibits a reasonable energy dependence at intermediate energies but overestimates the quantity Hp(10) by a factor of 3.5 at 16 keV. Therefore, it is necessary to assess the feasibility of applying correction factors, considering their level of complexity, or alternatively, the addition of filters, taking into account both the manufacturing costs of the filters and the potential design modifications to the monitor.

For future work, it is recommended to simulate the monitor's exposure to neutron fields and evaluate the photon contribution generated in these fields to the dose equivalent and simulate some filters combinations.

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Quantification of Gamma Radiation on Drilling Cuttings Through HPGe Detector: Comparative Study With Well Logs Spectrometry Data

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Abstract

Geophysical well logging plays a crucial role in geological monitoring, employing a suite of tools to measure the physical properties of penetrated rock formations. Among these tools, the gamma ray log stands out for its robustness in lithological differentiation and its support in porosity estimation. However, this tool is sensitive to conditions such as borehole diameter variations, washouts, caving, and signal attenuation caused by the presence of drilling fluid, factors that do not influence measurements conducted in laboratory environments. This study aimed to assess the viability of high-resolution gamma spectrometry using a High-Purity Germanium (HPGe) detector applied to drilling cuttings samples, correlating laboratory-acquired spectra with well log data. Photopic windows for K-40, U-238, and Th-232 were defined based on spectral counts acquired from the laboratory, with the objective of identifying the most representative energy peaks that align with downhole log data. Additionally, X-ray fluorescence (XRF) analysis was performed to determine the total concentrations of these elements, allowing comparison between isotopic data (HPGe) and total elemental content (XRF). Caliper, density (RHOB), neutron porosity (NPHI), and resistivity logs were analyzed to control for variables that interfere with the gamma log response, such as borehole enlargement, the presence of conductive drilling fluid, and deviations in borehole geometry. Results indicated that the 238 keV energy peak (Th-232) exhibited the strongest correlation with well log data, followed by the U-238 peak near 295.2 keV, although the latter showed greater oscillation in unstable environments. A moving average was applied to the high-resolution well log data to align it with the lower-resolution, 3-meter intervals of the drilling cuttings samples, which improved the overall correlation. These findings demonstrate that gamma spectrometry with HPGe is an effective tool for validating and complementing gamma ray well logs, providing enhanced insight into the distribution of radioactive elements and the environmental factors that attenuate their signals during drilling operations.

Keywords: Gamma Ray Spectrometry; HPGe; Well Logging; Drilling Cuttings, Campos Basin.

1 Introduction

Gamma-Ray log is part of the basic suite of tools in well logging, used for the identification of different lithologies and measurement of shale content in porous rocks, which underscores the importance of gamma-ray logging in a well environment. However operation in a drilling environment is expensive, and while gamma-ray data is standard in most operations, spectral gamma-ray (measurement of eTh, eU and K individually) is not, occurring in significant additional expenses for its operation. What in combination with the difficulties inherent to the well environment like the presence of drilling fluid and fractures, allow gamma-ray analyses in a laboratory environment through HPGe measurements on drilling cutting samples to be an alternative for spectral gamma-ray well logging tools.

The study was conducted in the distal portion of the Quissamã Formation, part of the Macaé Group, from the lower Albian to the base of the Outeiro Formation, located in the offshore area of the Campos Basin (RJ). These deposits reflect maximum flooding surfaces that mark the transition between shallow and distal marine environments.

For this study, drilling cuttings samples were selected for gamma-ray analysis on a HPGe detector, representing the interval of well 1-BRSA-1007-RJS that covers the transition from distal carbonate platform members to the onset of siliciclastic sedimentation.

Based on the obtained spectra, the best photopeak windows were defined for the elements K-40, Th-232, and U-238. These peaks were then compared with the well gamma log data to identify which components show the highest correlation between laboratory and downhole measurements. Additionally, X-ray fluorescence (XRF) analyses were performed on portions of each sample to determine the total elemental concentrations of K, Th and U, allowing a comparison between the isotopic data (HPGe) and the total elemental content (XRF).

To account for variables inherent to the borehole environment, caliper, density (RHOB), neutron porosity (NPHI), and resistivity logs were analyzed. This information was essential to assess the effects of borehole diameter variation, fluid presence, and lithological heterogeneities on the propagation and attenuation of gamma signals during well logging.

2 Materials and Methods

Through the measurement of radioelement activity in drilling cuttings using a High-Purity Germanium (HPGe) detector, it was possible to generate gamma ray curves from geological samples in a laboratory environment, using the well log gamma tool as a reference. Additional well logging tools were also employed to observe and control variables inherent to the borehole environment, such as borehole diameter, fluid presence, and formation compaction conditions. Furthermore, X-ray fluorescence (XRF) data were used as a complementary method to analyze the behavior of potassium (K), uranium (U), and thorium (Th) throughout the stratigraphy, contributing to improved geochemical characterization and correlation between laboratory data and well log measurements.

In this study, 35 drilling cuttings samples were used, representing the interval between 4449 m and 4551 m from well 1-BRSA-1007-RJS, with each sample corresponding to a 3-meter section of the borehole. For gamma spectrometry acquisition with the HPGe detector, portions of each sample were placed in polypropylene containers with a standardized volume of 140 mL and subjected to secular equilibrium.

2.1 Study Area

The study area comprises of the interval between 4449 m and 4551 m of well 1-BRSA-1007-RJS, located at the upper boundary of the Quissamã Formation, which corresponds to the distal portion of the Macaé Member, in the offshore sector of the Campos Basin.

This study focuses on the transition from a carbonate platform to a deeper marine environment, marked by the Maximum Flooding Surface (MFS), commonly referred to as the "Beta Marker." This stratigraphic boundary represents the shift from a chemically controlled, carbonate-dominated phase to a zone with greater continental influence and increased water column thickness. As a result of this deepening, there is a significant reduction in calcium concentrations, particularly in its fixed forms, as recorded in the Outeiro Formation. The transition captured within this unit reflects paleoenvironmental changes and variations in sedimentary dynamics, providing important insights into the evolution of depositional conditions in the region.

The Quissamã Formation is characterized by carbonate sediments composed of oncoids, ooids, peloids, and bioclasts, formed in high- to moderate-energy shallow marine settings. In its more distal portions, an increase in clay content (Albian shales) is observed, with a gradational transition from calcisiltites and calcilutites to marlstones and shales. The lower boundary of this formation is marked by an unconformity with the evaporites of the Retiro Formation, while its upper boundary is defined by the Beta Marker, an anomaly recorded in gamma ray and resistivity logs, which marks the contact with the overlying Outeiro Formation (Winter, 2007; Dias, 1990; Guardado, 2000).

2.2 Drilling Cuttings

Cuttings samples are rock fragments broken down by the action of the drill bit during well drilling operations. These fragments are transported to the surface by the drilling fluid, propelled by the pressure differential within the wellbore (ANP, 2005).

2.3 Drilling Fluid

Drilling fluid is the additive used during the well drilling process. It enables the upward transport of rock fragments (cuttings samples) through the annular space, the region between the drill bit and the wall of the rock formation. Components found in water-based or oil-based drilling fluids, such as bentonite, lime, potassium chloride, emulsifiers, caustic soda, among others (Growcock & Harvey, 2005), can alter the physical properties of the surrounding rocks. These alterations directly impact the response of geophysical tools, such as the gamma ray logging tool, thereby influencing the interpretation of well log data.

2.4 High Purity Germanium (HPGe)

To obtain the gamma radiation spectrum of natural radionuclides, a coaxial High-Purity Germanium (HPGe) semiconductor detector with a relative efficiency of 30% (model GC3020 by Canberra Industries) was used. A low-noise preamplifier (model 2002 CSL) was connected to a Digital Spectrum Analyzer (DSA 1000) with 8192 channels, along with the Genie 2000 software. The energy range analyzed spanned from 50 keV to 2 MeV. The activity concentration is calculated using the following equation (IAEA, 1989):

$$A = \frac{N_{ei}}{\varepsilon \cdot I_{\gamma} \cdot t \cdot m} \quad (1)$$

the activity concentration of the radionuclide under consideration (Bq/kg); Nei the total net photopeak area for the chosen transition (ROI counts), determined experimentally with the gamma spectrometry system by the software Genie-2000; ε is the counting efficiency for the specific energy considered ($\varepsilon \leq 1$); t is the counting time (s); Iγ is the gamma abundance of the radionuclide under consideration ($I\gamma \leq 1$); and is the mass of the sample analyzed (kg).

2.5 X-ray Fluorescence

The XRF analysis was performed using a Malvern Panalytical Epsilon 1 instrument, which includes an integrated computer. The samples were prepared in 3-gram portions, ground in a tungsten ball mill at a speed of 30 revolutions per minute (rpm). After grinding, the samples were sieved using a stainless-steel mesh with a 106-micrometer opening.

The relative contents of major elements SiO_2, K_2O, CaO, MgO, Al_2O_3, TiO_2, Fe_2O_3, MnO, and P_2O_5 were determined over a 900s measurement period, and the contents of trace elements U and Th were measured over a period of 660s. The obtained values were normalized to a total of 100%, then used for the calculation of the efficiency for each sample and to compare the Th, U and K contents with that of the HPGe detector.

2.6 Well Logs

Well logs are profiles generated by tools used to measure the physical properties of rocks in the well environment. These tools are essential not only for the geological study of the drilling environment but also for monitoring risk factors in the drilling process, such as determining the ideal drilling fluid density and detecting the presence of faults and washouts. Different well tools are instrumental in identifying and controlling factors that affect gamma radiation attenuation, such as the invasion of the drilling fluid and thickness of the formed mudcake, as well as borehole diameter variation.

2.6.1 Caliper

The gamma radiation emitted by the density tool is directly affected by the distance between the equipment and the wellbore wall, making caliper measurements essential for assessing the reliability of data obtained from gamma logging. This profile serves as an indicator of casing presence (reduction in diameter) or washouts (increase in wellbore diameter) (Rider, 2002).

2.6.2 Gamma-Ray (GR)

The gamma ray log measures the natural radiation emitted by the rock, originating from the radioelements 40K, which emits at 1.46 MeV, as well as those from the Uranium and Thorium families, which occur across a series of energy ranges, with emissions in the spectrum from 0 to 3 MeV (Rider, 2002). Gamma rays have various uses in well log analysis, including lithological identification, well correlation, clay content measurement in porous reservoir rocks, and even the determination of shale type based on the contribution of each radioelement to the gamma ray API (Ellis & Singer, 2008).

The emissions distributed across various energy channels through the spectrum are called photopic windows. For this study, the photopic windows of Uranium and Thorium at all their energies were evaluated to replicate the well log data in a laboratory environment, while controlling the variables inherent to the drilling environment. This includes the importance of drilling fluid density in gamma ray attenuation (Florenzano et al., 2024; Wahl, 1983), which can generate experimental differences in the data, as expressed by attenuation due to variations in borehole diameter and source-to-detector distance (Ellis & Singer, 2008).

$$N = N_{0*\ \exp(-\ \mu x \rho b)} \qquad (2)$$

Exponential attenuation law, where the attenuation of gamma-rays depends on ρ_b , that is the medium bulk density, μ is the mass absorption coefficient and x is the distance over which attenuation takes place.

2.6.3 Electrical Resistivity (RT)

Electrical resistivity is measured using tools that pass an electric current either directly or indirectly through the formation. The resistivity is strongly influenced by the interstitial pore waters (formation water), which contain dissociated salts that carry current when there is good pore connectivity. However, when there is poor connectivity between pores, the rock matrix itself has a greater influence on the resistivity value (Rider, 2002).

2.6.4 Bulk Density (RHOB)

The density tool operates based on the principle of gamma radiation emission directly into the formation. The intensity of attenuation caused by the rock, or the difference between the emitted and received radiation by the detector, generates what is known as gamma density, represented by the RHOB parameter. (Rider, 2002).

2.6.5 Neutron Porosity (NPHI)

The Neutron Porosity tool (NPHI) estimates rock porosity by measuring the interaction between fast neutrons and hydrogen nuclei, which are primarily found in pore fluids such as water. When fast neutrons collide with hydrogen atoms, they lose energy, a process known as neutron scattering. The tool detects these slowed (thermalized) neutrons, and the amount detected is proportional to the hydrogen content within the rock. Since hydrogen is abundant in water, higher neutron counts generally indicate greater porosity (Ellis & Singer, 2008).

3. Results

To allow the comparison between HPGe and XRF analysis on drilling cuttings with well log data, the centimeter scale resolution of the well log data was converted into the same resolution of the drilling cuttings by averaging.

It was obtained the isotopic concentration for eTh in the photopeaks 238, 338.3, 583, 727.2, 911.6 and 969.1 keV, eU in the photopeaks 186.2, 295.2, 351.9, 609.3, 1120 and 1764.5 keV and K in the photopeak 1460.8 keV (IAEA, 2003), they were then analyzed qualitatively and quantitatively through Spearman's Coefficient to measure which photopeaks had the best monotonic correlation with the Gamma-Ray well data. Those best photopeaks were then averaged in accordance with their radioelement emission probability (Damascena, 2015), with a Savitzky-Golay filter applied to smooth the data, suppressing high-frequency noise, allowing for a better correlation between HPGe and well log data.

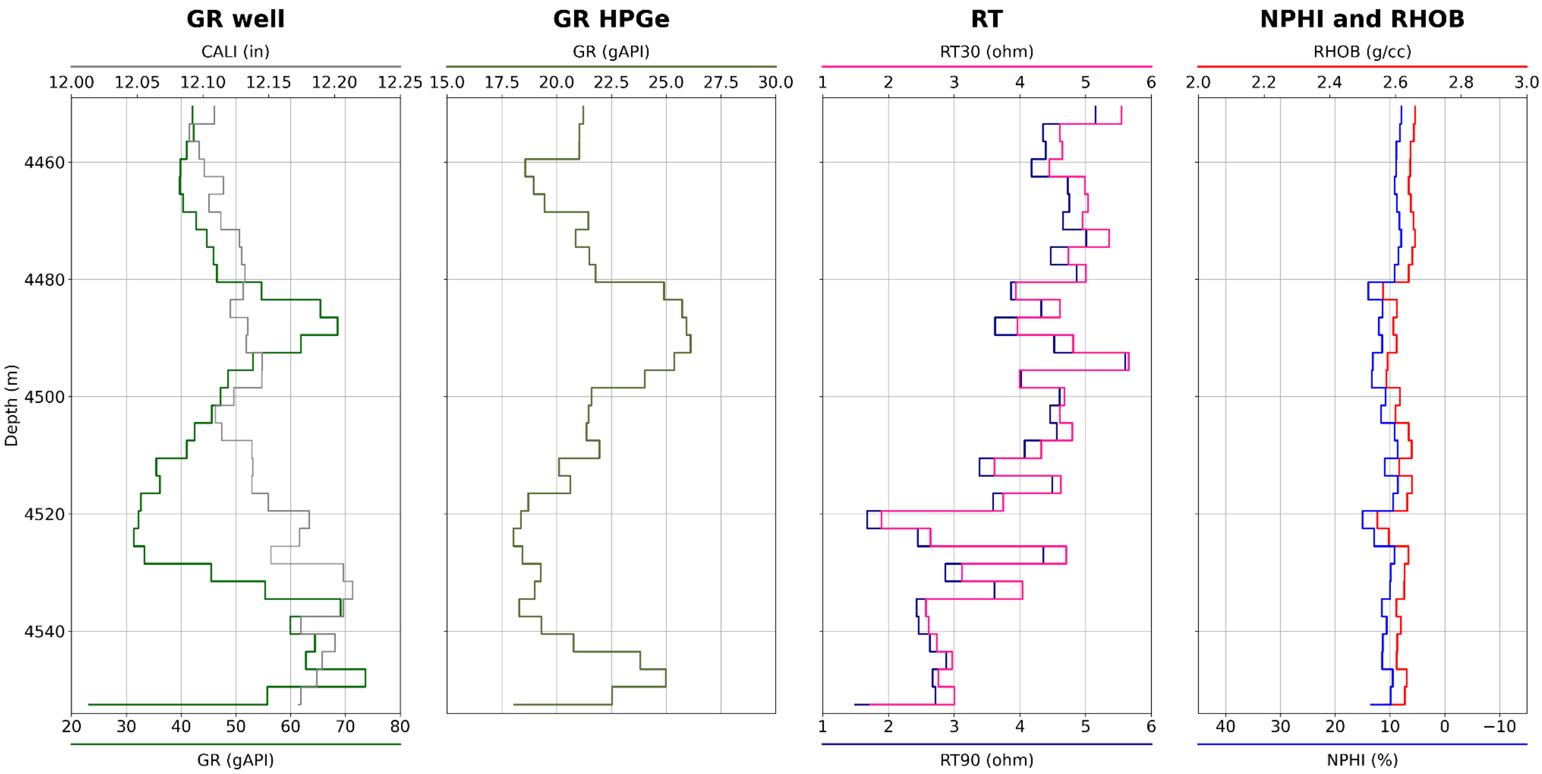

Figure 1: Well Logs properties GR and CAL (track 1), HPGe gamma through (Ellis & Singer, 2008) formula (track 2), Resistivity from 30 ohm to 90 ohm (track 3), Density and porosity correlated (track 4).

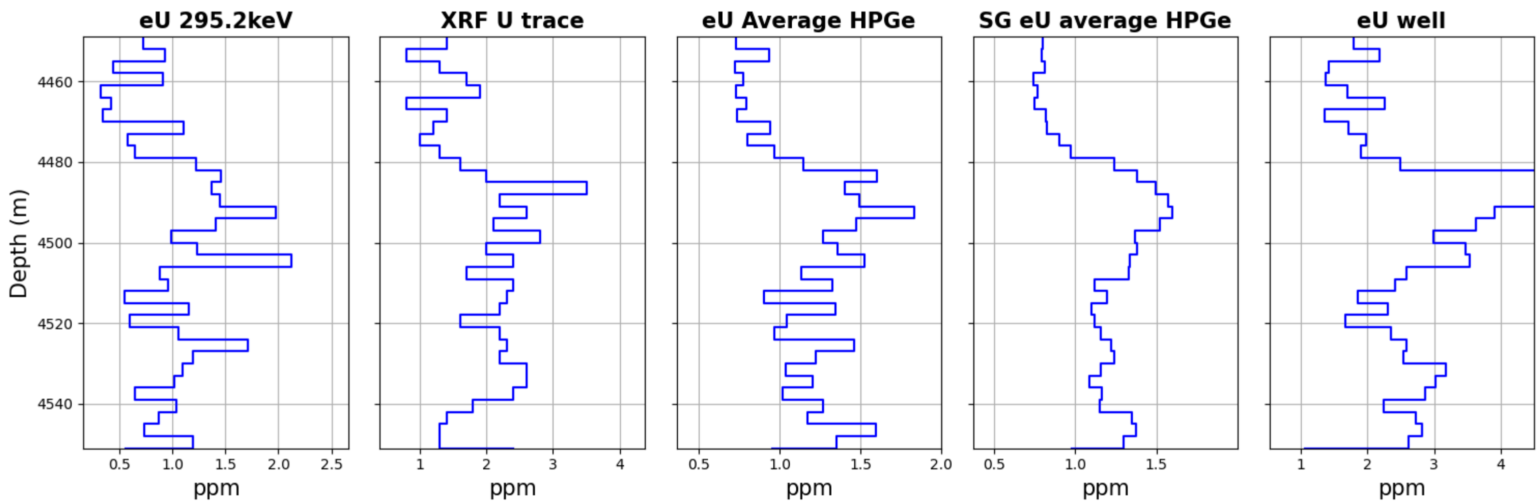

Figure 2: Concentration of eU calculated through 295 keV photopeak (track 1), Concentration of U acquired on XRF through the measurement of trace elements (track 2), Concentration calculated through the average of selected peaks of eU (track 3), Concentration calculated through the average of selected peaks of eU after application of a Savitzky-Golay filter (track 4), eU well log in well 1-BRSA-1007-RJS in the studied interval (track 5).

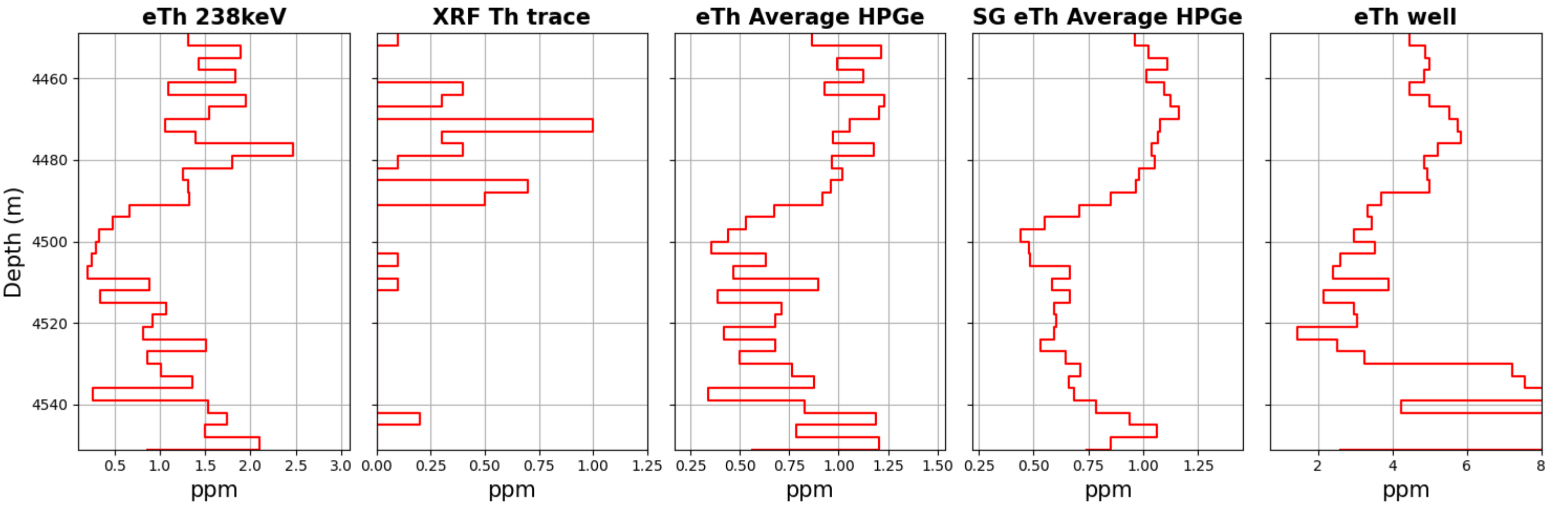

Figure 3: Concentration of eTh calculated through 238 keV photopeak (track 1), Concentration of Th acquired on XRF through the measurement of trace elements (track 2), Concentration calculated through the average of selected peaks of eTh (track 3), Concentration calculated through the average of selected peaks of eTh after application of a Savitzky-Golay filter (track 4), eTh well log in well 1-BRSA-1007-RJS in the studied interval (track 5).

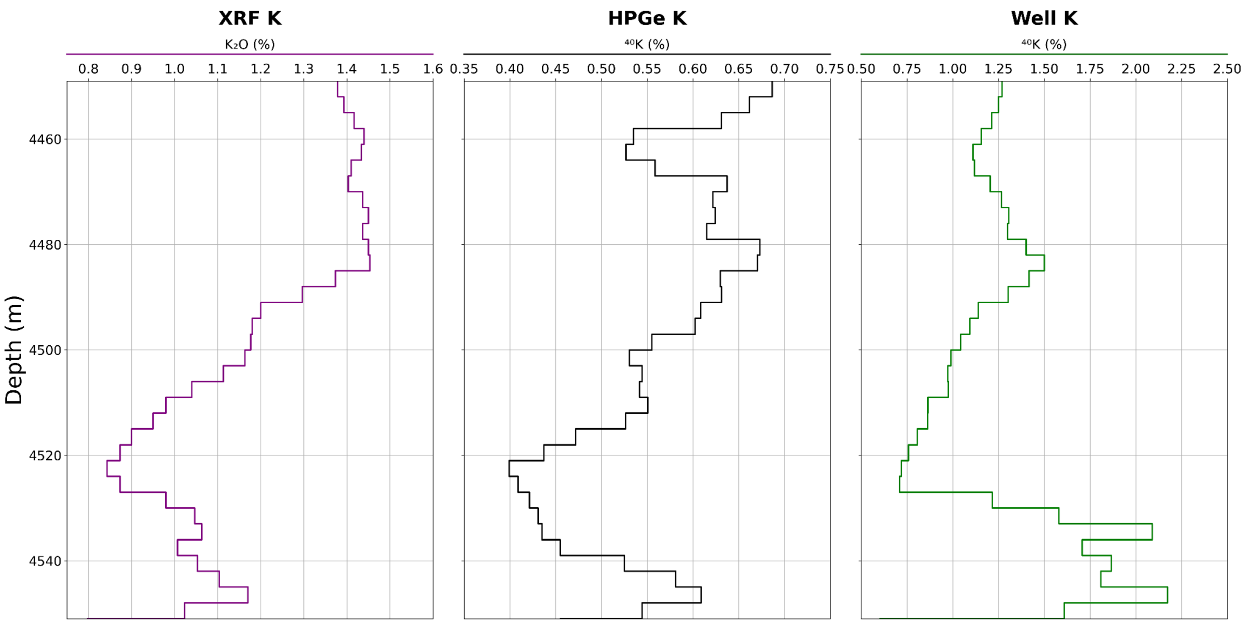

Figure 4: K content acquired on XRF through measurement of major elements (track 1), K content acquired from photopeak 1460.8 keV from HPGe (track 2), K well log from well 1-BRSA-1007-RJS in the studied interval (track 3).

4. Discussion

Analyzing the eTh measurements, the 238 keV photopeak showed the best correlation with the well log thorium data, especially in the upper half of the studied interval. For eU, the 295.2 keV photopeak had the best individual correlation, but the averages of the most correlated photopeaks generally provide more consistent correlations and values closer to the well log data, the average for eTh was calculated using 238 and 583 keV, while for eU it was used the 186.2, 295.2, 351.9 and 609.3 keV photopeaks.

The data from the well's gamma-ray tool, along with the caliper information (Figure 1), show consistency with the K, eU, and eTh values obtained by HPGe, already converted to gAPI units. The behavior of the different radioactive elements appears to be well aligned throughout the studied interval. However, between 4520 m and 4540 m, a divergence in the behavior of the logs is observed for all the analyzed radioisotopes, which is also reflected in the calculated gAPI profile (Figure 1).

The behavior of the density and neutron curves, along with the resistivity logs, appears well defined and stable, likely reflecting the absence of notable drilling fluid invasions, as well as fractures.

The analyses performed by XRF for thorium are found to be near the detection limit, likely due to the low concentration of this element in the studied interval. This is consistent with a predominantly carbonate lithology, with minimal siliciclastic contribution, indicating that the X-ray fluorescence equipment is operating near its detection threshold for this element. The obtained HPGe thorium curves (Figure 3) appear more smoothed, with lower numerical amplitude, reflecting a more natural pattern when compared to the well log data, which show higher and more irregular values around 4540 m, this was further reinforced by the utilization of a Savitky-Golay filter on the logs of concentration achieved by the average of the best photopeaks.

The same pattern is observed in the measurements of uranium (Figure 2) and potassium (Figure 4), which show strong correlations according to Spearman's coefficient and maintain compatible scales between the XRF and HPGe data. This behavior contrasts with that of thorium, whose response is limited due to the proximity to the detection limit of the XRF equipment.

5. Conclusion

This study demonstrated a strong correlation between HPGe gamma-ray spectral analyses of drilling cuttings and well log data, particularly for potassium, uranium, and thorium. The photopeaks at 238 keV (thorium), 295.2 keV (uranium), and 1460.8 keV (potassium) stood out as the most reliable indicators. Weighting these peaks by their emission probabilities, combined with applying a Savitzky-Golay filter, resulted in smoother and more consistent profiles.

Observed discrepancies between 4520 m and 4540 m suggest that factors like wellbore geometry and mudcake thickness affect measurements and should be considered in future interpretations. Additionally, XRF analyses were consistent for potassium and uranium but limited for thorium due to detection limits.

Overall, this work highlights the value of integrating laboratory and well log methods for detailed radioelement characterization in challenging downhole environments, presenting HPGe as a valid alternative for Spectrometric gamma-ray well log tools as well as total content analysis (XRF).

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Optimization of dose rate map acquisition time for internal dosimetry in nuclear medicine using 3D U-Net convolutional neural networks

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Abstract

Personalized internal dosimetry plays a crucial role in optimizing treatment planning and monitoring in radionuclide therapy. However, generating dose rate maps through Monte Carlo simulations remains a computationally demanding and time-intensive task, limiting its adoption in routine clinical practice. This work explores the application of convolutional neural networks (CNNs) to accelerate the generation of dose rate maps without compromising accuracy. A 3D UNet architecture is being trained on a synthetic dataset generated using SIMIND and GATE Monte Carlo simulation codes, modeling various biodistributions of Lutetium-177 (Lu-177). These simulations provide ground-truth dose distributions for training the neural network. The model aims to learn the mapping from SPECT like input data to corresponding 3D dose rate distributions, significantly reducing the time required for dose estimation. The next phase of the study involves model tuning to improve accuracy and generalization, followed by validation using real patient datasets. Key performance metrics will include dose distribution agreement and computational efficiency compared to standard Monte Carlo methods. Preliminary results suggest that CNN based prediction of dose rate maps may offer a reliable and rapid alternative to traditional simulations. This approach has the potential to streamline the dosimetry workflow, enhance clinical decisionmaking, and support the broader implementation of personalized dosimetry in nuclear medicine therapies.

Keywords: 3D U-Net; Dose Rate Maps; Clinical Internal Dosimetry

1 Introduction

Radionuclide Therapy (RNT), also referred to as metabolic radiotherapy or radiopharmaceutical therapy, is a well-established and continuously expanding therapeutic modality within the arsenal of nuclear medicine. It involves the administration of radiopharmaceuticals radioactive substances linked to pharmaceutical compounds that, due to their pharmacokinetic properties, are selectively targeted to specific cells or tissues, such as tumors or inflammatory sites. Once administered, the radionuclides emit ionizing radiation, depositing energy directly into pathological structures with the aim of inducing cellular damage and, consequently, eliciting a therapeutic response Sapienza and Willegaignon (2019); Siegel et al..

The effectiveness of RNT is tied to its ability to deliver a sufficient radiation dose to the target tissue to eradicate or control disease, while minimizing exposure and potential damage to surrounding healthy tissues and critical organs. In this context, personalized internal dosimetry emerges as a critical component, enabling the quantification of the absorbed dose and thus optimizing the balance between therapeutic benefit and toxicity Stabin (2017).

Understanding the dose–response relationship is fundamental to all forms of radiotherapy, and RNT is no exception. Tailoring treatment to the individual characteristics of each patient—through anatomical, physiological, and radiopharmaceutical biodistribution data is increasingly recognized as essential for optimizing clinical outcomes. Personalized internal dosimetry seeks to overcome fixed-activity administration schemes or those based solely on simplified parameters (e.g., body weight), advancing toward a treatment planning strategy adapted to individual needs Sapienza and Willegaignon (2019); Berdeguez; Massicano (2010).

Significant advances in medical imaging technologies such as Positron Emission Tomography (PET), Single-Photon Emission Computed Tomography (SPECT), Computed Tomography (CT), and Magnetic Resonance Imaging (MRI) have provided increasingly sophisticated tools for characterizing the biodistribution of radiopharmaceuticals and delineating patient anatomy information that serves as the basis for dosimetric calculations. However, converting this information into precise, personalized dose estimates typically demands computationally intensive methods. Robust techniques such as Monte Carlo simulations are considered the gold standard in terms of accuracy, but they require considerable processing time, especially when applied to complex geometries derived from medical images Berdeguez; Kim et al. (2025); Massicano (2010).

This high computational cost poses a major bottleneck to the routine clinical implementation of personalized internal dosimetry. The need to optimize computation time without compromising dosimetric accuracy represents an urgent challenge. In recent years, Artificial Intelligence (AI), particularly Deep Learning (DL) techniques such as Convolutional Neural Networks (CNNs), has emerged as a promising approach to tackle complex problems across several domains, including nuclear medicine and dosimetry LeCun et al. (2015); Goodfellow et al. (2016); Yang et al. (2025); Kim et al. (2025); Lee et al. (2019). The capacity of these techniques to learn intricate patterns from large volumes of data opens new perspectives for process optimization and the extraction of meaningful information from medical images Litjens et al. (2017). In dosimetry, AI has been investigated for tasks such as automatic organ segmentation, prediction of radiopharmaceutical biodistribution, and, crucially, for accelerating dose calculation Kim et al. (2025); Yang et al. (2025); Clement David-Olawade et al. (2025); Vali (2024); He et al. (2016); Lee et al. (2019); Seo et al. (2020).

This research aims to develop a methodology based on convolutional neural networks to significantly reduce the time required to generate absorbed dose rate maps in personalized internal dosimetry from SPECT/CT images. The objective is to assess whether the model can reliably reproduce the results obtained from Monte Carlo simulations.

2 Methodology

This section describes the procedures adopted to develop a convolutional neural network model for predicting dose rate maps in radionuclide therapy (RNT). The methodology encompasses the generation of simulated data based on SPECT/CT images and dose maps obtained via Monte Carlo simulations, as well as the training and evaluation of the model.

2.1 Data Generation for Simulation and Training

A convolutional neural network (CNN) requires data that represent different clinical scenarios. To this end, ten patient profiles were designed, each with distinct biodistributions of the radionuclide ^{177}Lu, commonly used in targeted therapies for patients with metastatic castration-resistant prostate cancer Hennrich and Eder (2022).

A range of biological half-lives (T_{bio}) was defined between 25 h and 34 h, with 1-hour increments. These values were informed by studies such as that of Kurth et al. Kurth et al. (2018), which reported T_{eff} values ranging from 21.7 h to 85.7 h. The effective half-life (T_{eff}) was calculated according to the equation:

$$\frac{1}{T_{\text{eff}}} = \frac{1}{T_{\text{bio}}} + \frac{1}{T_{\text{phys}}} \tag{1}$$

where $T_{\text{phys}} = 161.52$ h for ^{177}Lu. To compute the residual activity over time, the effective decay constant was used, defined by $\lambda_{\text{eff}} = \ln(2)/T_{\text{eff}}$.

The Zubal torso phantom Zubal et al. (1994) was employed for patient simulation. This model is frequently used in internal dosimetry simulations and medical image reconstruction studies.

The time points selected for SPECT image simulation were: 4 h, 24 h, 96 h, and 168 h after ^{177}Lu administration. The choice of these time points was based on guidelines from *MIRD Pamphlet No. 26* Ljungberg et al. (2016) and the EANM Sjögreen Gleisner et al. (2022).

Monte Carlo Simulation of SPECT Images Using SIMIND

The SPECT acquisition simulations and density map generation were performed using the Monte Carlo code *SIMIND*, version 8.0 Ljungberg (2012). This software enables a detailed modeling of the image acquisition process, including the physical interactions of radiation with matter and the response of the gamma camera detector.

Each simulation was configured using the `CHANGE` program, which is responsible for generating the input files with the `.smc` extension. For each of the ten biodistribution profiles and four time points, a text file named `phantom.zub` was created to define the activity values for each organ in the phantom. SIMIND was set up to use this file as both the density and activity source model.

A tumor lesion was incorporated into the liver using the file `tumor_liver.inp`. As described in the SIMIND manual Ljungberg (2012), `.inp` files allow for the insertion of volumes with specific geometries and defined activity levels at predetermined locations within the phantom.

The energy windows adopted, defined in the `spect.win` file, were configured for the radionuclide ^{177}Lu, using the scoring routine `Scattwin` (index 84 from menu 2 in `CHANGE`) for scatter correction via the Triple Energy Window (TEW) method. Table 1 presents the energy limits employed.

Table 1: Energy windows defined for TEW correction.

Window	Lower Limit (keV)	Upper Limit (keV)	208 keV Peak %
Lower Scatter Window	166.4	187.2	20
Main Photopeak Window	187.2	228.8	10
Upper Scatter Window	228.8	249.6	10

The parameters used for simulating the projection data at each time point and the attenuation maps are summarized in Table 2.

Table 2: Main simulation parameters.

Parameter	Value/Description
Radionuclide	^{177}Lu
Lesion	Liver tumor
Phantom image file	vox_man
Source image file	vox_man
Phantom type	-2 (Zubal vox_man)
Source distribution type	-2 (Zubal vox_man)
Collimator	MEGP
Scoring routine	1 (scattwin)
Number of histories per view	1×10^9

SPECT Reconstruction and Activity Map Generation

Tomographic Reconstruction with OSEM

After the SIMIND simulations, the SPECT projections were reconstructed into three-dimensional images using the Python library `Pytomography`. The *Ordered Subsets Expectation Maximization* (OSEM) algorithm was employed with 10 iterations and 4 subsets.

The reconstruction was performed with the following corrections:

- **Attenuation Correction**: using attenuation maps generated by SIMIND;
- **CDR Correction**: estimated from metadata provided by SIMIND; and
- **Scatter Correction**: applied using the Triple Energy Window (TEW) method;

Activity Map Generation

To convert the counts into activity values (MBq/voxel), a calibration factor was applied:

$$\mathrm{CF} = \frac{1}{S \cdot t} \tag{2}$$

where S is the system sensitivity (cps/MBq) and t is the acquisition time per projection (s).

2.2 Dose Rate Map Generation with OPENGATE/Geant4

To generate the dose rate maps, simulations were performed using GATE 10, which provides a Python interface for the Geant4 toolkit.

The procedure followed the structure defined in the `dose_rate.py` script, made available by the OpenGATE collaboration.

The `dose_rate.py` script, distributed with GATE 10, simplifies the configuration and execution of dosimetry simulations for radionuclide therapy.

Simulation Configuration

The simulation input is defined by a JSON file containing essential parameters, including:

- `ct_image`: the density distribution image generated by SIMIND;
- `activity_image`: the reconstructed activity map;
- `table_mat` and `table_density`: HU-to-material and HU-to-density conversion tables provided with GATE 10;
- `radionuclide`: `Lu177`;
- `activity_bq`: specific to each patient and time point;
- `density_tolerance_gcm3`: density tolerance for mapping; and
- `number_of_threads`: 10;

The following command line is used to start the simulation:

```
python dose_rate.py dose_rate.json -o output/
```

where:

- `python dose_rate.py` executes the simulation script;
- `dose_rate.json` is the configuration file containing the simulation parameters;
- `-o output/` defines the output directory where the generated files will be saved.

2.3 Convolutional Neural Network

This section describes the methodology used to develop a three-dimensional convolutional neural network capable of predicting dose rate maps from activity maps, providing a fast alternative to Monte Carlo simulations.

3D UNet Architecture

The network structure follows the 3D UNet model Ronneberger et al. (2015), composed of an encoder (contracting path) and a decoder (expanding path) connected by *skip connections*. Each block consists of 3D convolutions followed by ReLU activation. Downsampling operations are performed using `MaxPool3D`, and upsampling with `ConvTranspose3D`. The final layer applies a convolution to produce the predicted volume with a single output channel representing the dose rate map.

Data Preparation and Augmentation

The training dataset consists of pairs of 3D volumes: activity maps (input) and dose rate maps (label). All images have a resolution of $128 \times 128 \times 128$ voxels.

Preprocessing: activity maps are normalized to the range $[0, 1]$.

Data Augmentation: applied only to the training set using the TorchIO library. The transformations include:

- Random rotations (`RandomAffine`);
- Flipping (`RandomFlip`);
- Elastic deformations (`RandomElasticDeformation`);
- Gaussian noise (`RandomNoise`).

Training

The model was trained using the PyTorch framework with the following parameters:

Table 3: Training configurations

Parameter	**Value**
Loss function	MSE, SSIM
Optimizer	Adam
Learning rate	10^{-4} (adjusted with scheduler)
Number of epochs	Defined by convergence
Batch size	Adjusted according to GPU memory
Monitoring	Save best-performing model

Performance Evaluation

The model performance was evaluated on the test set using quantitative metrics that assess error, structure, and correlation, as shown in Table 4.

Table 4: CNN model evaluation metrics

Metric	**Description**
MAE	Mean Absolute Error
MSE	Mean Squared Error
SSIM	Structural Similarity Index between maps

3 Preliminary Results

This section presents the results obtained thus far using the proposed methodology.

3.1 Data Generated for Simulation

The input data used for the simulations conducted in this study are presented below. These datasets were designed to represent different profiles of radiopharmaceutical uptake and clearance, simulating realistic clinical scenarios for ^{177}Lu.

Table 5 shows the activity values (in MBq) for each of the ten simulated patients at 4 h, 24 h, 96 h, and 168 h post-administration.

Table 5: Generated data for 10 patients with different biological half-lives.

Patient ID	T_{bio} **(h)**	**Activity (MBq) in the image at:**			
		4 h	**24 h**	**96 h**	**168 h**
1	25	6509	3427	341	34
2	26	6537	3516	377	40
3	27	6563	3601	415	48
4	28	6587	3681	453	56
5	29	6609	3757	492	64
6	30	6631	3829	531	74
7	31	6650	3899	570	83
8	32	6669	3965	610	94
9	33	6686	4027	649	105
10	34	6703	4088	689	116

4 SPECT Simulation with SIMIND

SPECT images were generated through Monte Carlo simulations using the SIMIND software with the parameters defined in Table **??**.

To adequately represent the distribution of the radiopharmaceutical ^{177}Lu, the data from Table 5 were used, resulting in a total of 40 SPECT projection datasets. The definition of the energy windows was also included for subsequent scatter correction using the TEW method, according to the parameters listed in Table **??**.

The simulations yielded projection data representative of the spatial distribution of activity within the phantom. From the metadata in the SIMIND output files, relevant information was extracted for the reconstruction step. Figure 1 presents a set of simulated projections.

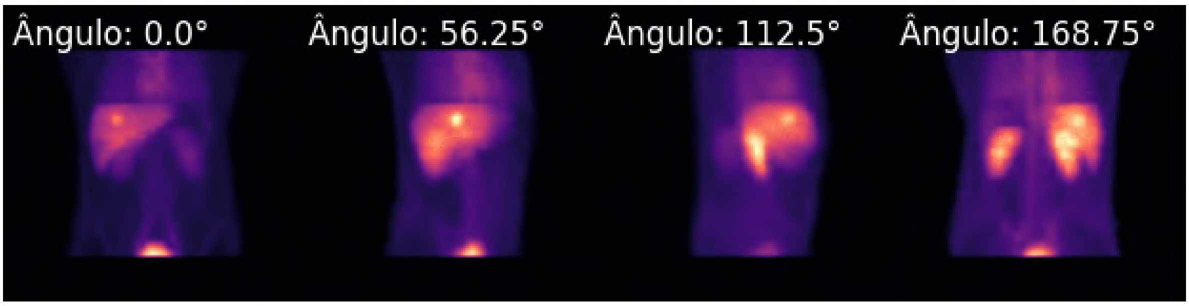

Figure 1: Simulated projections with SIMIND in the photopeak window at different acquisition angles.

5 SPECT Reconstruction and Activity Map Generation

The simulated projections were reconstructed to obtain three-dimensional activity distribution maps. The reconstruction process included corrections for physical effects that degrade image quality, such as attenuation, scatter, and collimator–detector response.

The image output from SIMIND, shown in Figure 1, is expressed in units of *counts/second/MBq*, since each simulation is normalized to 1 MBq with an acquisition time of 1 second per projection. To proceed with the reconstruction, these normalized projections were converted to *counts* by scaling the data with the administered activity and acquisition time per projection, yielding the expected average number of counts. To simulate the statistical variability inherent to the detection process, Poisson noise was applied. Figure 2 shows the adjusted image.

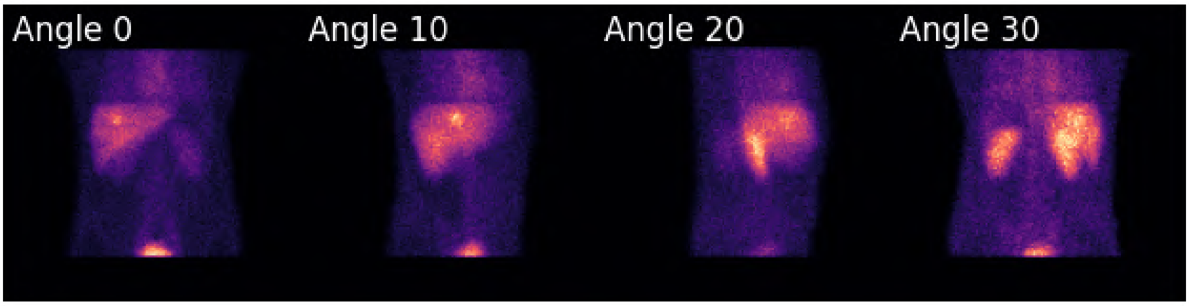

Figure 2: Projections converted to *counts*.
Source: Author

Attenuation correction was performed using the attenuation map generated by SIMIND, as shown in Figure 3.

Figure 3: Transverse slice of the attenuation map used in SPECT image correction.
Source: Author

Figure 4 shows the estimated Compton scatter, calculated from the energy windows adjacent to the photopeak.

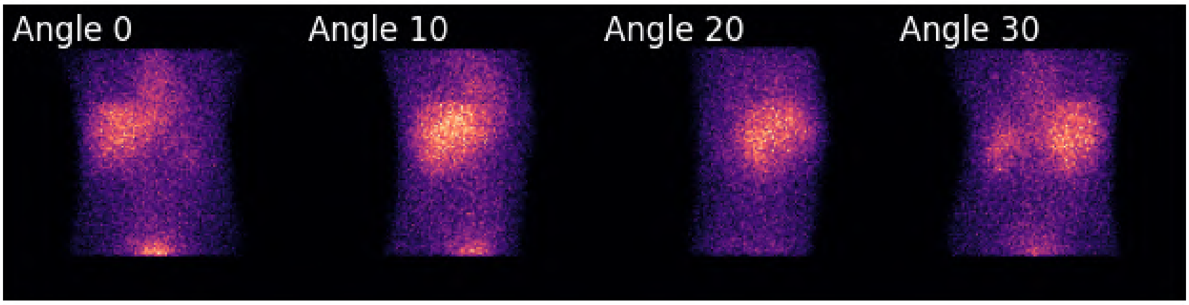

Figure 4: Images showing the estimated Compton scatter for various projection angles.
Source: Author

Due to this effect, the image contains an increased number of photons registered in the photopeak window. To correct for this, the TEW method was applied. Figure 5 displays the projection at angle 0 before and after the correction.

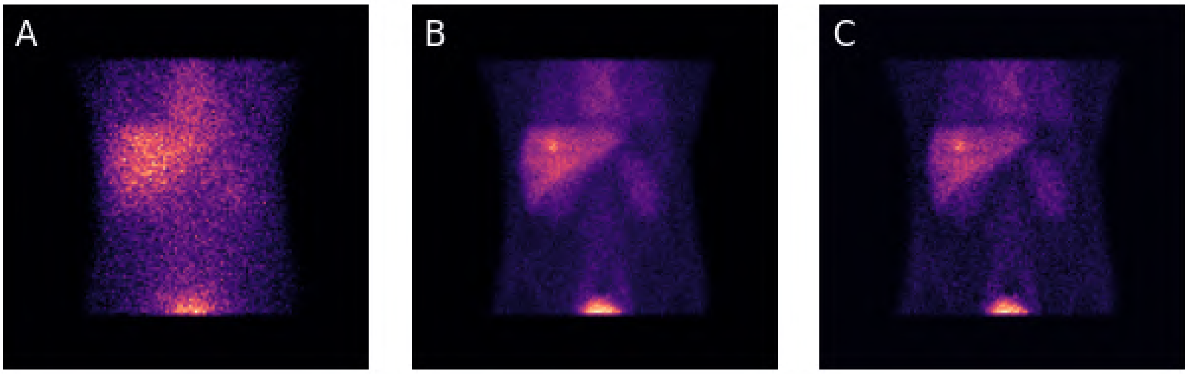

Figure 5: A - Compton scatter image, B - image without scatter correction, and C - image after TEW correction.
Source: Author

After applying these corrections, the reconstruction was performed using the *OSEM* algorithm with 10 iterations and 4 subsets.

The activity maps were obtained after reconstruction by converting the image units from *counts* to MBq using equation 2. Figure 6 illustrates the reconstructed SPECT image slices transverse, coronal, and sagittal, showing the simulated radiopharmaceutical distribution and the liver lesion region.

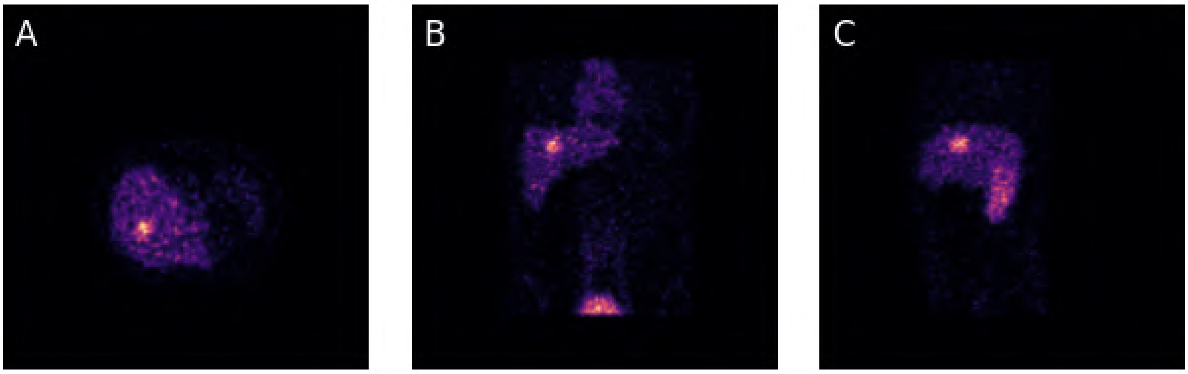

Figure 6: Slices A - transverse, B - coronal, and C - sagittal of the reconstructed SPECT image.
Source: Author

Figure 7 shows the SPECT image fused with the attenuation map.

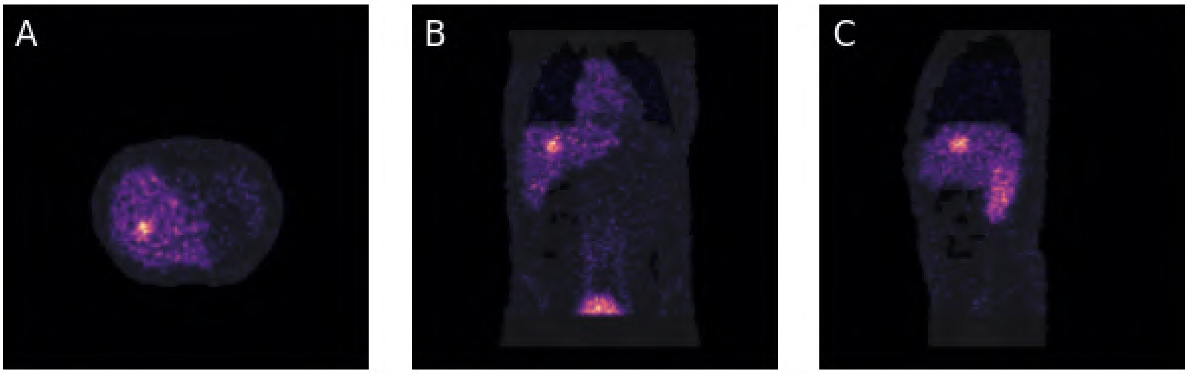

Figure 7: Fused SPECT/CT image slices in planes A - transverse, B - coronal, and C - sagittal.
Source: Author

5.1 Dose Rate Maps

Dose rate maps were generated using GATE 10 with the *dose_rate* script. These simulations used as input the reconstructed SPECT images converted into activity maps. Figures 8 and 9 show the dose rate maps for the first patient at 24 hours and 168 hours post-injection, respectively.

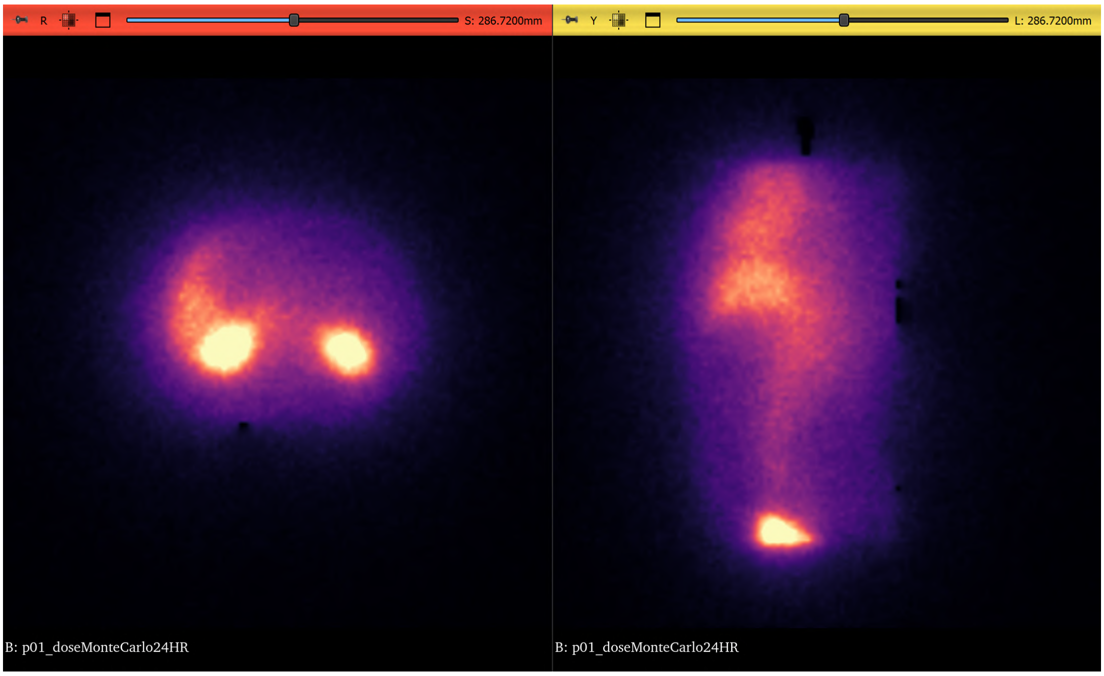

Figure 8: Dose rate map at 24 hours post-injection: transverse slice on the left and coronal slice on the right.
Source: Author

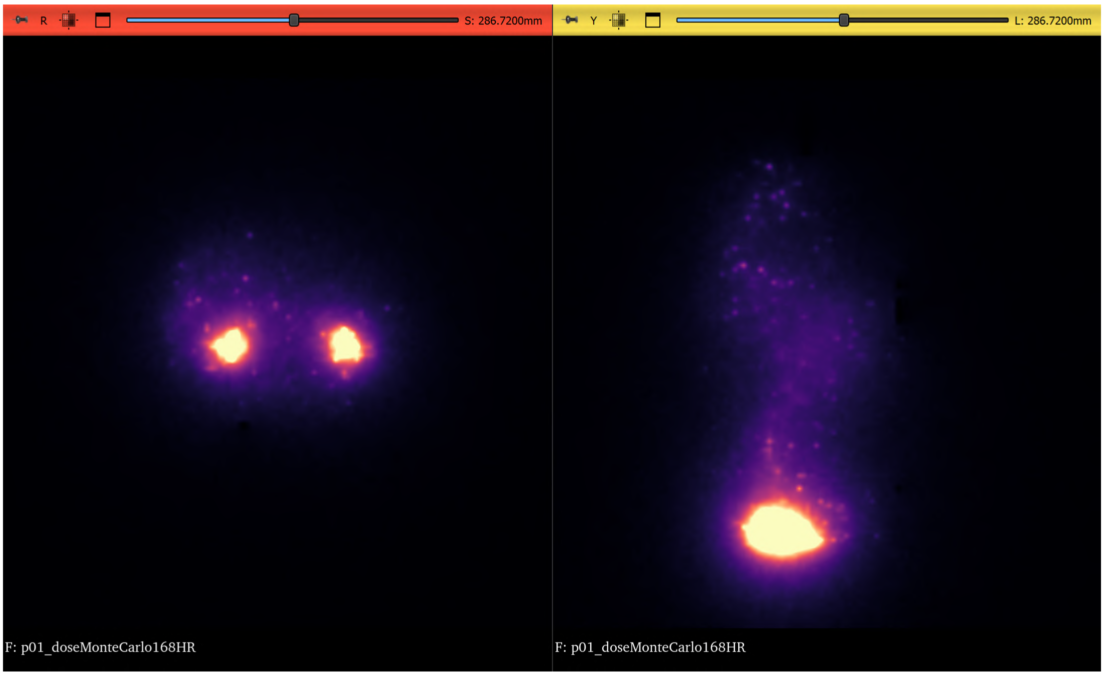

Figure 9: Dose rate map at 168 hours post-injection: transverse slice on the left and coronal slice on the right.
Source: Author

6 Convolutional Neural Network

The following presents the evaluation results of the CNN model. Performance analysis was conducted through 3-Fold Cross-Validation, using quantitative metrics and observations on the model's behavior during training and inference.

6.1 Training and Validation Performance

The loss function evolution was monitored for both training and validation data throughout the epochs for each fold. Figure 10 shows the loss curves for all three folds, revealing a rapid decrease in the cost function during the initial iterations. The proximity between training and validation curves across epochs indicates good generalization capability of the network, with no evidence of overfitting.

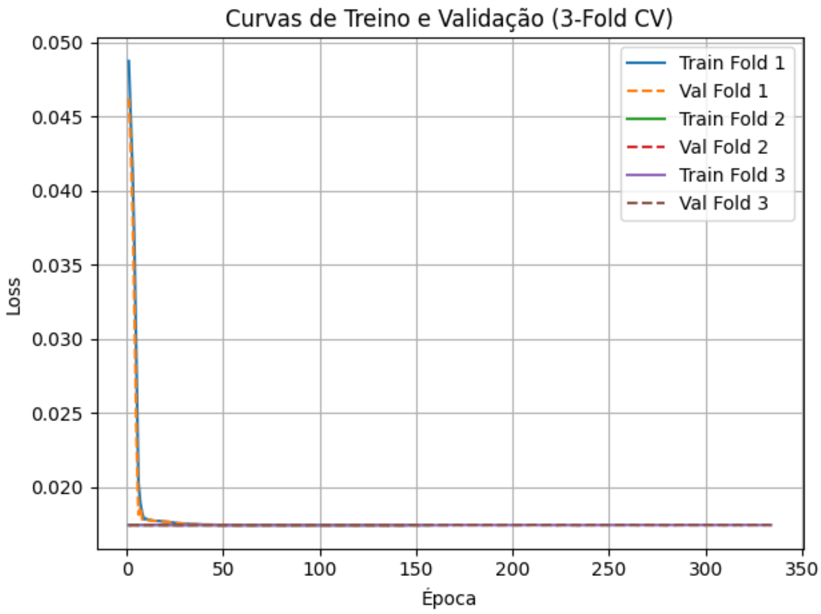

Figure 10: Loss curves for training and validation sets over the epochs.

6.2 Quantitative Evaluation

Model accuracy was assessed using the Mean Absolute Error (MAE), Mean Squared Error (MSE), and Structural Similarity Index (SSIM Approx). These metrics were computed per patient within each fold of the cross-validation.

The MAE values showed low dispersion and consistency across patients and folds, with an average around 0.0399. This stability demonstrates that the network produces dose estimations compatible with those generated by Monte Carlo simulation. Figure 11 presents a heatmap of the MAE distribution, enabling intra- and inter-fold variability analysis.

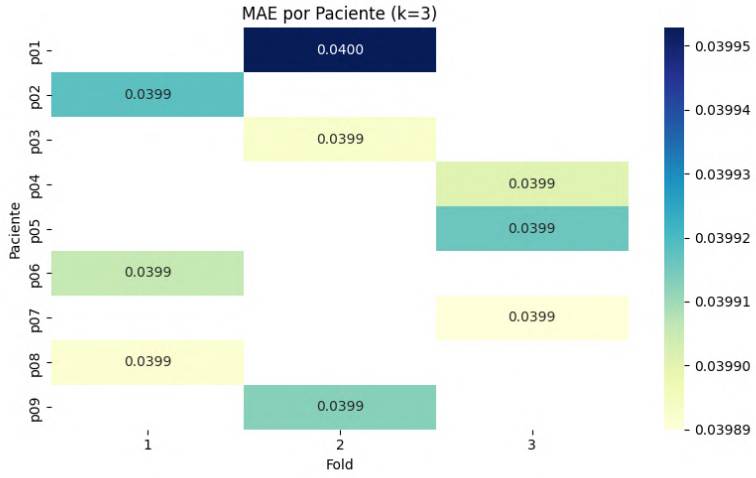

Figure 11: MAE distribution per patient and fold.

Figure 12 illustrates MSE values per patient. MSE remained low (around 0.0025), indicating high predictive accuracy.

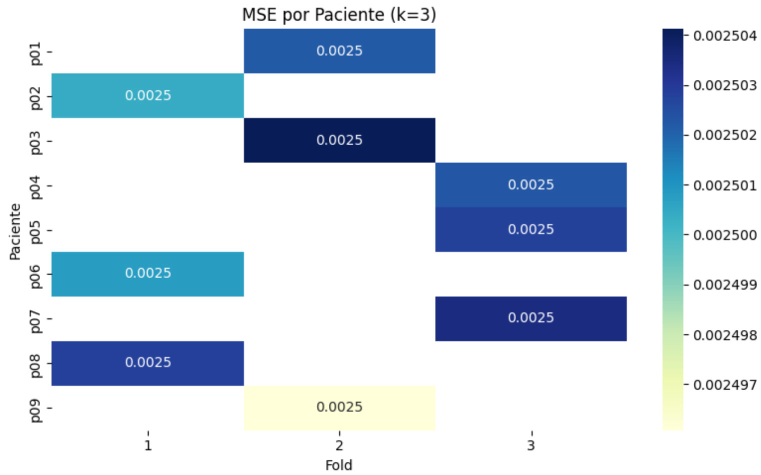

Figure 12: MSE heatmap per patient and fold.
Source: Author

Figure 13 shows the SSIM metric distributed by patient and fold. Values near zero, oscillating between positive and negative, suggest limitations of this metric in capturing subtle structural differences in three-dimensional maps. This observation highlights the importance of critically evaluating the suitability of this metric for the problem at hand.

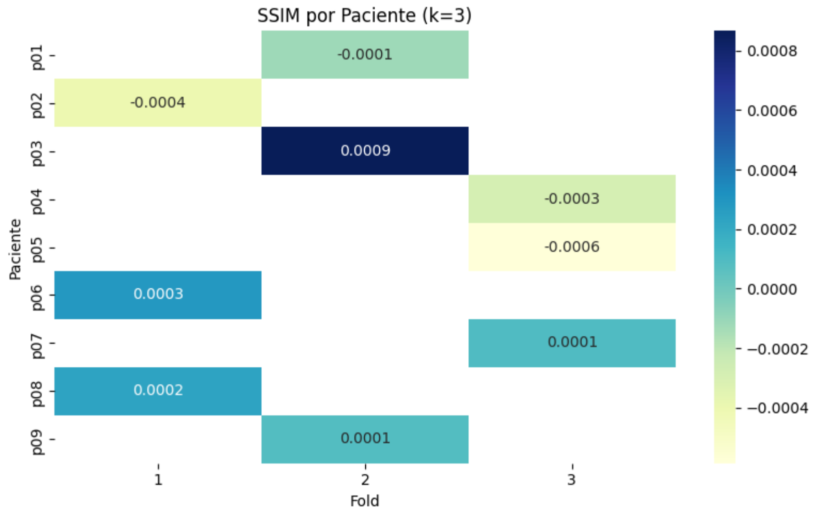

Figure 13: SSIM Approx heatmap per patient and fold.
Source: Author

6.3 Qualitative Evaluation of Predictions

In addition to the quantitative analysis, a qualitative inspection of the predictions was performed to assess spatial and morphological fidelity. As an illustrative example, patient 10 was selected, for whom both predicted and reference dose maps were generated. A visual comparison of axial, sagittal, and coronal slices allows evaluation of the network's ability to reproduce spatial dose distribution and capture anatomical variations.

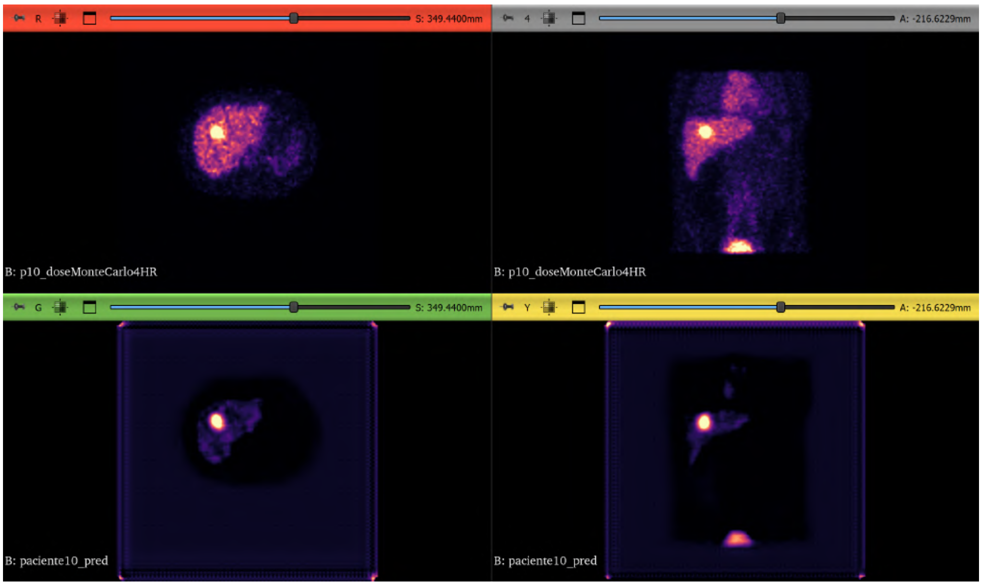

Figure 14: Comparison between the reference (Ground Truth) and CNN-predicted dose rate maps for patient 10.
Source: Author

7 Conclusions

This work presented the development of an approach based on three-dimensional convolutional neural networks for predicting absorbed dose rate maps, with the aim of accelerating the process of personalized internal dosimetry in radionuclide therapy.

Thus far, the results obtained demonstrate that the neural network based on the 3D UNet architecture is capable of reproducing with good fidelity the spatial structure of the simulated dose rate maps, with low visual discrepancy between the predictions and the references. The network was trained and validated using simulated SPECT images and dose rate maps generated with GATE 10. Quantitative metrics indicate stability and generalization capability even with a reduced dataset, enhanced by the application of data augmentation techniques.

However, it should be emphasized that the results presented are preliminary. Although the spatial structure of the predicted maps is well represented, the dose values estimated by the CNN still show inconsistencies when compared to the Monte Carlo references.

Next Steps

The next step involves refining the predictive model, focusing on the quantitative adjustment of the estimated absorbed dose values, aiming for higher numerical accuracy.

After completing this phase, a new network will be trained using the same parameters, but with real patient images provided by the Hospital do Amor in Barretos. Data from 20 patients treated with ^{131}I will be used, each with two acquisition time points.

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Mass Analysis of Computational Mouse Models Developed at CDTN: A Comparative Study of Male and Female Versions

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Abstract

Voxel-based computational models play a critical role in preclinical dosimetry, providing realistic anatomical representations that enhance the accuracy of internal dose estimation in animal studies. This work presents the development of a male voxel-based mouse model (ML_BRA 01), constructed from MRI images and segmented using open-source tools (ImageJ and GIMP). It compares it with an improved version of a female model (FM_BRA 04). The anatomical structures were manually delineated and processed to separate internal contents and walls, enabling more precise mass quantification. Mass values were computed using voxel volume and standardized tissue density values from ICRP publications (ICRP, 2009). The study identifies anatomical and physiological differences in sex and age by comparing organ mass values between the two models. Results show significant mass discrepancies in organs like the intestines, liver, and bladder, highlighting the influence of biological variability and segmentation criteria. The findings emphasize the importance of using anatomically accurate models that reflect sex and developmental differences for reliable preclinical dosimetry applications.

Keywords: Voxel-based model; Preclinical Dosimetry; Mouse model; Segmentation;

1. Introduction

Animal models have been fundamental in preclinical research for developing and evaluating new therapeutic strategies, imaging techniques, and investigating exposure–response relationships to different ionizing and non-ionizing radiation (Xie; Zaidi, 2015). In this context, internal dosimetry plays a central role in assessing the risk-benefit ratio, especially given the increasing number of studies involving radiopharmaceuticals (Gupta et al., 2019).

Mathematical models have been developed to estimate energy deposition in mouse organs, aiming to improve the accuracy of dosimetric estimates. Initially, these models employed simplified geometric shapes to represent animal anatomy (Hui et al., 1994). Subsequently, advances such as preclinical imaging enabled the construction of voxelized models with more realistic anatomical representations (Bitar et al., 2007; Kolbert et al., 2003).

Studies have shown that parameters such as organ mass, shape, and distance between source and target organs significantly influence S-values calculation, essential for absorbed dose estimation (Hindorf et al., 2004). In therapeutic scenarios, where injected activities may induce deterministic effects, dosimetry can be based on the MIRD formalism, which integrates pharmacokinetic data and specific S-values (Mauxion et al., 2013)

Using codes such as MCNP and Geant4, Monte Carlo simulations have been widely employed to model radiation transport and energy deposition in voxelized anatomical models, accounting for the complex physical interactions between radiation and tissues (Bitar et al., 2007)

Given this context, the development of a voxelized male model, complementary to a previously developed and improved female model, aims to improve the accuracy of dosimetric estimates by enabling the analysis of anatomical differences related to sex and age, which is crucial for personalized and efficient preclinical assessments. This study aims to develop a male voxelized mouse model to complement a previously constructed female model and analyze anatomical differences related to sex and age, enhancing the accuracy of preclinical dosimetric evaluations.

2. Methodology

2.1.Development of the male voxel-based computational model ML_BRA

The male computational model ML_BRA 01 was developed based on a magnetic resonance imaging (MRI) scan of a male C57BL/6 mouse, provided by the *Turku Center for Disease Modeling* at the University of Turku. The original image had a matrix size of 110 × 110 × 156 mm and voxel dimensions of 0.13 × 0.13 × 0.16 mm³. The DICOM file was processed using ImageJ (Schneider et al., n.d.) software and converted from 16-bit to 8-bit format to comply with a 256 grayscale level range. The number of slices was decreased to reduce file size, and a new field of view (FOV) was defined, ensuring complete preservation of the mouse's anatomy. As a result, the adjusted matrix was resized to 238 × 731 × 52 mm, with voxel dimensions of 0.13 × 0.13 × 0.48 mm³.

Organ segmentation was performed using anatomical reference tools, including IMAIOS Vet (2025), which provides a female mouse's computed tomography (CT) scan. Additionally, scientific literature, such as the study by Knoblaugh et al. (2018), was essential to ensure a detailed and anatomically accurate representation within the computational model, as represented in figure 1.

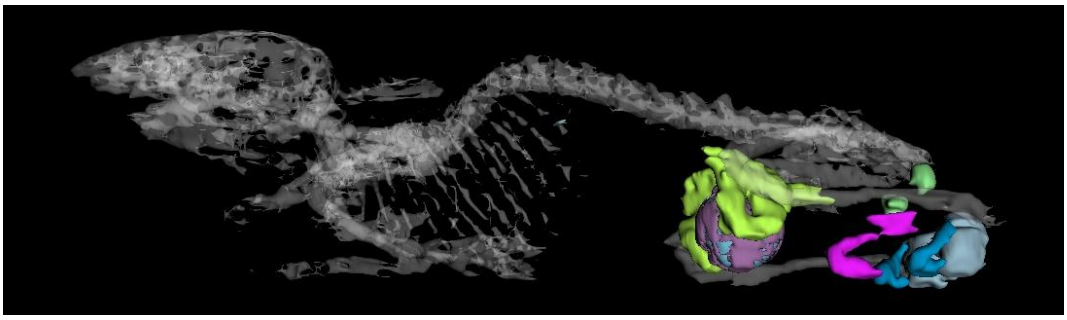

Figure 1: ML_BRA 01 model highlighting the male reproductive system and skeletal structure.

Segmentations were performed using GIMP® software **(**n.d.), based on the Z-axis of the matrix, which consisted of 52 slices. After identifying the structure to be segmented, each slice was individually imported into the software, where a mask was manually drawn around the entire region corresponding to the structure, as illustrated in figure 2. Once the mask was defined, the elements belonging to the structure were assigned a grayscale value of 255 (white), while external regions were assigned a value of 0 (black), as shown in figure 2.

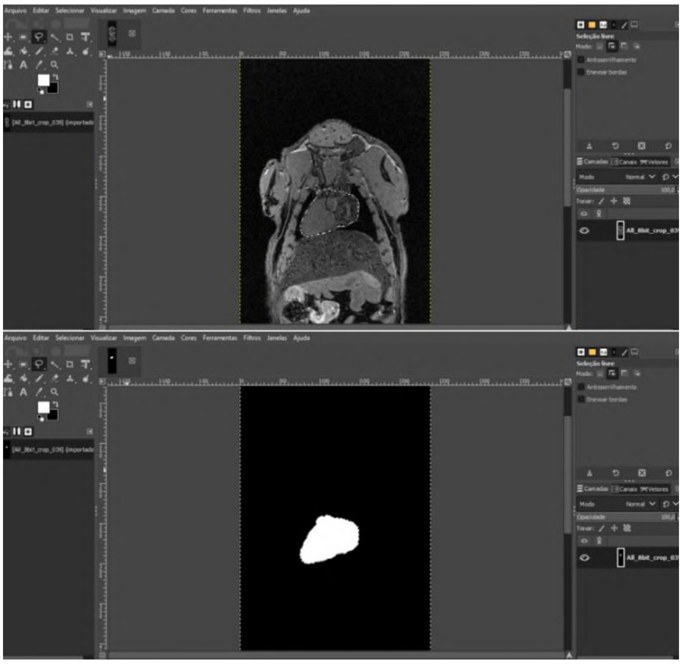

Figure 2: GIMP interface highlighting the segmentation and masking of the heart.

Organs with internal cavities were refined using ImageJ software (Schneider et al., n.d.). After anatomical definition using the masks generated in GIMP® **(**n.d.), the tools "*Erode*", "*Image Calculator*", and "*Math*" functions within ImageJ (Schneider et al., n.d.) were applied to separate the organ wall from its internal contents. This approach ensures greater uniformity in structural mass values compared to other models. In the ML_BRA 01 model, this technique was applied to the intestines and skin, as illustrated in figure 3. Separation was performed manually for other structures, such as the bladder and stomach.

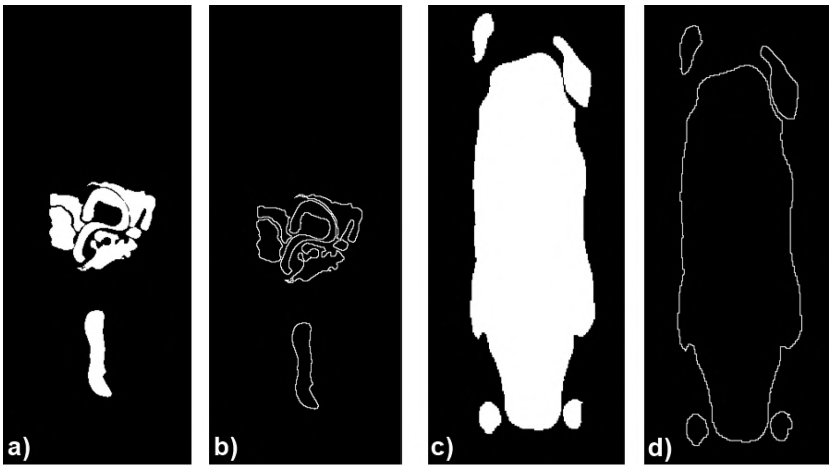

Figure 3: Method for separating wall and content: (a) Complete intestine; (b) Segmented intestinal wall; (c) Whole body; (d) Segmented skin.

After segmenting all structures, the computational model was constructed in ImageJ (Schneider et al., n.d.)using the "*Math*" and "*Image Calculator*" tools. To ensure anatomical differentiation, voxels in each region were assigned distinct values within the 8-bit grayscale range, providing a unique code for each structure. During the integration of the segmented structures, unwanted overlaps between adjacent organs and tissues were identified and corrected using the “*Voxel Phantom Tools*” plugin, an ImageJ extension designed for volumetric adjustments. As a result, a complete and encoded three-dimensional computational model was obtained, with all anatomical structures correctly identified, as illustrated in figure 4.

Figure 4: Segmented and coded structures of the ML_BRA 01 model.

2.2. Improvement of the Female Voxel Computational Model from FM_BRA 03 to FM_BRA 04

During the development of the FM_BRA 04 model, corrections were made to specific structures that showed inconsistencies in the previous version (FM_BRA 03). Additionally, incorporating the actual tissue thickness was a significant improvement, which had not been considered before. This approach allowed for a more accurate representation of the physical dimensions of organs and structures such as the trachea, intestines, and skin, resulting in mass estimates that better reflect their real anatomical representation. These modifications enhanced the model’s quality, making it more suitable for preclinical applications and dosimetric studies.

2.3. Age estimation of computational models

A digital analysis approach was employed through segmentation to estimate the approximate age of the ML_BRA 01 model. In the GIMP® software (n.d.), a body mask covering the entire region of the mouse body was drawn and filled with pixel values of 255 (white). This mask was then processed in the AMIDE software (Loening, n.d.), allowing the calculation of the model's mass and a more accurate estimate of its age. The mass values obtained for the respective models were compared with a reference dataset provided by The Jackson Laboratory (n.d.), which correlates body weight with age. This comparison enabled a close estimation of the models' ages and allowed further comparisons.

3. Results and discussion

3.1. Complete models and comparisons of mass values

The male voxelized computational model ML_BRA 01 was finalized with a comprehensive set of segmented anatomical structures. The model can be visualized along the three orthogonal axes and through its three-dimensional reconstruction, as illustrated in figure 5.

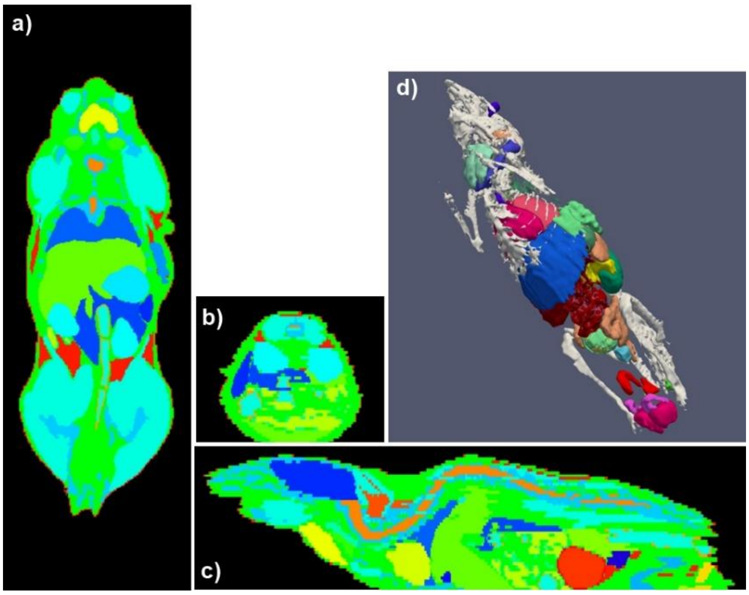

Figure 5: Representations of the male voxel computational model ML_BRA 01.
(a) Axial view, displaying the transverse slice with details of internal organs in the horizontal plane
(b) Coronal view, showing the anatomical distribution of organs in the frontal section of the body.
(c) Sagittal view, presenting the anatomical structure in a lateral longitudinal section.
(d) Three-dimensional (3D) visualization of the model, highlighting the location and spatial relationships of the constituent organs.

The enhanced female voxelized computational model FM_BRA 04 can also be observed along the three orthogonal axes and through its three-dimensional reconstruction, as shown in figure 6.

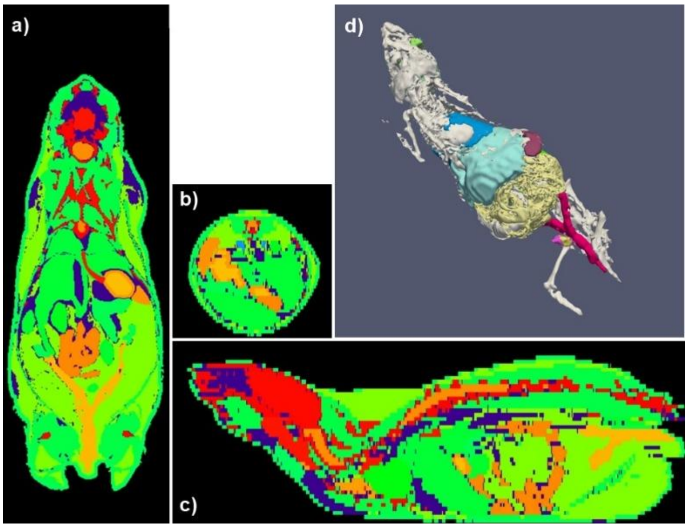

Figure 6: Representations of the female voxel computational model FM_BRA 04.
(a) Axial view, displaying the transverse slice with details of internal organs in the horizontal plane
(b) Coronal view, showing the anatomical distribution of organs in the frontal section of the body.
(c) Sagittal view, presenting the anatomical structure in a lateral longitudinal section.
(d) Three-dimensional (3D) visualization of the model, highlighting the location and spatial relationships of the constituent organs.

Table 1 presents the mass values calculated for the ML_BRA 01 model, derived using density values based on ICRP Publication 110 (2009) data for voxelized human phantoms, since specific animal data are unavailable in the literature. In addition to the mass values of ML_BRA 01, the table also includes data for the enhanced female model FM_BRA 03 and a comparison between this model and the updated FM_BRA 04 version.

Table 1: Organs, mass, and density values; Comparison between versions FM_BRA 03 and 04, and comparison of model FM_BRA 04 vs. ML_BRA 01.

Organ	ρ (g.cm⁻³)	FM_BRA 03	FM_BRA 04	ML_BRA 01	FM_BRA 04 vs. ML_BRA 01
		mass (g)			
Bladder	1,040	0,029	0,029	0,218	**-87%***
Bladder - Content	1,040	0,020	0,020	0,165	**-88%***
Bladder - Wall	1,040	0,009	0,009	0,053	**-83%***
Bones	1,040	2,701	1,412	1,361	4%
Brain	1,050	0,447	0,459	0,494	-7%
Esophagus	1,040	0,039	0,028	-	-
Eye - Right	1,050	-	0,014	0,012	13%
Eye -Left	1,050	-	0,013	0,011	19%
Eyes	1,050	0,026	0,027	0,023	16%
Eyes - Bulb	1,050	0,022	-	-	-
Eyes - Crystalline	1,050	0,004	0,005	-	-
G. adrenal	1,030	0,005	0,005	-	-
G. Left Suprarenal	1,03	-	-	-	-
G. Right Suprarenal	1,030	-	-	-	-
Gallbladder	1,030	0,016	0,016	0,015	9%
Harder's Gland	1,000	0,030	0,030	-	-
Heart	1,050	0,281	0,282	0,322	-12%
Intestines	1,040	1,706	1,704	1,247	-
Kidney - Right	1,050	0,168	0,168	0,225	-
Kidney -Left	1,050	0,158	0,158	0,224	-
Kidneys	1,050	0,326	0,326	0,448	-27%
Large Intestine	1,040	0,750	0,750	0,500	50%
Large Intestine - Contents	1,040	0,468	0,468	0,399	**17%***
Large Intestine - Wall	1,040	0,282	0,282	0,101	**178%***
Liver	1,050	1,855	1,894	1,937	**-2%**
Lung	0,380	0,185	0,182	0,225	-19%
Muscles	1,050	7,650	7,791	6,975	12%
Ovaries	1,040	0,017	0,017	-	-
Pancreas	1,050	0,237	0,245	0,194	26%
Salivary gland	1,030	0,192	0,191	0,179	7%
Skin	1,090	1,170	4,189	0,722	480%
Small Intestine	1,040	0,956	0,954	0,747	**28%***
Small Intestine - Content	1,040	0,459	0,457	0,416	**10%***
Small Intestine - Wall	1,040	0,497	0,497	0,331	**50%***
Spinal Cord	1,030	0,121	0,132	0,130	1%
Spleen	1,040	0,096	0,096	0,080	20%
Stomach	1,040	0,197	0,200	0,313	**-36%***

Stomach - Content	1,040	0,100	0,100	0,169	**-40%***
Stomach - Wall	1,040	0,097	0,100	0,144	**-30%***
Thymus	1,030	0,065	0,068	0,046	47%
Thyroid	1,040	0,017	-	0,015	-
Trachea	1,030	0,064	0,117	0,017	**596%***
Uterus	1,030	0,129	0,129	-	-
Waste Tissues	0,950	15,507	15,491	9,298	**66%***
Brown fat	1,000	-	-	0,062	-
Epididymis	1,050	-	-	0,031	-
Male reproductive organ	1,050	-	-	0,030	-
Preputial gland	1,050	-	-	0,021	-
Prostate	1,030	-	-	0,018	-
Seminal vesicle	1,050	-	-	0,271	-
Testis	1,040	-	-	0,153	-
Thymus	1,030	-	-	0,046	-
Thyroid	1,040	-	-	0,015	-
Trachea	1,040	-	-	0,017	-
Tumor	1,000	-	-	0,074	-

* These organs may exhibit significant variations due to physiological diversity among individuals during preclinical image acquisition used as the basis for model construction, or due to different segmentation approaches, such as considering structural thickness.

3.2. Anatomical Mass Differences Between Models

In comparing the computational model versions FM_BRA 03 and FM_BRA 04, notable differences were observed, particularly in the estimated masses of specific organs and tissues. The updated mass calculations in FM_BRA 04 led to substantial structural corrections, such as bones and eyes. At the same time, tissues like the brain, heart, liver, muscles, pancreas, salivary glands, and spinal cord showed more subtle adjustments. These differences reflect improvements in both segmentation criteria and volumetric quantification methodology. A key enhancement introduced in FM_BRA 04 was the consideration of actual tissue thickness. Unlike FM_BRA 03, which relied on more simplified anatomical assumptions, FM_BRA 04 implemented refinements to represent better the physical dimensions of structures such as the trachea, intestines, and skin, contributing to more accurate organ mass estimations.

The comparison between the adjusted female model (FM_BRA 04) and the male model (ML_BRA 01), as presented in figure 5, highlights relevant differences in organ masses, primarily attributable to the age gap between the animals, 26 weeks for the female and 25 weeks for the male.

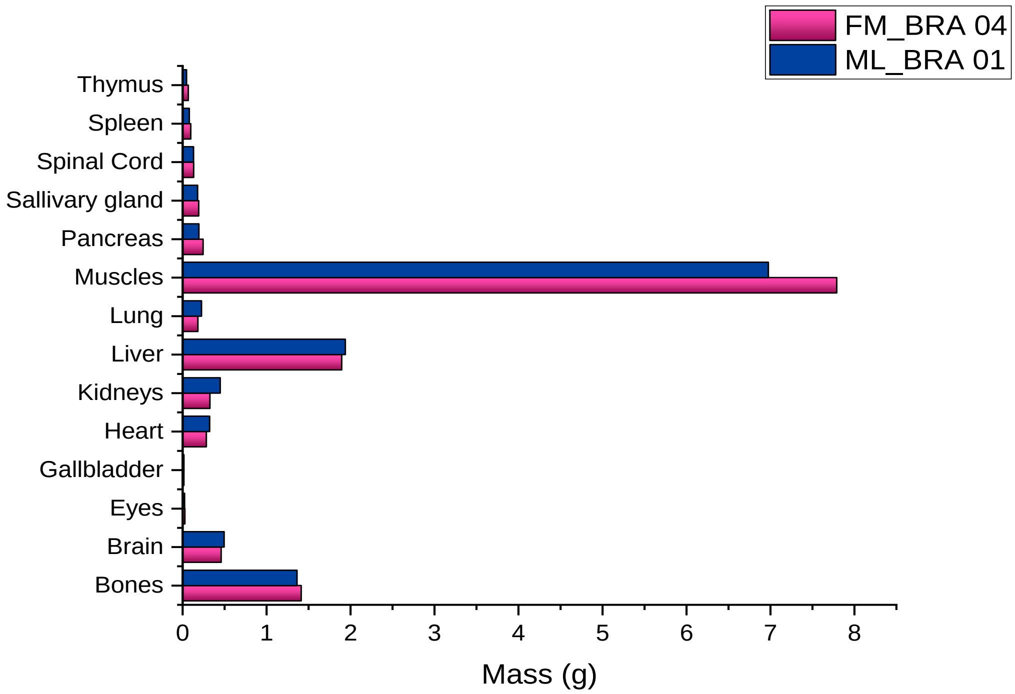

Figure 5: Comparison of mass values between models ML_BRA 01 and FM_BRA 04.

The organs selected for the construction of graph 1 were those of solid nature and minimal physiological variation, aiming to ensure a fairer comparison. This difference likely accounts for the increased mass observed in skeletal muscle and bone in the female model, suggesting a more advanced stage of somatic development. In contrast, the liver presented a higher mass in the male model, possibly due to sex-specific physiological traits or differing growth dynamics. Other organs, such as the lungs, spleen, and heart, exhibited less pronounced differences, remaining within the expected range of individual biological variability.

Conclusions

The analysis of different versions of computational models of C57BL/6 mice highlights the importance of biological and methodological variables in defining the morphology of segmented organs. Factors such as sex, age, and segmentation techniques were found to be decisive in determining the final mass values of organs and tissues. Subtle physiological and developmental differences can significantly impact the anatomical parameters used in dosimetric applications, even among individuals of the same strain. These findings underscore the need for anatomically representative models incorporating biological variability to improve the accuracy of absorbed dose estimations in preclinical studies. The development of ML_BRA 01 and the improved segmentation applied to FM_BRA 04 exemplify how refined modeling approaches can support more reliable internal dosimetry calculations.

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Analysis of spatial resolution in two different phantoms as a quality parameter in MRI pediatric pathology

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Abstract

Magnetic Resonance Imaging (MRI) has been extensively utilized in pediatric diagnostics in recent years due to its non-ionizing nature. In pediatrics, the evolution of imaging technologies, along with advances in the specialty, highlights the need for optimization of exams. This optimization aims to follow more appropriate protocols for specific pathologies and to ensure optimal scan times using increasingly sophisticated equipment. Therefore, ensuring the quality of pediatric images, this enables the implementation of acquisition optimizations, allowing for a reduced number of scans within an examination, while maintaining sufficient quality for medical report issuance without the need to repeat the exam. The aim of this study was to analyze the spatial resolution of images obtained from two different phantoms: a cylindrical Magphan 170 model from The Phantom Laboratory® filled with a copper sulfate solution; and a cylindrical and ACR type, 180mm from Konex filled with a Nickel and Sodium Chloride solution. The images acquired from both phantoms were analyzed based on protocols from the American Association of Physicists in Medicine (AAPM) and the American College of Radiology (ACR). The quality parameters (spatial resolution and assessment of the presence of occasional artifacts) were verified using two different software programs: RadiAnt Dicom Viewer© and ImageJ©. The obtained results showed agreement between the two programs. The spatial resolution achieved in this study allows for the high-precision visualization of small structures, which is essential for the diagnosis of pediatric and newborn pathologies.

Keywords: Magnetic Resonance; Spatial Resolution; Pediatric Pathology.

1 Introduction

Magnetic Resonance Imaging (MRI) has been extensively utilized in pediatric diagnostics in recent years due to its non-ionizing nature. In pediatrics, the evolution of imaging technologies and advances in the specialty have highlighted the need to optimize MRI exams — to follow protocols better suited for specific pathologies and to ensure optimal scan times using increasingly sophisticated equipment. (Sobol, 2012), (AAPM, 2010).

Thus, there is a demand for equipment that ensures both image quality and safety for patients and operators. (ACR, 2015), (Hsieh et al., 2024) In structural or functional pathologies of the central nervous system (CNS) in newborns, such as myelomeningocele, ultrasound exam alone may be insufficient for diagnosis. In such cases, high-resolution MRI images are necessary, as they allow for the visualization of small structures with high precision, aiding in the diagnosis of fetal or pediatric malformations. (Secker et al.,2023)

Spatial resolution is defined as the ability to distinguish between two distinct, nearby structures. It is typically measured in line pairs per millimeter. In some phantoms, objects are presented by size, which can be converted to line pairs per centimeter (lp/cm) using equation 1, where 'object size' is in centimeters.

$$\frac{line\ pairs}{cm} = \frac{1}{2x(object\ size)} \qquad (1)$$

By ensuring the quality of pediatric images, we can implement acquisition optimizations that reduce the number of scans in an examination, while maintaining sufficient quality for medical report issuance without the need for a repeat exam. (Hsieh, 2024), (Storey et al.,2006), (Hendricks et al., 2024), (Buschong,2009).

The aim of this study was to analyze the spatial resolution of images obtained from two different phantoms. Both images were acquired using a 3T MRI scanner (Achieva Philips).

2 Material and Methods

Quality control tests were performed on a Philips Achieva 3T MRI scanner using an 8-channel head coil (SENSE_HEAD_8). The tests were conducted on three separate days. Images were acquired from two phantoms: a cylindrical type, Magphan 170 model (The Phantom Laboratory®) filled with a copper sulfate solution; and a cylindrical ACR type, 180mm model (Konex) filled with a Nickel and Sodium Chloride solution.

Figure 1: ACR Phantom.

Figure 2: Magphan Phantom.

The ACR protocol was used to acquire images from the two phantoms with the parameters presented in Table 1.

Table 1: Sequence parameter used in image acquisitions for quality control

Sequence	TR (ms)	TE (ms)	FOV (mm)	MATRIX (pixels)	NEX	Bandwidth (Hz/pixel)	Flip angle	Slice thickness (mm)
Spin Echo	500	20	256 x 256	256 x 256	1.0	364	90	5.0

a: repetition time; b: echo time; c: field of view; d: number of excitations; e: bandwidth.

Since the internal geometries of the phantoms are distinct, the slice spacing parameter used in the acquisition is also different for each phantom. Thus, the spacing used for the Magphan 170 was 8.2 mm and for the ACR was 7.5 mm.

3 Results

The development of automated, artificial intelligence-driven tools for the objective evaluation of quality control tests is a current focus for many authors. Nevertheless, a comprehensive assessment of phantom performance is critical, primarily to determine their efficiency and ensure the quality of medical images for an accurate diagnosis, bridging the fields of physics and medicine.

While the literature compares various phantoms, these comparisons don't typically involve images from high-field scanners (3.0 Tesla). The use of different protocols with adapted parameters for each phantom type is also observed. For this study, a single image acquisition protocol was applied to both phantoms, with only the slice spacing adjusted for each to replicate a clinical routine. (Hendriks et al., 2024; Shang et al., 2022; Dupuis et al.,2024; Alvarez et al.,2022).

The analysis of quality parameters (spatial resolution and assessment of the presence of occasional artifacts) was performed based on the quality protocols suggested by the American College of Radiology (ACR, 2015). The images acquired from the phantoms were verified on a Barco® clinical monitor model E-3620 with 3MP resolution and with two different software programs: RadiAnt Dicom Viewer© and ImageJ©.

The images (figures 3 and 4) were viewed using the software described previously, and the results were compared with the tolerance recommended in the ACR protocol (2015). The tolerance in this study is 1 mm for the theoretical pixel, according to equation 2.

The analysis of quality parameters, specifically spatial resolution and the assessment of occasional artifacts, was performed based on the quality protocols suggested by the American College of Radiology (ACR, 2015). The images acquired from the phantoms were verified on a Barco® E-3620 clinical monitor with a 3MP resolution and with two different software programs: RadiAnt Dicom Viewer© and ImageJ©.

The images (Figures 3 and 4) were viewed using the previously described software, and the results were compared with the tolerance recommended in the ACR protocol (2015). The tolerance in this study is 1 mm for the theoretical pixel, according to Equation 2.

$$theoretical\ pixel = \frac{Matrix}{FOV} \qquad (2)$$

In the ACR phantom, the spatial resolution standard is the of hole array pairs, as shown in figure 3. Therefore, to obtain the resolution in line pairs, equation 1 was used.

Figure 3: Image of the ACR Phantom, showing the holes in the spatial resolution analysis region obtained during the test.

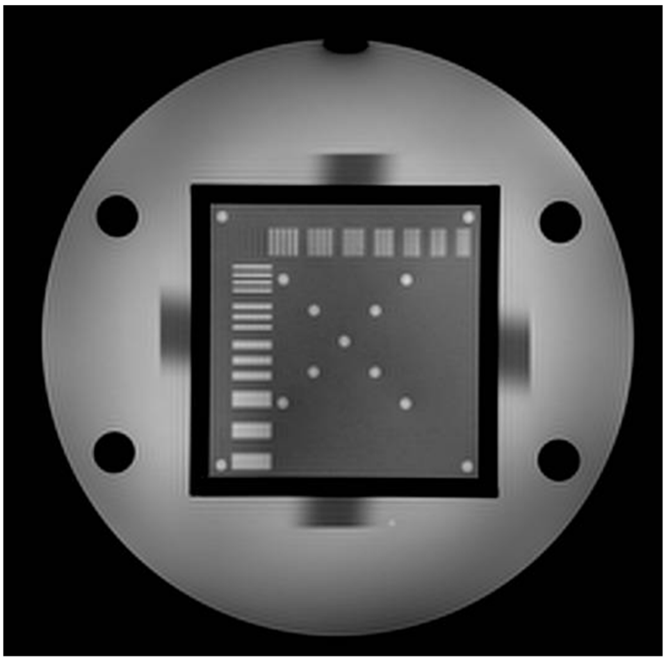

Figure 4: Image of the Magphan Phantom, showing the structures in line pairs per centimeter in the spatial resolution analysis region obtained during the test.

For quality control, spatial resolution testing establishes the minimum limit of a region that can be visualized. This is directly applicable to the resolution of pediatric MRI, particularly for fetal cases that require intrauterine surgical intervention and for prenatal diagnosis. On T1-weighted images acquired with phantoms, sets of lines or holes are observed as hypointense (dark) contrast regions at 1.0 mm resolution. This finding can be correlated with the contrast and resolution of cerebrospinal fluid (CSF) in clinical patient scans. Such resolution is critical for the clinical and surgical management of patients with myelomeningocele, as MRI findings of spinal anomalies and dysraphism necessitate an anatomical resolution in the 1.0 mm range. Ultimately, high-definition MRI with fine spatial resolution is indispensable for distinguishing subtle clinical findings in both pediatric and fetal diagnostics. (Araujo, 2022), (Walsh and Adzick, 2000), (Dair, 2014).

The results obtained are presented in Table 2. The findings for both phantoms were identical, irrespective of the software utilized.

Table 2: Results of the analyses of both phantoms in line pairs per centimeter (lp/cm)

Phantom	Day	ImageJ (lp/cm)	RadiAnt (lp/cm)
	01	5	5
ACR	02	5	5
	03	5	5
	01	5	5
Magpham	02	5	5
	03	5	5

In this study, a theoretical pixel of 1.0 mm was obtained using equation 2. This value is consistent with that typically obtained in clinical protocols for newborns, which use a FOV of 180 mm and an acquisition matrix of 180 x 144 pixels. In pediatric protocols, the theoretical pixel is 1.34 mm, as the parameters have approximate values of a 200 mm FOV and a matrix of 268 x 155 pixels. The results presented in Table 2 are therefore consistent with the spatial resolution required for visualization of small structures. (Och et al., 1992), (Bushong ,2010), (Secker et al.2023), (D'Arco et al,2022).

Figures 3 and 4 show that the phantoms' internal structures, in terms of spatial resolution, are arranged with distinct geometries. The Magphan 170 phantom has 11 line-pair structures, while the ACR phantom has three hole-array structures. It was noted that the analysis of spatial resolution using the Magphan phantom was more optimized due to the ease of visualizing the line pairs.

4 Conclusion

The images acquired with both phantoms were analyzed based on the protocols of the American Association of Physicists in Medicine (AAPM) and the American College of Radiology (ACR). The results showed strong agreement. The spatial resolution achieved in this study allows for the high-precision visualization of small structures, which is essential for the diagnosis of pediatric and neonatal pathologies.

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APPLICATION OF STEAM METHODOLOGY IN SCIENCE TEACHING: Prototyping a cloud chamber

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Abstract

The objective of this work is to integrate engineering and chemistry concepts into science education at the basic education level through a practical, interdisciplinary approach using the STEAM methodology (Science, Technology, Engineering, Arts, and Mathematics). The proposed project involves creating a Wilson (cloud) chamber prototype utilizing Peltier cells to observe radiation emitted by natural radioactive materials in the classroom. This accessible and innovative educational tool allows students to directly visualize ionizing particle trajectories, enhancing their understanding of physical and chemical phenomena. The use of Peltier cells is a strategic choice due to their ability to create the necessary conditions for alcohol vapor cloud formation, enabling the observation of radiation tracks. The construction process will be thoroughly documented, providing material lists, assembly diagrams, and practical tips for teachers and students in basic education. The work aims to improve science teaching by facilitating the understanding of complex concepts through hands-on experimentation, encouraging greater student engagement with topics such as radiation, ionization, and the integration of engineering and chemistry principles in science education.

Keywords: STEAM; Cloud Chamber; Education.

1 Introduction

In 2022, the MCMEG research group at the Federal University of Minas Gerais (UFMG) developed a low-cost and easy-to-assemble cloud chamber (Fonseca et al., 2022) for use in classrooms. This initial model, shown in Figure 1, required dry ice to reach a low operational temperature of approximately $-78°C$. The group is currently working on a new version of the chamber that aims to improve the previous design by addressing key limitations. In particular, the new version is using Peltier elements and other components to achieve and maintain the required low temperatures, making the system more practical and accessible for educational environments.

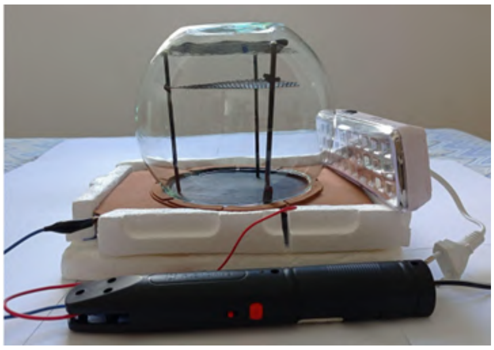


Figure 1: Cloud Chamber MCMEG. Source: Fonseca et al. (2022)

A similar approach was proposed by Barrett (2024) in the project "Cloud Chamber Using Peltier Cooling" developed at the University of Liverpool. Their objective was the same: to reduce costs and replace dry ice with Peltier modules. The chamber, shown in Figure 2, was successfully constructed for under £40 and was capable of visualizing radiation trails from naturally radioactive materials using only Peltier cooling. This reinforces the viability of using thermoelectric components in the development of affordable and effective cloud chambers.

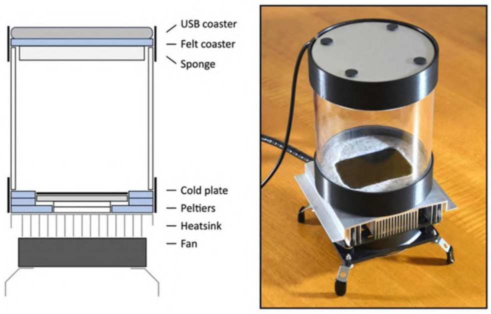


Figure 2: Barret's Peltier Cloud Chamber. Source: Barrett (2024)

1.1 Importance of New Learning Methodologies

The integration of engineering concepts in science education at the basic education level is a topic that has gained attention in the search for enhancing student learning. This approach, innovative and practical in nature, is essential for stimulating interest and understanding of chemical and physical phenomena (Quigley et al., 2017).

Interdisciplinarity can be understood as a fundamental condition for teaching and research (at both the university and the high school level) in contemporary society (Leis, 2005). This concept implies collaboration between different areas of knowledge, enabling a more comprehensive and integrated understanding of phenomena. In this context, the integration of engineering and science concepts represents a unique opportunity to promote interdisciplinarity, enriching students' education and preparing them to tackle complex challenges that require the integration of knowledge from various disciplines (Arpaci et al., 2023).

Throughout the historical evolution of science education at the basic education level, the integration of concepts from Science, Technology, Engineering, Arts, and Mathematics (STEAM) has stood out as an innovative and transformative approach. Defined as a methodology that seeks to articulate and apply knowledge from school subjects, the central goal of STEAM is to create an effective connection between these disciplines, allowing them to gain meaning in concrete situations when integrated into an individual's knowledge structure (Lorenzin, 2017).

Thus, considering the topic chosen for this work, STEAM emerges as an important element in the evolution of integrating engineering concepts into science education at the basic education level. It plays a crucial role by providing an interdisciplinary approach that combines essential elements from the aforementioned areas. It is worth noting that this approach not only enriches students' education but also prepares them to face complex challenges that require the practical application of scientific and engineering concepts in real-world situations(Yulianti et al., 2024).

1.2 The Wilson Chamber

The cloud chamber was invented by Charles T. R. Wilson in 1897, who was originally studying how clouds form in the atmosphere. During his experiments, he noticed that ionizing radiation helped vapor condense into droplets. This led him to create a device that could make the paths of charged particles visible. The cloud chamber works by filling a sealed container with vapor-saturated gas, usually alcohol, and cooling it quickly to create a supersaturated environment. When a charged particle passes through, it ionizes the gas molecules along its path. These ions act as centers where the vapor condenses, forming tiny droplets. The droplets create a visible track that shows the particle's path. The shape and thickness of the track depend on the type and energy of the particle. This invention was important in particle physics, helping scientists discover new particles like the positron and the muon, and allowing them to study radiation in a direct and visual way (Wisconsin Madison Department Physics; Cabral et al., 2025). Figure 3 illustrates the formation of these trails.

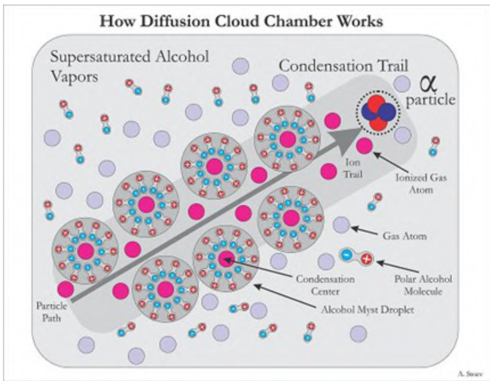


Figure 3: How Difussion on Cloud Chamber Works. Soucer: Wikipédia (2025)

2 Methodology

2.1 Construction of the Wilson Chamber

In order to make the project replicable in schools, all materials were selected for their low cost, availability, and potential for reuse. The aim was to allow students from elementary and high school to build a working Wilson cloud chamber using accessible resources. The project began with research into commercially available models. However, only one similar product was found on a business website during an online search, with a total cost of approximately R$1,781.67 due to importation fees.

In this project to construct the cloud chamber the following materials were used:

Power supply: An ATX power supply from an unused desktop computer was repurposed to provide the necessary current for the system. This choice significantly reduced the cost and ensured stable voltage for the Peltier modules.

Peltier modules: TEC1-12706 thermoelectric plates were used for cooling (Mallmann et al., 2016). When current flows through these modules, one side becomes cold while the other heats up. For optimal heat dissipation, an initial test was conducted using a standard CPU heat sink, which proved insufficient. Consequently, a more efficient cooler with heat pipes was selected to improve thermal performance.

Chamber structure: A small glass aquarium was used as the transparent dome of the cloud chamber, allowing for clear visibility of the particle tracks. An acrylic plate served as the base, supporting the Peltier modules and creating the cold region necessary for vapor condensation.

Alcohol source: A sponge soaked in 99% isopropyl alcohol was placed at the top of the chamber using small neodymium magnets. Isopropyl alcohol was chosen because it is readily available with high purity and has a relatively high operational temperature compared to other alcohols. It starts to function effectively at around $-26°C$, while ethanol requires $-31°C$, and methanol $-44°C$.

Cooling alternatives: Although traditional cloud chambers often use dry ice, this project investigated the use of Peltier modules as a safer and more accessible alternative for school settings.

Each component was chosen not only for its technical role but also for its pedagogical potential. The structure allows students to understand the physical principles of radiation and phase change through hands-on construction and observation. The modularity of the setup also encourages experimentation with different cooling methods, vapor sources, and chamber shapes. The pre-model is shown in Figure 4, illustrating the initial design and structure developed during the early stages of the project.

Figure 4: Peltier Cloud Chamber. Soucer: Author

2.2 Peltier Plates Testings

Peltier plates are used to generate the cooling effect, required to create a supersaturated vapor inside the chamber (Silva, 2021). The TEC-12706 Peltier module was chosen due to its affordability and ease of implementation in an educational setting. These thermoelectric devices create a temperature difference between their two sides when current flows through them, making one side cold and the other side hot. In this study, different configurations of Peltier plates were tested to evaluate their cooling efficiency for the cloud chamber.

The following configurations were tested:

- **Three Peltier plates in series**: This arrangement increases the total voltage but maintains constant current.
- **Three Peltier plates in parallel**: This arrangement maintains a constant voltage while increasing the total current.
- **Hybrid configuration (two plates in parallel, one in series)**: This combines the benefits of both, series and parallel configurations. A schematic diagram of this configuration is presented in Figure 5.

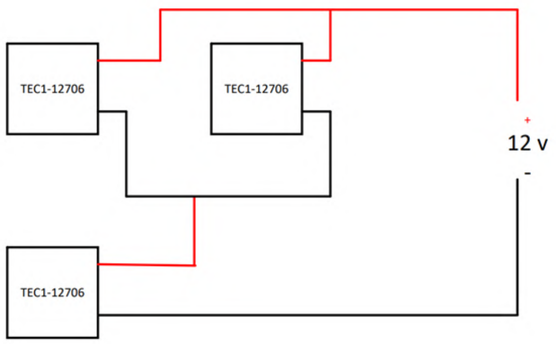
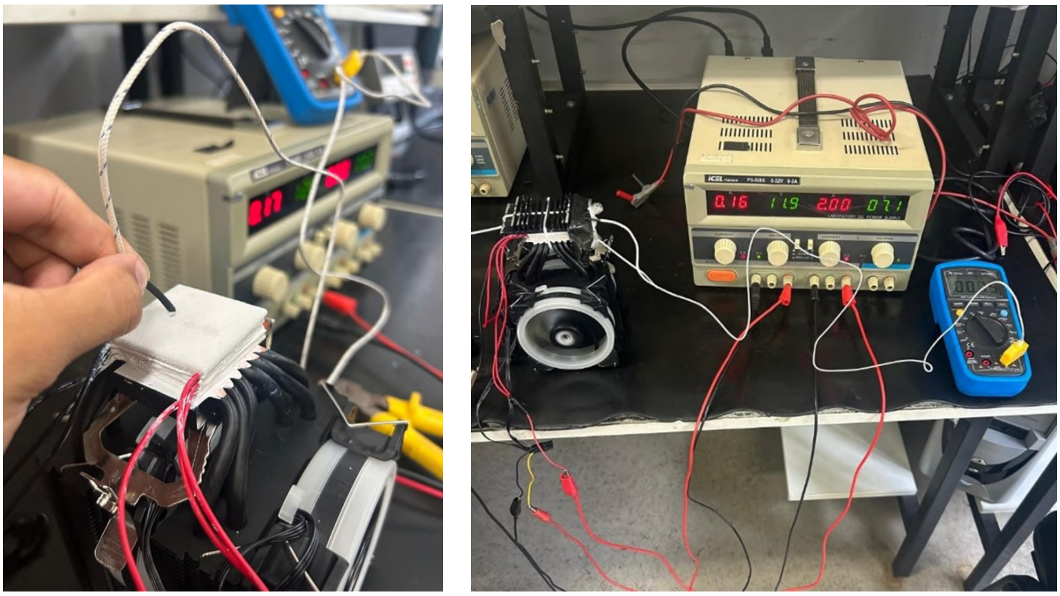


Figure 5: Hibrid configuration - two plates in parallel and in serie. Soucer: Author

The Peltier plates were tested at the Department of Electrical Engineering of the Federal University of Minas Gerais (UFMG). The objective was to evaluate the thermal behavior and cooling efficiency of the Peltier modules under different combinations of current and voltage. During the experiments, a thermocouple was used to monitor the plate temperatures in real time. Figures 6(a) and 6(b) present the test setups and temperature readings obtained.

The experiments involved applying different voltages using a bench power supply, while configuring the Peltier modules in series, parallel, and hybrid arrangements. Temperature data were collected systematically to assess the efficiency and thermal stability of each configuration. These results were essential in determining the most suitable setup for the final version of the cloud chamber. Figure 6(a) and 6(b) show the equipment and the prototype chamber with the Peltier modules under testing.



(a) (b)

Figure 6: Test of Peltier Cells. Soucer: Author (a) Test Peltier (b) Test Bench.

2.3 Refrigeration Cooler

A cooling system, equipped with heat pipes (Werner et al., 2024), was used to dissipate the heat generated by the Peltier plates. The heat pipes ensure efficient thermal transfer, maintaining the required temperature differential between the hot and cold sides of the Peltier plates. The efficiency of the cooling system was tested with different coolers, and those with heat pipes proved to be significantly more efficient.

3 Results

This section presents the results obtained from the tests conducted with the Peltier plates and the cloud chamber prototype, along with the performance of the system under different configurations. The data provided here reflects the effectiveness of the cooling system and the visibility of particle tracks formed by the ionizing radiation.

3.1 Peltier Plate Testing Results

The Peltier plate configurations were tested to determine their thermal efficiency and the cooling capabilities of the cloud chamber prototype. The results for each configuration, including the current, voltage, and temperature achieved, are presented in Table 1.

Configuration	**Current (A)**	**Voltage (V)**	**Temperature (°C)**
Three Peltier plates in series	1.7	12	15
Three Peltier plates in parallel	4.0	3.75	11
Hybrid configuration (2 parallel, 1 series)	3.5	12	4

Table 1: Peltier plate configurations, current, voltage, and temperature data. Soucer: Author

From the results in Table 1, it can be observed that the hybrid configuration (two plates in parallel and one in series) achieved the lowest temperature, reaching 4°C. The series configuration resulted in a higher temperature of 15°C, while the parallel configuration showed moderate cooling at 11°C. The hybrid configuration appears to offer the best balance between energy efficiency and thermal performance.

Based on the results obtained from the bench tests, the Peltier setup reached a minimum temperature of approximately 4°C. In contrast, the work conducted by S. D. Barrett reported temperatures between −20°C and −25°C, indicating a significant difference in cooling performance. This discrepancy is likely due to the use of more efficient Peltier modules in Barrett's work, as well as the implementation of multiple modules in optimized configurations. In the present study, three Peltier plates were used, arranged in the configuration that showed the highest efficiency during the tests, with two plates connected in parallel and one in series. These findings highlight a potential improvement for future developments: the use of higher-amperage or multiple Peltier elements to achieve lower temperatures, which would enhance the visibility of radiation tracks inside the cloud chamber.

3.2 Cloud Chamber with Radioactive Materials

To assess the functionality of the cloud chamber prototype, tests were conducted using natural radioactive materials, such as monazite and sand from Guarapari beach. These materials emit alpha and beta radiation, which allowed for the observation of particle tracks in the cloud chamber.

The results of the tests with the radioactive samples are shown in Figures 7 and 8. Figures 7 demonstrates the tracks formed in a cloud chamber using dry ice and 8 shows the tracks in the prototype using Peltier plates.

Figure 7: Monazite rock in a cloud chamber using dry ice. Tracks are clearly visible. Source: Fonseca et al. (2022)

Figure 8: Monazite rock in a cloud chamber using Peltier plates. Tracks are visible, but less defined compared to dry ice. Source: Author.

Figures 7 and 8, the tracks formed with the dry ice-based chamber are significantly more visible compared to the prototype using Peltier plates. This can be attributed to the lower cooling capacity of the Peltier plates, which resulted in a less dense condensation cloud. Nevertheless, the tracks were still visible, demonstrating the potential of the Peltier-based chamber for educational purposes, albeit with some limitations.

In the work by S. D. Barrett, a monazite sample was not used for direct comparison; however, a sample of boltwoodite was employed. The results obtained with the boltwoodite sample show a striking similarity to those observed in the initial cloud chamber developed by the MCMEG group, as illustrated in Figure 9.

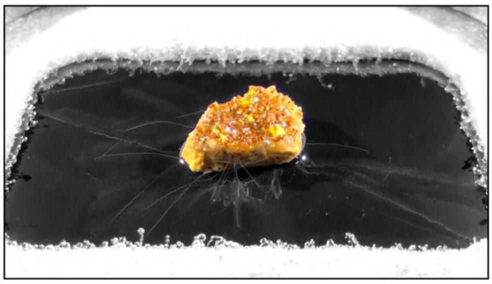


Figure 9: A sample of boltwoodite, a mineral containing some uranium. Soucer: Barrett (2024)

3.3 Results from Guarapari Sand Test

Additionally, a test was performed using sand from Guarapari Beach, a region known for its naturally occurring radioactive minerals (Xavier et al., 2007). A Geiger counter was used to measure the radiation levels emitted by the sand, while particle tracks were simultaneously observed in the cloud chamber prototype.

In both samples, the primary naturally radioactive mineral was monazite, which is characteristic of the sand from Guarapari. An important observation was the close agreement between the radiation levels measured by the Geiger counter and the density of particle tracks visualized in the chamber. The monazite crystal sample, showed in Figure 10(a), registered a radiation level of 2.031 μSv/h, while the monazitic sand sample showed in Figure 10(b) registered 2.044 μSv/h, indicating consistency between the two measurement methods.

(a) (b)

Figure 10: Geiger counter measuring radiation exposure from Guarapari sand. The exposure rate is similar to that of monazite. Soucer: Author (a) Monazite (b) Guarapari Sand.

The Geiger counter readings, shown in Figure 10(b) and Figure 10(b), confirmed that the radiation exposure from the Guarapari sand was similar to that of monazite (Souza et al., 2023), indicating the presence of natural radioactive materials in the sand. Figure 11 shows the particle tracks captured in the cloud chamber, where the tracks are visible, although less defined than those observed with the dry ice chamber. This result confirms that the cloud chamber prototype with Peltier plates can effectively detect radiation from natural sources, albeit with some limitations in track visibility.

The construction and testing of the cloud chamber prototype using Peltier plates have demonstrated the feasibility of creating a low-cost and accessible educational tool for teaching basic physics concepts, particularly ionizing radiation and particle detection. While the Peltier-based cloud chamber was less efficient than traditional dry ice-based chambers, it still proved effective in visualizing particle tracks and can serve as a valuable resource in the classroom.

The hybrid configuration of Peltier plates (two in parallel and one in series) provided the best cooling performance, achieving the lowest temperature of 4°C. Although the cloud chamber prototype using Peltier plates was less efficient than the dry ice-based chamber, it was still capable of visualizing particle tracks, albeit less defined and with a less dense condensation cloud. The use of natural radioactive materials, such as monazite and sand from Guarapari beach, allowed for successful demonstration of radiation detection in the cloud chamber, confirming the viability of the Peltier-based design for educational purposes.

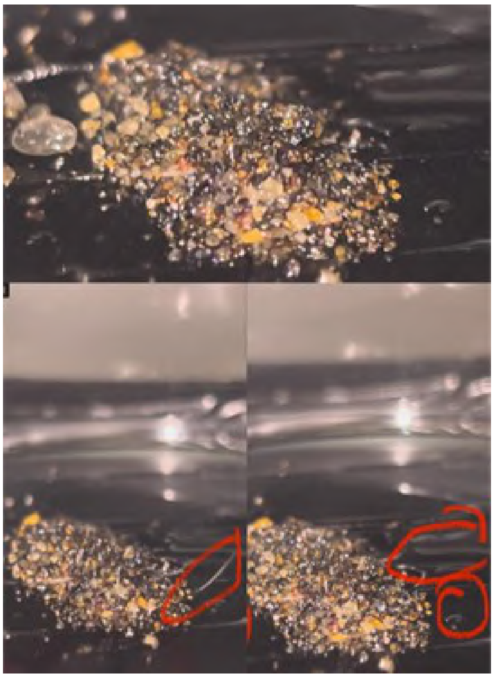


Figure 11: Particle tracks observed in the cloud chamber using Guarapari sand. Tracks are visible but less dense compared to the dry ice chamber. Soucer: Author

The development of a low-cost cloud chamber using Peltier plates provides an opportunity to integrate engineering and physics concepts in a practical and accessible way. This device aligns with the principles of STEAM education, which encourages interdisciplinary learning and hands-on experimentation. By building and using the cloud chamber, students can directly observe the effects of ionizing radiation and gain a deeper understanding of fundamental physics concepts such as particle detection, ionization, and supersaturation (Fonseca et al., 2022).

Figure 12: Peltier Cloud Chamber. Soucer: Author

The project showed promising results, but the Peltier cells' cooling capacity limited performance, especially when compared to the clearer tracks in (Barrett, 2024). Future improvements, such as adding a heated surface and exploring ultrasonic mist dispensers, could enhance efficiency.

This work highlights the value of STEAM (Science, Technology, Engineering, Arts, and Mathematics) in science education. Building a cloud chamber with Peltier modules allowed students to directly observe radiation, encouraging experimentation and critical thinking. Despite some limitations, the prototype successfully displayed particle tracks, confirming its potential for educational use and reinforcing the effectiveness of STEAM in teaching real-world concepts.

4 Conclusions

In conclusion, the Peltier-based cloud chamber represents a promising and cost-effective educational tool for teaching complex scientific concepts. With further development, this prototype has the potential to become a valuable resource for enhancing science education, providing a hands-on learning experience in the classroom. By integrating engineering, chemistry, and physics, the project exemplifies the power of STEAM methodologies to foster interdisciplinary thinking, increase student engagement, and improve understanding of challenging scientific topics. This approach demonstrates the effectiveness of active, practical learning in promoting deeper connections between theory and real-world applications. As a prospect for future developments, the implementation of a closed-loop temperature control system is proposed, incorporating sensors, current control for the Peltier plates, and a microcontroller. It is also recommended to perform tests with Peltier plates with higher current capacity, which may potentially improve cooling performance. Such improvements could broaden the range of knowledge areas addressed in the project, further expanding the learning opportunities it offers.

Acknowledgements

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Effect of repolymerization and gamma irradiation on the morphology of styrene-divinylbenzene composites with nanomagnetite

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Abstract

The development of polymeric composites incorporating magnetic nanoparticles has gained increasing attention for applications in environmental remediation and radiation-resistant systems. In this study, we investigated the effects of repolymerization and gamma irradiation on the morphology and structural stability of styrene-divinylbenzene (ST-DVB) composites doped with magnetite. Two samples were prepared: one synthesized in a single polymerization step (ST-DVB-Mag) and another subjected to an additional repolymerization cycle with a new addition of monomers and magnetite (Repol). Both samples were exposed to a 100 kGy dose of gamma radiation at the Gamma Irradiation Laboratory of CDTN. Morphological characterization by scanning electron microscopy (SEM) revealed that the repolymerized sample exhibited a more uniform and continuous surface, with magnetite particles better encapsulated within the polymer matrix. In contrast, the single-step sample showed surface irregularities and exposed magnetite clusters, which became more pronounced after irradiation. Energy Dispersive Spectroscopy (EDS) confirmed these observations, with the single-step sample displaying more intense iron peaks, indicating greater surface exposure of magnetite, while the repolymerized sample showed reduced Fe signal intensity due to improved encapsulation. Post-irradiation analysis demonstrated that the repolymerized composite maintained its structural integrity, whereas the single-step sample exhibited signs of surface degradation. These results suggest that repolymerization enhances both the morphological cohesion and radiological stability of the composite. The findings are supported by literature reports that emphasize the importance of nanoparticle dispersion and matrix reinforcement in magnetic polymer systems. Overall, the study highlights repolymerization as a simple yet effective strategy to improve the performance of ST-DVB/magnetite composites, particularly for use in high-risk or radioactive environments where mechanical robustness and magnetic recoverability are essential.

Keywords: Gamma irradiation; Polymeric composites; Magnetite; Repolymerization.

1 Introduction

Gamma irradiation is widely used to evaluate the structural resilience of materials under extreme conditions. In nuclear facilities, polypropylene (PP) filters are employed to remove cobalt-60 (Co-60) from radioactive liquid waste. These filters undergo mechanical degradation when exposed to gamma doses above 10 kGy, especially in air, where radiolytic decomposition of C–C bonds lead to significant loss of tensile strength and elongation (Chang *et al.*, 2021). In the medical field, polylactic acid (PLA) packaging materials also degrade under gamma irradiation. Doses between 1 and 30 kGy cause reductions in molecular weight, melting temperature, and mechanical strength, compromising their performance in sterilization contexts (Benyathiar *et al*., 2021). For space and nuclear applications, polymer nanocomposites have been irradiated with doses up to 158 kGy to simulate long-term exposure. These materials are designed for radiation shielding and must maintain structural integrity under high-energy ionizing radiation (More *et al.,* 2021).

In this study, a dose of 100 kGy was selected to simulate severe irradiation stress. This level exceeds typical industrial sterilization doses (10–30 kGy) and aligns with accelerated aging protocols. Lower doses were not tested. Literature suggests that resistance at 100 kGy implies stability across a wide range of lower doses, including those commonly used in industrial and environmental applications. The goal is to assess the composite's performance under extreme conditions and predict its behavior in more moderate scenarios.

Polymer-based composites incorporating magnetic nanoparticles offer multifunctionality for environmental remediation, catalysis, and radiation-resistant technologies. Among these, styrene-divinylbenzene (ST-DVB) copolymers have been extensively studied due to their chemical stability, ease of functionalization, and compatibility with various fillers (Mendes *et al.,* 2022; Reis *et al*., 2022; Recio-Colmenares *et al.,* 2022). The addition of magnetite (Fe_3O_4) introduces magnetic properties that enable separation and recovery in contaminated aqueous systems.

However, integrating magnetite into polymer matrices presents challenges. The high surface energy and magnetic dipole interactions of the nanoparticles often lead to aggregation, which compromises composite uniformity and reduces access to functional sites (Ghanooni *et al.,* 2022; Recio-Colmenares *et al.,* 2022). Magnetite is also sensitive to acidic and oxidative environments, which can degrade its structure. These limitations demand synthesis strategies that improve nanoparticle dispersion and encapsulation.

Repolymerization is one such strategy. A second polymerization step reinforces the polymer matrix and promotes more effective encapsulation of magnetite. This approach enhances both mechanical strength and radiation resistance. This study represents an initial step in evaluating the effects of repolymerization and gamma irradiation on ST-DVB/magnetite composites. Although exploratory, the research addresses a relevant gap in the development of radiation-resistant magnetic materials. By comparing samples synthesized with and without repolymerization, before and after exposure to high-dose gamma radiation, we assess structural improvements and their implications for high-risk applications.

2 Materials and Methods

To evaluate the influence of synthesis strategy and gamma irradiation on the structural and morphological properties of ST-DVB/magnetite composites, a series of experimental procedures were carried out. The methodology was designed to compare two distinct synthesis approaches—single-step polymerization and repolymerization—and to assess the effects of high-dose gamma radiation on each material. The preparation of the composites followed established protocols for suspension polymerization, while the irradiation and characterization steps were conducted using specialized facilities and instrumentation. The following sections detail the synthesis conditions, irradiation parameters, and analytical techniques employed in this study.

2.1 Synthesis of ST-DVB/Magnetite Composites

The synthesis of styrene-divinylbenzene (ST-DVB) composites doped with magnetite nanoparticles was performed using the suspension polymerization method described by Reis et al. (2022). The organic phase (OP) was composed of styrene (Sty, ≥99%) and divinylbenzene (DVB, 55%) in a 25:1 volume ratio, with benzoyl peroxide (BPO) as the initiator. Monomers were pre-washed with 10% NaOH and deionized water to remove inhibitors. The aqueous phase (AP) included trisodium phosphate (Na_3PO_4), calcium chloride ($CaCl_2$), and sodium dodecyl sulfate (SDS), which were dissolved separately and then combined to form a suspension medium. The OP and AP were mixed in a 1:4 volume ratio and introduced into a thermostated reactor (OptiMax™ 1001, Mettler Toledo). The polymerization followed three steps:

1. Initial stirring at 900 rpm at 25 °C for 3 hours;
2. Heating phase to 82 °C while maintaining 900 rpm;
3. Polymerization at 82 °C with reduced stirring (500 rpm) for 5 hours.

After polymerization, the product was filtered, washed with hot water, treated with 2 N HCl, rinsed with ethanol, and dried at 40 °C for 24 hours.

For the ST-DVB-Mag sample, 3.3 g of magnetite nanoparticles (10% w/w of OP, 50–100 nm, Sigma-Aldrich®) were dispersed in the OP using an ultrasonic bath for 3 minutes before polymerization. For the Repolymerized sample, the same procedure was repeated: the previously synthesized ST-DVB-Mag composite was subjected to a second polymerization cycle with a fresh addition of monomers and magnetite, following the same synthesis protocol.

In Figure 1 schematic representation of the synthesis methodology for ST-DVB/magnetite composites, including the repolymerization step. The process begins with the preparation of the organic phase (styrene, divinylbenzene, benzoyl peroxide, and magnetite) and the aqueous phase (Na_3PO_4, $CaCl_2$, and SDS), followed by suspension polymerization in a thermostated reactor. The initial polymerization yields the ST-DVB-Mag composite, in which magnetite nanoparticles are partially encapsulated. In the repolymerization step, fresh monomers and magnetite are added to the preformed composite, resulting in a second polymerization cycle that enhances nanoparticle encapsulation and structural uniformity.

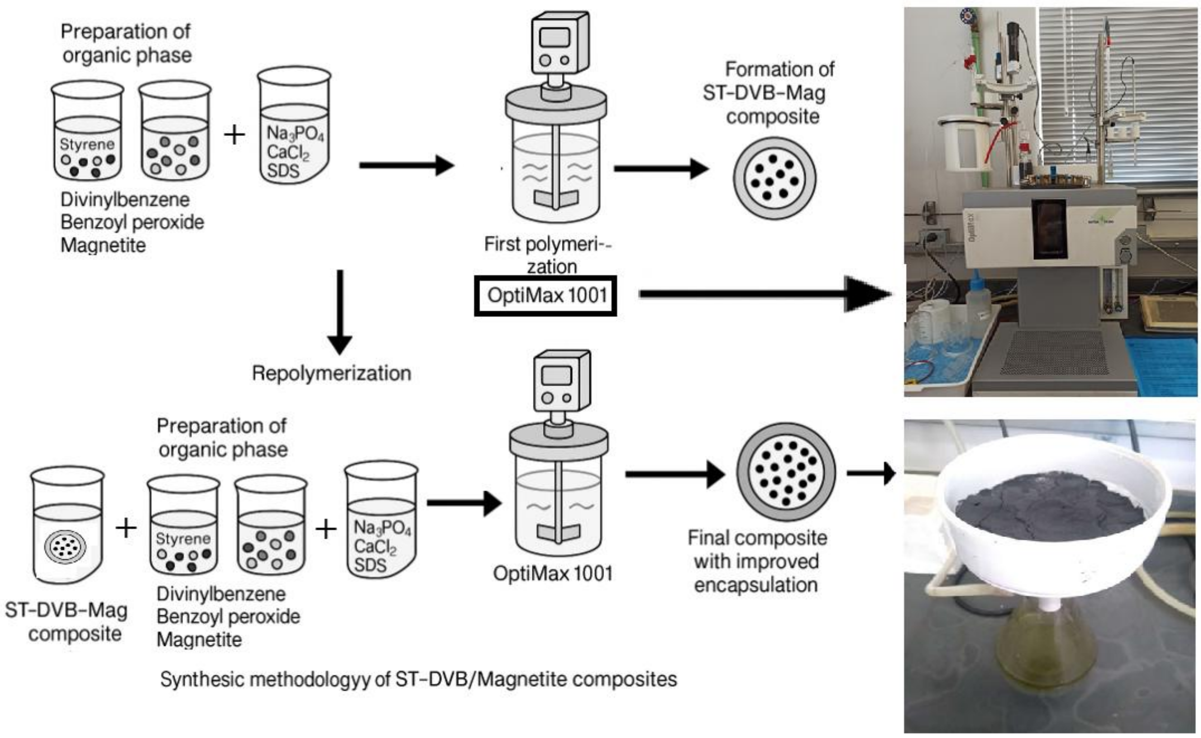

Figure 1: Schematic illustration of the synthesis methodology for ST-DVB/magnetite composites, including the repolymerization step. This figure was generated using artificial intelligence as a visual aid to complement the experimental methodology.

2.2 Gamma Irradiation and Morphological Characterization

Both composite samples were exposed to gamma radiation at a dose of 100 kGy using a Cobalt-60 source at the Gamma Irradiation Laboratory of CDTN (Centro de Desenvolvimento da Tecnologia Nuclear). This facility provides controlled irradiation environments for testing the radiological stability of materials under high-dose conditions, simulating extreme operational environments.

Morphological analysis was conducted using Scanning Electron Microscopy (SEM) with a JEOL JSM-IT300 microscope, located at the Electron Microscopy Laboratory of CDTN. This analysis enabled the evaluation of surface morphology, particle distribution, and the encapsulation efficiency of magnetite within the polymer matrix, both before and after gamma irradiation.

3 Results and Discussion

To evaluate the influence of synthesis strategy and gamma irradiation on the structural and morphological characteristics of ST-DVB/magnetite composites, a series of comparative analyses were conducted. The results are presented in a sequence that highlights the differences between the samples synthesized in a single step and those subjected to repolymerization, both before and after exposure to gamma radiation. Scanning Electron Microscopy (SEM) and Energy Dispersive Spectroscopy (EDS) were employed to assess surface morphology and elemental distribution, respectively, providing information into the encapsulation efficiency of magnetite and the structural integrity of the composites. The following subsections present and discuss the findings in detail, supported by representative micrographs and spectral data.

3.1 Morphological Characterization of Non-Irradiated Composites

The morphological characteristics of the ST-DVB/magnetite composites were evaluated by scanning electron microscopy (SEM), allowing a direct comparison between the sample synthesized in a single polymerization step and the one subjected to repolymerization. This analysis aimed to assess the effectiveness of the repolymerization strategy in improving the encapsulation of magnetite nanoparticles and enhancing the structural uniformity of the composite. The SEM images, presented in Figure 2, reveal distinct differences in surface structure, particle distribution, and overall homogeneity between the two materials.

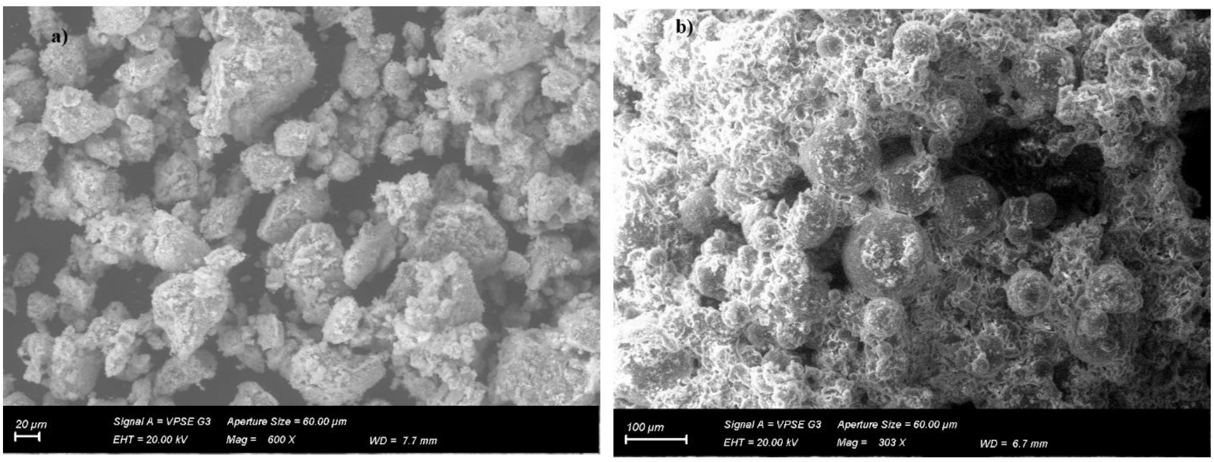

Figure 2: Scanning Electron Microscopy (SEM) images of ST-DVB/magnetite composites. a) Sample synthesized in a single polymerization step, showing a heterogeneous surface with exposed magnetite particles and signs of surface stress. b) Repolymerized sample, exhibiting a more uniform and continuous morphology, with better encapsulation of magnetite within the polymer matrix.

The single-step synthesized sample exhibits a rougher surface with visible clusters of magnetite, suggesting incomplete integration of the nanoparticles into the polymer matrix. In contrast, the repolymerized sample displays a smoother and more continuous morphology, with fewer exposed particles and a more cohesive structure. These visual differences provide initial evidence that the additional polymerization step contributes to a more effective encapsulation of the magnetic phase, which is critical for ensuring the composite's mechanical integrity and functional performance in demanding environments. In the sample obtained through a single synthesis step, the surface appears irregular and heterogeneous, with visible clusters of magnetite particles partially exposed on the surface of the polymer matrix. The morphology suggests that the encapsulation of the magnetic nanoparticles was incomplete, leading to a less cohesive structure. Additionally, signs of surface stress and microfractures are evident, which may compromise the mechanical and radiological stability of the material, especially under high-dose gamma irradiation.

In contrast, the repolymerized sample exhibits a markedly more uniform and continuous surface. The magnetite particles appear to be more effectively embedded within the polymer matrix, with fewer exposed regions and a smoother overall texture. This improved morphology is likely a result of the second polymerization step, which promotes better dispersion of the nanoparticles and reinforces the structural integrity of the composite. The increased homogeneity observed in the repolymerized sample indicates a more efficient interaction between the polymer and the magnetic filler, which is essential for applications requiring mechanical robustness and resistance to radiation-induced degradation. These observations support the hypothesis that repolymerization enhances the encapsulation of magnetite and contributes to the formation of a more stable and functional composite material. The morphological improvements observed in the repolymerized sample suggest its greater suitability for use in environments where structural integrity and magnetic recovery are critical, such as in the treatment of radioactive or hazardous aqueous waste. To complement the morphological analysis, Figure 3 presents the Energy Dispersive Spectroscopy (EDS) spectra for the same samples previously examined by SEM. Spectrum a) corresponds to the sample synthesized in a single polymerization step, while spectrum b) refers to the repolymerized sample.

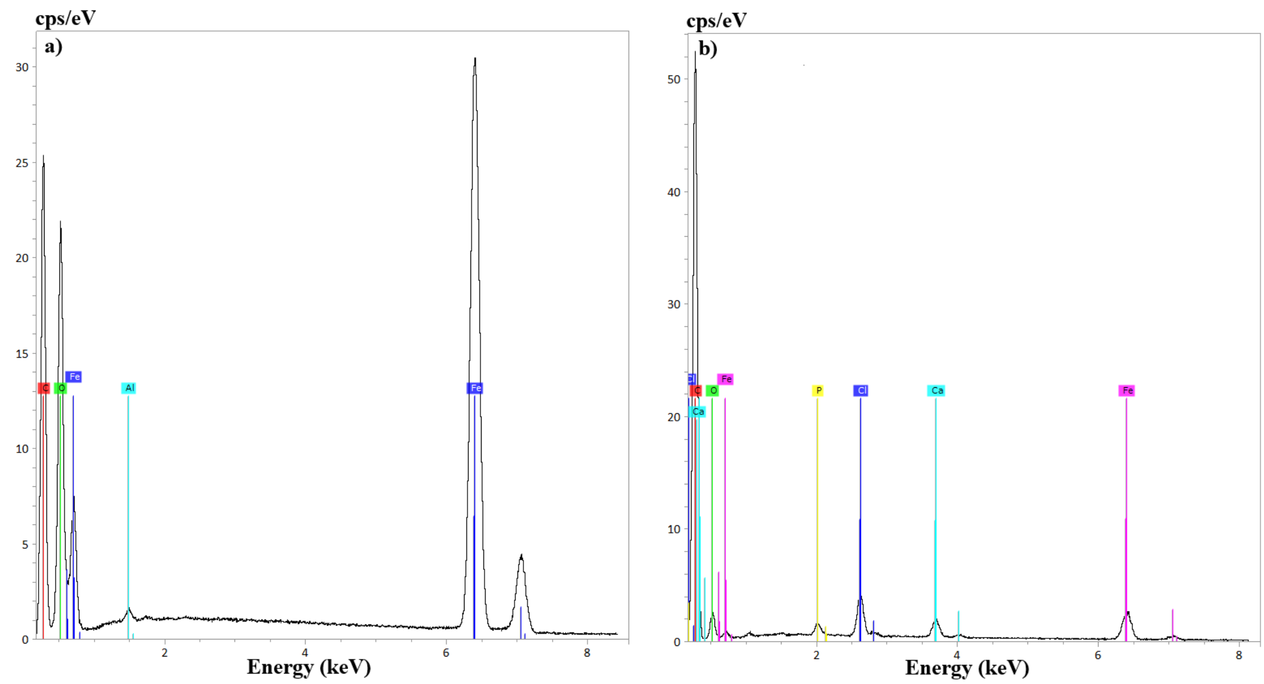

Figure 3: Energy Dispersive Spectroscopy (EDS) spectra of ST-DVB/magnetite composites. a) Spectrum of the sample synthesized in a single step, showing more intense iron (Fe) peaks, indicating greater surface exposure of magnetite. b) Spectrum of the repolymerized sample, with less intense Fe peaks, suggesting more effective encapsulation of magnetite within the polymer matrix.

In spectrum a, the iron (Fe) peaks are notably more intense, with prominent signals around 6.4 keV and 7.1 keV. This pronounced presence of iron suggests that the magnetite nanoparticles are more exposed on the surface of the polymer matrix. The higher surface concentration of iron aligns with the SEM observations, where magnetite clusters were visibly protruding from the polymer, indicating incomplete encapsulation during the initial synthesis.

In contrast, spectrum b shows significantly lower intensity for the iron peaks. This reduction in signal strength is indicative of more effective encapsulation of the magnetite within the polymer matrix. The repolymerization process appears to promote a more homogeneous distribution and deeper integration of the magnetic nanoparticles, thereby reducing their direct exposure to the electron beam during EDS analysis. These findings reinforce the conclusion that repolymerization enhances the structural integration of magnetite into the polymer network. The improved encapsulation not only contributes to a more uniform morphology but also suggests greater protection of the magnetic phase, which is essential for maintaining functionality in environments subject to radiation or chemical stress.

3.2 Comparative Morphological Analysis After Gamma Irradiation

Figure 4 presents SEM micrographs of the ST-DVB/magnetite composites before and after exposure to gamma radiation. Images a) and b) correspond to the non-irradiated samples synthesized in a single step and via repolymerization, respectively. Images c) and d) show the same samples after irradiation with a 100 kGy dose.

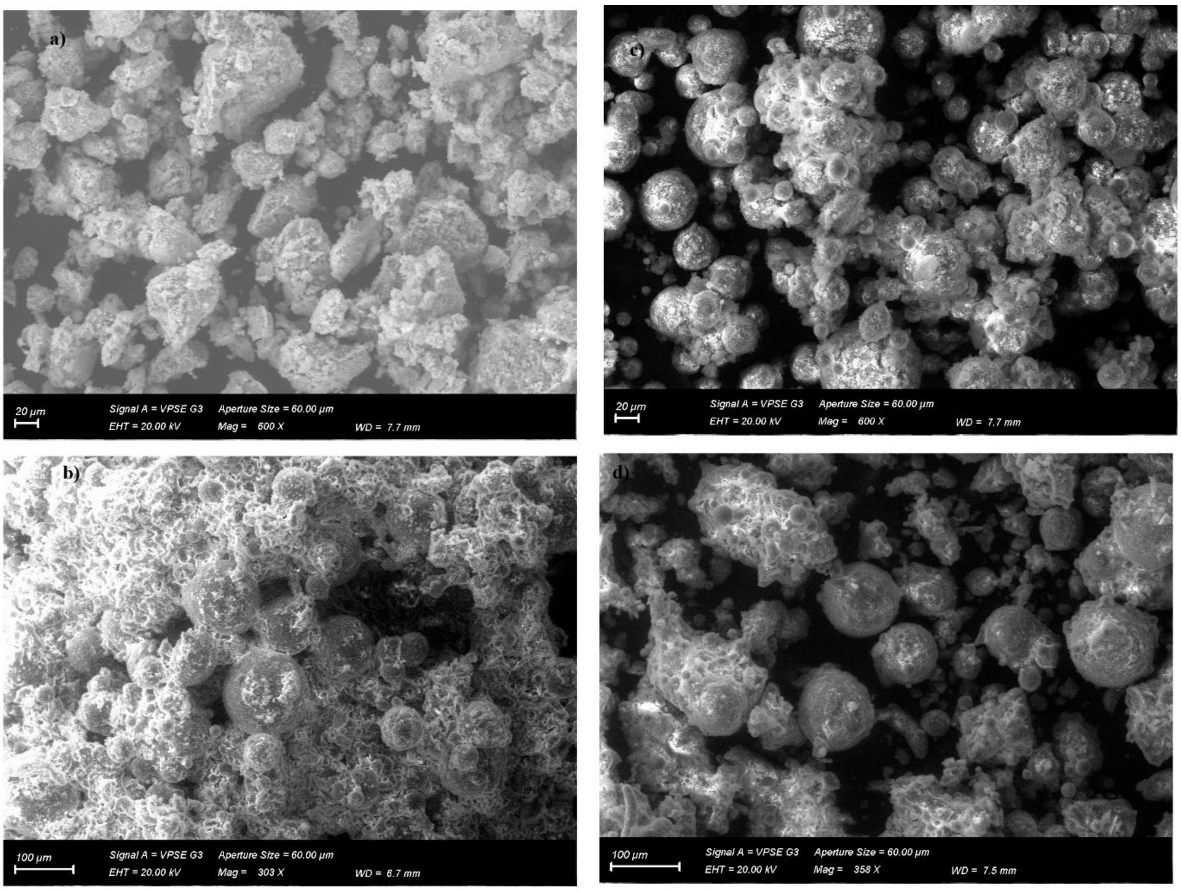

Figure 4: Scanning Electron Microscopy (SEM) images of ST-DVB/magnetite composites before and after gamma irradiation. a) Non-irradiated sample synthesized in a single polymerization step. b) Non-irradiated repolymerized sample. c) Irradiated version of sample a), showing smoother surfaces and signs of surface modification. d) Irradiated version of sample b), maintaining structural integrity and exhibiting more spherical and homogeneous particles.

Comparing images, a) and c), it is evident that the sample synthesized in a single step undergoes noticeable surface changes after irradiation. While the general structure is preserved, the surface appears smoother and less porous, suggesting superficial fusion or reorganization of the polymer matrix. These changes may compromise the mechanical stability of the material, especially considering the already limited encapsulation of magnetite observed in the non-irradiated state.

In contrast, the repolymerized sample (images b) and d)) demonstrates greater morphological stability. After irradiation, the particles become more spherical and homogeneous, but the overall structure remains cohesive and intact. The fibrous or filamentous network observed in the non-irradiated sample is less pronounced or absent, indicating some surface reorganization. However, this does not appear to compromise the integrity of the composite. These results highlight the superior structural resilience of the repolymerized composite under high-dose gamma irradiation, reinforcing the benefits of the additional polymerization step in enhancing both encapsulation and radiological stability.

The schematic illustration in Figure 5 compares the structural differences in magnetite incorporation between the single-step polymerization and the repolymerization processes, as observed in the SEM images presented in Figures 2 and 4. On the left side, labeled Single-Step Polymerization, the image depicts magnetite nanoparticles partially embedded in the polymer matrix. This representation is based on the SEM micrograph in Figure 2a, which shows a heterogeneous surface with visible magnetite clusters protruding from the polymer. The irregular morphology and surface stress observed in the SEM image are reflected in the schematic by the rough polymer boundaries and exposed nanoparticles. This incomplete encapsulation is further supported by the EDS spectrum in Figure 3a, which shows intense iron peaks, indicating surface exposure of magnetite.

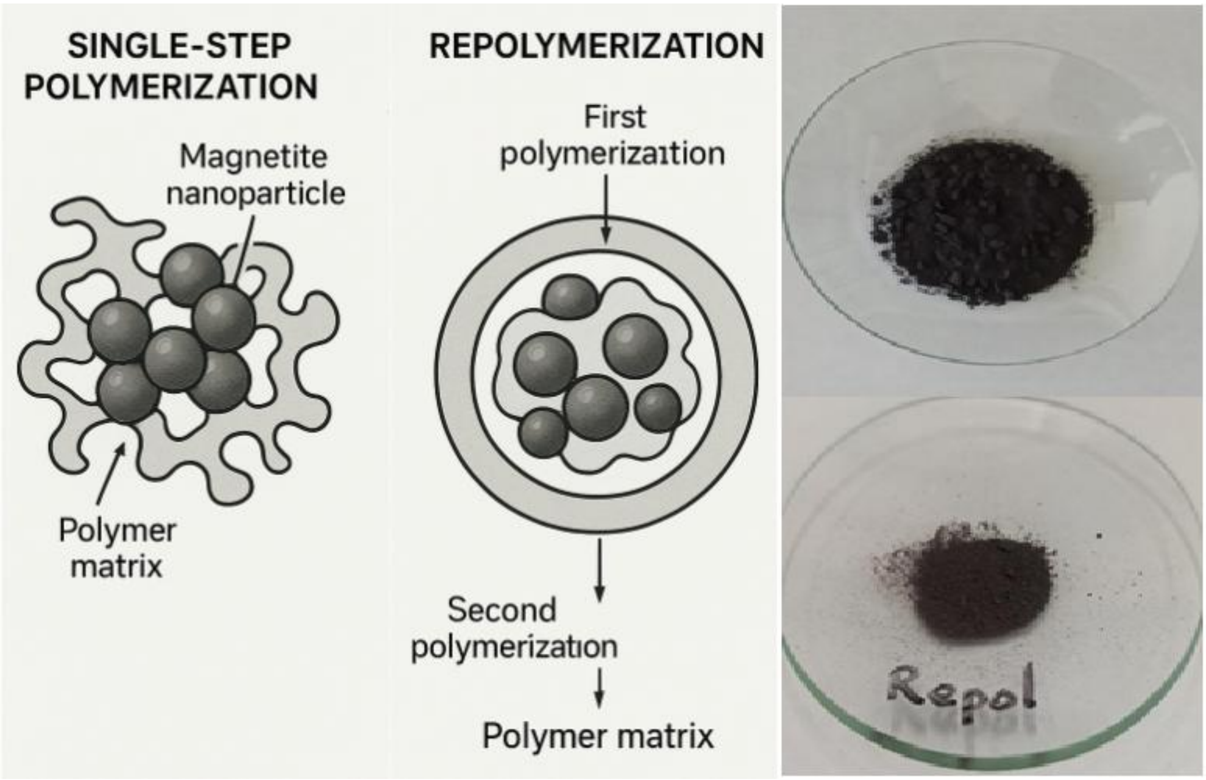

Figure 5: Schematic representation of magnetite nanoparticle incorporation in ST-DVB composites synthesized via single-step polymerization (left) and repolymerization (right). The illustration highlights the improved encapsulation and morphological uniformity achieved through the repolymerization process, as supported by SEM observations (Figures 1 and 3). This figure was created using artificial intelligence as a visual aid to complement the experimental findings.

On the right side, labeled Repolymerization, the schematic illustrates a more uniform and continuous polymer matrix with magnetite nanoparticles fully encapsulated. This visual is inspired by Figure 2b, where the SEM image shows a smoother surface and more cohesive structure, suggesting improved dispersion and integration of the magnetic phase. The second polymerization layer is shown enveloping the initial composite, reinforcing the matrix and reducing nanoparticle exposure. This is corroborated by Figure 3b, where the EDS spectrum shows reduced iron signal intensity, indicating better encapsulation. Additionally, Figure 4d supports this interpretation by showing that the repolymerized sample maintains its structural integrity even after gamma irradiation, with more spherical and homogeneous particles. Overall, the illustration visually reinforces the morphological improvements achieved through repolymerization, aligning with the experimental observations and analytical data.

3.3 Discussion

The incorporation of magnetic nanoparticles into polymeric matrices has been widely explored for the development of functional materials with applications in environmental remediation and radiation-resistant systems. This study demonstrates that repolymerization significantly improves the morphological integrity and encapsulation efficiency of magnetite in ST-DVB composites, especially under gamma irradiation.

SEM analysis revealed that the repolymerized sample exhibited a smoother and more continuous surface, with fewer exposed magnetite clusters. These observations were supported by EDS spectra, which showed reduced Fe peak intensity in the repolymerized sample, indicating deeper integration of magnetite into the polymer matrix. While these findings are qualitatively robust, future studies should incorporate quantitative metrics to strengthen the conclusions. For example:

- Particle size distribution from SEM images could be analyzed using image processing software to compare the dispersion and aggregation levels between samples.

- Surface roughness could be quantified using profilometry or atomic force microscopy (AFM), providing numerical evidence of morphological improvements.

- Iron content (%Fe) from EDS could be normalized and statistically compared across samples to quantify the degree of surface exposure.

Such metrics would allow for a more precise evaluation of encapsulation efficiency and structural uniformity. Quantifying these parameters would also facilitate comparisons with other composite systems reported in the literature and support predictive modeling for performance under radiation.

The observed reduction in surface-exposed iron in the repolymerized sample supports the hypothesis that a denser and more continuous polymer layer forms around the magnetite particles. This encapsulation enhances mechanical stability and protects the magnetic core from environmental degradation. Similar conclusions were drawn by Mendes *et al.* (2022), who reported that improved dispersion and encapsulation of maghemite in ST-DVB copolymers led to better performance in adsorption applications.

Furthermore, the resistance of the repolymerized sample to gamma irradiation aligns with findings by Recio-Colmenares et al. (2022), who demonstrated that macroporous magnetic ST-DVB composites maintained their structural integrity under harsh conditions when magnetite was well integrated into the polymer network. The smoother and more homogeneous morphology observed post-irradiation in the repolymerized sample suggests that the additional polymer layer acts as a protective barrier, mitigating radiation-induced damage.

These results collectively reinforce the notion that synthesis strategies aimed at improving nanoparticle encapsulation—whether through chemical modification, surfactant use, or structural reinforcement via repolymerization—are critical for the development of robust magnetic composites. The present study contributes to this body of knowledge by demonstrating that a relatively simple modification in the synthesis protocol can yield significant improvements in both morphological and radiological stability.

3.4 Future Perspectives

Building upon the promising results obtained in this study, several avenues for future research can be envisioned to further enhance the performance and applicability of ST-DVB/magnetite composites. One potential direction involves the chemical functionalization of the polymer matrix to introduce specific binding sites, thereby improving the selectivity and efficiency of the material in adsorption-based applications. Additionally, testing the composites in real-world scenarios—such as industrial effluents or radioactive wastewater—would provide valuable understanding of their practical performance under complex environmental conditions.

Investigating the reusability and magnetic recovery of the composites over multiple operational cycles could also shed light on their long-term stability and economic viability. Another promising approach would be the integration of these materials into filtration membranes or continuous flow systems, enabling their deployment in scalable water treatment technologies. Furthermore, exploring the effects of other types of ionizing radiation, such as electron beams or ultraviolet light, could broaden the understanding of the composite's resilience in diverse radiation environments. Finally, computational modeling and molecular simulations may help predict the structural behavior of the composites, guiding the design of next-generation materials with optimized encapsulation and mechanical properties.

4 Conclusions

This study demonstrated that the repolymerization of ST-DVB/magnetite composites significantly enhances their morphological integrity and radiological stability. Comparative SEM and EDS analyses revealed that repolymerized samples exhibit more uniform surfaces and improved encapsulation of magnetite nanoparticles, resulting in reduced surface exposure of iron and greater resistance to gamma irradiation. In contrast, the single-step synthesized sample showed signs of morphological degradation after irradiation, highlighting the limitations of conventional synthesis methods in ensuring long-term structural stability. These findings confirm that the additional polymerization step acts as a protective reinforcement, mitigating the effects of high-energy radiation. The results are consistent with previous literature emphasizing the importance of nanoparticle stabilization and matrix reinforcement in the development of functional magnetic materials. As a practical implication, the improved encapsulation and radiation resistance achieved through repolymerization suggest that these composites could be effectively employed in radioactive wastewater treatment systems, radiation shielding components, or magnetic separation devices used in nuclear and industrial environments. Their enhanced durability under high-dose gamma exposure supports their potential for long-term use in high-risk applications where mechanical robustness and magnetic recoverability are essential.

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Speknife: open-source software for X-ray spectra correction in solid state dosimetry

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Abstract

The accurate measurement of X-ray spectra is fundamental for determining beam characteristics and extracting quantitative dosimetric quantities such as mean energy, spectral resolution, fluence, energy fluence, air kerma, absorbed dose, and dose equivalents. Reliable dosimetry in solid-state detectors requires correction of systematic errors inherent to spectrometer instrumentation, a critical step for ensuring radiation safety and compliance with international standards. Despite the importance of this process, there has been a lack of open-source, freely distributed, and collaborative tools specifically designed for X-ray spectra correction in the context of solid-state spectrometers. Addressing this gap, we present Speknife, coded in Python 3 and based on the Stripping method, allowing users to correct their spectra without requiring knowledge of programming languages. Users define the required variables in a text file, and the software outputs the corrected spectrum, along with uncertainties for each energy–count pair and the mean energy. Speknife was validated using a CdTe spectrometer and ISO 4037 narrow beam qualities (N10 to N100), achieving mean energy errors below 2% and uncertainties under 1% across all tested beam qualities. This tool supports batch processing of multiple spectra and is available for download on GitHub for both local use and also an adapted version for Google Colab. By providing an accessible and validated solution for X-ray spectra correction, Speknife software supports the use of spectrometry for solid-state dosimetry and radiation quality characterization, aligning with collaborative scientific progress.

Keywords: X-ray spectrometry; spectra correction; Stripping method; ISO 4037; CdTe spectrometer; Speknife; solid-state dosimetry.

1 Introduction

The X-ray spectrometry is an experimental technique with protagonims on dosimetry applications, being used to determine materials' linear attenuation coefficients or enabling the obtaining of estimations of mean energy values, spectral resolution, air kerma, conversion coefficients from air kerma to dose equivalents and also being useful as a tool of non-invasive x-ray tube voltage (Huy and Van Dung, 2023; Frimaio et al., 2019; ISO, 2019; Silva et al., 2000; Terini et al., 2004; Nascimento et al., 2021).

However, the use of this kind of instrumentation requires that corrections be made in the measured spectra, since physical and electronic process that involves the interaction of radiation in the detector can impact the final result of the measurement (Nascimento et al., 2021; Gilmore, 2008). Especially in solid state detectors, this impact is significant in CdTe and CZT diodes due to their high atomic number (Castro et al., 1984; Seelentag and Panzer W., 1979; Tomal et al., 2015; Redus et al., 2009).

Although X-ray spectrometry posses a diversity of papers published and a diverse amplitude of approaches, the uncertainty evaluation associated with the measured spectra and the procedure of the spectra correction is not usually done, and there is a lack of information related to it (Moralles et al., 2007; Santoro-Fernandes et al., 2020). Besides this, it is worth mentioning that there is a lack of open-source and free distribution software that proposes to correct the spectra and evaluate their uncertainty, so that obligate researchers who work with X-ray spectrometry to reinvent the wheel.

It is in this context that Speknife software is proposed, in order to not be necessary to reinvent the wheel, but to just keep it spinning and eventually make some small balancing.

The Speknife software is an open-source, free-distribution, collaborative, and expandable code that is useful to correct spectral distortions in a CdTe solid state detector of 1 mm thick and will evaluate the uncertainties in the process of spectra correction, being validated for N series ISO 4037-1, from N10 to N100.

Although it seems to be a specific software for a particular case, the stripping logic implemented could be expanded for other solid state detectors and different thicknesses of depleted layer according to demand, once it is open-source, freely distributed, and collaborative.

1.1 ISO 4037-1 X-ray qualities

The ISO 4037 is a norm that establishes reference radiation fields for the calibration of dosimeters and dose rate meters and the energy response of X and gamma rays in radiation protection. This norm has four parts, being the first one being the scope of this paper. This part one presents the characteristics of the radiation fields and their production methods (ISO, 2019).

It is worth mentioning that there is also other literature that establishes radiation reference fields for X-rays as the BIPM qualities for inter-comparisons and IAEA diagnostics qualities from TRS-457 (IAEA, 2007; Peixoto and et. al, 2016).

The X-ray part qualities from ISO 4037-1 have four groups of reference radiation fields: the narrow spectrum series (N), the wide spectrum series (W), the high kerma rate series (H), and the low kerma rate series (L). Each one of the groups mentioned is separated according to its spectral resolution. The implementation of each one of the spectra series is related to the additional and inherent filtration used, besides fulfilling X-ray tube characteristics requirements (ISO, 2019).

1.2 Uncertainty evaluation

The uncertainty of each energy value of the spectra, E_i, is a combination of the calibration uncertainty, $\sigma_{E_{cali}}$, with the resolution of each energy bin, ($\sigma_{E_{res}}$), as represented in Equation 1.

$$\sigma_{E_i} = \sqrt{\sigma^2_{E_{cali}} + \sigma^2_{E_{res}}} \tag{1}$$

The uncertainty due to resolution could be considered as rectangular, since the spectra is a histogram and the calibration attaches each channel from the histogram to the middle energy that represents it, E_i. The calibration component comes from the propagation of the straight line coefficients, $a;b$, to the energy, considering the correlation between them (BIPM, 2008).

The uncertainties related to the counting of each energy bin are also a combination of the uncertainty of Poisson statistics, (σ_{Pois}), with the uncertainty of the spectra corrections (σ_{corr}), as represented in Equation 2.

$$\sigma_{C_i} = \sqrt{\sigma^2_{Pois} + \sigma^2_{corr}} \tag{2}$$

The Poisson uncertainty is due to the independence of the photon emission from the X-ray tube, being accounted as $\sqrt{n}$ where n is the counts in channel i (Santoro-Fernandes et al., 2020). In turn, the uncertainty related to the spectra correction is a sum in quadrature of the procedure of the escape correction, Compton correction, and efficiency correction.

2 Method

2.1 The Speknife software

It was used the Python 3.8.10 in the development of the Speknife software and which possesses one basic requirement for its operation, and a second optional requirement. If it is desired to do only the correction without exhibiting graphs, it is only required to have Numpy installed. However, if it is desired to plot graphs, it is necessary to have Matplotlib.pyplot installed.

The software was developed to keep the user distant from the need of programming knowledge in Python, and because of that, the person who will make use of the Speknife will only need to fill a *input.txt* file with some parameters. There are four obligatory arguments to be passed to the input file if the user wants only a simple use of the correction algorithm. However, if is desired a full experience in Speknife there are other 16 arguments to declare.

The obligatory arguments are the electric voltage of the tube, the total number of channels, and the coefficients a and b of the calibration straight line.

In turn, the optional arguments are if the file to be corrected is of the type *.mca* or a simple file with energy column and counts column; if uncertainty will be evaluated and the uncertainty of the coefficients of the line and the Pearson correlation; if will be displayed any kind of plot and what of the six types it will be and what type of normalization will be used; if the plots will be compared to a external source, being possible to use the N series from PTB that already is available, or give a own reference and it base name (N, W, RQR, BIPM); if it will be calculated the tube voltage by spectrometry and its necessary parameter. In Figure 1 could be seen an example of the parameters to be chosen in a full use of the Speknife software.

As mentioned before, Speknife could calculate tube voltage by spectrometry. This is done using the last 15 points of the spectra in a linear regression with weights, considering the uncertainty of the energy and counts. This information requires that the user give a parameter named 'shift' to manually choose which points of the high-energy side of the spectra will be used. In a future implementation is desired that IA could be used to automatically fit this line in the last 15 appropriate points of the spectra.

input.txt

```
#Basic requirements
tube_voltage                100 #Must be an integer
channel_number              2048
a                           0.07599
a_uncertainty               0.0000072
b                           -0.08387
b_uncertainty               0.0054
r_pearson                   -0.984
mca                         True
uncertainty_analysis        True

#Plot control
plot                        True #show or not the plot
plot_type                   2
normalization_number        1 #plot_type 4 only a normalization 0 is possible
save_plot                   True

#Tube voltage analysis
tube_voltage_plot           True
tube_voltage_measurement    True
shift                       -12 #kv plot shift

#Display comparisons plot
other_database              True
base_name                   N
reference_path              reference_radiation/PTB_spectra/narrow_beam/ #Path to the reference data to be compared
reference_name              PTB #Reference name for the quality to display on the plot
```

Figure 1: Exemple of the full use of *input.txt* file

When running the code, at the end of the run it exhibit in the terminal the values of the quantities calculated and their uncertainties, when required, and save a file with the corrected spectra in a text file also containing some basic information of the quantities calculated and some parameters chosen by the user in the input file to keep the traceability of the information used to made the corrections.

The software is available to download at GitHub (https://github.com/matheusrebe/Speknife) for a local use version and also a version adapted to the Google Colab environment. The Colab version was built in a way that the Jupyter Notebook does not show any source code to the user. In order to run it, it will be only necessary to place the Colab paste in the personal Google Drive root directory, specify the arguments listed in the input file, and press three buttons in the Jupyter notebook.

The Speknife makes use of the stripping method to correct the three main phenomena that distort the spectra that are the escape of characteristic X-rays, the Compton plateau, and the intrinsic efficiency loss (Nascimento et al., 2021; Castro et al., 1984; Seelentag and Panzer W., 1979; Tomal et al., 2015).

The corrections are made from the higher energy channel regressively until channel zero in a loop. The algorithm also attributes zero values in counts in channels higher than 2 keV of the maximum nominal energy produced by the X-ray tube in order to avoid using pile-up data in the corrections. Besides that, at the end of the channel-by-channel correction process, negative count values are automatically attributed a zero value to avoid using them in the mean energy calculation.

Regarding the uncertainty calculation, it is done continuously during the interactions of the software. The code receives the file *input.txt* with the parameters necessary to begin the calculations based on the calibration data, given by the user, and creates an uncertainty list for the energy and for the counts, and keeps updating it at each correction interaction made inside the software.

2.2 Validation criteria

The corrections made by the Speknife were evaluated quantitatively by the mean energy error, which shall not be higher than 2% of the nominal value. The choice of this value is due to the ISO 4037-1 statement that characterized radiation fields do not differ their mean energy values by more than 2% of the nominal value (ISO, 2019).

2.3 X-ray spectra measurement

The measurements were done with the Be window of the spectrometer positioned at a distance of 100 cm from the focal spot of the X-ray tube (ISO, 2019). The general scheme used in the measurements is shown in Figure 2.

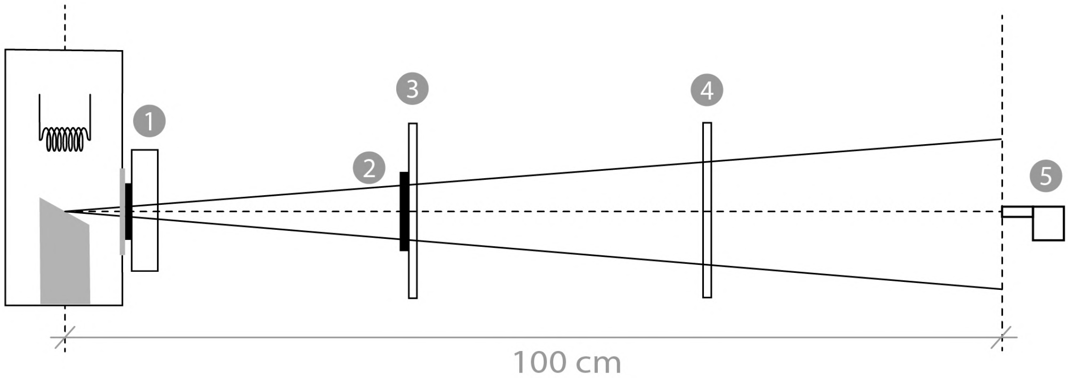

Figure 2: Experimental scheme of the measurements.

In Figure 2, the number one represents the shutter, which, if it is open, allows the passage of the X-ray; the number two is the additional filtration, and the numbers three and four are collimators. In that way, it is ensured that the diameter of the beam is 10 cm at 100 cm from the focal spot.

The spectrometer, which is element five, is an XR100-T CdTe detector from manufacturer Ampetk and was configured by the DPPMCA software using the CdTe default PX5 (Amptek, 2019). Seeking to obtain reasonable counting statistics, a 600 s acquisition time was defined for all the spectra. The reference qualities were measured, each one, five times, and the detector was calibrated using a sealed ^{241}Am source, measured once during 600 s.

3 Results

3.1 Validating Speknife

The results obtained for the mean energy of raw spectra, Speknife corrected spectra, and ISO nominal values are presented in Figure 3. It could be observed that only N20 and N25 meet the 2% criteria without correction, while N10 and N15 have an error of 3%, and the other qualities show an error of 4%, 13%, 8%, 5%, and 7%, respectively. However, when using the Speknife to correct the spectra, all the qualities met the 2% criteria.

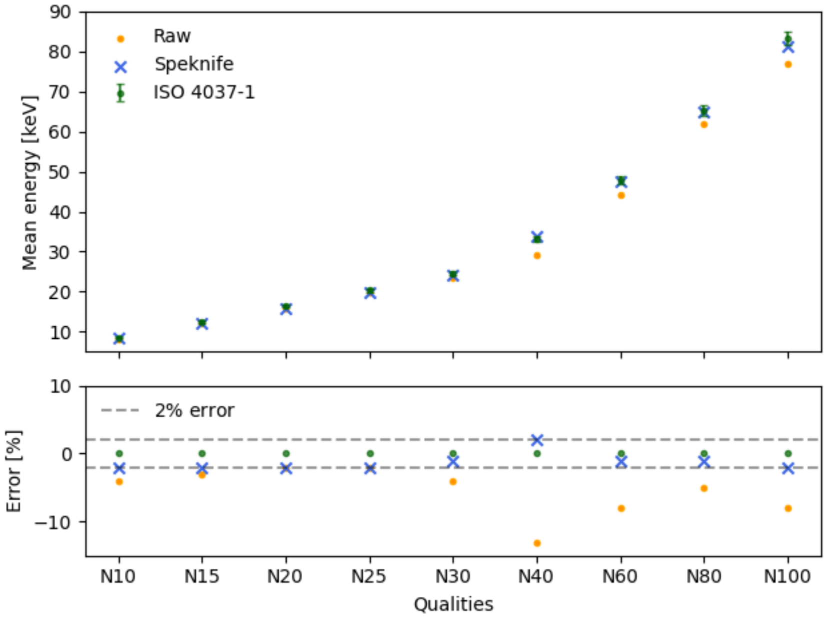

Figure 3: Mean energy comparison for the qualities N10 to N100

In Figure 4, the N60 and N100 spectra corrected by Speknife are compared to the raw spectra and the impact of each distortion effect.

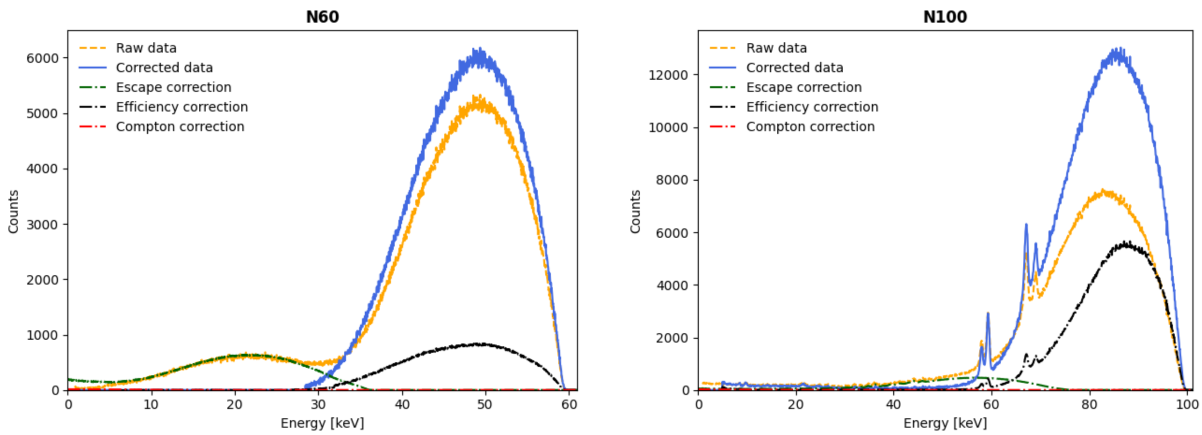

Figure 4: N60 and N100 corrected by Speknife.

3.2 Uncertainty evaluation

The uncertainty evaluation of the mean energy was obtained by the uncertainty propagation of the mean energy equation with the repeatability and tube voltage by spectrometry. The percentage uncertainty of each component and the standard combined uncertainty for each quality are presented in Table 1. It is worth mentioning that only repeatability is a type A uncertainty.

Table 1: Mean energy uncertainties of the radiation qualities [%] - N10 to N100.

	Mean energy	Voltage	Repeatability	Combined uncertainty
N10	0.015	0.082	0.0030	0.083
N15	0.011	0.080	0.0020	0.081
N20	0.0084	0.060	0.0030	0.061
N25	0.0086	0.064	0.018	0.067
N30	0.0080	0.17	0.0033	0.17
N40	0.0074	0.15	0.012	0.15
N60	0.0083	0.72	0.0030	0.72
N80	0.0077	0.076	0.0022	0.076
N100	0.0088	0.083	0.0020	0.083

4 Conclusions

The Speknife software was developed in Python 3 for local use on a computer, but can also be used on the Google Colab version, both available on GitHub for download. It is an open-source, freely distributed, and collaborative software built for users who do not necessarily know Python, and with the idea to be continuously developed by users, making it possible to build it for other detectors and other methodologies of correction algorithms.

The software allows the user, through an input text file, to declare if they desire a simple correction of the spectra without graphs and uncertainties, but it is also possible to do so with six graph options. It is also possible to do batch processing of multiple spectra in sequence, if they share the same number of bins and calibration coefficients.

In the validation of the software, N10 to N100 ISO 4037 spectra were measured, and the raw spectra showed some errors in the mean energy higher than 2%, but when using Speknife, all errors became below the maximum accepted value.

It was also implemented the uncertainty evaluation in the correction of the spectra, a subject not fully discussed in literature, being found values lower than 0.01% for the repeatability and mean energy, and lower than 0.1% for the voltage determined by spectrometry, being this component the one that most contributes to the standard uncertainty. Thus, the standard uncertainty for the mean energy of the qualities measured was lower than 1% for the X-ray spectrometry using a CdTe detector in the chosen electronic conditions.

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Automated Classification and Manual Segmentation of Pediatric Brain Tumors Using Deep Learning on MRI Data

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Abstract

Introduction: The world Health Organization (WHO) has set a target to achieve a 60 % survival rate for pediatric cancer patients by 2030. However, currently, 95% of childhood cancer deaths occur in low and middle-income countries. Among pediatric tumors, brain neoplasms-such as gliomas and medulloblastomas require precise diagnosis through magnetic resonance imaging (MRI). This study aimed to develop a deep learning model to classify and segment brain tumors in pediatric MRI scans (relaxometry *T1*, *T2* and diffusion-weighted sequences), using data from patients with and without tumor diagnoses treated at the Federico Gomez Children`s Hospital. Methods: MRI scans from 60 pediatric patients (20 with tumors and 40 without tumors) were analyzed. For positive cases, manual segmentation was performed using 3D Slicer to delineate regions of active tumor, necrosis, edema, and healthy tissue. The results revealed distinct quantitative profiles: **tumor:** volume = 248 mm³; diffusion intensity = 1990.4 $\pm$ 405.9 a.u., **edema:** volume = 25.6 mm³; diffusion intensity = 974$\pm$753a.u. The significant interpatient variability highlighted the importance of personalized analysis approaches. Conclusion: Manual segmentation improved precision in tumor characterization, supporting early diagnosis and treatment planning in pediatric oncology.

Keywords: Convolutional neural network, MRI, relaxometry *T1*.

1 Introduction

Childhood cancer represents the second leading cause of mortality among children and adolescents aged 4 to 15 worldwide. According to recent projections, by 2025, over 20 million new cancer cases will be diagnosed globally, with approximately 80% concentrated in low- and middle-income countries (Roberto Rivera Luna, 2024). In this context, early and accurate diagnosis through magnetic resonance imaging

(MRI) has become a critical factor in clinical management, as it enables timely initiation of therapies that improve survival rates, enhance oncological outcomes, and reduce the economic burden of treatment. Studies demonstrate that early detection can substantially improve pediatric patients' quality of life (Resende & Alves, 2021).

Despite advances in automated methods for brain tumor segmentation and classification in adults, their application in pediatric populations remains limited. This disparity stems from fundamental differences in the anatomical and pathological characteristics of pediatric tumors, which require specialized therapeutic strategies. Furthermore, traditional image processing techniques face challenges in sensitivity and accuracy when applied to this age group, underscoring the need to develop tools tailored to the unique aspects of pediatric oncology.

1.1 State of the Art

Recent advances in artificial intelligence have driven the development of tools for automated brain tumor detection and segmentation, though with a predominant focus on adult patients. In 2020, a convolutional neural network (CNN) architecture achieved 93% accuracy in binary classification (tumor vs. non-tumor), using both accuracy and loss metrics across training and validation sets. This work emphasized the importance of optimizing filtering parameters across different network configurations (Febrianto et al., 2020).

Subsequently in 2023, Gómez-Guzmán et al. evaluated pretrained models (InceptionV3, ResNet50, among others) for adult tumor classification, comparing metrics including accuracy, recall, precision, AUC, and specificity. InceptionV3 demonstrated: accuracy 97%, precision 97%, recall 96%, specificity 99%, and AUC 0.99, while ResNet50 yielded: accuracy 96%, precision 97%, recall 96%, specificity 99%, and AUC 0.99. Their results revealed significant performance variations across architectures, highlighting the need for model adaptation to dataset-specific characteristics (Gómez-Guzmán et al., 2023).

A landmark achievement in this field was the 2013 establishment of the BRATS (Brain Tumor Segmentation) initiative, which provides a standardized MRI dataset containing *T1*, contrast-enhanced *T1*, *T2*, and T2-FLAIR sequences for segmentation algorithm training. While this resource has been crucial for methodological reproducibility and progress, its application to pediatric populations remains limited due to anatomical and pathological differences (Dai et al., 2018).

Recent studies (Bengtsson et al., 2025), (Griffiths-King et al., 2025), (Kharaji et al., 2024) on automatic segmentation in the pediatric population have shown significant advances in the development of neural networks, especially U-Net, which has been progressively improved to make segmentation more accurate. However, major challenges remain, such as the localization of the tumor, edema, and necrotic tissue. Unlike these approaches, our method first proposes to manually segment the images so that when automatic segmentation is later developed, it can achieve higher accuracy in three regions of interest: tumor, edema, necrotic tissue, and healthy tissue.

1.2 Nuclear Magnetic Resonance (NMR)

Nuclear Magnetic Resonance (NMR) is a widely used medical technique for generating detailed, non-invasive images of the human body. This method employs a strong magnetic field and radio waves to produce high-resolution images of internal tissues and organs. Beyond clinical applications, NMR also plays a pivotal role in chemistry and physics, where it is used to study molecular and material structure and dynamics (Darío & Torres, 2007). NMR operates through the interaction between an external magnetic field and atomic nuclei possessing spin – a quantum property associated with angular momentum and an intrinsic magnetic moment (I), determined by nuclear structure. From a classical perspective, spin can be conceptualized as the rotation of the nucleus about its axis at a specific frequency.

A rigorous mathematical approach is essential for understanding this technique's fundamentals, requiring the description of nuclear behavior and environmental interactions through quantum mechanical principles (Brown, 2003). The human body consists predominantly of water- and fat-containing tissues, both rich in hydrogen atoms. The hydrogen nucleus' magnetic moment aligns parallel to its rotation axis. Under an external magnetic field, spin manipulation generates signals forming the basis for magnetic resonance imaging. When atomic nuclei are exposed to an external magnetic field, they absorb energy at a specific frequency (Larmor frequency). Subsequent energy release produces detectable electrical signals that provide structural and functional sample information. Following emission, nuclei return to equilibrium, ready for renewed excitation (Haacken, 1999).

1.2.1 Relaxometry *T1*

The longitudinal relaxation constant (T1), referred to as spin-lattice relaxation, is characterized by an exponential decay process defined by its relaxation time, measured in milliseconds (ms.). A reduced T1 value indicates the system returns to equilibrium more rapidly, corresponding to faster relaxation. In other words, a shorter T1 implies more immediate energy release. This relaxation process is mathematically described by an exponential function (Gili & Alonso, n.d.):

$$M_z = M(1 - k\, exp(-t/T1)). \tag{1}$$

A quantitative understanding of *T1* enables optimization of acquisition protocols and facilitates the development of novel molecular contrast agents for advanced imaging applications.

1.2.2 Relaxometry *T2*

The transverse relaxation (*T2*) results from the loss of phase coherence in transverse magnetization, manifesting as exponential decay of its perpendicular component (*xy*-plane) relative to the static magnetic field (B_0). This phenomenon occurs because nuclear spins experience local magnetic fields arising from the superposition of the applied external field and fields generated by neighboring spins.

Local field variations induce differences in spin precession frequencies, causing gradual dephasing of individual spins over time, as spins lose phase coherence, the net transverse magnetization vector undergoes exponential decay and geometrically, this process resembles progressive fanning-out of individual spin vectors. The total transverse magnetization $M_{x\gamma}$ corresponds to the vector sum of individual spin components in the *xy*-plane. Its temporal evolution is governed by:

$$\frac{dM_{xy}}{dt} = -\frac{M_{xy}}{T2}, \tag{2}$$

the term $-\frac{1}{T2}$ quantifies the decay rate due to spin-spin relaxation processes.

1.2.3 Diffusion

Magnetic Resonance Imaging (MRI) is currently the only non-invasive method capable of detecting and quantifying molecular diffusion *in vivo*, that is, the translational displacement of molecules. In human tissues, the diffusion signal originates from molecular motion within three primary compartments: **(a)** the extracellular space, **(b)** the intracellular space, **(c)** the intravascular space (Le Bihan, Denis, 1988).

Among these, the intravascular compartment exhibits the highest diffusion due to blood flow dynamics. Consequently, highly vascularized structures, such as certain tumors, often display an increased signal on diffusion-weighted imaging (Thoeny et al., 2005).

Tissues with high cellular density (e.g., many tumors) restrict the free movement of water molecules, resulting in a reduced diffusion signal. This property enables microstructural characterization of lesions based on cellular organization. Unlike stationary water molecules, those in motion do not fully recover their phase after the application of the second diffusion gradient, leading to signal attenuation. This attenuation is directly proportional to the amplitude of the applied gradients, thereby allowing quantification of molecular diffusion.

1.3 Artificial Intelligence

Artificial Intelligence (AI) is a computer science discipline focused on developing systems capable of performing tasks that would typically require human intelligence (Zabala Leal, 2023). This technology can: Acquire information through human-like cognitive processes, execute actions based on: observational learning, pattern imitation, evolutionary algorithms (Gopinath, 2023).

1.3.1 Convolutional Neural Network

A Convolutional Neural Network (CNN) is a type of deep learning algorithm specifically designed for tasks involving object recognition, such as image classification and detection (Gelvez, 2019). This type of network includes at least one convolutional layer; from which it derives its name.

Convolution is a commutative mathematical operation involving two functions that enables operations with real values (Beinat, 2022). In the context of Convolutional Neural Networks (CNNs), this operation is performed between an input—such as an image—and a filter or kernel, producing an output known as a feature map. This operation, also referred to as cross-correlation, highlights patterns or specific features within an image, thereby facilitating the automatic learning of relevant spatial representations.

Convolutional Neural Networks are designed to process two-dimensional data, making them particularly effective for image recognition and voice signal processing. However, they can be adapted to work with one-dimensional or multidimensional data, broadening their applicability across various domains.

Currently, there are CNNs specialized in tasks such as edge detection in images—a capability that is especially useful in the analysis and classification of medical images. However, a limited understanding of how these networks function can hinder their fine-tuning during training, particularly when dealing with complex anatomical structures such as the brain.

The convolution operation can be mathematically expressed as:

$$s(t) = (x * w)(t) = \int x(a)w(t-a)da. \tag{3}$$

In neural networks, this operation is applied discretely, where $s(t)$ represents the input, $(x * w)(t)$ the kernel or filter, and the output is a feature map that highlights certain aspects of the input image.

1.4 Manual Segmentation

In this section, we delve into the meticulous process of manually segmenting tumors in brain magnetic resonance imaging (MRI) (Figure 1). We acknowledge the special attention this procedure requires, given its time-consuming nature and susceptibility to errors. Precision and efficiency are essential, as any deviation could have significant consequences for patient diagnosis and treatment (Daimary et al., 2020).

Aware of the inherent challenges of manual segmentation, we have chosen to work closely with a highly experienced neuroradiologist, whose specialized knowledge—acquired over two decades of clinical practice—is invaluable. Their expertise not only provides a deep understanding of brain anatomy and tumor characteristics but also ensures the skill and accuracy necessary to perform this task with excellence (Tirivangani Magadza, 2021).

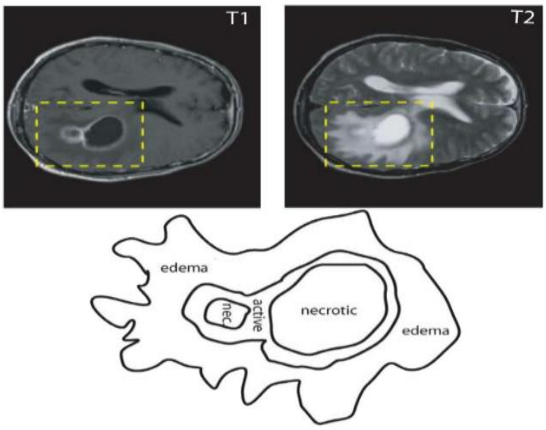

Figure 1: Imagen of tumor, necrotic and edema (Tirivangani Magadza, 2021).

2 Methodology

Approximately 4,200 magnetic resonance imaging (MRI) scans were collected from 60 pediatric patients, including diffusion sequences and relaxometry sequences in *T1* and *T2*. All scans were acquired under standardized clinical protocols, with acquisition parameters including a repetition time (*TR*) between 15 ms. and 1000 ms., echo time (*TE*) between 2.38 ms. and 14 ms., and a resolution matrix of 240 × 240 pixels. The cohort was classified into several diagnostic subgroups validated by both clinical and histopathological studies. These subgroups included common pediatric brain tumors such as ependymoma, medulloblastoma, brainstem glioma, astrocytoma, among others.

Image Selection and Preparation:

From the total available data, 140 MRI images were selected for the classification stage during model training. This selection included 60 images from patients with tumors and 80 images from patients without tumors. Subsequently, the network was validated using new images, both with and without tumors. Selection criteria prioritized images with a high signal-to-noise ratio, free of motion artifacts, and with complete anatomical coverage. This study was approved by the Safety and Ethics Committee of the Federico Gómez Children’s Hospital. Only *T1*-weighted scans without contrast were considered to ensure uniformity in signal characteristics. All selected cases were confirmed as tumor-positive or tumor-negative through histopathological analysis and radiological evaluation performed by a certified pediatric neuroradiologist with over 15 years of experience. Training and validation images were obtained under standardized clinical protocols. Only studies from patients with or without tumors were included, excluding those with significant artifacts.

Image Classification:

In this work, we proposed a basic convolutional neural network (CNN) model and used it to process enhanced MRI images with an RGB color channel and a batch size of 32 images.

We started with a convolutional layer of 16 filters, each with a filter size of 3×3×3. The goal of using just 16 filters in this initial stage was to identify basic patterns such as corners, edges, and lines. Then, we added a max pooling layer with a filter size of 2×2×2 to extract the maximum value within each region of the image.

After this, we expanded the convolutional layers, increasing the number of filters to 32, then 64, and finally 128, while maintaining a similar filter size (3×3×3). As the number of filters grows, the model can combine simple patterns to form more complex ones, such as circles, squares, etc.

To maximize this structure, we added max pooling layers after the convolutional layers. Additionally, we incorporated a fully connected dense layer of 256 neurons, along with a softmax output layer that computes the probability of each class and outputs a final classification decision as Yes/No, depending on whether the MRI image contains a tumor.

Image Segmentation:

Once the classification stage was completed, segmentation was performed on images from 20 patients with confirmed brain tumor diagnoses. For each case, segmentation masks were generated that delineated three key anatomical regions: (i) the active tumor region, (ii) the peripheral zone including edema and necrotic tissue, and (iii) normal brain tissue. These regions were manually segmented using the open-source software 3D Slicer, following a slice-by-slice methodology. The segmentations were validated by a pediatric neuroradiologist with over 15 years of clinical experience, thereby ensuring the quality and consistency of the generated masks.

Subsequently, quantitative metrics were extracted from each segmented region, including volume, mean signal intensity, and standard deviation, based on *T1* and *T2* relaxometry sequences as well as diffusion images. These values were used to characterize the properties of each region.

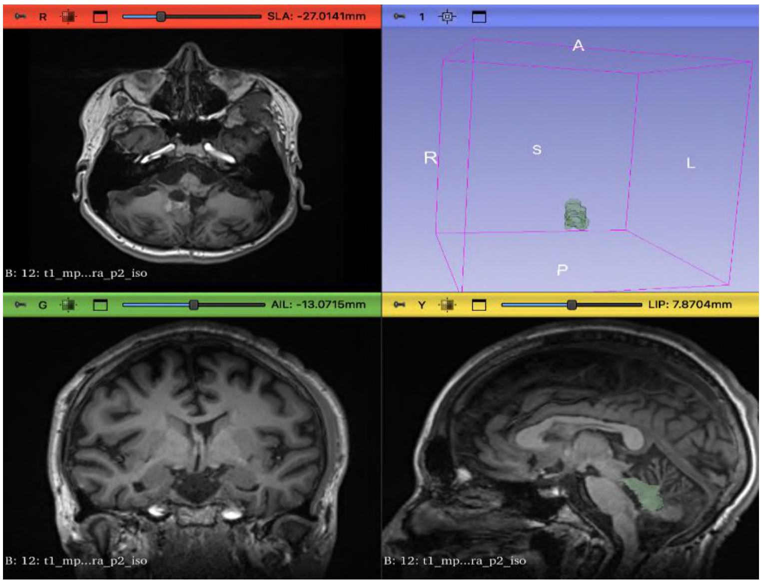

Figure 2: Segmentation of tumor using 3D slicer.

3 Results

Evaluation

The main expected results of this research are, firstly, to deepen the understanding and development of a Convolutional Neural Network (CNN) applied to magnetic resonance imaging (MRI) in pediatric patients. The research is structured around two primary objectives:

Image Classification:

A CNN model was trained to classify MRI images obtained through *T1* relaxometry, aiming to detect the presence or absence of brain tumors in pediatric patients. The model achieved an area under the ROC curve (AUC) of 0.75, with a sensitivity of 0.91 (95% CI: 0.825–0.877) and specificity of 0.92 (95% CI: 0.953–0.980). External validation confirmed the model's generalizability, obtaining an AUC of 0.70, with sensitivity of 0.91 (95% CI: 0.807–0.827) and specificity of 0.93 (95% CI: 0.900–0.910). These findings suggest that the proposed model is reliably capable of distinguishing between MRI images positive and negative for tumors in pediatric patients, even at an early stage of development.

These results suggest that it is feasible to achieve robust and generalizable performance in tumor classification and segmentation models applied to pediatric MRI using deep learning techniques.

Segmentation and Qualitative Analysis

Once classification was completed, segmentation was performed on 20 tumor-positive cases using manually generated masks validated by an experienced pediatric neuroradiologist. Three anatomical regions were delineated: (i) active tumor tissue, (ii) surrounding tissue including edema and necrotic areas, and (iii) adjacent healthy brain tissue.

For each region, descriptive metrics such as volume and signal intensity were extracted from the following sequences: diffusion, *T1* and *T2* relaxometry, as well as diffusion-weighted images. The table below shows the results obtained from a single patient:

Table 1.- Volume and Signal Intensity values vs SD of a patient.

Zone	**Volume (mm^3)**	**Signal Intensity (u.a.) vs SD diffusion**	**Signal Intensity (u.a.) vs SD *T1***	**Signal Intensity (u.a.) vs SD *T2***
Tumor	248	1990.4±405.9	40.5±45.2	805.05±148
Necrotic	0	0	0	0
Edema	25.6	974±753	108.1±104.3	526.3±339.9
Healthy tissue	2.017e+04	989.96±481.4	200.5±42.5	400.6±121.3

4 Discussion

The first stage of this research involved the classification of magnetic resonance imaging (MRI) scans from pediatric patients using a Convolutional Neural Network (CNN) based on *T1* relaxometry images. The model successfully identified 20 patients with active tumors and 40 patients with no evidence of tumors, achieving high accuracy in both the training and validation sets.

In parallel, progress was made in the manual segmentation of images from patients diagnosed with brain tumors. This stage is crucial, as it provides the masks required for the pretraining of the future automatic segmentation network. Unlike classification, segmentation aims to distinguish three specific brain tissue regions (active tumor, edema/necrosis, and healthy tissue), using not only T1 images but also T2 and diffusion sequences. The use of diffusion imaging—still not widely described in the literature for this purpose—represents an important innovation in this project and is expected to contribute to more accurate segmentation. The use of MRI images combined with *T1*, *T2*, and diffusion relaxometry techniques enabled a detailed analysis of brain pathologies in pediatric patients with tumors. These techniques were essential for accurately identifying tumor regions, clearly distinguishing affected tissue from surrounding healthy tissue, and improving the quality of brain structure segmentation. In addition, quantitative values of volume, signal intensity, and standard deviations were obtained across different regions of interest: active tumor, necrotic tissue, edema, and adjacent healthy tissue—providing a more comprehensive characterization of the tumor environment.

Qualitative Observations Revealed Consistent Patterns Across Cases:

Tumor regions exhibited elevated signal intensities on diffusion-weighted imaging, consistent with the high cellularity and restricted diffusion typically observed in malignant tissues.

Edematous areas demonstrated heterogeneous signal characteristics, particularly on *T2*-weighted sequences, reflecting variable water content and tissue response in the peritumoral region. Healthy brain tissue showed lower and more uniform signal intensities, with reduced standard deviation across imaging sequences, serving as a stable internal reference. Necrotic regions, when present, predominantly displayed low or absent signal, indicative of the lack of viable tissue structure. These qualitative differences substantiate the clinical relevance of multimodal MRI in tumor characterization and provide a foundation for the future refinement of segmentation models. Although statistical analyses were not conducted due to constraints in sample size, the reproducibility of these patterns across multiple patients suggests their potential utility as reliable features for automated tissue classification in subsequent implementations.

Limitations of the study:

This study presents several limitations that should be considered when interpreting the obtained results. Firstly, although approximately 4,200 magnetic resonance images were collected, only 140 images were used for the initial training of the classification model. The small sample size, especially in the training set, constitutes a limitation that may affect the generalizability of the results. This restriction was due to the inclusion of only new patients without any intervention, meaning those who had neither undergone surgical resection nor received any treatment. Future research with larger samples will help validate and extend these findings.

5 Conclusions

The developed convolutional neural network can be useful in improving accessibility and enabling early detection of tumors using magnetic resonance imaging techniques. With a larger sample set, the neural network’s processing will become more effective and help reduce false positives in tumor diagnosis.

Additionally, the network can provide insights into the patient’s brain before and after treatment if a tumor is detected.

Manual segmentation is being performed using the 3D Slicer program to analyze images of patients with tumors and to identify the three regions that need to be analyzed. Once the images of tumor patients are segmented, another network will be created using the manually pre-trained data.

The importance of manual segmentation lies in identifying areas of interest to determine which regions of the brain are affected in pediatric patients with tumors, which can be crucial for early diagnosis and effective treatment.

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Comparative Monitoring of Lens, Thyroid, and Whole Body in Interventional Cardiology Procedures

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Abstract

The International Commission on Radiological Protection (ICRP) recommended a substantial reduction in the equivalent dose limit for the eye lens, following the revised threshold for radiation-induced cataracts. In response, new regulatory guidelines were issued for the radiological protection of the lens in planned exposures, which were adopted immediately by the Brazilian National Nuclear Energy Commission (CNEN). Several studies have investigated alternative methods to estimate lens dose. Some approaches rely on correlations between lens dose and readings from dosimeters placed on other body sites. However, patient anatomy, positioning, and the clinical team's experience can influence interventional practice and impact these estimates. This study aimed to compare individual monitoring procedures using the EYE-D™ holder with MCP-N thermoluminescent detectors (LiF: Mg, Cu, P) for measuring the personal dose equivalent Hp(3) at the lens, and the electronic dosimeter PM1630 for measuring Hp(10**)** at the thyroid and whole body in interventional cardiology staff. A total of 22 procedures were analyzed, involving physicians and nurses, with dosimeters placed on the thorax, thyroid, and eye lens. For physicians, the mean doses per procedure were 160.8 μSv to the thorax, 110.8 μSv to the thyroid, and 116.7 μSv to the eye lens. The mean ratios between lens dose and those measured at the thyroid and thorax were 1.54 and 1.70, respectively. For nursing staff, the mean per-procedure doses were 11.4 μSv at the thorax and 6.8 μSv at the lens. This is a preliminary study, still in progress, with plans to expand the sample size and perform a stratified analysis by professional role and procedure type.

Keywords: Interventional Cardiology; Occupational dosimetry; Lens dose; Interventional procedures.

1 Introduction

With advances in medical and imaging technologies, minimally invasive procedures have become central to the diagnosis and treatment of numerous diseases. In this context, interventional radiology stands out by offering clinical effectiveness with reduced invasiveness, leading to shorter hospital stays and lower healthcare costs [2,6]. Within this field, interventional cardiology has emerged as a leading subspecialty, employing cutting-edge techniques for diagnostic and therapeutic interventions in cardiovascular diseases, the leading cause of mortality in Brazil, accounting for approximately 28–30% of annual deaths according to data from the Global Burden of Disease and the Ministry of Health [5,8].

In Brazil, this recommendation was incorporated into the National Nuclear Energy Commission (CNEN) Standard NN 3.01, which requires monitoring of the personal dose equivalent to the eye lens (Hp(3)) for workers exposed to ionizing radiation. However, clinical practice still faces challenges, as conventional dosimetry based on Hp(10) at the thorax or Hp(0.07) at the neck does not accurately reflect lens dose. This underscores the need for specific devices calibrated for Hp(3) [6].

Although protective measures such as lead glasses and suspended shields are commonly used, their effectiveness varies depending on procedure type, staff positioning, room configuration, and adherence to proper use. Therefore, direct monitoring of eye lens dose remains both a practical and technical challenge in interventional environments.

Against this background, the present study proposes comparative monitoring of personal dose equivalents in different anatomical regions, the lens (Hp(3)), thyroid, and whole body (Hp(10)) during interventional cardiology procedures. Using thermoluminescent and electronic dosimeters, the study aims to quantify occupational exposure among healthcare professionals and to investigate possible correlations between measured doses, thereby contributing to the evaluation of conventional monitoring practices and protection strategies in light of current occupational radiation protection recommendations.

2 Materials and Methods

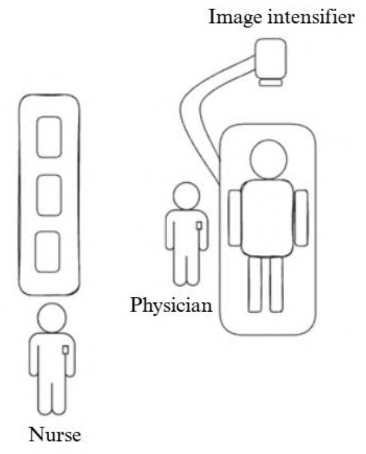
This study aimed to evaluate occupational exposure to ionizing radiation among healthcare professionals involved in interventional cardiology procedures. The research was carried out at a hospital in Belo Horizonte, Minas Gerais, Brazil, with prior authorization from the responsible department, and spanned two weeks. During this time, 22 procedures were monitored by the unit's routine and team availability. Participants included interventional physicians and nursing staff directly involved in the procedures, who were selected based on their potential for radiation exposure. The study sought to quantify, characterize, and compare absorbed doses according to professional role and anatomical region.

2.1 Occupational Exposure Monitoring

Occupational exposure was measured simultaneously at four anatomical sites: thorax, thyroid, and both eyes. For monitoring the equivalent dose to the eye lens, EYE-D™ holders equipped with MCP-N thermoluminescent detectors (LiF: Mg, Cu, P) were positioned laterally at the level of the professionals' eyes. Dose assessment at the thorax and thyroid was performed using PM1630 electronic dosimeters, placed externally on the lead apron. The thorax dosimeter was positioned on the anterior thorax, while the thyroid dosimeter was attached at the base of the neck; both were applied exclusively to physicians. For nursing staff, monitoring was limited to the thorax region.

Team positioning followed the standard configuration commonly observed during procedures: the physician stood to the right of the patient's head, facing the left side of the X-ray tube, while the nurse remained near the base of the table to assist with materials. The image intensifier was positioned above the patient and adjusted according to the requirements of each examination. Figure 1 illustrates the spatial arrangement of the room and staff during the procedures.

Figure 1 – Schematic representation of the room during procedures, indicating the positioning of the team throughout the procedures.



Source: The author, 2025.

All doses recorded during the procedures were collected individually for each monitored professional and subsequently organized into electronic spreadsheets. These spreadsheets included information on professional category, procedure type, personal dose equivalents Hp(3) and Hp(10) measured at different anatomical sites, and X-ray equipment parameters such as fluoroscopy time, cumulative air Kerma, and dose–area product (DAP), which may be used for complementary analyses.

2.2 Statistical Analysis

The collected data were organized and processed using RStudio® (version 2024.06.1), with R employed for data manipulation and graph generation. Descriptive statistics, including mean, median, and standard deviation, were first applied to the personal dose equivalents recorded at different anatomical regions (thorax, thyroid, and eye lens), analyzed separately for physicians and nurses, and stratified by procedure type (catheterization or angioplasty). Results were summarized in tables and complemented by graphical representations to facilitate visualization of exposure patterns across professional groups.

To evaluate the feasibility of estimating eye lens dose from other anatomical sites, ratios were calculated between doses measured at the eyes and those obtained at the thorax and thyroid, considering the right and left eyes separately. This analysis allowed the assessment of proportionalities that may serve as indirect estimates of ocular exposure. To examine the association between fluoroscopy time and recorded doses, Spearman's rank correlation test was applied, chosen because it does not assume normality and is more robust in the presence of outliers, which were identified during data processing. Correlation coefficients (ρ) were calculated for each anatomical region and professional group, with p-values < 0.05 considered statistically significant.

3 Results

3.1 Analysis of Occupational Doses by Anatomical Region

Equivalent doses recorded by dosimeters placed on the thorax, thyroid, and both eyes of healthcare professionals during interventional cardiology procedures were analyzed. Tables 1 and 2 present descriptive statistics of these doses, stratified by procedure type and professional category. The category "All" refers to the combined analysis of both procedure types, providing an overall view of exposure by anatomical region.

During the data collection period, 22 procedures were monitored by room access authorization and team availability. It was noted, however, that physicians typically perform up to 10 procedures per day on average, highlighting the importance of considering cumulative exposure over the month. Since the values presented in the tables correspond to the average dose per procedure, the actual monthly dose for these professionals may be considerably higher, particularly in high-demand settings.

Table 1: Recorded doses for physicians.

Professional	Procedure	Anatomical Region	Mean (µSv)	Median (µSv)	Standard Deviation (µSv)
Physicians	All	Thorax	160.8	54.8	459.3
		Thyroid	11.8	33.8	291.7
		Eye lens	116.7	55.0	242.5
	Catheterization	Thorax	217.3	52.7	599.0
		Thyroid	140.72	35.2	378.6
		Eye lens	136.18	53.0	313.0
	Angioplasty	Thorax	79.2	66.3	54.0
		Thyroid	67.5	32.5	69.3
		Eye lens	88.57	88.8	76.7

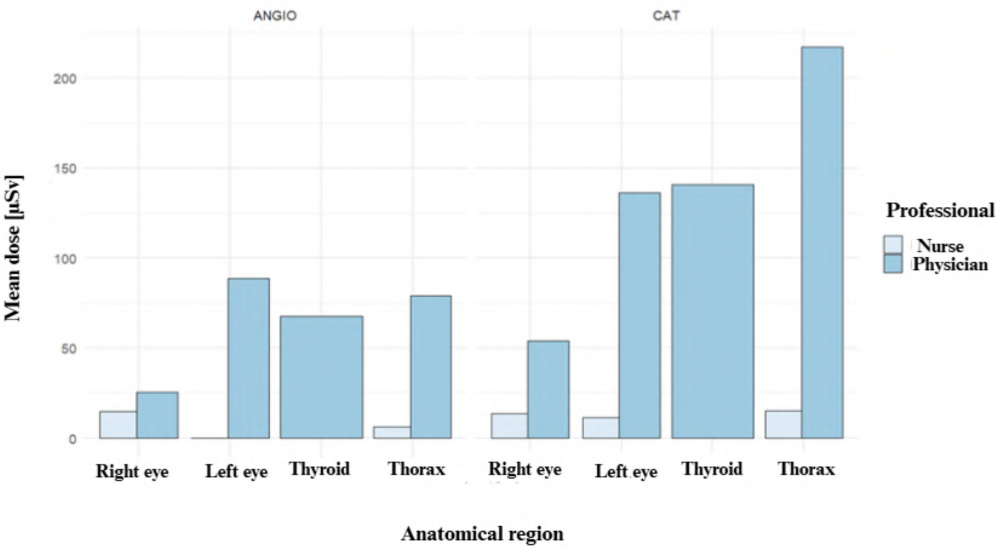
Table 2: Recorded doses for the nursing staff.

Professional	Procedure	Anatomical Region	Mean (µSv)	Median (µSv)	Standard Deviation (µSv)
Nurse	All	Thorax	11.44	1.9	31.2
		Left eye lens	6.8	0.0	25.5
		Right eye lens	14.3	0.0	37.2
	Catheterization	Thorax	15.02	1.73	40.51
		Left eye lens	11.63	0.0	32.89
		Right eye lens	13.84	0.0	42.94
	Angioplasty	Thorax	6.27	2.24	6.85
		Left eye lens	0.0	0.0	0.0
		Right eye lens	14.77	0.0	29.45

The data reveal notable differences in radiation exposure between procedure types and professional categories. Overall, physicians received substantially higher average doses than nursing staff, particularly during catheterization procedures, which exhibited the highest values across all analyzed anatomical regions. For instance, the mean dose to the thorax for physicians was 160.8 µSv, compared to 11.4 µSv for nurses. For the left eye lens, physicians had an average dose of 116.7 µSv, over 17 times higher than that measured for nursing staff (6.8 µSv). A finding of particular interest was the asymmetry in ocular doses: among physicians, the highest dose was recorded in the left eye, reflecting the typical C-arm positioning and predominant direction of scattered radiation. In contrast, for nursing staff, the right eye received a higher dose, likely related to the position occupied during procedures (Figure 1), with the left side being relatively more shielded and the right side more exposed.

Furthermore, the analysis revealed very low or even zero median doses for nursing staff, indicating minimal occupational exposure in many procedures. This may be explained by greater distance from the radiation source, shorter time spent in the procedure room, or effective use of protective barriers. In contrast, physicians exhibited high variability in recorded doses, as reflected by large standard deviations, suggesting that factors such as fluoroscopy time, procedure complexity, and positioning relative to the X-ray tube substantially influence exposure. To complement this analysis, Figure 2 shows the distribution of doses by anatomical region, procedure type, and professional category, corroborating the patterns observed in the previously presented data.

Figure 2: Distribution of the average doses recorded by anatomical region, procedure type, and professional category.



Source: The author, 2025.

3.2 Ratios between doses in the lens and other regions

To evaluate the feasibility of estimating eye lens dose based on measurements from other anatomical sites, ratios were calculated between doses recorded at the eyes and those obtained from dosimeters positioned on the thorax and thyroid. Table 3 presents the mean, median, and standard deviation of these ratios for physicians and nurses.

Table 3: Ratios between ocular and body doses by professional category

Professional	Ratio	Mean	Median	Standard Deviation
Physician	Right eye lens/ thorax	0.77	0.05	1.85
	Left eye lens/ thorax	1.07	0.94	1.04
	Right eye lens/ thyroid	0.97	0.09	2.13
	Left eye lens/ thyroid	1.54	1.27	1.43
Nurse	Right eye lens/ thorax	10.73	0.0	46.38
	Left eye lens/ thorax	2.20	0.0	6.78

Source: The author, 2025.

Among physicians, the ratio between the dose to the left eye lens and the dose to the thorax showed the most consistent values, with a mean of 1.07 and a median of 0.94, suggesting a possible proportionality between these regions. This finding indicates that, under this configuration, doses measured on the thorax may, in some cases, reflect the magnitude of exposure to the left eye lens. However, the other ratios for both physicians and nursing staff displayed greater variability, as indicated by high standard deviations and discrepancies between mean and median, highlighting a current limitation in the practical application of these ratios for estimating ocular exposure.

3.2 Association Between Fluoroscopy Time and Dose in Different Anatomical Regions

In addition to the analysis of absolute doses, the correlation between fluoroscopy time and doses recorded in each anatomical region was evaluated using Spearman's rank correlation test, due to the non-parametric nature of the data and the presence of outliers. The results indicate a general trend of increasing doses with longer fluoroscopy times, although the strength of these correlations varied depending on the anatomical region and professional category.

Table 4 presents the Spearman correlation coefficients (ρ) between fluoroscopy time and doses measured at the thorax, thyroid, and eye lens (right and left), along with the corresponding p-values. The highest and statistically significant correlations ($p < 0.05$) were observed for the physicians' eyes ($\rho = 0.43$ for the right eye lens and $\rho = 0.47$ for the left eye lens) and for the nurses' thorax ($\rho = 0.45$), indicating moderate and significant associations under these conditions.

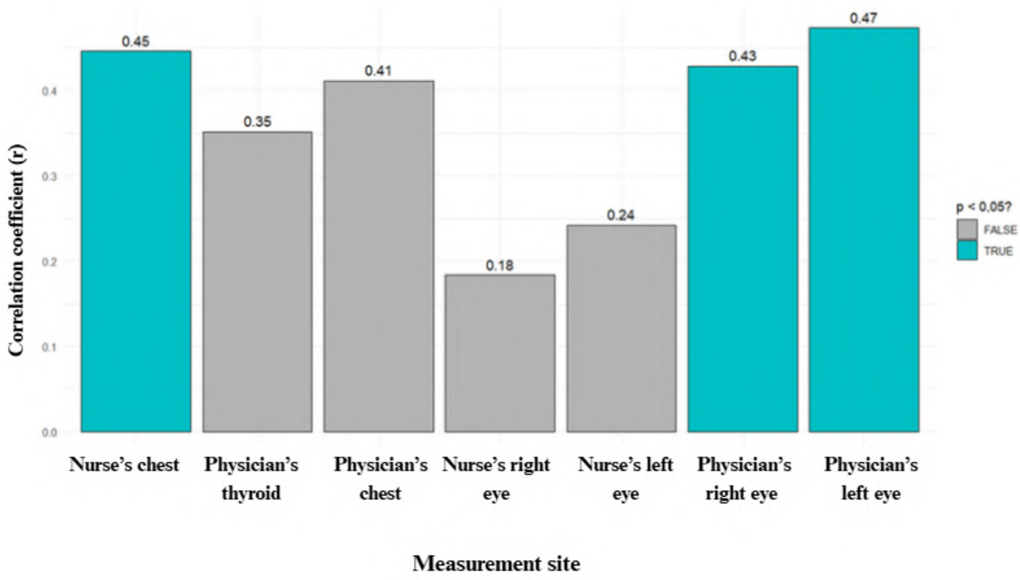
Table 4: Spearman correlation coefficients (ρ) between fluoroscopy time and occupational doses, by anatomical region and professional category.

Professional	**Anatomical Region**	**Spearman's ρ**	***p*-value**	**Interpretation**
Physician	Hp(10) thorax	0.412	0.057	Moderate, quasi-significant
	Hp(10) thyroid	0.352	0.108	Moderate, not significant
	Hp(3) right eye lens	0.429	0.047	Moderate, significant
	Hp(3) left eye lens	0.474	0.026	Moderate to strong, significant
Nurse	Hp(10) thorax	0.446	0.037	Moderate, significant
	Hp(3) left eye lens	0.242	0.277	weak, not significant
	Hp(3) right eye lens	0.184	0.413	weak, not significant

Source: The author, 2025.

Figure 3 provides a visual summary of the correlation coefficients, facilitating comparison across different measurement sites. Bars highlighted in blue indicate statistically significant results ($p < 0.05$), emphasizing the observed correlations between fluoroscopy time and eye lens doses in physicians, as well as thorax doses in nurses.

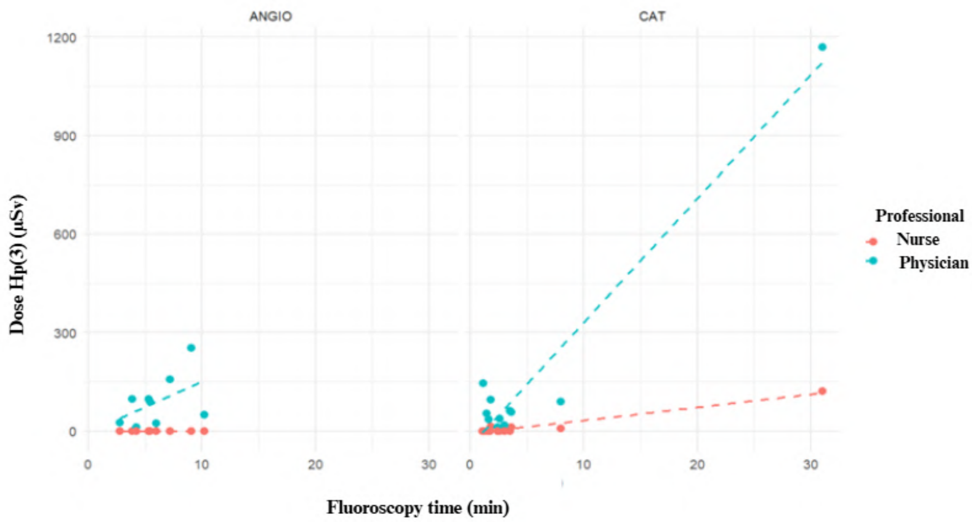
Figure 3: Correlation between fluoroscopy time and occupational dose.



Source: The author, 2025.

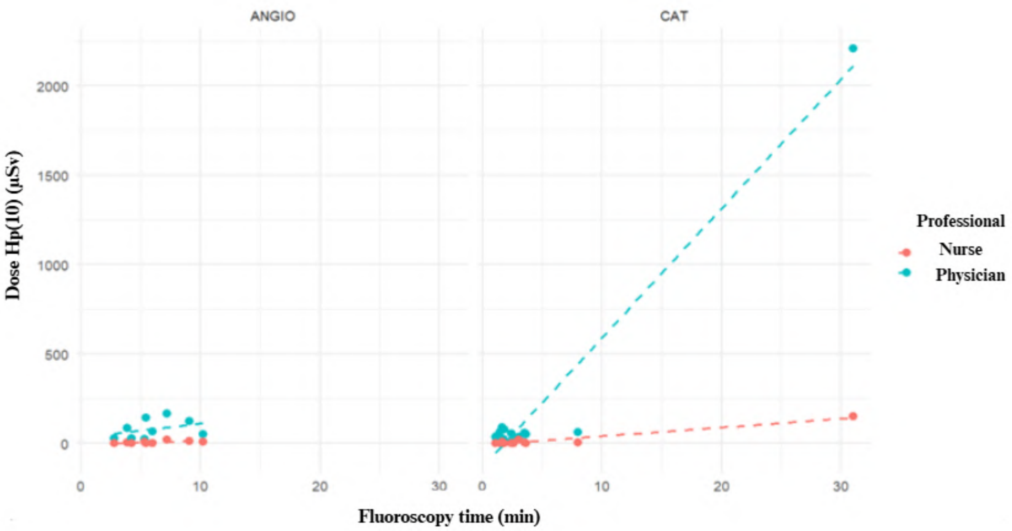
Additionally, Figures 4a and 4b present scatter plots designed to provide a clearer visualization of the distribution patterns of the collected data, highlighting the relationship between fluoroscopy time and doses recorded in the different anatomical regions.

Figure 4a: Scatter plot of fluoroscopy time versus dose in the left eye lens.



Source: The author, 2025.

Figure 4b: Scatter plot of fluoroscopy time versus dose in the thorax.



Source: The author, 2025.

4 Conclusions

This study enabled a comparative assessment of occupational radiation exposure across different anatomical regions in physicians and nursing staff during interventional cardiology procedures, with a focus on catheterization and angioplasty. The results demonstrated marked differences between professional categories, with physicians consistently exhibiting higher average doses, particularly during catheterization procedures, which were associated with greater exposure. The highest dose was recorded in the left eye lens of physicians, reflecting the typical positioning of the primary operator on the patient's left side, close to the C-arm. In contrast, nursing staff presented substantially lower dose levels, with very low or even zero median values in some regions, indicating reduced exposure, likely due to greater distance from the radiation source and more protected positioning. Nevertheless, an asymmetry in ocular dose was also observed among nurses, with slightly higher doses recorded in the right eye lens.

Analysis by anatomical region revealed that, for both professional groups, the thorax was the area of highest exposure, followed by the thyroid and the eyes. Spearman's correlations indicated a significant association between fluoroscopy time and cumulative doses to the eyes in physicians, suggesting that longer exposure times lead to higher doses to this radiosensitive region. Among nursing staff, most correlations were weak and non-significant, reinforcing the lower exposure levels observed in this group.

Despite the relevant contributions of this study, several limitations should be acknowledged, particularly the small sample size, which limits the generalizability of the findings. Therefore, future studies are expected to expand data collection, incorporating additional technical parameters of the equipment, as well as patient characteristics such as body mass index (BMI), which may directly influence the generation of scattered radiation and, consequently, occupational doses.

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